Cryocooler Fundamentals and Space Applications

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Sponsored by
The Cryogenic Society of America

Short Course
Cryogenic Engineering Conference
Tucson, Arizona
June 28, 2015

Lecturers:
Ray Radebaugh, NIST
R. G. Ross, Jr., JPL

Cryocooler Fundamentals and Space Applications

Contents

Morning (8:00 – 12:00): One 15 min. break

Session 1: Information sources, definitions, history, applications, thermodynamics, cryocooler types, Joule-Thomson (JT) coolers, Brayton cryocoolers, Claude systems, heat exchangers

Session 2: Regenerative cycles, Stirling cryocoolers, Gifford-McMahon (GM) cryocoolers, 4 K regenerators, pulse tube cryocoolers, types, examples, modeling, cryocooler comparisons, new research

Afternoon (1:00 – 5:00): One 15 min. break

Session 3: Space cryocoolers, history, applications, long life, Stirling coolers, pulse tube coolers, JT systems, Brayton coolers, system design, qualification, vacuum, conduction and radiation thermal loads
Ron Ross

Session 4: Performance requirements, thermal measurements, orientation, generated vibration and supression, launch vibration, EMI, AIRS example, sizing, temperature stability, integration
Ron Ross

Course Goals

- Provide information to a variety of students
- Learn about various types of cryocoolers
- Learn about space applications of cryocoolers
- Provide information resources for future use
- Answer questions

Outline

Session 1

- Information sources
- Definitions
- History
- Applications and problems
- Thermodynamics
- Cryocooler types and operating principles
- Joule-Thomson (JT) cryocoolers
- Brayton cryocoolers
- Claude systems
- Heat exchangers

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Cryogenic Information, p. 1

- Government Web Sites
  - –NIST Cryogenic Technologies Group
    - •www.cryogenics.nist.gov
    - •Material properties, fluid properties, copies of papers
  - –Los Alamos National Lab. Thermoacoustics
    - •www.lanl.gov/projects/thermoacoustics
    - •DeltaE thermoacoustics computer model
  - –NASA
    - •http://www2.jpl.nasa.gov/adv_tech/
    - •http://cryo.gsfc.nasa.gov/index.html

Cryogenic Information, p. 2

- Non-Profit Technical Societies
  - Cryogenic Society of America (www.cryogenicsociety.org)
    - Cold Facts Newsletter (4 times/year)
    - Sponsors short courses and workshops
  - Superconductors.org (http://superconductors.org)
    - Superconductor information and news for the beginner
    - Links to other superconductor information
    - Beware of high T_c claims by author
  - ASME, AChE, AIAA, SPIE, IEEE

Cryogenic Information, p. 3

- Cryogenic Fluid Properties
  - NIST Scientific Databases (www.nist.gov/srd)
    - REFPROP, fluids & mix: NIST 23 (85 fluids,75 prop. $200)
    - Steam: NIST 10, ASME and IAPWS (25 prop., $50)
    - NIST Chemistry Webbook (34 fluids, 12 prop., free online)
  - Cryodata (www.cryodata.com)
    - Pure fluids, GASPACK(36 fluids, 28 properties, $595)
    - Helium, HEPAK (0.8 to 5000 K, 33 properties, $595)
    - Water and steam, STEAMPAK (273 to 2000 K, $595)

Cryogenic Information, p. 4

- Cryogenic Material Properties
  - NIST Cryogenic Technologies Group
    - www.cryogenics.nist.gov
    - Material properties equations on web, >80 materials (1-300 K)
  - Cryodata (www.cryodata.com)
    - METALPAK, 14 metals, 4 properties, $550
    - CPPAK & EXPAK, 21 & 28 additional mat., $325 each
    - CRYOCOMP, 61 materials, 7 prop., $990

Cryogenic Information, p. 5

- Material Properties (cont.)
  - JAHM Software (www.jahm.com)
    - MPDB, 1000 materials, 19 properties, $495
    - Temperature dependent, including much cryogenic
  - MatWeb (www.matweb.com)
    - 24,000 materials, 10-20 room temp. prop., free online
  - Electro Optical Industries, Inc. (www.electro-optical.com)
    - Emissivities of many materials at room temp., free online

Cryogenic Information, p. 6

- Cryocooler Publications
  - Journals
    - Cryogenics
    - Cryogenic Engineering (Japan) (English abstract & figures)
    - Review of Scientific Instruments
  - Conference Proceedings
    - International Cryocooler Conference (ICC); 2 yrs., even
    - Cryogenic Engineering Conference (CEC); 2 yrs., odd
    - International Cryogenic Engineering Conference (ICEC)
    - International Cryogenic Materials Conference (ICMC)
    - International Institute of Refrigeration (IIR)
    - International Society for Optical Engineering (SPIE)
    - Applied Superconductivity Conference (ASC); 2 yrs., even
    - American Society of Mechanical Engineers (ASME trans.)

Cryogenic Information, p. 7

- Suggested Reading
  - Books
    - G. Walker, *Cryocoolers*, Plenum Press, NY (1983)
    - G. Walker and E. R. Bingham, *Low-Capacity Cryogenic Refrigeration*, Oxford Press (1994)
    - R. Barron, Cryog*enic Systems*, Oxford Press, (1985)
    - T. Flynn, *Cryogenic Engineering*, Marcel Dekker, NY (1997)
    - J. Ekin, *Experimental Techniques for Low Temperature Measurements*, Oxford Press, (October, 2006)
    - J. G. Weisend II, ed., *Handbook of Cryogenic Engineering*, Taylor & Francis, Philadelphia (1998)
    - Bernd Seeber, ed., *Handbook of Applied Superconductivity,* Institute of Physics Publishing, Bristol (1998)

Cryogenic Information, p. 8

- Suggested Reading (continued)
  - Books (continued)
    - M. Elwenspoek and H. Hansen,*Silicon Micromachining*, Cambridge Univ. Press (1998)
    - N. Mulaf and K. Williams, *An Introduction to Micromechanical Systems Engineering*, Artech House (2004)
    - W. M. Kays and A. L. London, *Compact Heat Exchangers, 3rd ed.*, McGraw-Hill (1984)
    - Neeraj Khare, *Handbook of High-Temperature Superconductor Electronics*, Marcel Dekker (2003)
    - S. Kakac, H. F. Smirnov, and M. R. Avelino, *Low Temperature and Cryogenic Refrigeration*, NATO Science Series, Kluwer Academic Publishers (2003)

Cryogenic Information, p. 9

- Suggested Reading (continued)
  - Articles
    - R. Radebaugh, “Cryocoolers: the state of the art and recent developments,” J. Phys.: Condens. Matter **21** (2009), 164219 (9pp)
    - A. Ravex, “Small Cryocoolers,” *Handbook of Applied Superconductivity,* Vol. 1, Inst. of Phys. (1998) pp. 721-746
    - R. Radebaugh, “Cryocoolers and High-T_c Devices,” *Handbook of High-Temperature Superconductor Electronics*, Marcel Dekker, New York (2003), pp. 379-424
    - R. Radebaugh, “Pulse Tube Cryocoolers,” *Low Temperature and Cryogenic Refrigeration*, NATO Science Series, Kluwer Academic Publishers (2003), pp. 415-434
    - R. Radebaugh, “Status and Recent Trends in Cryocooler Research,” *Cryogenics and Refrigeration* (Proc. ICCR 2008), Science Press, Beijing, (2008), pp. 11-22
    - R. Radebaugh, “Refrigeration for Superconductors,” *Proc. IEEE, Special Issue on Applications of Superconductivity*, Vol. 92 (2004), pp. 1719-1734

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Definitions

- **CRYOGENICS:** Science and technology of very low temperatures
- **CRYOGENIC TEMPERATURES:** Temperatures less than 120 K (common usage)
- **TEMPERATURE SCALES:**

Increment:

1 K = 1°C = 1.8°R = 1.8°F

Kelvin (K)	Celsius (°C)	Rankine (°R)	Fahrenheit (°F)
0	-273.15	0	-459.67
273.15	0	491.67	32
373.15	100	671.67	212

Cryogenic Fluids

- **CRYOGEN:** Fluid with normal boiling point < 120 K

Cryogen	Boiling point (K)	Triple point (K)
Krypton, Kr	119.78	115.77
Methane, CH_4	111.67	90.69
Oxygen, O_2	90.19	54.36
Argon, Ar	87.30	83.81
Fluorine, F	85.04	53.48
Carbon monoxide, CO	81.63	68.13
Air, $0.76N_2+0.23O_2+0.01Ar$	78.9 – 81.7	59.75
Nitrogen, N_2	77.36	63.15
Neon, Ne	27.10	24.56
Hydrogen (normal), H_2	20.39	13.96
Hydrogen (para), H_2	20.28	13.80
Helium-4, He^4	4.230	-
Helium-3, He^3	3.191	-

Source: NIST, REFPROP

Cryocoolers: Important Parameters

- Net refrigeration power, $\dot{Q}_c$: $\dot{Q}_c = dQ/dt$
- Power input, $\dot{W}$: $\dot{W} = dW/dt$
- Actual Coefficient of Performance, COP :

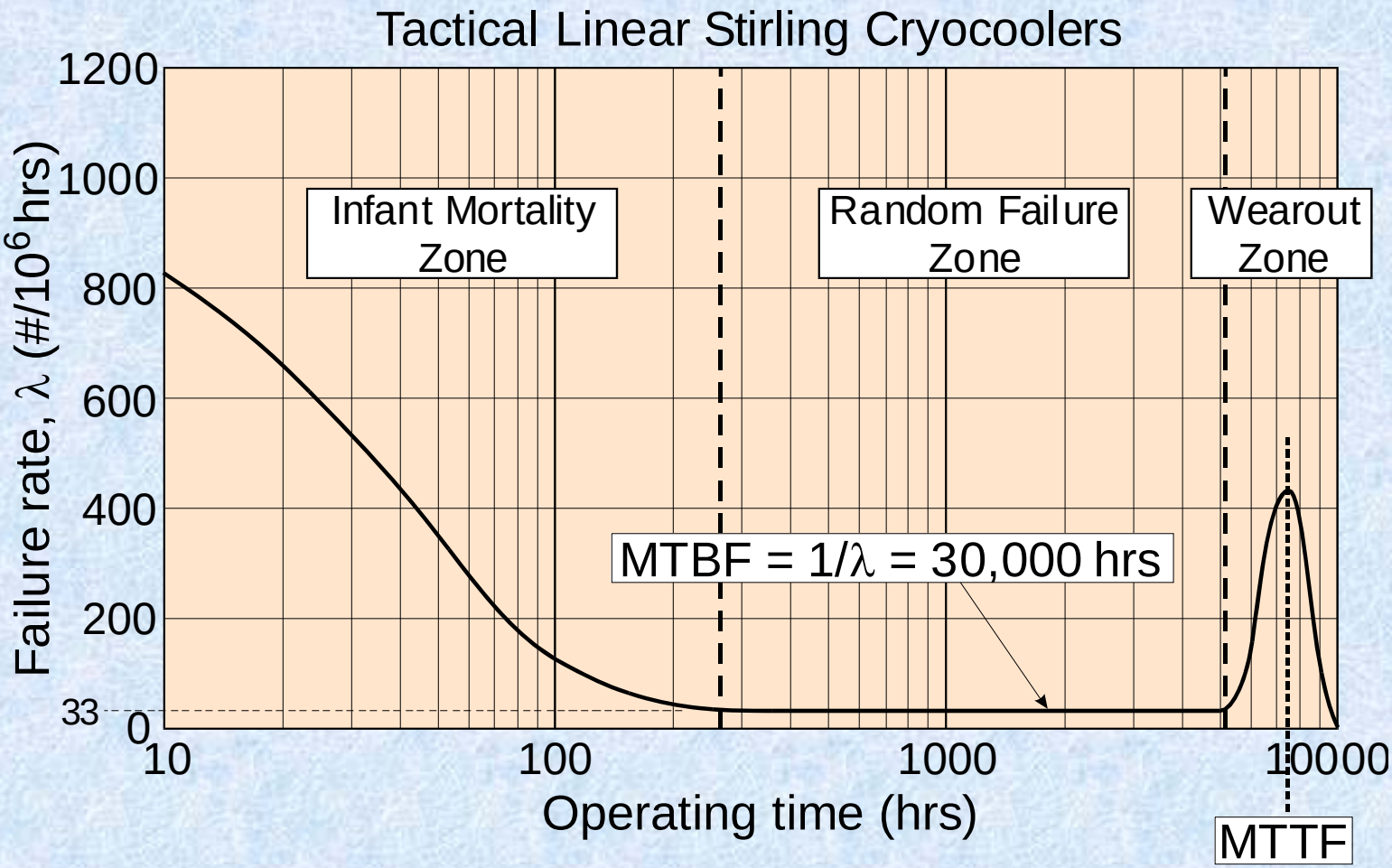
$$COP = \dot{Q}_c / \dot{W}$$

- Ideal (Carnot) COP : $COP_{Carnot} = T_c/(T_h - T_c)$
- Efficiency, η : $\eta = COP/COP_{Carnot}$
- Specific power, P_s : $P_s = 1/COP$

Important Parameters, p. 2

- Mass
- Volume
- Cool down time
- Vibration
- Acoustic noise
- Electromagnetic inerference (EMI)
- Operating life
  - Mean time between maintenance (MTBM)
  - Mean time between failure (MTBF)
  - Mean time to failure (MTTF)
- Reliability
- Shelf life
- Split or integral systems
- Cost
- Ambient cooling method
- Vibration tolerance
- Orientation
- Low/Zero gravity environment

Typical Failure Rate Curve



Hughes data

Failurec.cdr

Outline

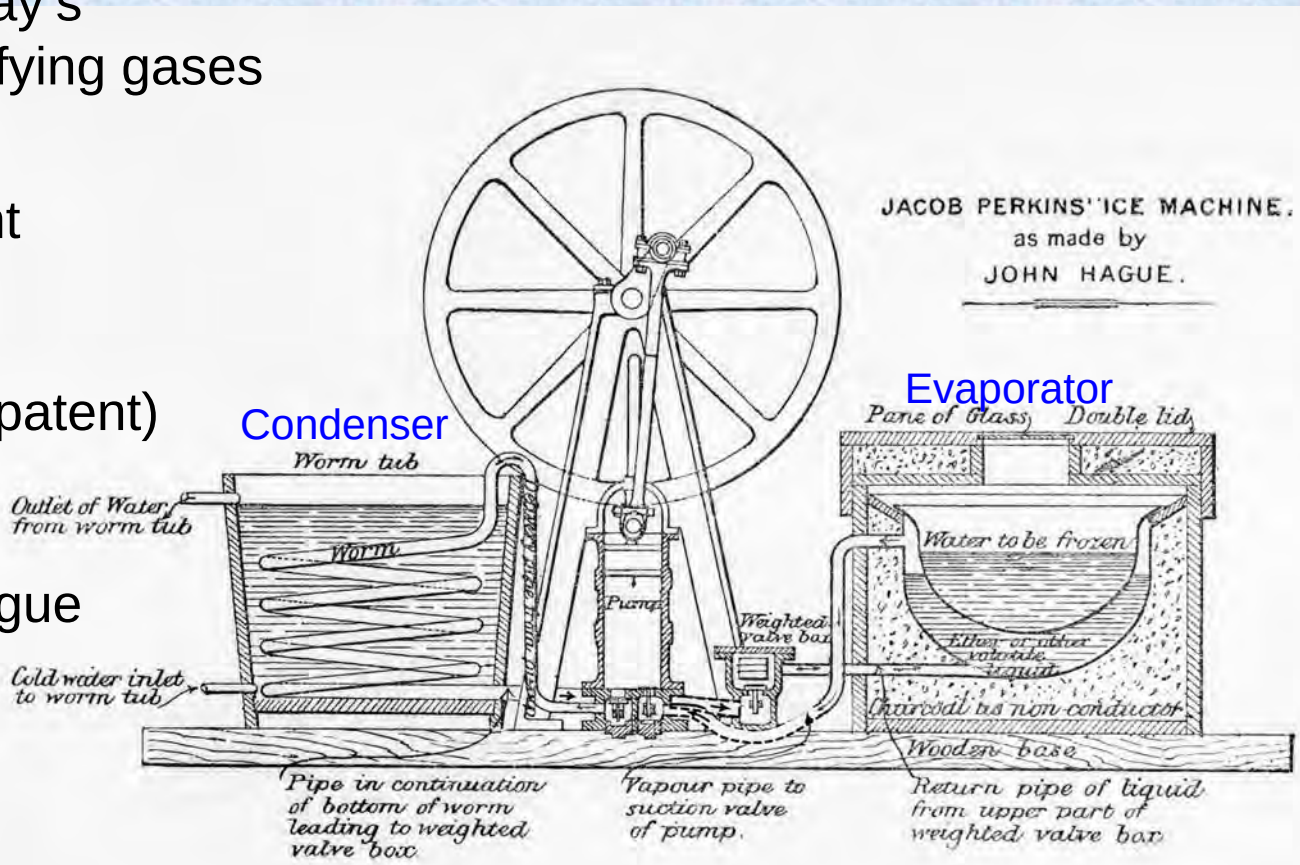
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First Ice Machine (1834)
Jacob Perkins

- Automated Faraday's technique of liquefying gases
- 1834 British patent by Jacob Perkins
- Ether refrigerant (patent) +35 °C NBP
- Made by John Hague in 1834

Vapor-compression cycle



Drawing courtesy: Science Museum/Science and Society Picture Library, London

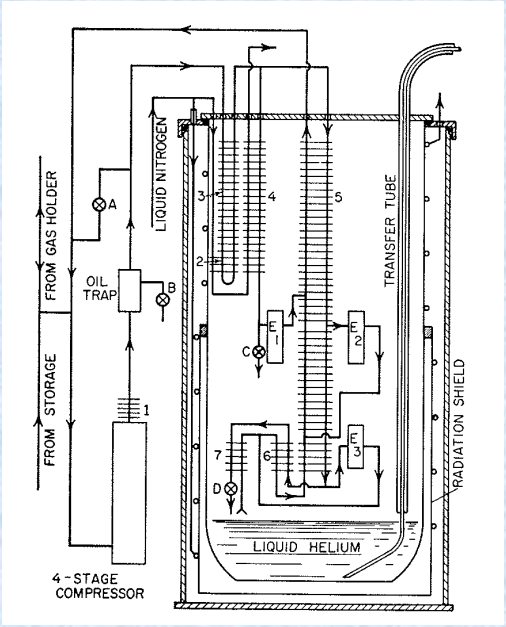
Helium Liquefaction (1950s)



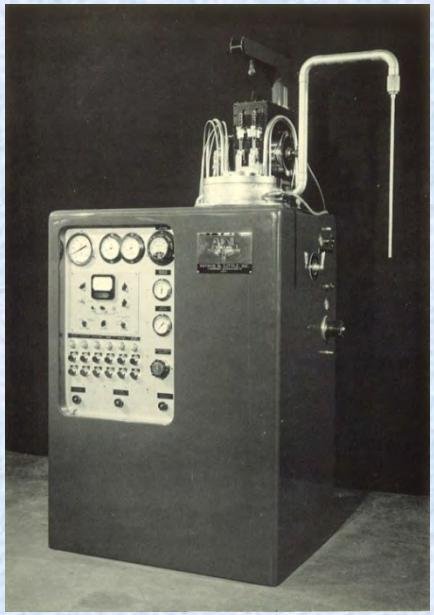
Collins2-1946.jpg

Collins helium liquefier

- Invented 1946; commercial 1947
- 4 L/h (original)
- 2 Expansion engines (Kapitza)
- 400 units worldwide (2005)
- Allowed 4 K research worldwide



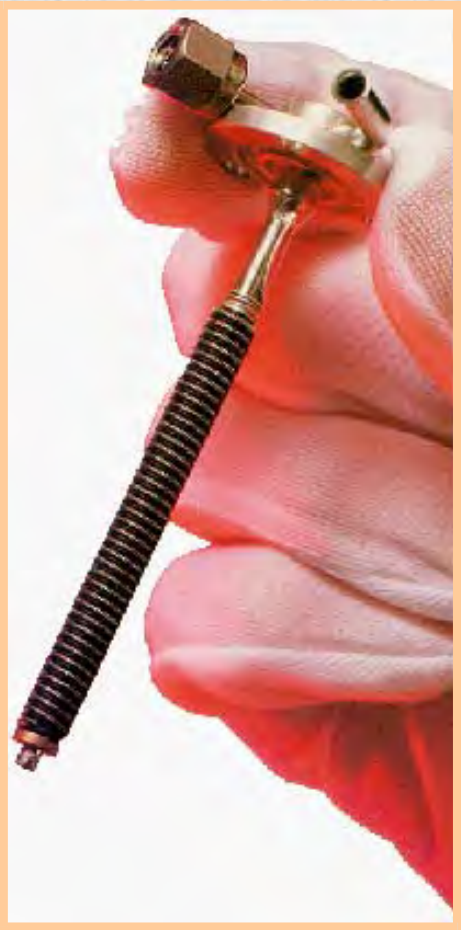
Collins He liq schematic.tif



Collins He liquefier-ADL.jpg

Miniature Joule-Thomson Cryocoolers
Open System





- Used for direct cooling of infrared sensors
- First direct coupling of cryocooler to device (late 1950s)

Air Products and APD Cryogenics

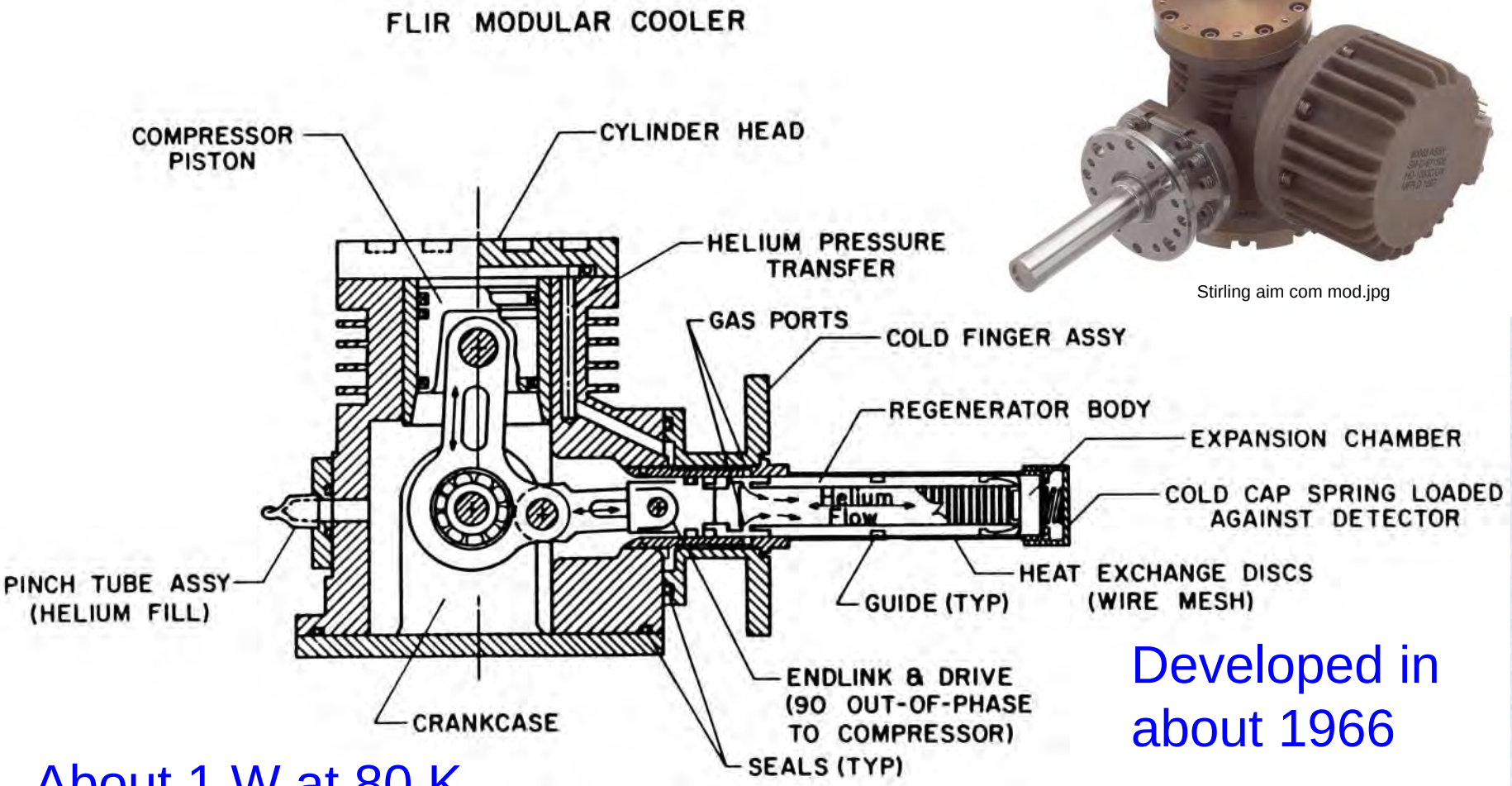
miniaturejt.cdr

First Coupling of Cryocooler to Application

Stirling Cryocooler for Cooling Infrared Detector



Stirling aim com mod.jpg



Developed in about 1966

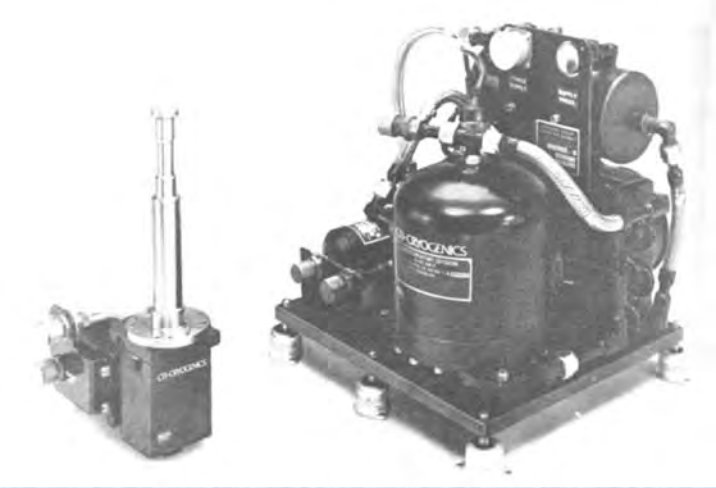
About 1 W at 80 K

Courtesy: G. Walker, *Cryocoolers*, Plenum (1983)

FLIRmodular-cooler.jpg

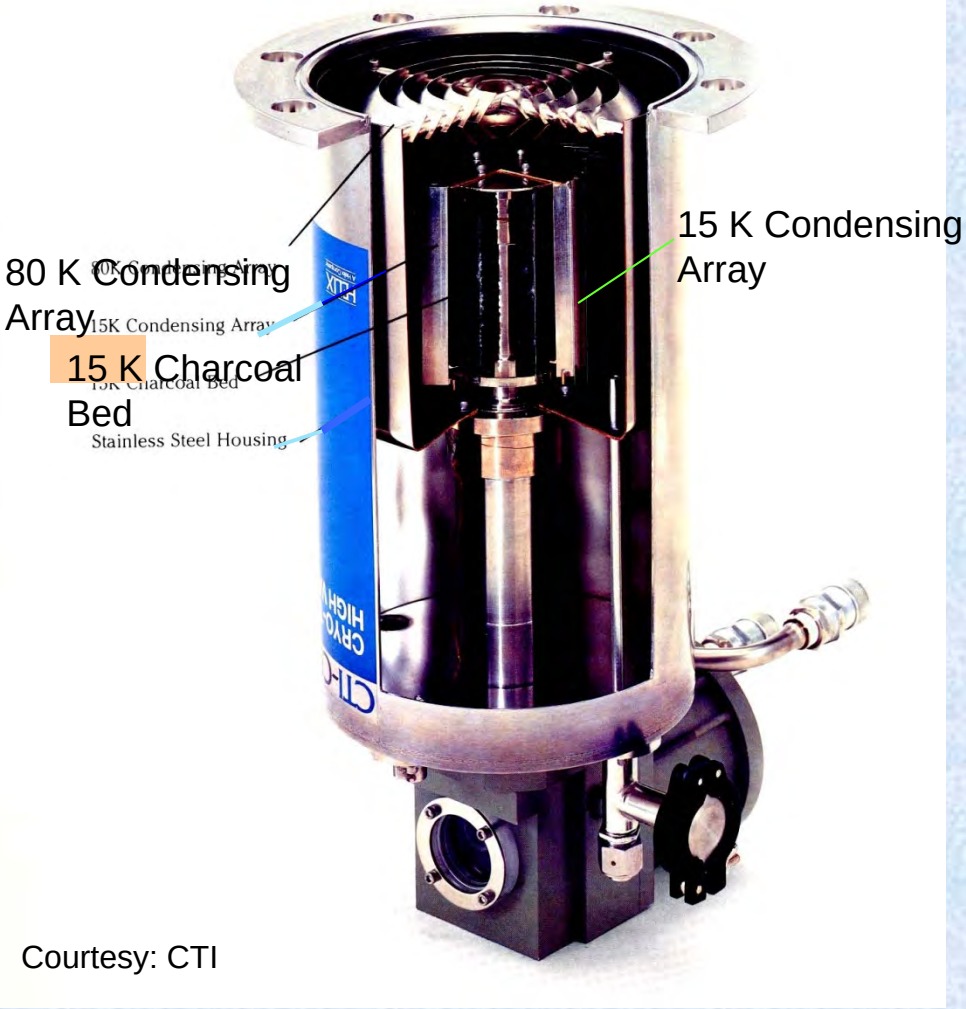
GM Cryocooler Early History

Gifford-McMahon (GM) Cryocooler Invented 1960



- 1970s 2-stage GM
- 1 W at 26 K
- CTI model 0120

1983 Cryopump with GM cryocooler



Courtesy: CTI

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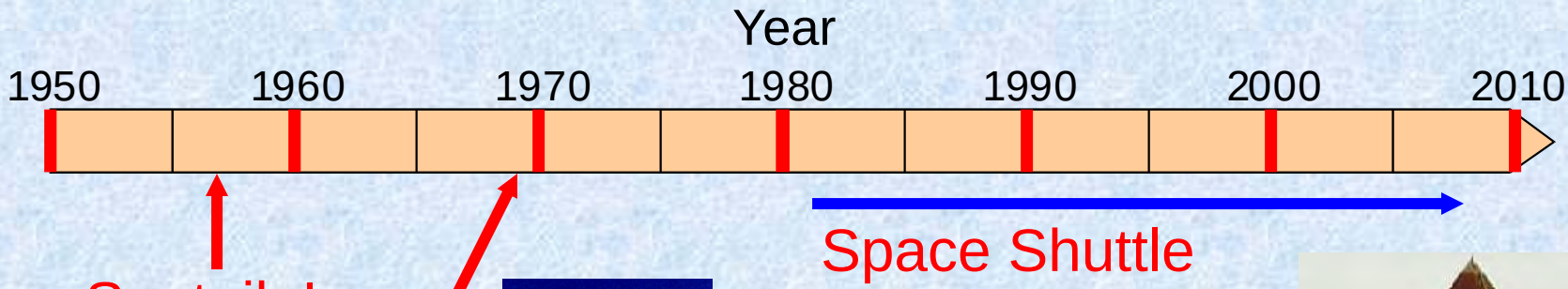
Benefits of Low Temperatures (Applications)

- Preservation of biological material and food
- Densification (liquefaction & separation)
- Quantum effects (fluids and superconductivity)
  - Low dissipation (superconductivity)
  - High-precision metrology (atomic parameters)
  - Action over a distance (fast response)
- Low thermal noise
  - Electromagnetic
  - Electronic
- Low vapor pressures (cryopumping)
- Property changes (permanent and temporary)
- Tissue ablation (cryosurgery)

Cryogenic Applications

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Oxygen and Hydrogen Rockets



Saturn-V

- Properties:
  H_2 and O_2
- <5% of US O_2
- 1.8×10^6 kg LOX
- 5.0×10^4 kg LH_2







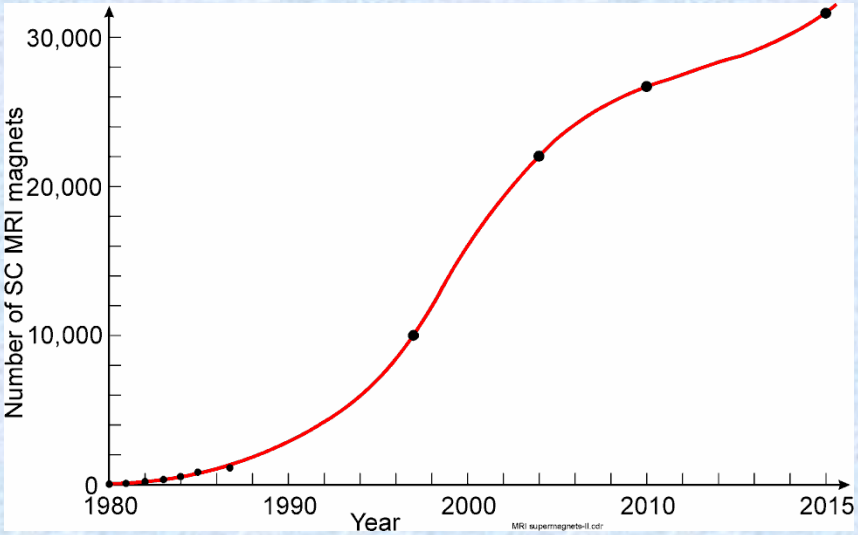
RL10
66.7 kN

Cryogenic Applications

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Magnetic Resonance Imaging (MRI)

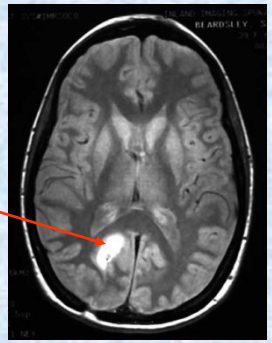




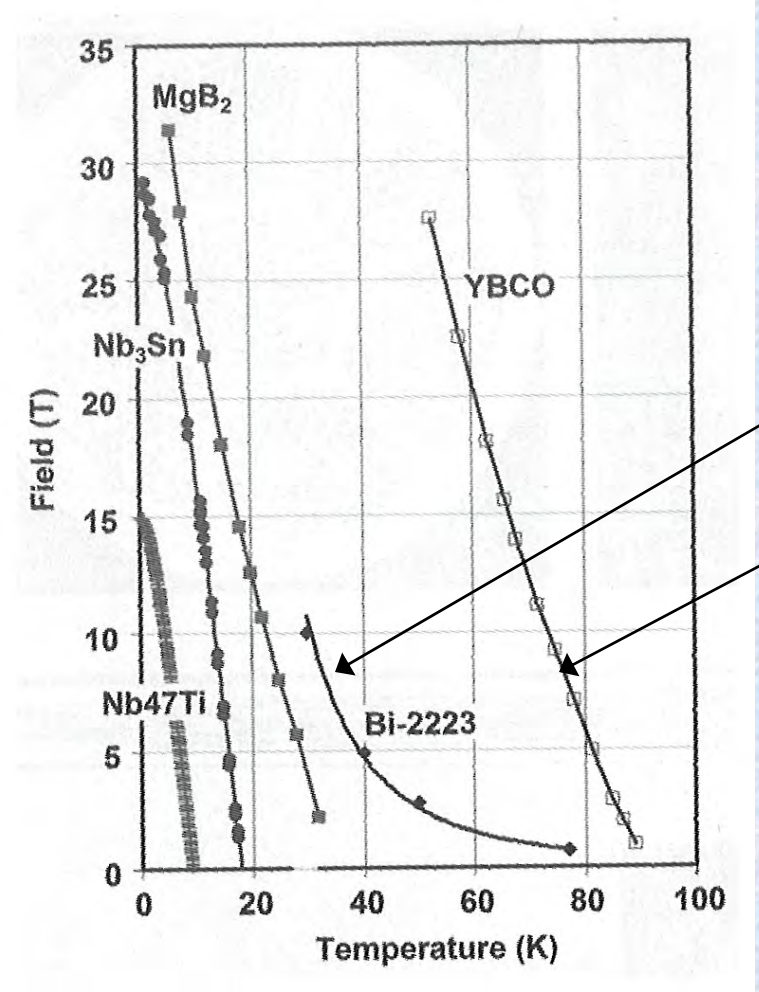
Cumulative number of MRI superconducting magnets sold

- 1.5 T Superconducting magnets
- 1 W at 4 K
- Non-magnetic regenerators
- >40,000 4 K GM cryocoolers since 1995

Tumor



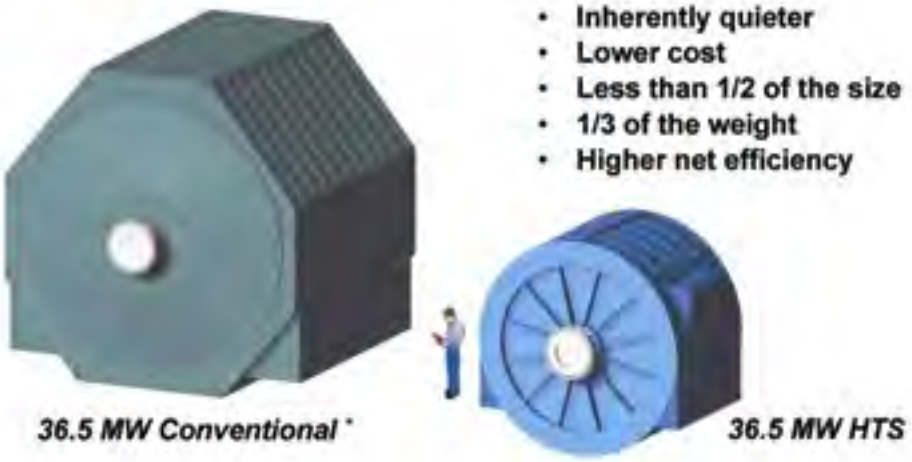
Critical Magnetic Fields



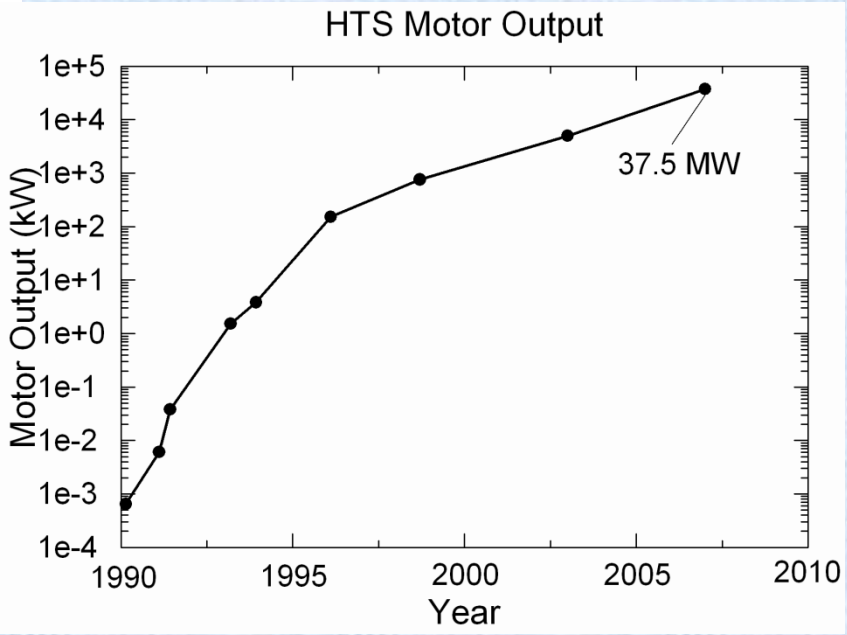
- 1st generation HTS wire
- 2nd generation HTS wire

Figure courtesy: Scanlan *et al.*
Proc. IEEE, vol. 92 (2004)

HTS Superconducting Motors



Drawing courtesy: American Superconductor Corporation



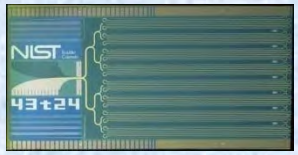
37.5 MW Ship Motor

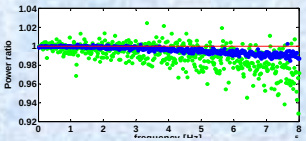


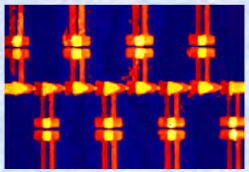
- Ship motor
- 150 rpm
- 2007
- AMSC

Superconducting Electronics: NIST Standards and Measurements

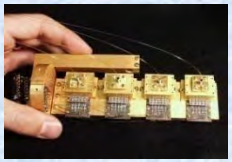
All based on condensed matter quantum phenomena











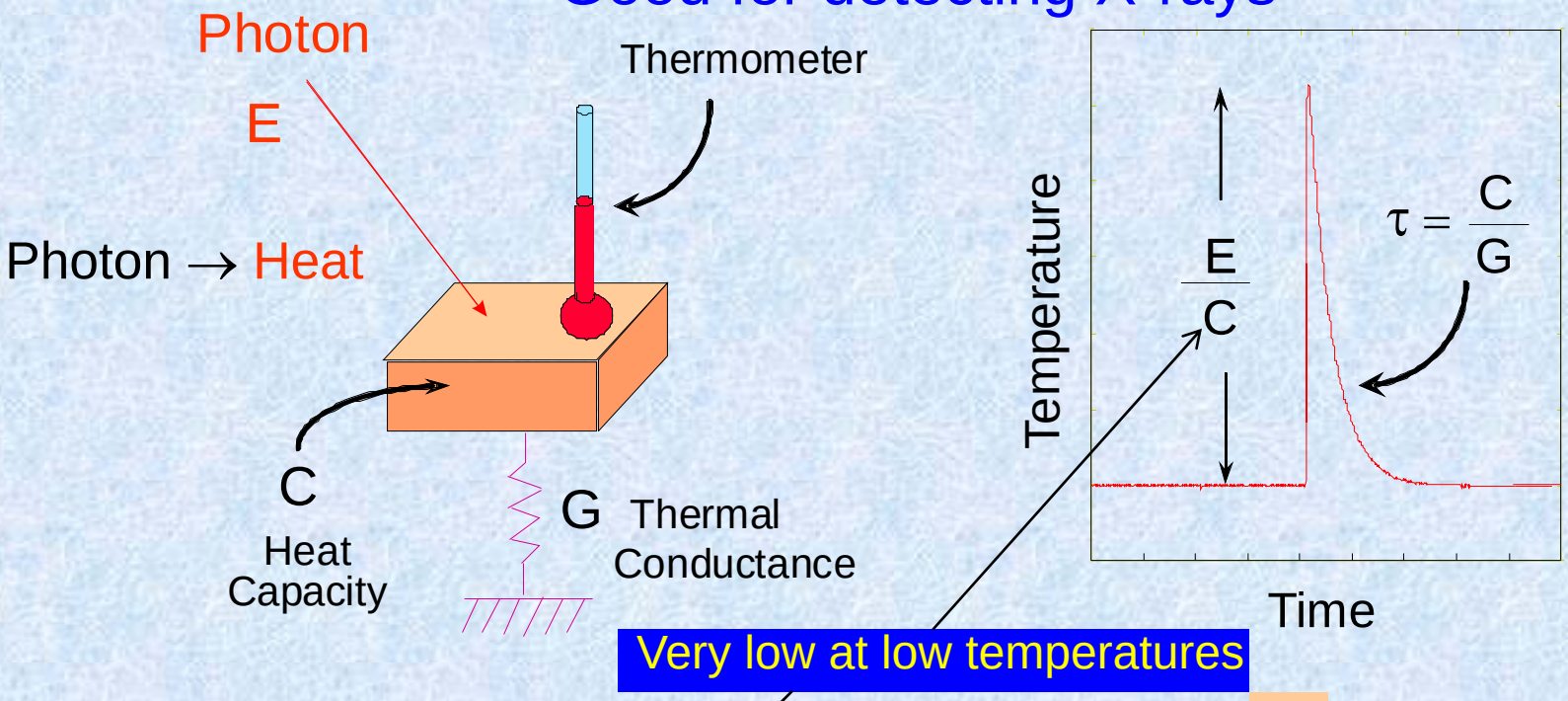
Voltage standards ★	Magnetic flux quantum
Temperature standards ★	Magnetic flux quantum
Capacitance standards ★	Electron charge
Infrared and x-ray imaging ★	Photon
Optical photon counting ★	Photon

★ Best-in-the-world

Courtesy: R. Harris, NIST

Low Thermal Noise Microcalorimeters (Bolometers)

Good for detecting X-rays



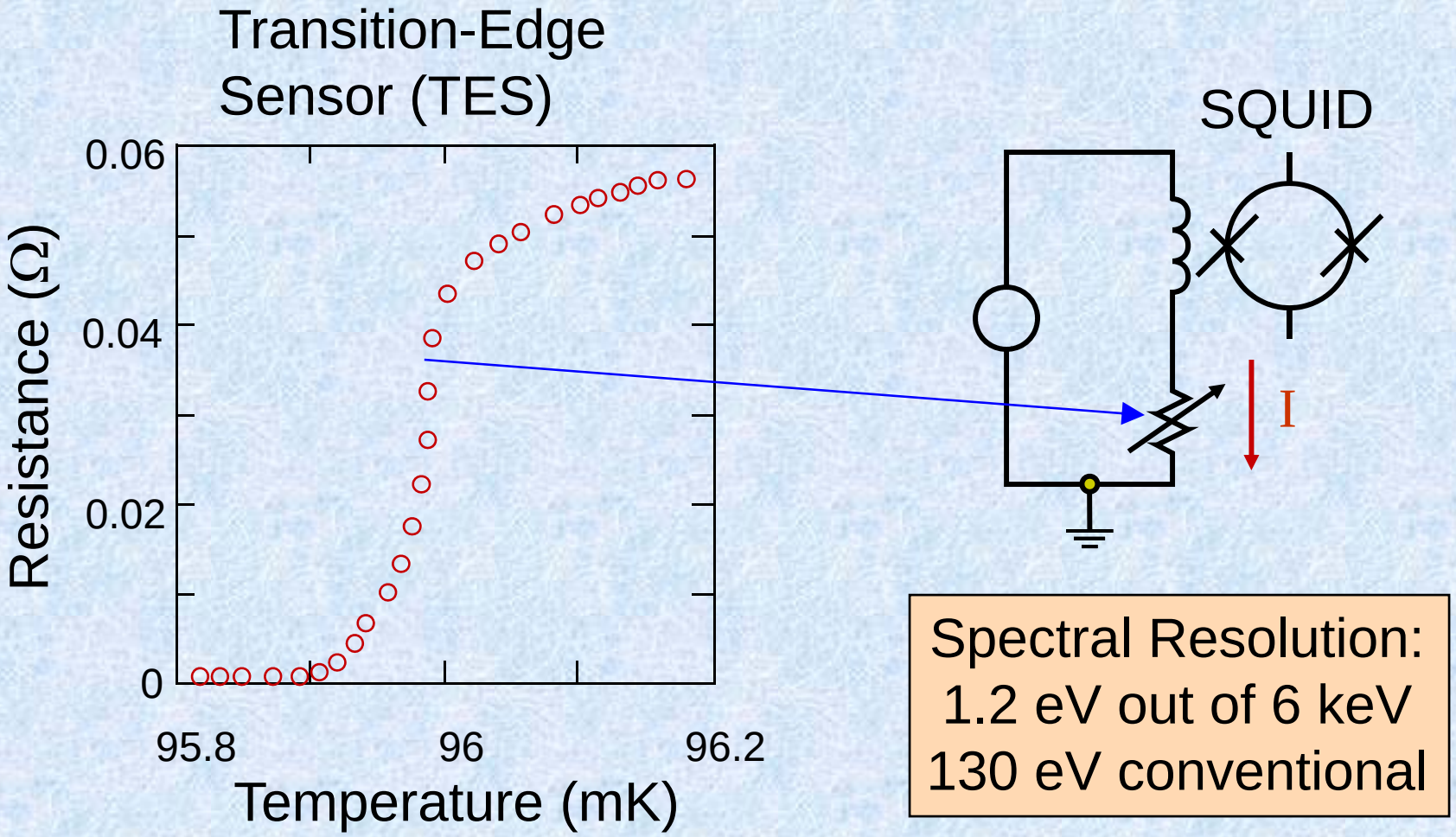
Very low at low temperatures

- Root-mean-square energy fluctuations ~ $(k_b T^2 C)^{1/2}$
- For T = 0.1 K, C = 10 pJ/K, ΔE ~ 10 eV
- Need to integrate a thermometer into C

several possibilities, such as transition-edge sensors

Courtesy: J. Ullom, NIST

Superconducting Transition-Edge Sensor



Spectral Resolution:
1.2 eV out of 6 keV
130 eV conventional

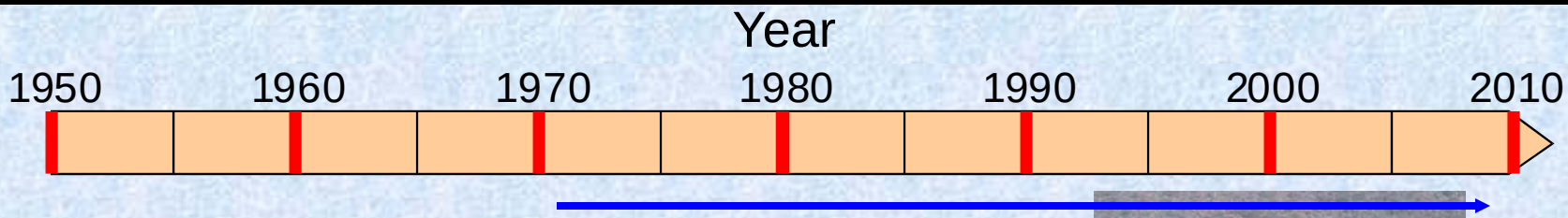
Photon → Heat → Resistance → Current

Courtesy:R. Harris, NIST

Cryogenic Applications

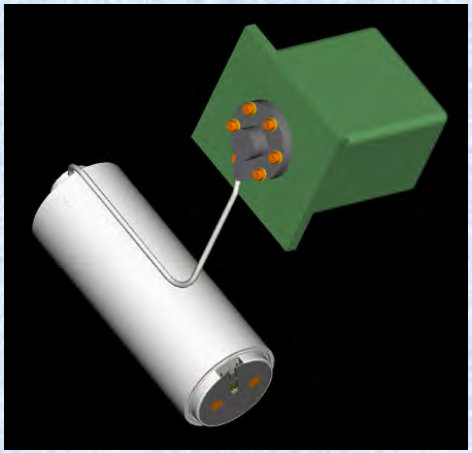
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Cooled Infrared Sensors







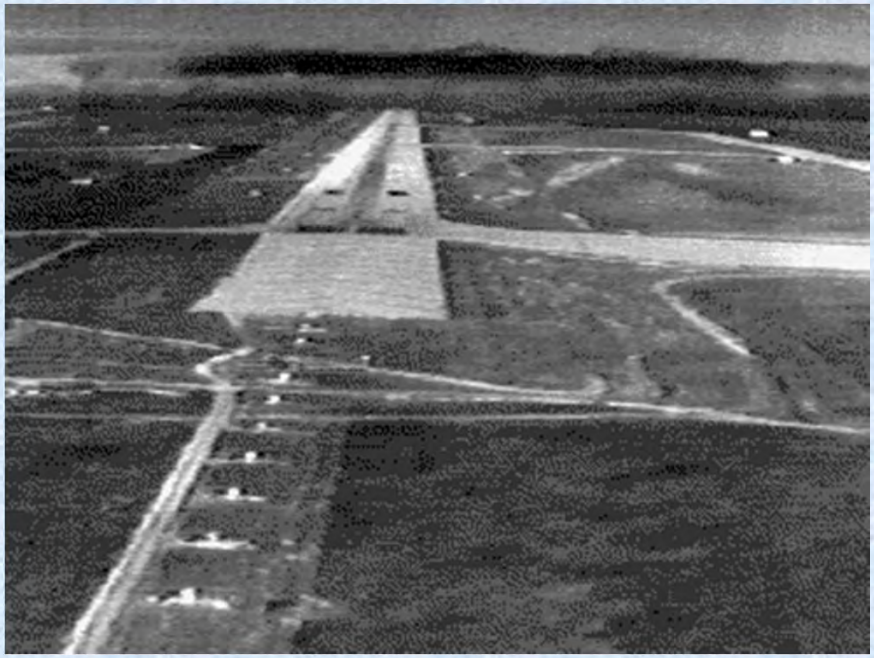


- Military tactical applications
- Night vision
- About 180,000 coolers to date since 1970s
- 0.3 to 1.75 W at 65 – 120 K

Military Advantage in Total Darkness



Selected readout: Black-hot



Selected readout: White-hot

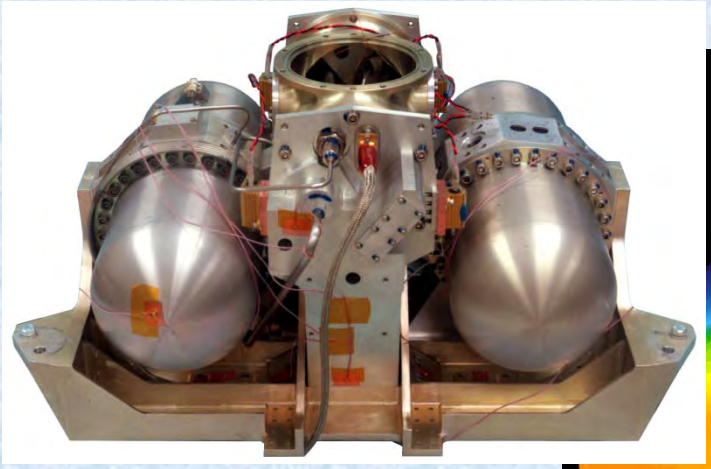
"Our night-vision capability provided the single greatest mismatch of the war."

General Barry McCaffrey, commander of the 24th Infantry Division, 1991 Persian Gulf War

Atmospheric Infrared Imaging

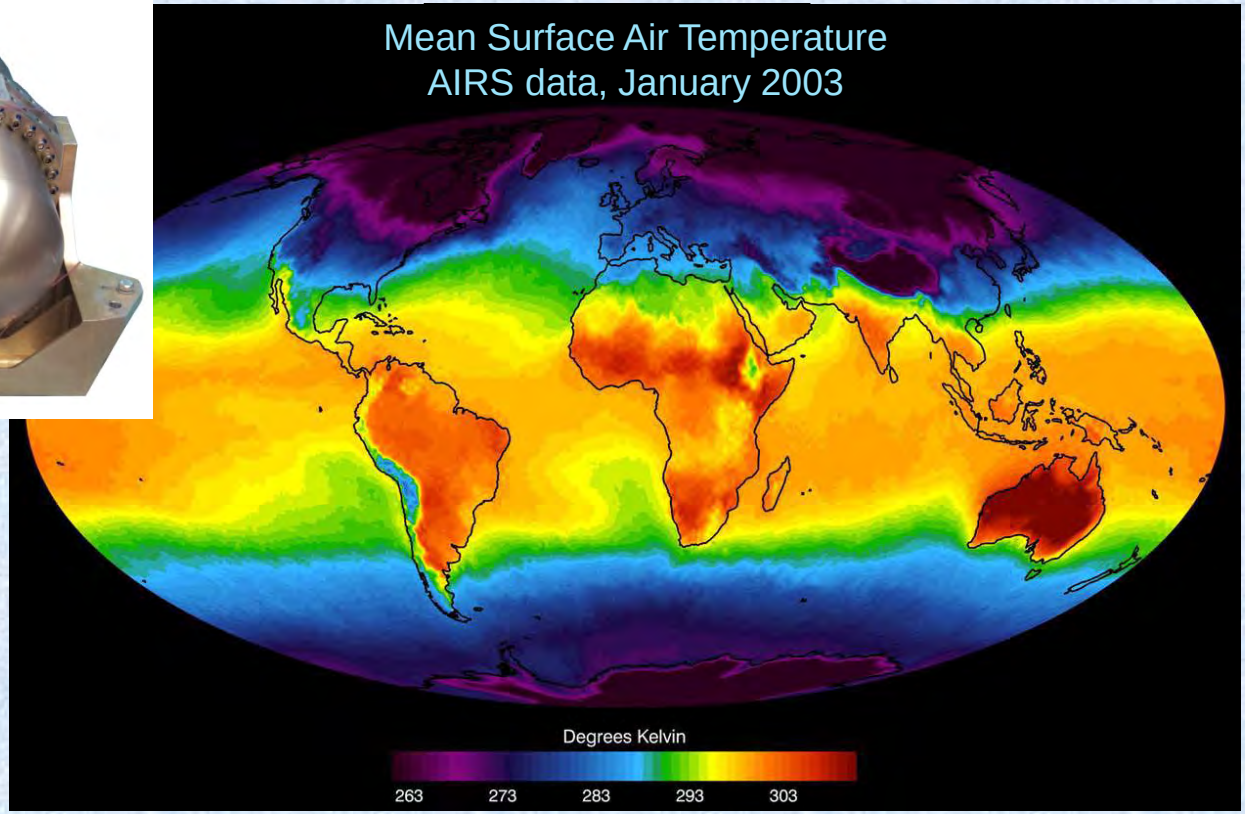
Earth Observing System (EOS)

Airs_cut.tif



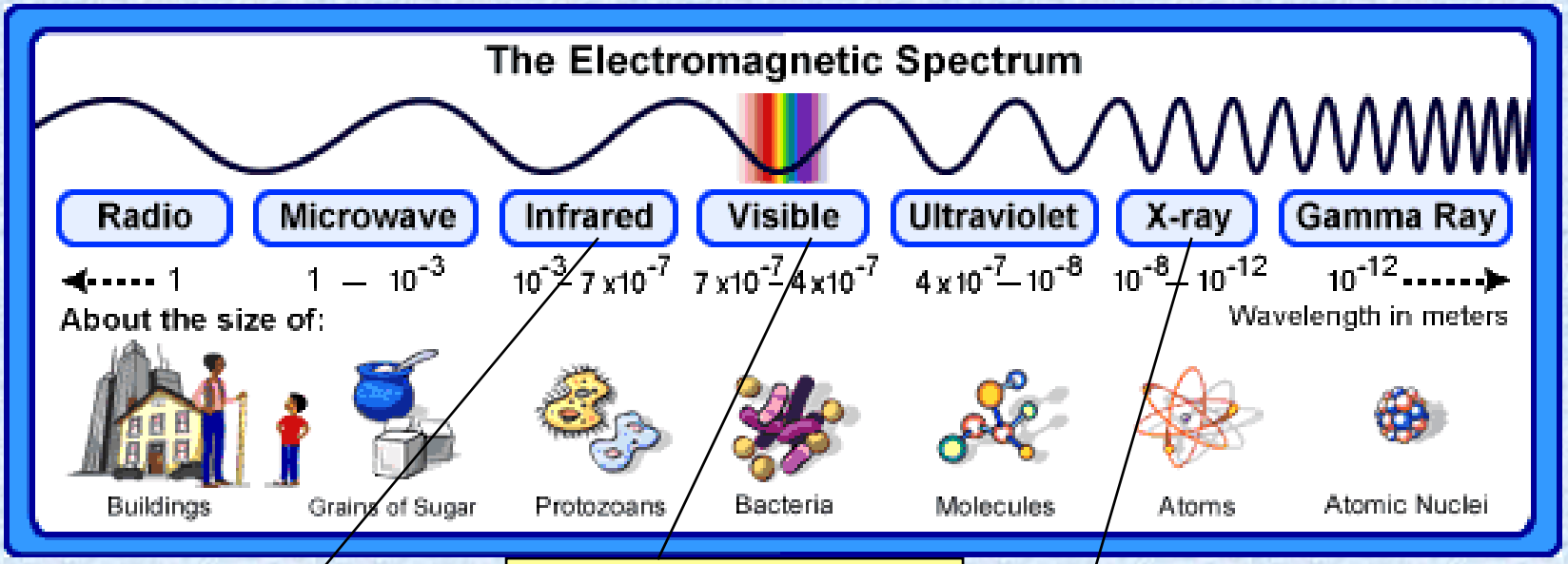
AIRS Pulse Tube Cryocoolers (2)

- 1.2 W @ 55 K
- 60 W input
- TRW
- May 2002 Launch



airs_surface_temp1_full Jan03.jpg

Infrared and X-Ray Space Telescope Missions



Only 4% of universe mass seen as visible

Why infrared?

- Cold universe
- Obscured regions
- Dust emissions
- Molecular spectral lines
- Highly redshifted universe

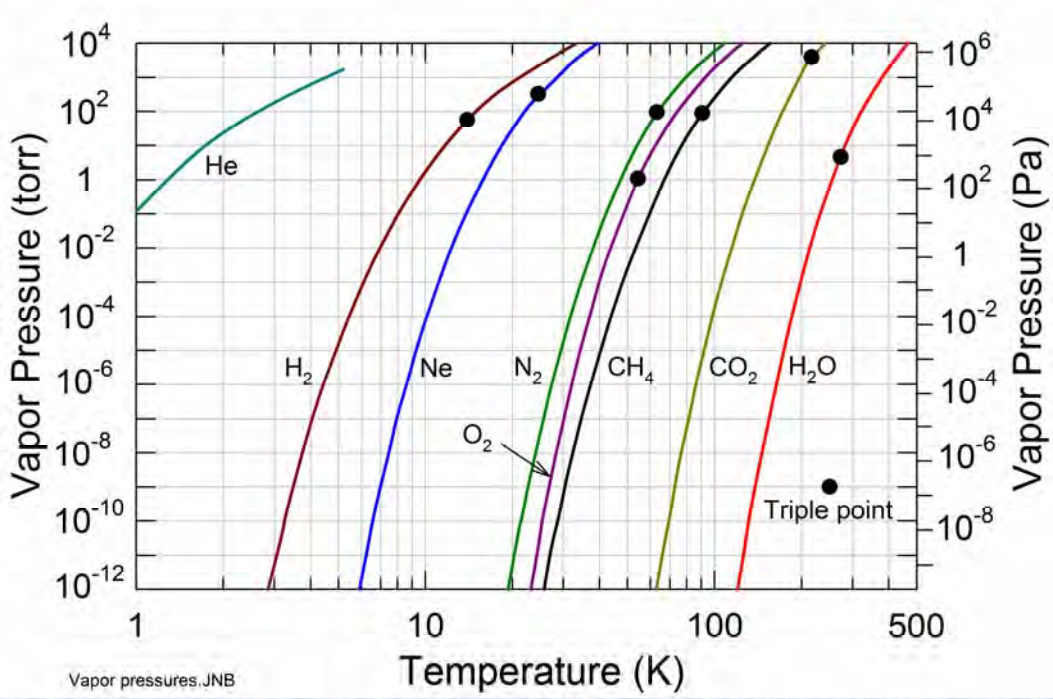
Why X-Ray?

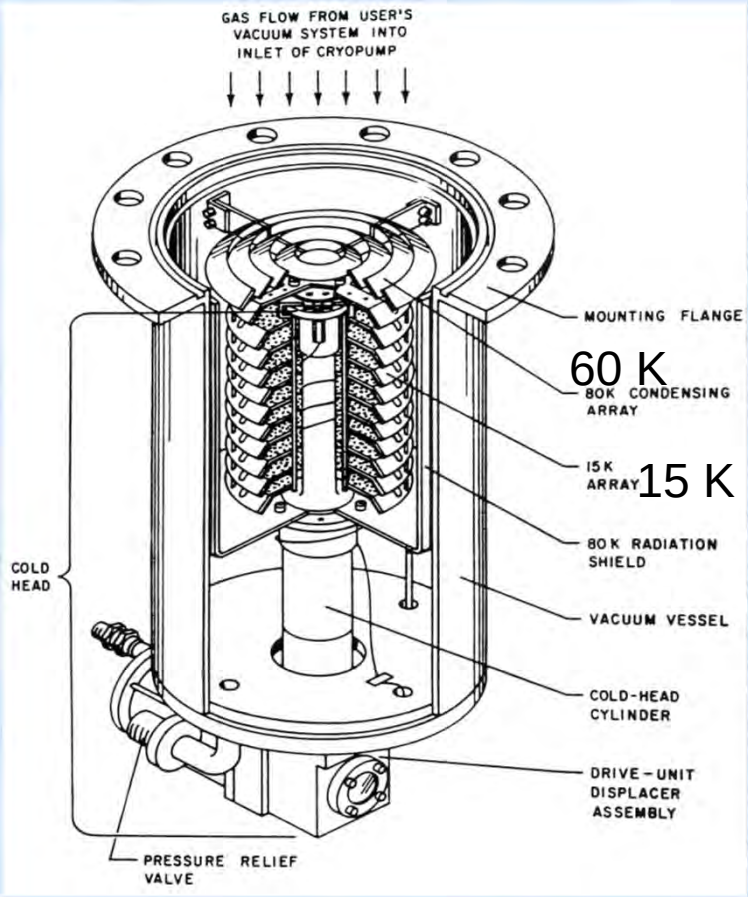
- Detect dark matter
- Test gravitational theory
- X-rays around black holes
- Elements in universe

Cryogenic Applications

- Preservation of biological material and food
- Densification (liquefaction & separation)
- Quantum effects (fluids and superconductivity)
  - Low dissipation (superconductivity)
  - High-precision metrology (atomic parameters)
  - Action over a distance (fast response)
- Low thermal noise
  - Electromagnetic
  - Electronic
- Low vapor pressures (cryopumping)
- Property changes (permanent and temporary)
- Tissue ablation (cryosurgery)

Cryopumps (15 K)





- Used to produce clean vacuum
- Required for semiconductor processing
- About 20,000/yr at peak of semiconductor business

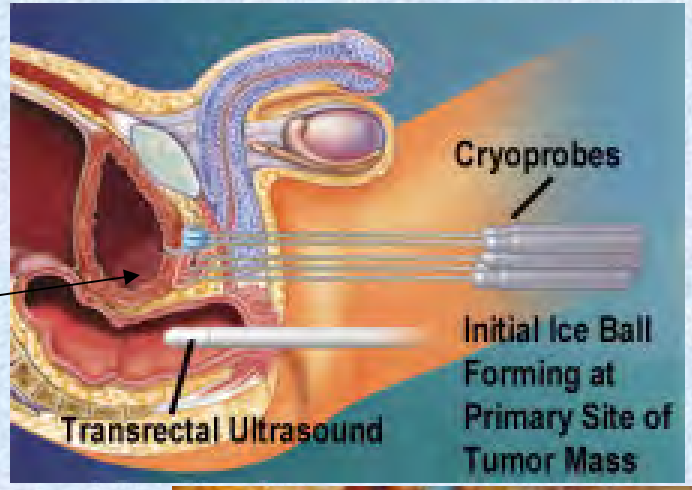
Cryogenic Applications

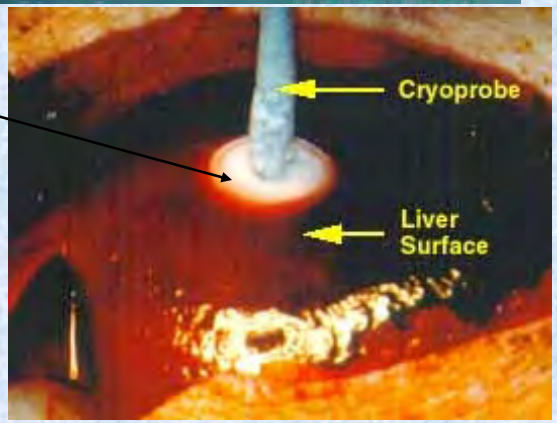
- Preservation of biological material and food
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  - Low dissipation (superconductivity)
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  - Action over a distance (fast response)
- Low thermal noise
  - Electromagnetic
  - Electronic
- Low vapor pressures (cryopumping)
- Property changes (permanent and temporary)
- Tissue ablation (cryosurgery)

Cryosurgery Applications and Markets

- Cancers
  - –Skin
  - –Cervix (precancerous)
  - –Prostate
  - –Liver
  - –Kidney
  - –Lung
  - –Brain
  - –Breast

Research





- Abnormal Uterine Bleeding
  - –120,000 women in U.S. each year get hysterectomies for this
  - –Recovery period is 6 weeks for hysterectomy
  - –Recovery period of 1 day for cryoablation
  - –Cryoablation of uterine lining received FDA approval in 2001

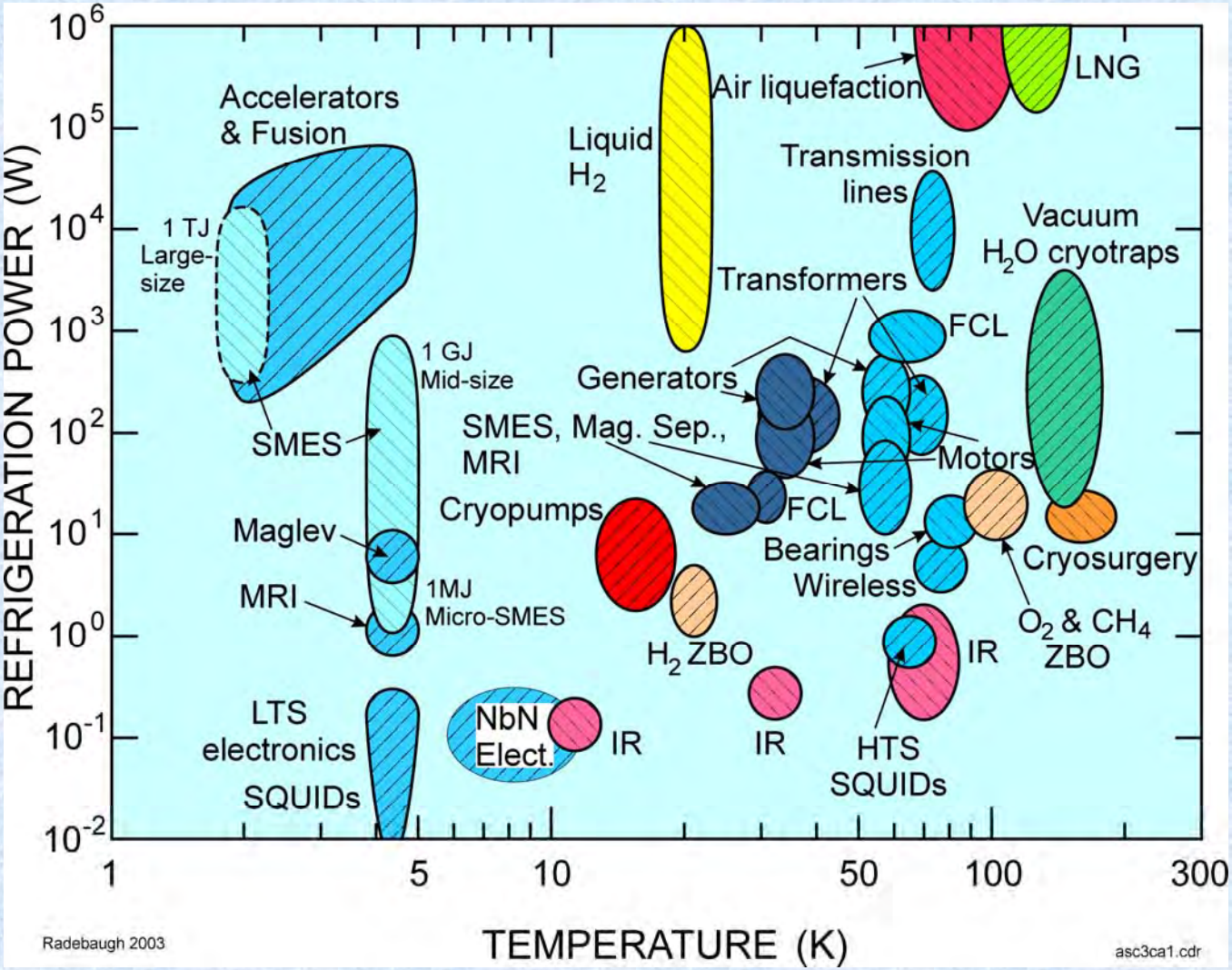
Cryosurgery Applications and Markets

- Cardiac Arrhythmias
  - –2.5 million in U.S. suffer from atrial fibrillation
  - –5 million in developed world
  - –Options include medication or rf catheter treatment
  - –Cryoablation received FDA approval in 2004 for clinical trials
  - –FDA approval in 2010 for clinical use

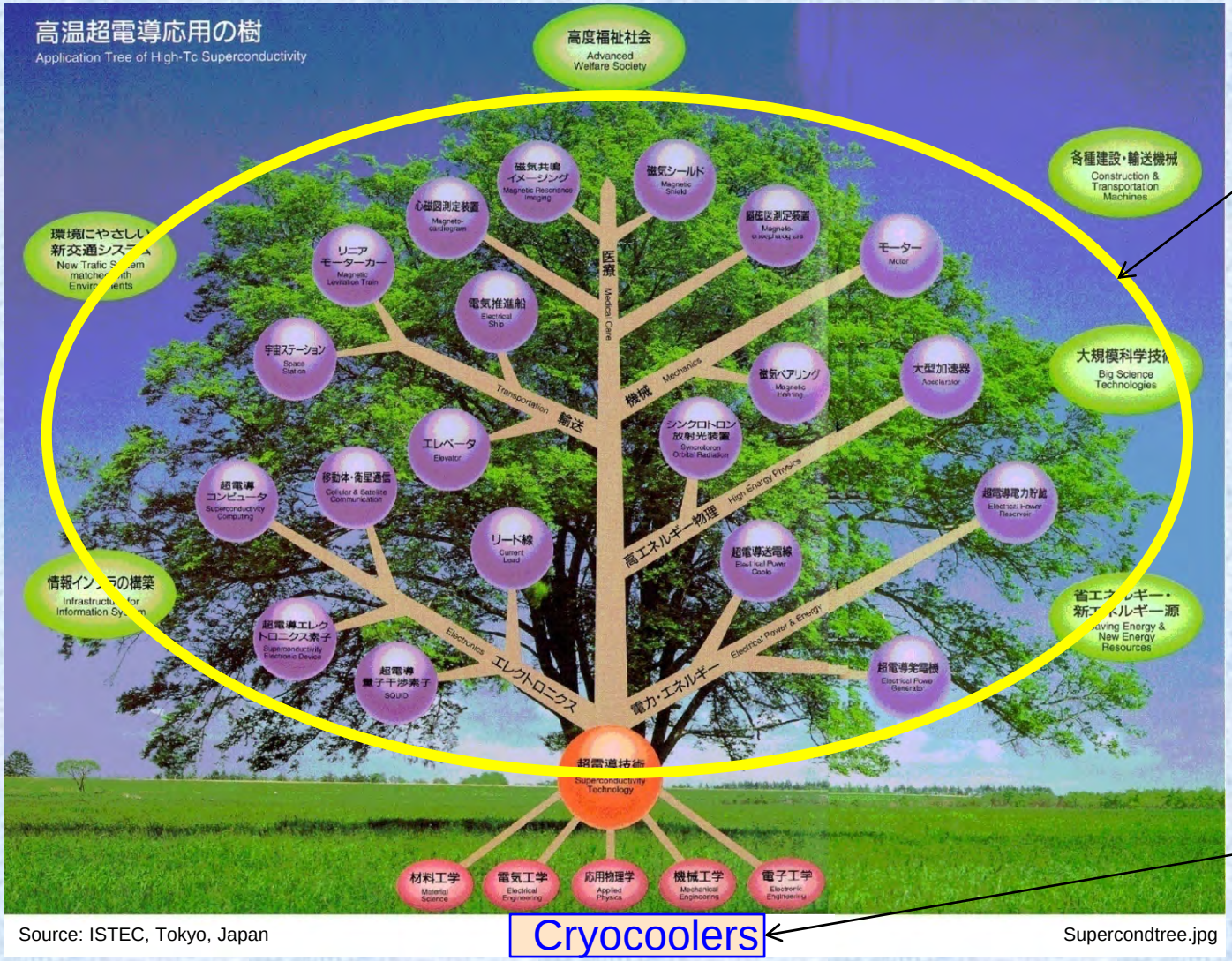
Cryo catheters vs. rf catheters: Comparison

	rf	Cryo
Held fixed by sticking to the wall	No	Yes
Can provide mapping	No	Yes
Free of blood clot formation	No	Yes
Pain-free	No	Yes

Cryocooler Applications



Cryocoolers – An Enabling Technology



Source: ISTEC, Tokyo, Japan

Supercondtree.jpg

Spectacular applications

Spectacular =

- high resolution
- high precision
- high sensitivity
- high power density
- low power
- low noise

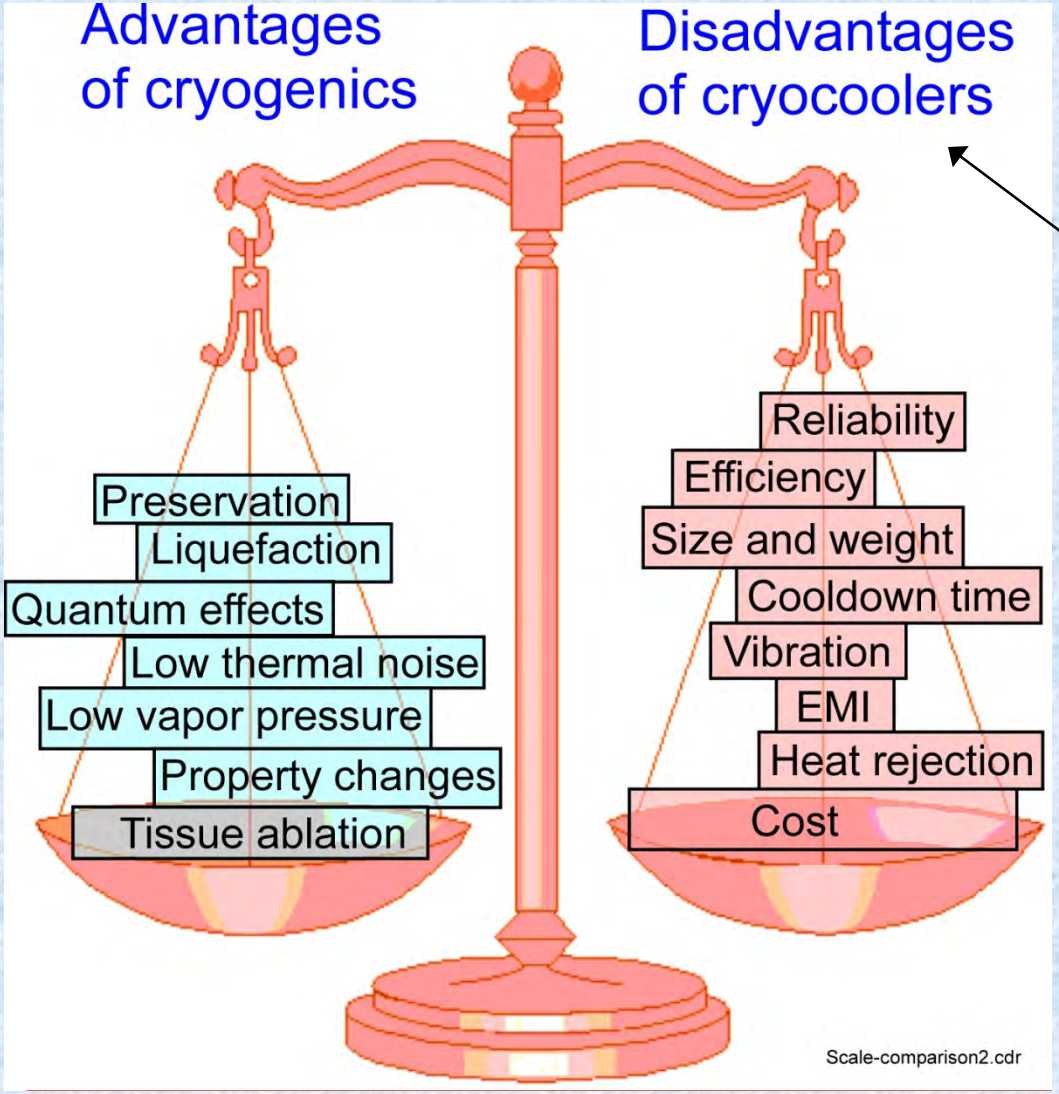
Cryocoolers

Enabling technology

Potential Cryocooler Problems

- Reliability
- Efficiency
- Size and weight
- Vibration
- Electromagnetic Interference (EMI)
- Heat rejection
- Cost

Balancing Act



Cryocooler Research

Goal

Reduce disadvantages

(Cryocooler invisible to user)

Outline

Session 1

- Information sources
- Definitions
- History
- Applications and problems
- Thermodynamics
- Cryocooler types and operating principles
- Joule-Thomson (JT) cryocoolers
- Brayton cryocoolers
- Claude systems
- Heat exchangers

Thermodynamic State Functions

	Intensive	Extensive
Pressure	P (Pa)	
Temperature	T (K)	
Volume	v (m^3/kg)	V (m^3)
Internal energy	u (J/kg)	U (J)
Enthalpy	h (J/kg)	H (J)
	$H = U + PV$	
Entropy	s (J/(kg·K))	S (J/K)
(Others)		

Two functions uniquely define all other functions

Equations of State

Ideal (Perfect) gas: $PV = mRT$ $PV = nR_0T$

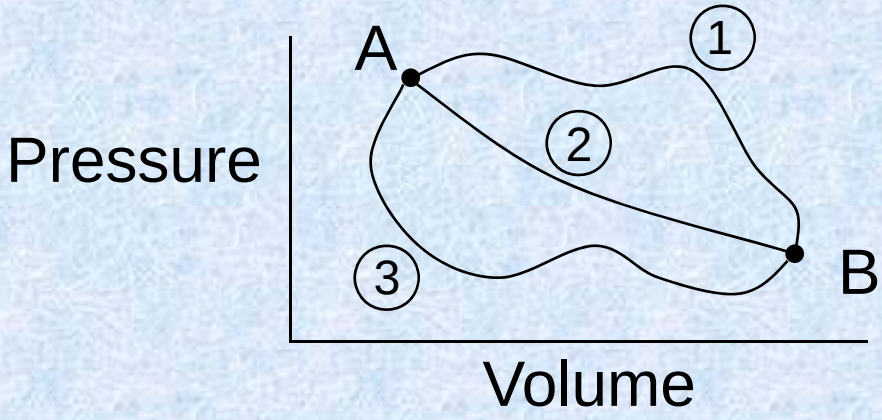
Gas constant, R (J/(kg·K)) R_0 (J/mol·K)

Van der Waals, Beattie-Bridgeman, etc.: $PV = ZmRT$

Two thermodynamic functions are sufficient to define the thermodynamic state of a fluid.

Thermodynamic Processes

Consider a change of thermodynamic state from A to B via three different paths.



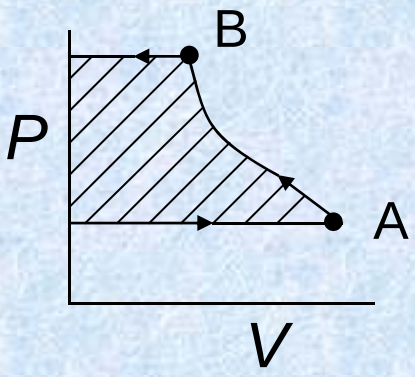
A change in, say, pressure and volume will cause the same change in the state functions (temperature, internal energy, enthalpy, entropy, no matter which process or path is followed,

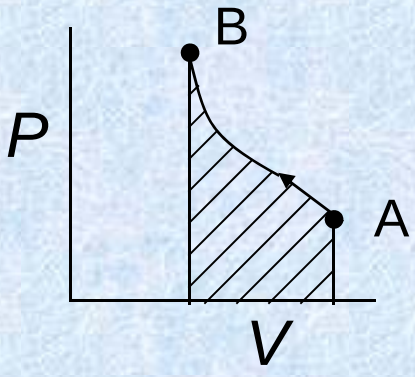
but

the work done (*W*) and the heat transferred (*Q*) do depend on the path or nature of the process.

Thermodynamic Processes, p. 2

Work done (*W*) is the area under the process on the *P*-*V* diagram



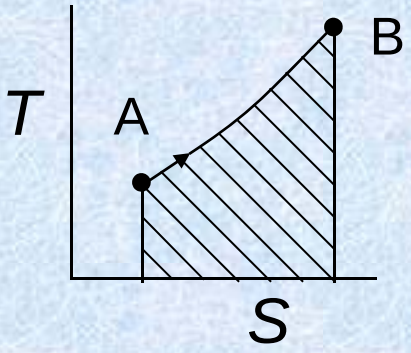


Flow process: $dW = \int_A^B V dP$

Non-flow process $dW = \int_A^B P dV$

Moving piston or rotating shaft

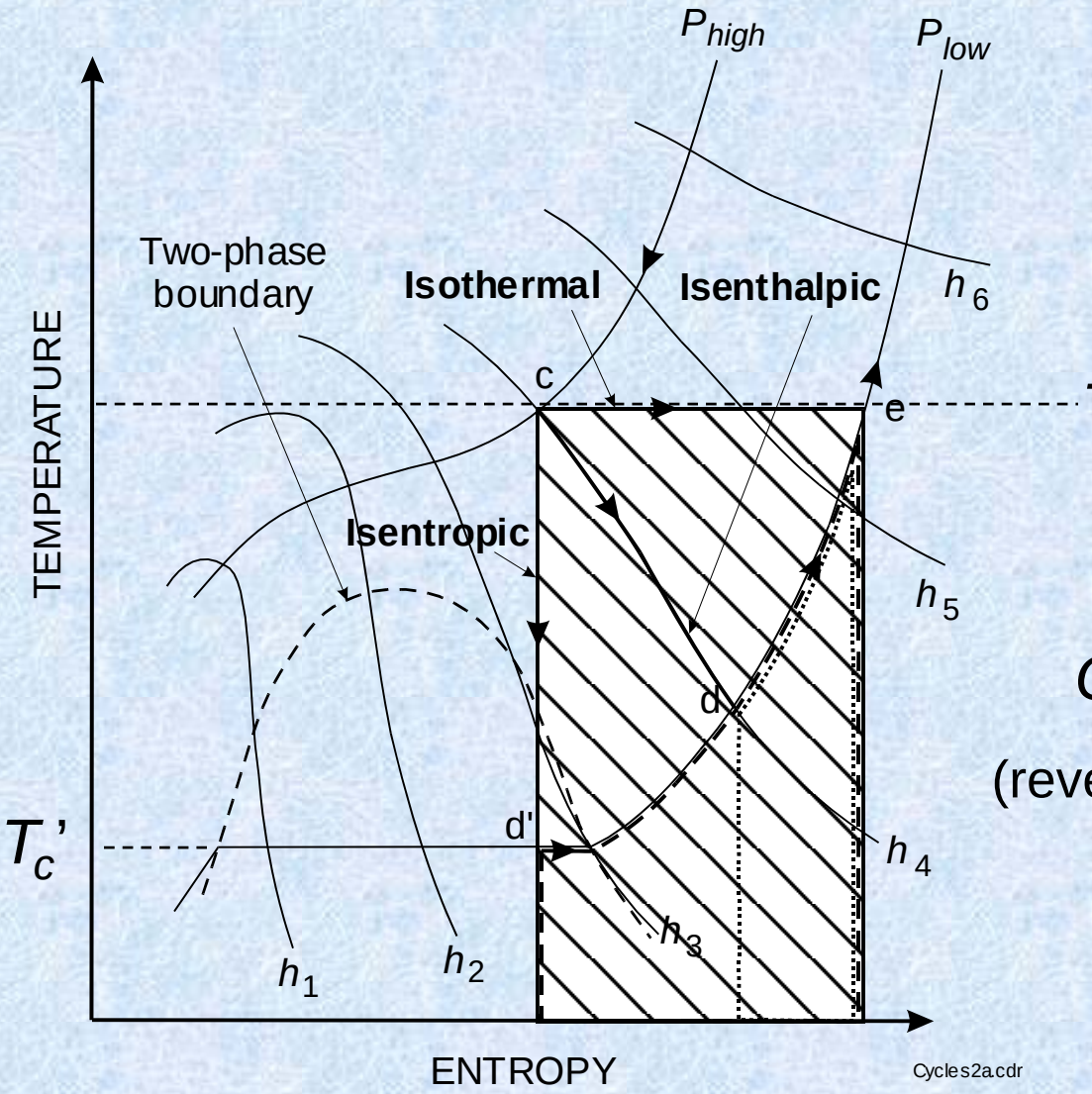
Heat transferred (*Q*) is the area under the process on the *T*-*S* diagram



$dQ \leq TdS$ (general)

$dQ = TdS$ (reversible)

Comparison of Expansion Processes



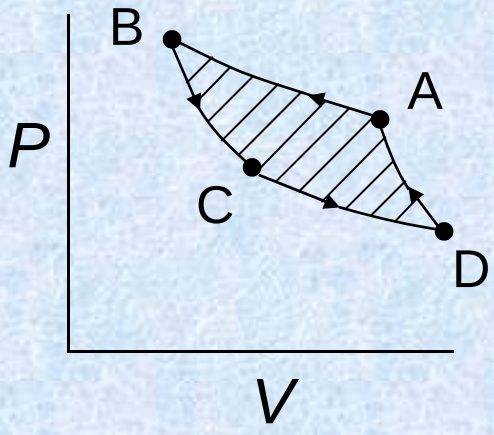
$$\dot{Q} = \int T dS$$

(reversible process)

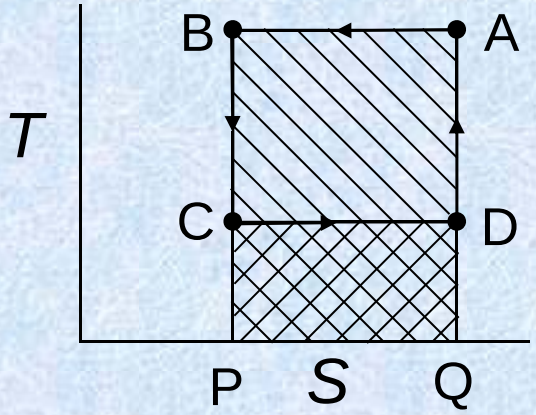
Thermodynamic Cycle

Sequence of thermodynamic processes followed repeatedly in the course of operation of a thermodynamic device (refrigerator)

Net work input: Area ABCD



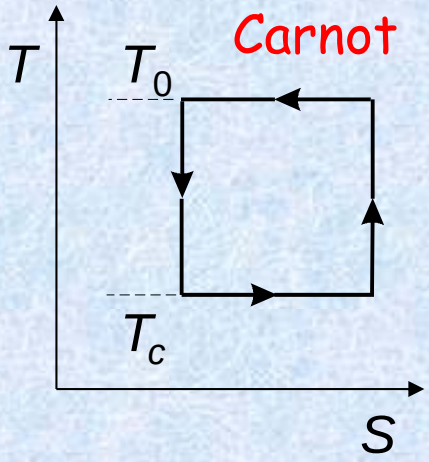
Carnot cycle

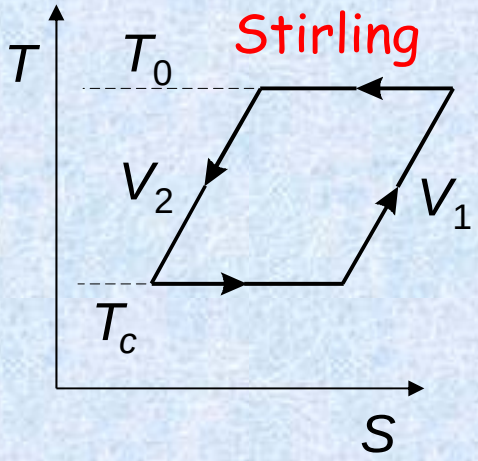


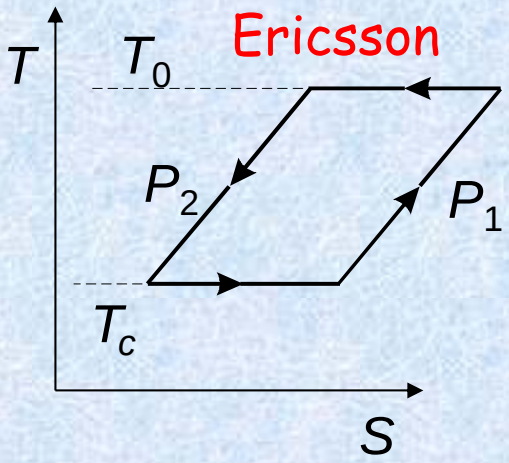
A-B: Isothermal compression
B-C: Isentropic expansion
C-D: Isothermal expansion
D-A: Isentropic compression

Heat supplied: area CDQP
Heat rejected: area ABPQ

Ideal Cycles

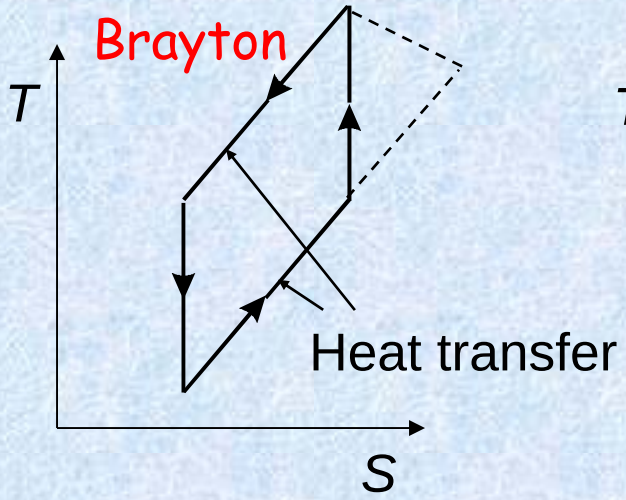


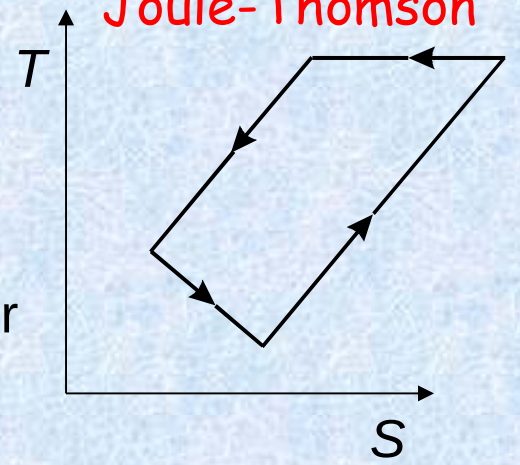




$dQ \leq TdS$

$dQ = TdS$ *(reversible)*

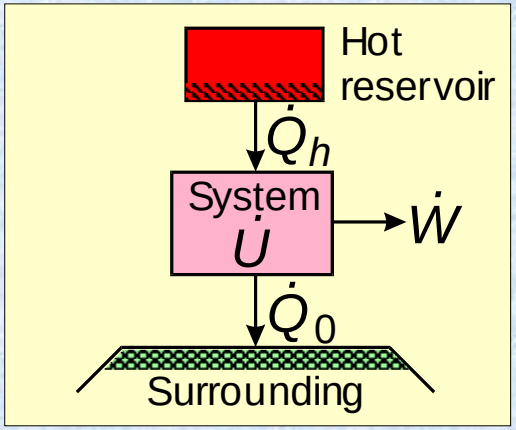




Thermodynamic Fundamentals

FIRST LAW OF THERMODYNAMICS
(Conservation of Energy)

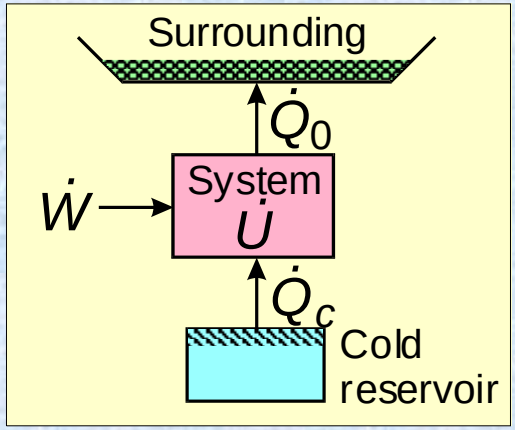
The net flow of energy across the boundary of a system is equal to the change in energy of the system.



Engine

$$\dot{W} = \dot{Q}_h - \dot{Q}_0 - \dot{U}$$

Closed System



Refrigerator

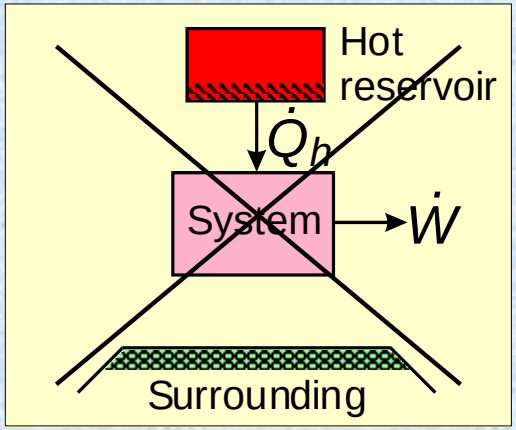
$$\dot{W} = \dot{Q}_0 - \dot{Q}_c + \dot{U}$$

Refrig8c.cdr

Thermodynamic Fundamentals, p. 2

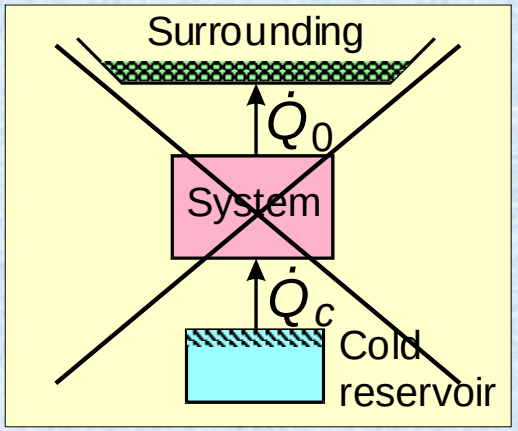
SECOND LAW OF THERMODYNAMICS
(Relation of work and heat)

Kelvin Statement



Of the heat supplied at a high temperature to a system not all can be converted to work; a fraction of it must be rejected as heat at a lower temperature.

Clausius Statement

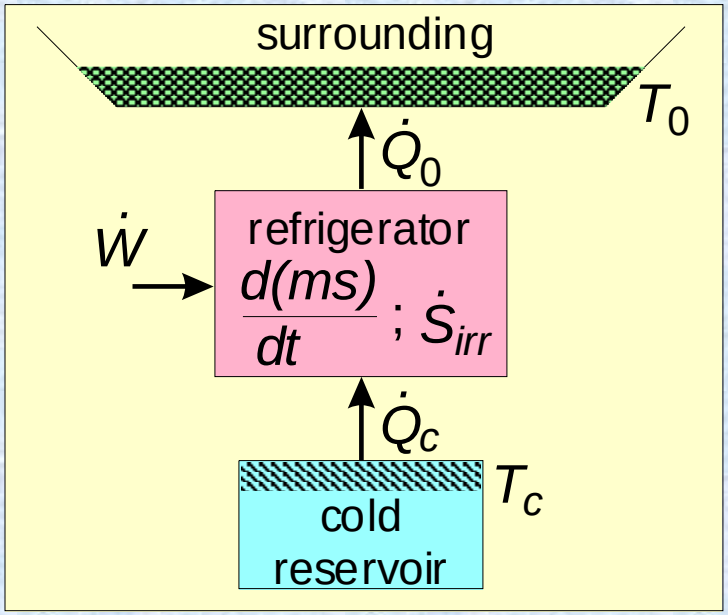


No process is possible whose sole result is the removal of heat from a reservoir at one temperature and the rejection of an equal quantity of heat to a reservoir at a higher temperature.

Refrig9c.cdr

Thermodynamic Fundamentals, p. 3

SECOND LAW OF THERMODYNAMICS
(Entropy Balance)



Closed System

$$\frac{\dot{Q}_c}{T_c} - \frac{\dot{Q}_0}{T_0} \leq \frac{d(ms)}{dt}$$

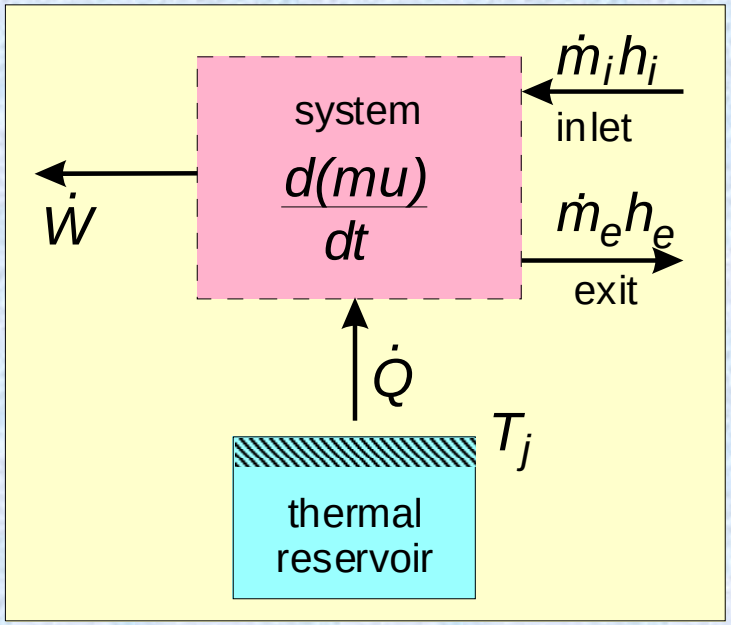
$$\frac{\dot{Q}_c}{T_c} - \frac{\dot{Q}_0}{T_0} + \dot{S}_{irr} = \frac{d(ms)}{dt}$$

$\dot{S}_{irr}$: Irreversible entropy production

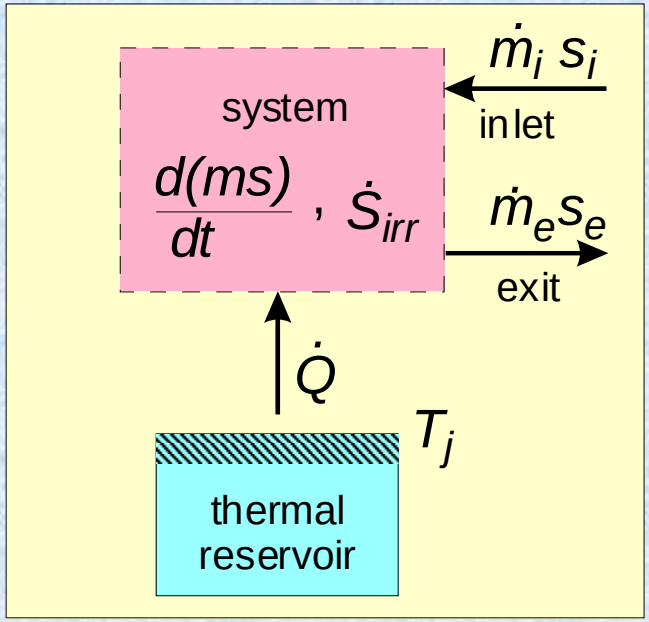
Refrig10c.cdr

Thermodynamic Fundamentals, p. 4

THERMODYNAMICS OF OPEN SYSTEMS
(Mass crosses system boundary)



(a) Energy Balance
(First Law)



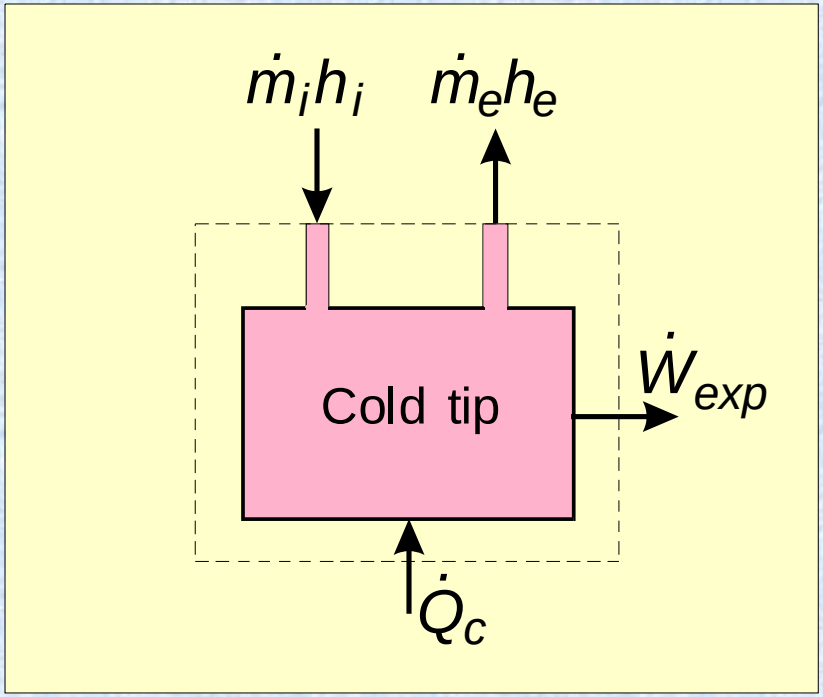
(b) Entropy Balance
(Second Law)

Open system useful for components

UCLA/Refrig11c.cdr

Thermodynamic Fundamentals, p. 5

Refrigerator Cold Tip



Open System Thermodynamics

First Law: (steady state)

$$\dot{Q}_c = \dot{W}_{exp} + \dot{m}_e h_e - \dot{m}_i h_i$$

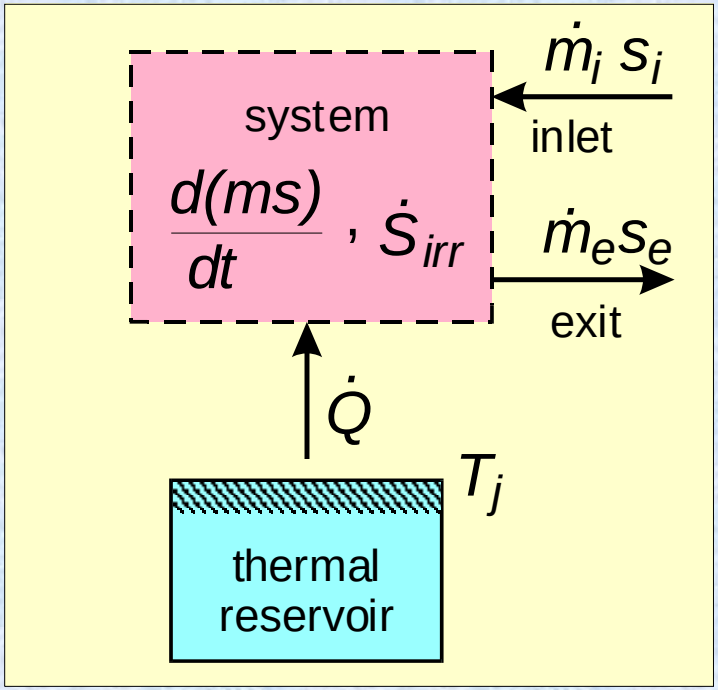
$$\dot{Q}_c = \dot{W}_{exp} + \dot{m}(h_e - h_i)$$

(h = specific enthalpy)

h is independent of pressure for an ideal gas

Refrig5c.cdr

Thermodynamic Fundamentals, p. 6



(s = specific entropy)

s is a function of pressure for an ideal gas

Open System Thermodynamics

Second Law: (steady state)

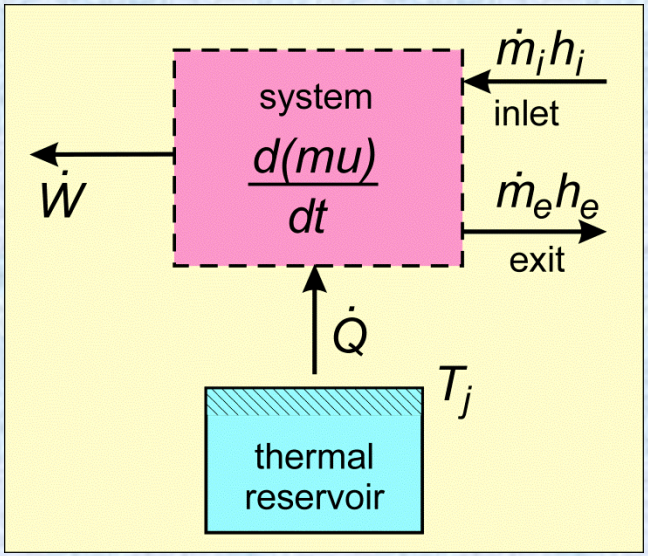
$$\dot{m}(s_e - s_i) = \frac{\dot{Q}}{T_j} + \dot{S}_{irr}$$

$$\dot{Q} = \dot{m}\int_a^b T ds - \int_a^b T d\dot{S}_{irr}$$

For temperature distribution

Refrig13c.cdr

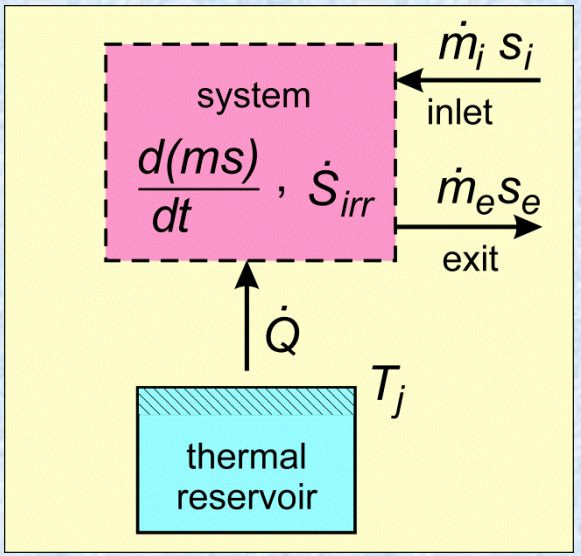
Thermodynamic Fundamentals, p. 7



Open System Thermodynamics

1st Law

2nd Law



$$\dot{Q} = \dot{W} + \dot{m}(h_e - h_i)$$

Steady state

$$\frac{\dot{Q}}{T} + S_{irr} = \dot{m}(s_e - s_i)$$

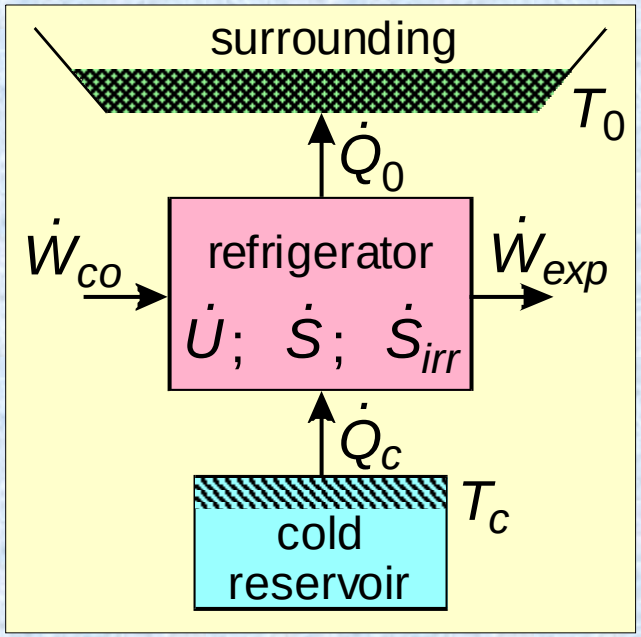
Combined: $$\dot{W} = \dot{m}\left[(h_i - Ts_i) - (h_e - Ts_e)\right] - TS_{irr}$$

Isothermal

$h - Ts = g$ = Gibbs free energy

Reversible process: $$\dot{W}_{rev} = \dot{m}(g_i - g_e) = \dot{m}\Delta g$$

Thermodynamic Fundamentals, p. 8



Definitions

Leaving system

Coefficient of Performance *COP*

$$COP = \frac{\dot{Q}_c}{\dot{W}_{co}} = \frac{T_c}{T_0 - T_c}\left[1 - \frac{(T_0 \dot{S}_{irr} + \dot{W}_{exp})}{\dot{W}_{co}}\right]$$

Specific Power

$$p_s \equiv \frac{1}{COP} = \frac{\dot{W}_{co}}{\dot{Q}_c}$$

First Law: (steady state)

$$\dot{W}_{co} - \dot{W}_{exp} = \dot{Q}_0 - \dot{Q}_c$$

Second Law: (steady state)

$$\frac{\dot{Q}_0}{T_0} = \frac{\dot{Q}_c}{T_c} + \dot{S}_{irr}$$

Ideal (Carnot) *COP*

$$COP_{Carnot} = \frac{T_c}{T_0 - T_c}$$

$S_{irr} = 0$
$W_{exp} = 0$

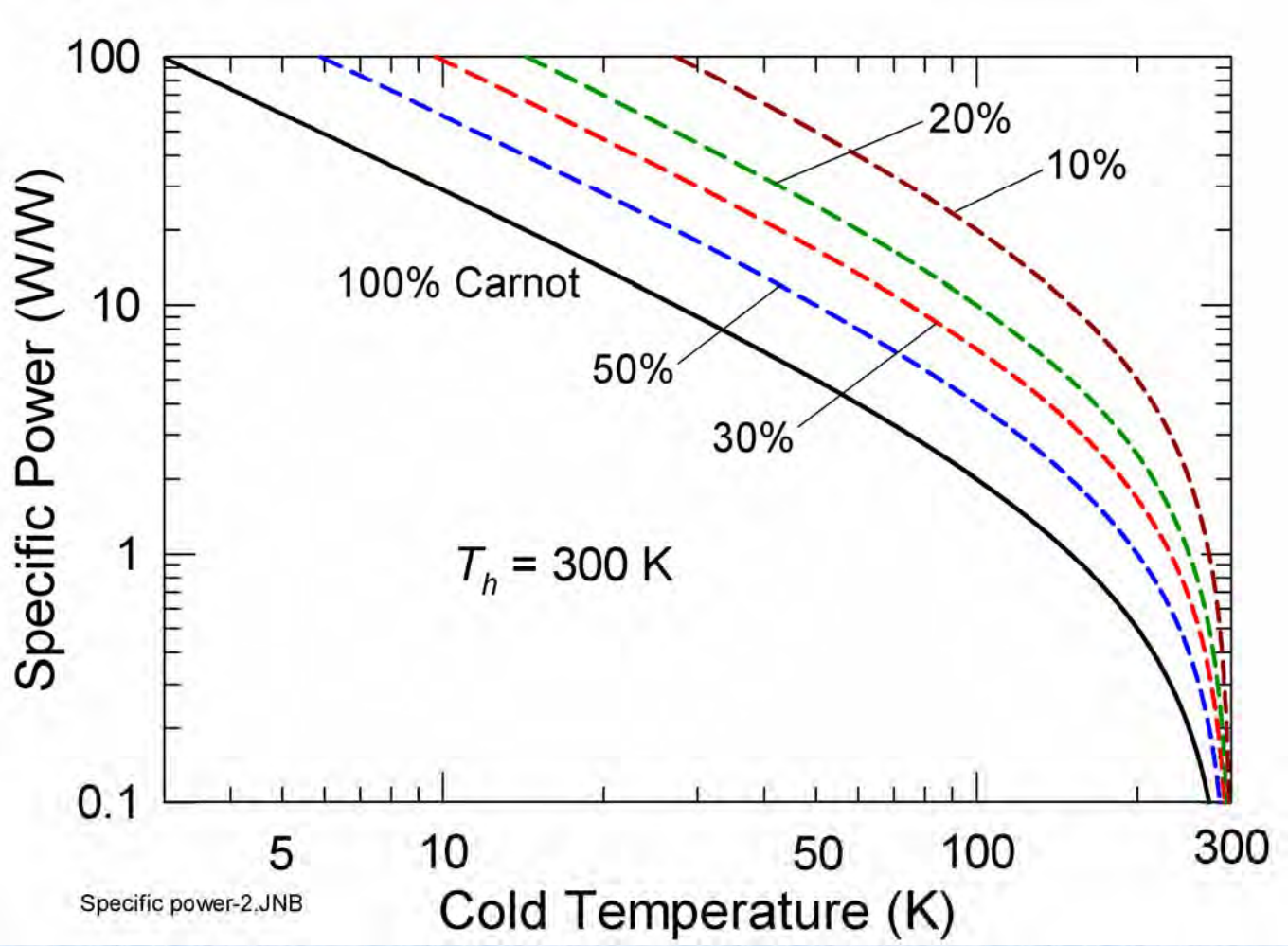
Efficiency

$$\eta = COP / COP_{Carnot}$$

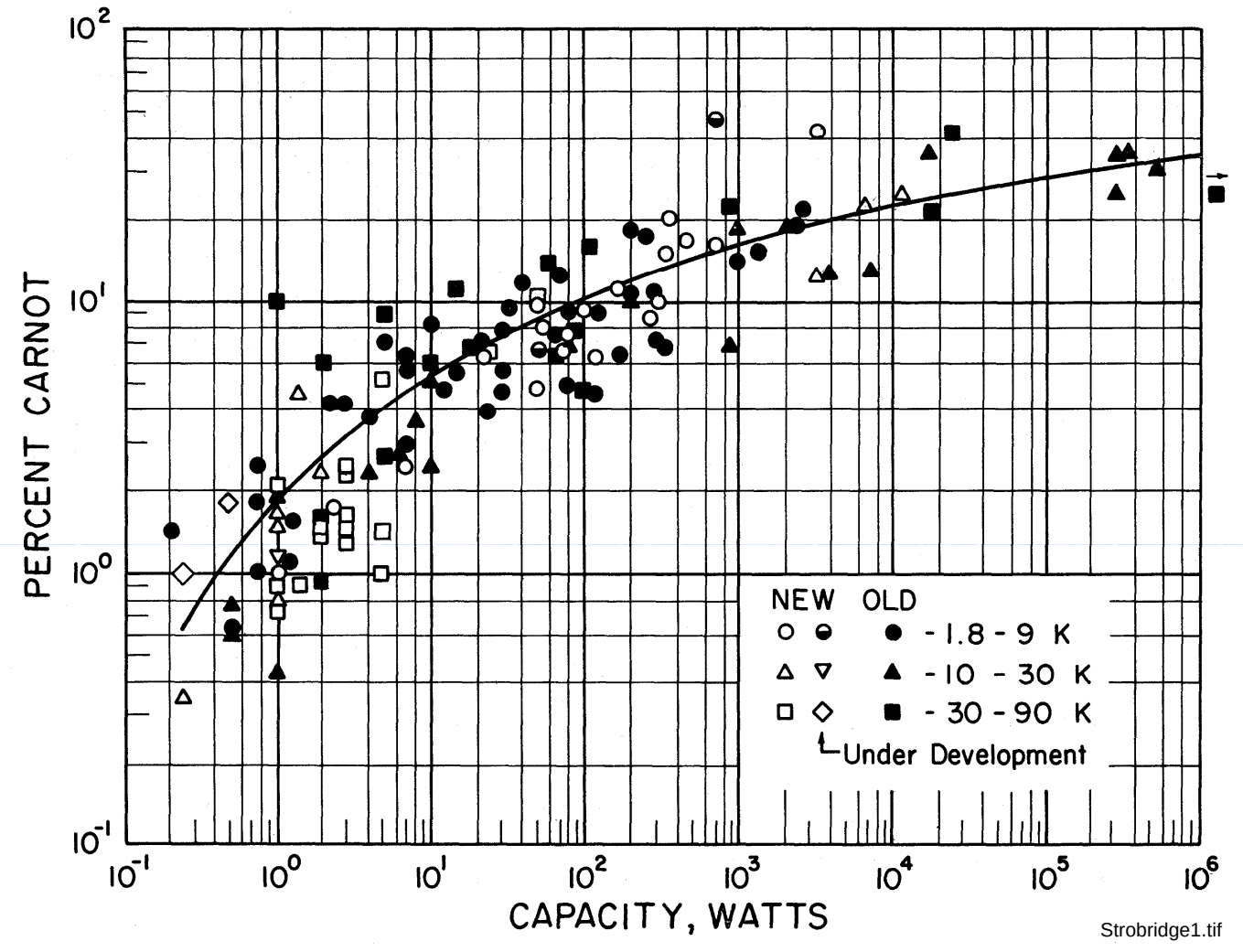
Refrig4ca.cdr

Specific Power

Watts of input power per watt of refrigeration power

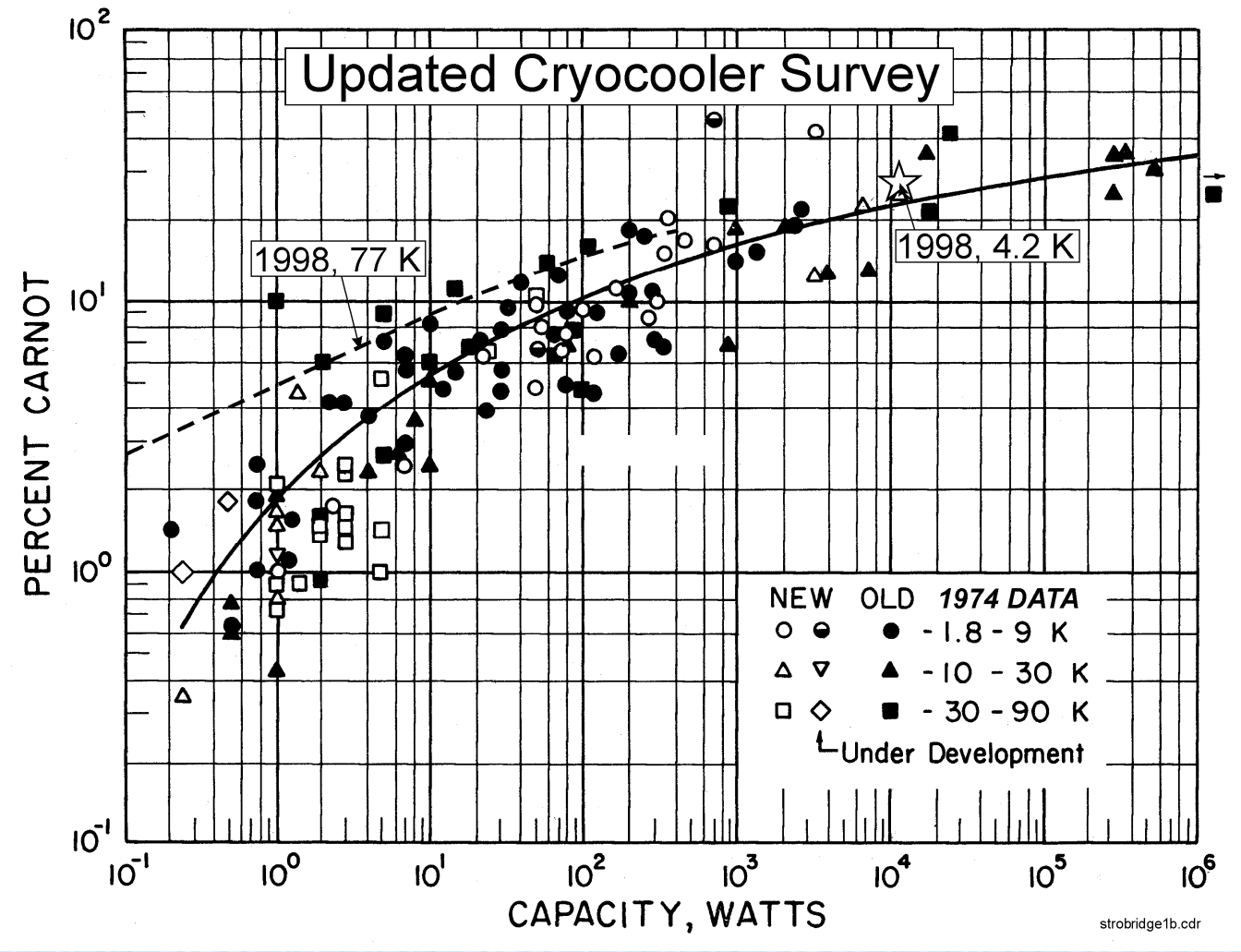


Strobridge Survey (Efficiency) 1974



Source: T. R. Strobridge, “Cryogenic Refrigeration – An Updated Survey,” NBS Tech. Note 655 (1974)

Updated Cryocooler Survey



Source: Bruning, *et al.*, "Survey of Cryocoolers for Electronic Applications", *Cryocoolers 10* (1999)

CILTEC Cryocooler Survey

CILTEC – Center for Interfacing Low Temperature Electronics and Coolers

H. J. M. ter Brake and G. F. M. Wiegerinck,
University of Twente, The Netherlands
www.CILTEC.org

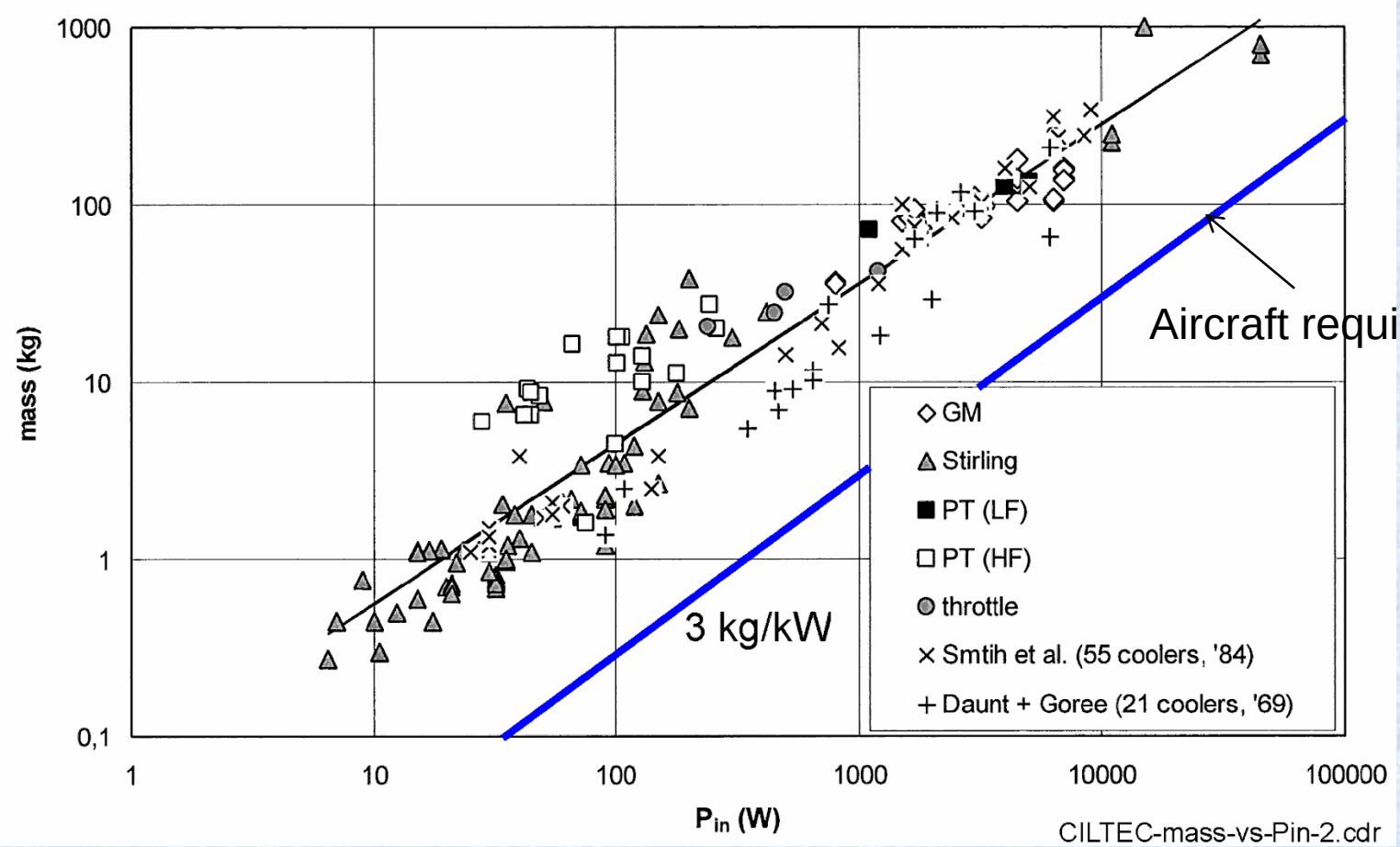
- Excel spread sheet on 27 parameters for 235 cryocoolers from 35 vendors

See: *Cryogenics*, vol. 42, pp. 705-718 (2002)

Efficiency trend about the same as the 1999 Bruning survey

Specific Mass of Cryocoolers

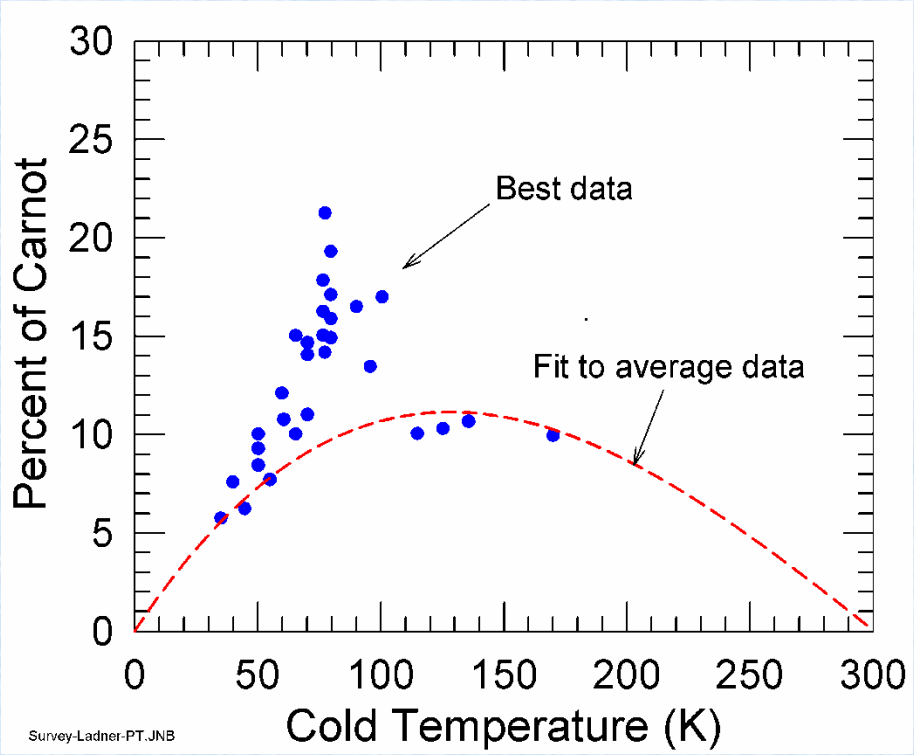
CILTEC 2002 Survey



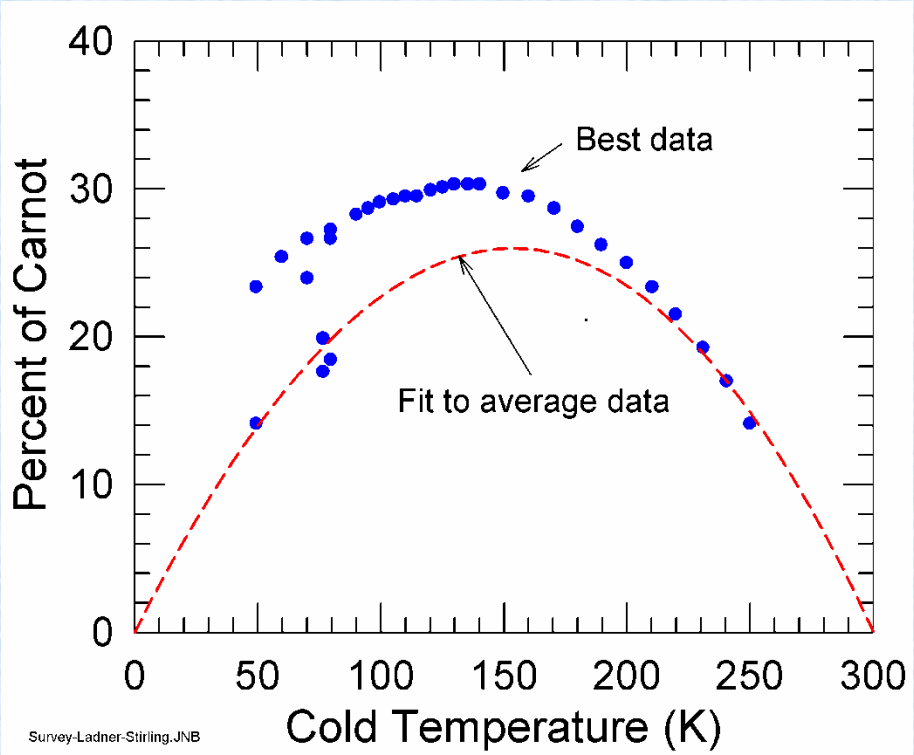
Temperature Dependence of Efficiency

Ladner Survey (>120 Data Sets)

Pulse tube



Stirling



Cryocoolers 16 (2011) p. 633-644

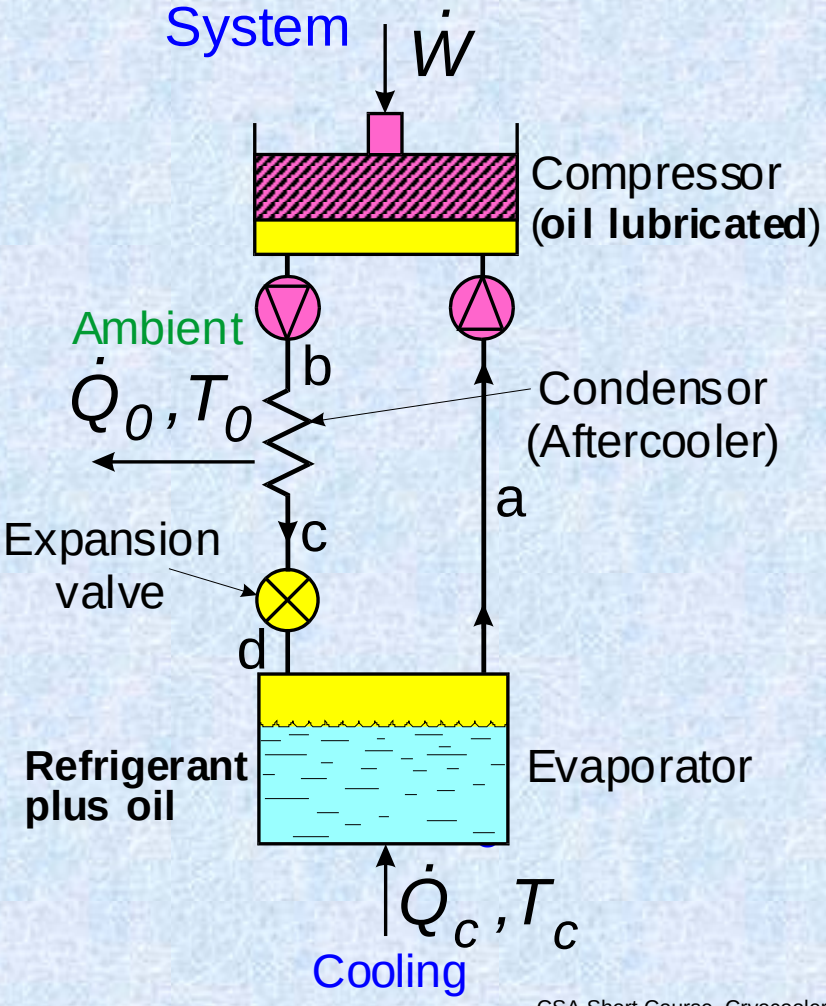
Outline

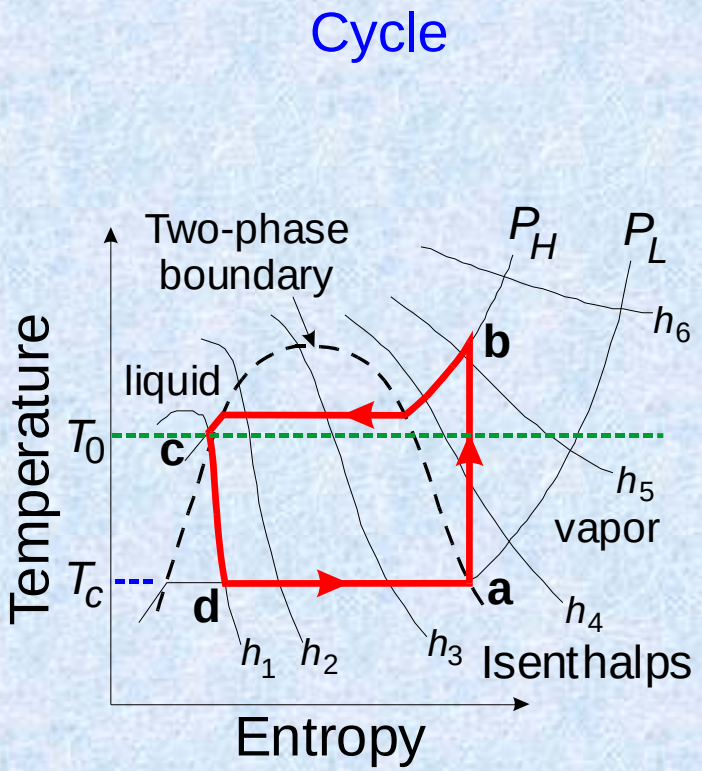
Session 1

- Information sources
- Definitions
- History
- Applications and problems
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- Cryocooler types and operating principles
- Joule-Thomson (JT) cryocoolers
- Brayton cryocoolers
- Claude systems
- Heat exchangers

Vapor-Compression Cycle

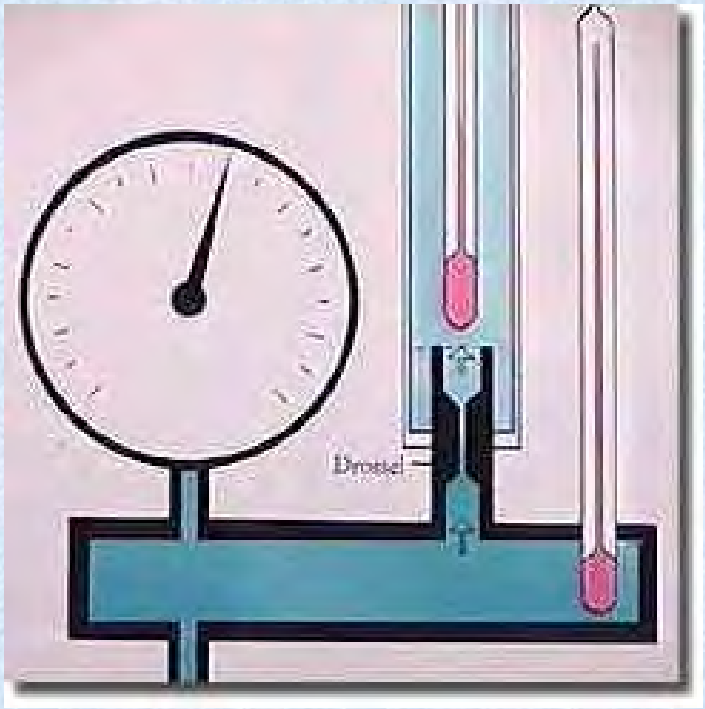
Domestic Refrigeration





Cycles18ca.cdr

Joule-Thomson Effect

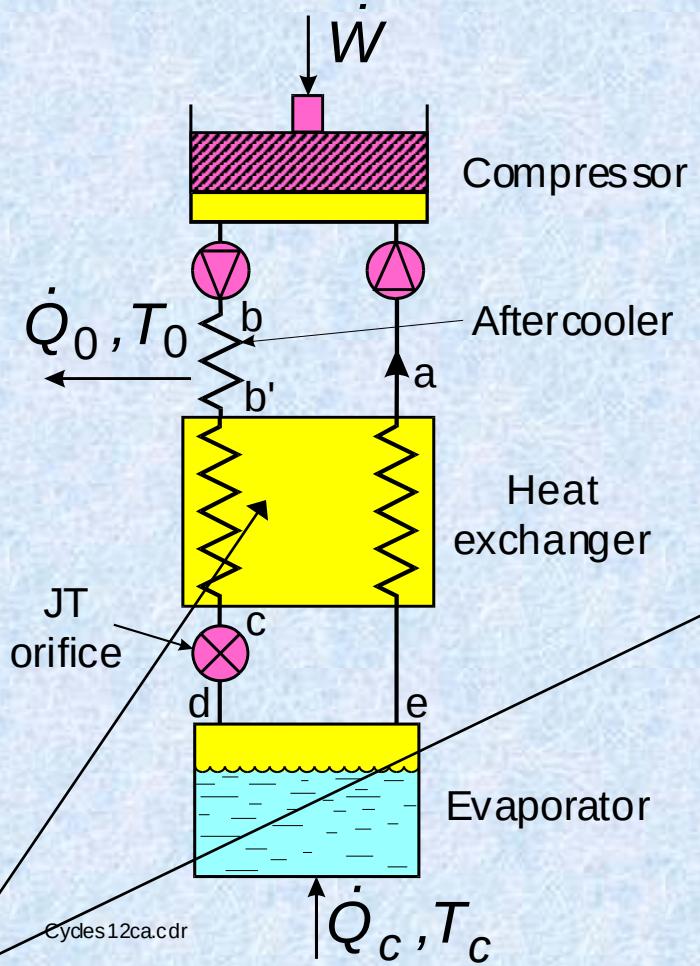


Joule-Thomson cooling (1852)

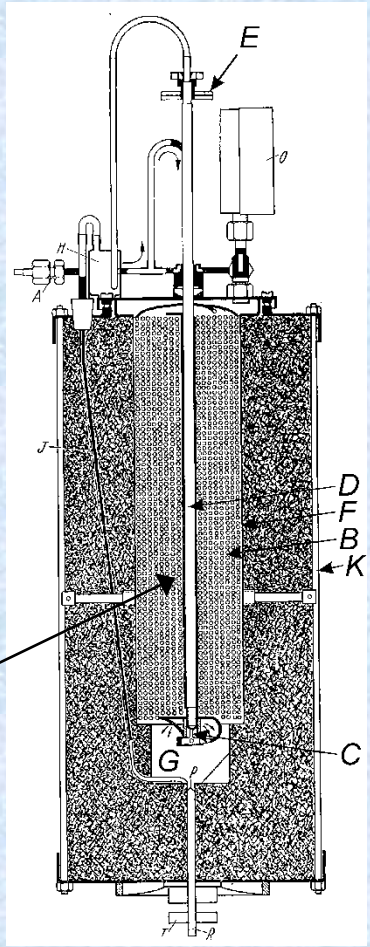
20 MPa → 0.1 MPa

300 K → 269.2 K

Recuperative heat exchanger
Siemens (1857)

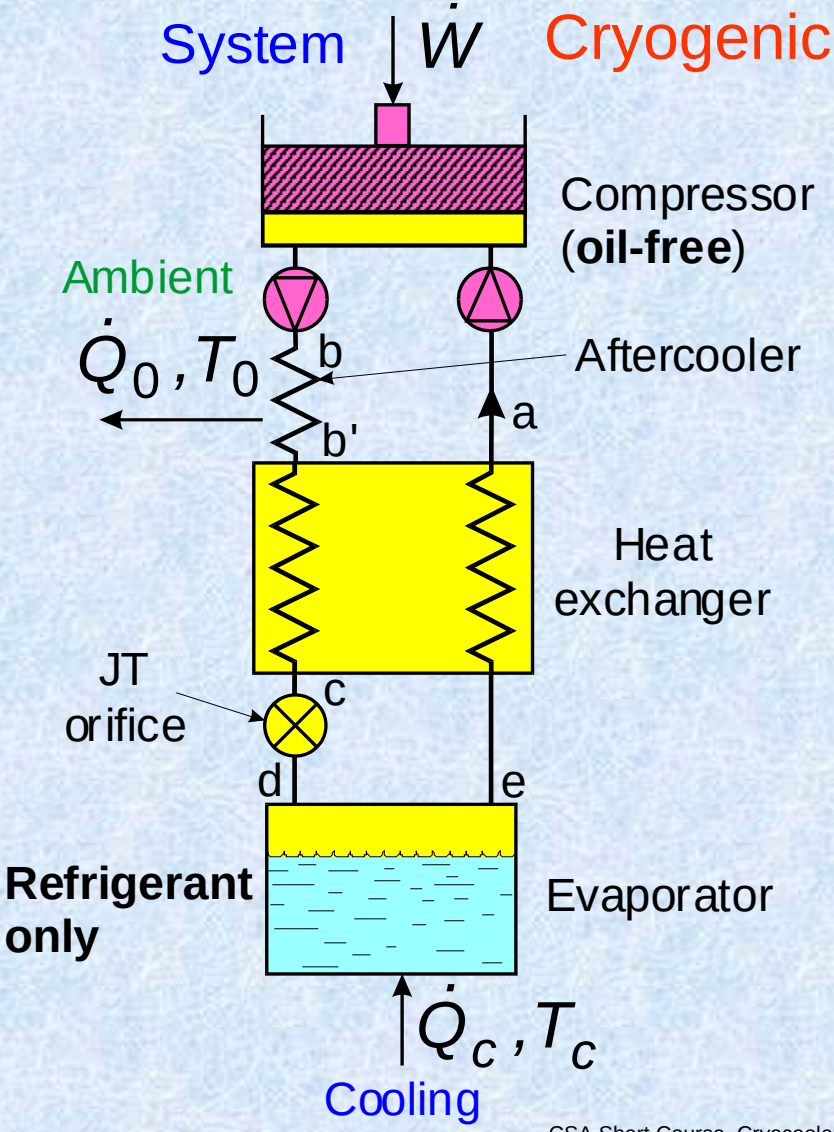


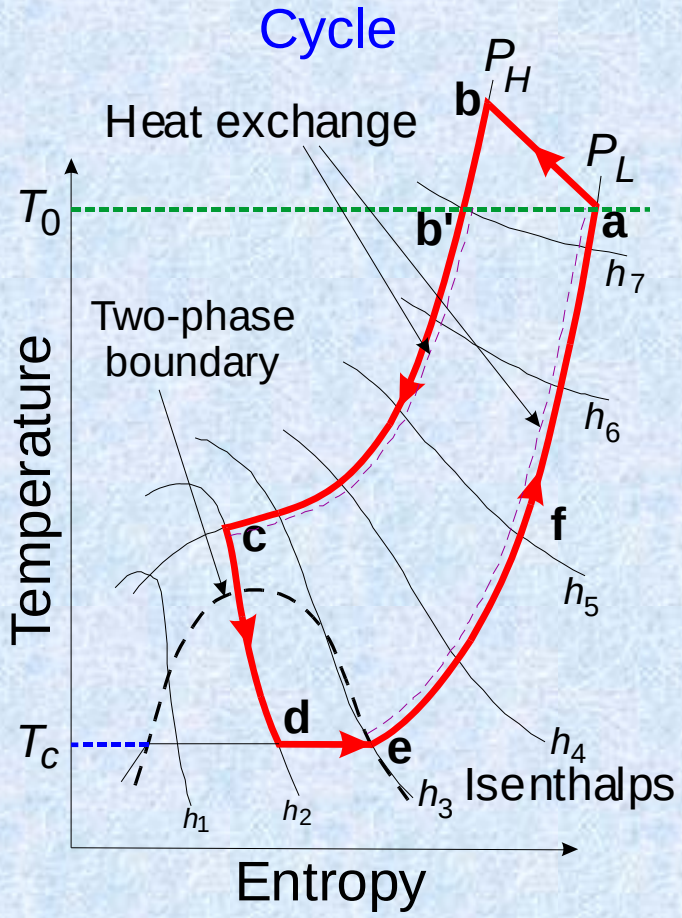
Joule-Thomson cryocooler
(Linde-Hampson)



Hampson O_2
Liquefier (1895)
20 MPa (3 L/hr)

Joule-Thomson (JT) Cycle
(Linde-Hampson Cycle)



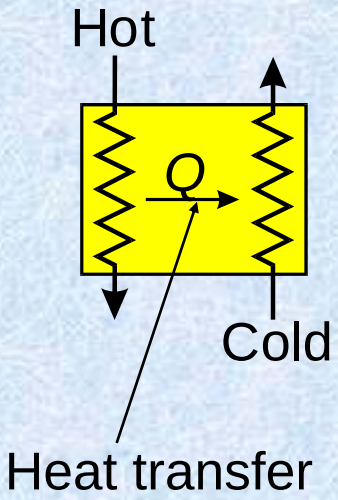


Cycles19ca.cdr

Heat Exchangers
Two Types

RECUPERATIVE

Separate channels with solid walls separatingthe continuously flowing hot and cold fluids. Fluids usually in counterflow.



Siemens: 1857

REGENERATIVE

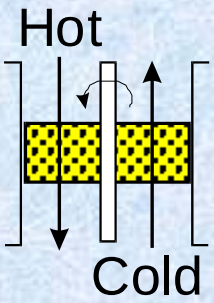
A single flow channel filled with a matrix of finely divided material subject to alternating flows of hot and cold fluids.



For continuous flow one must have two identical matrices with periodic switching

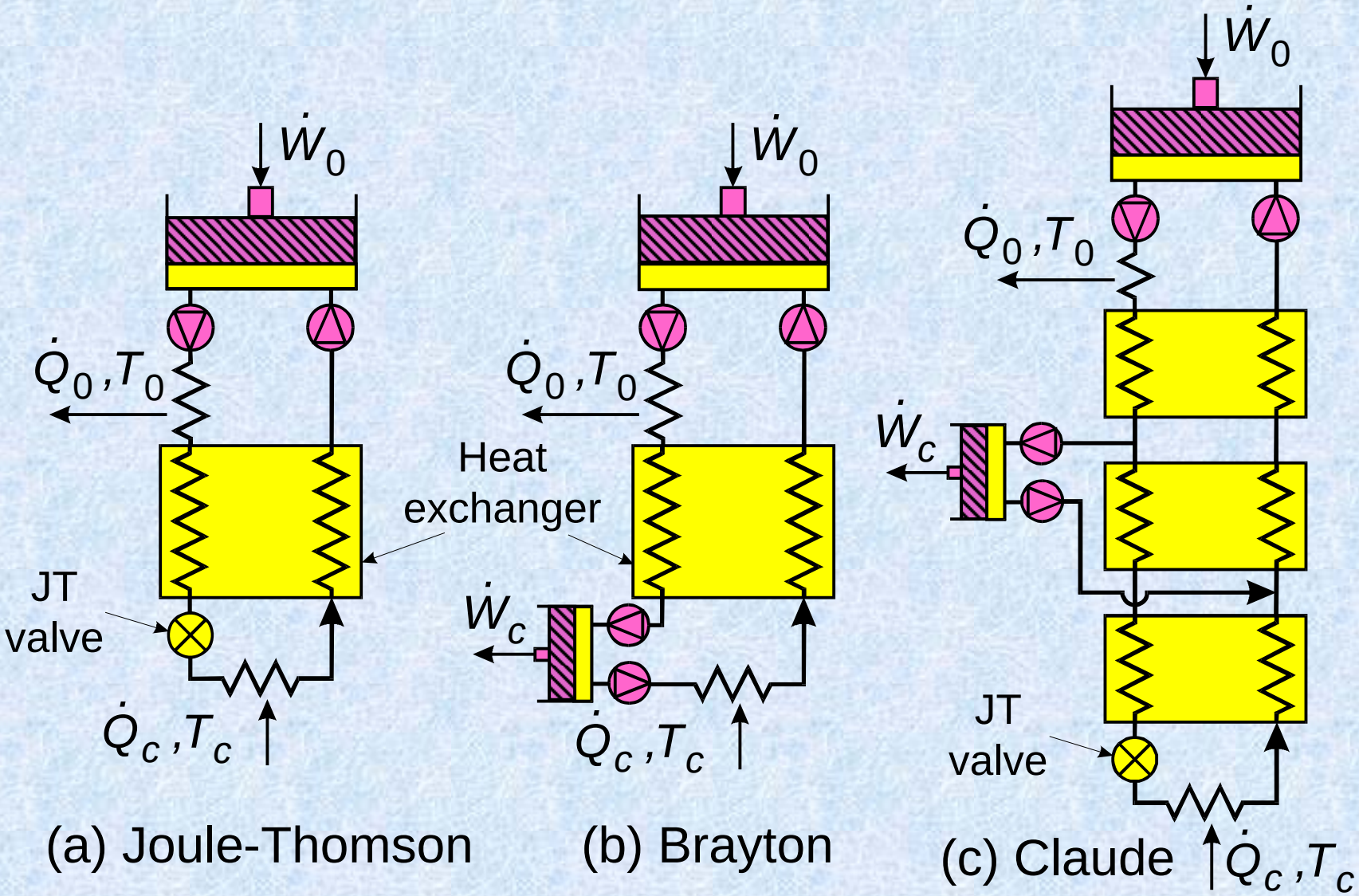
or

a disc rotating slowly (thermal wheel) and passing alternately through the hot and cold fluid ducts

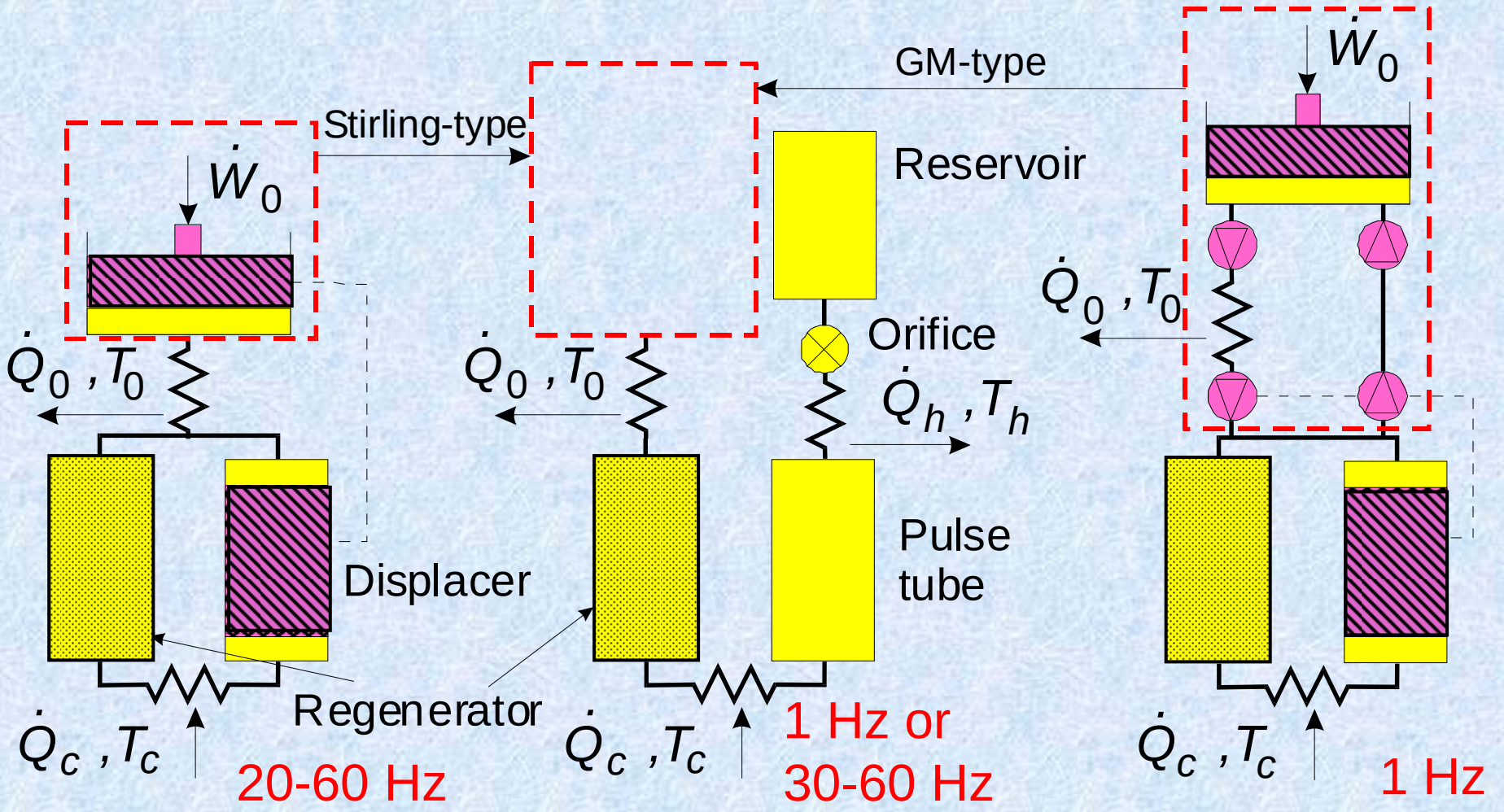


Stirling: 1816

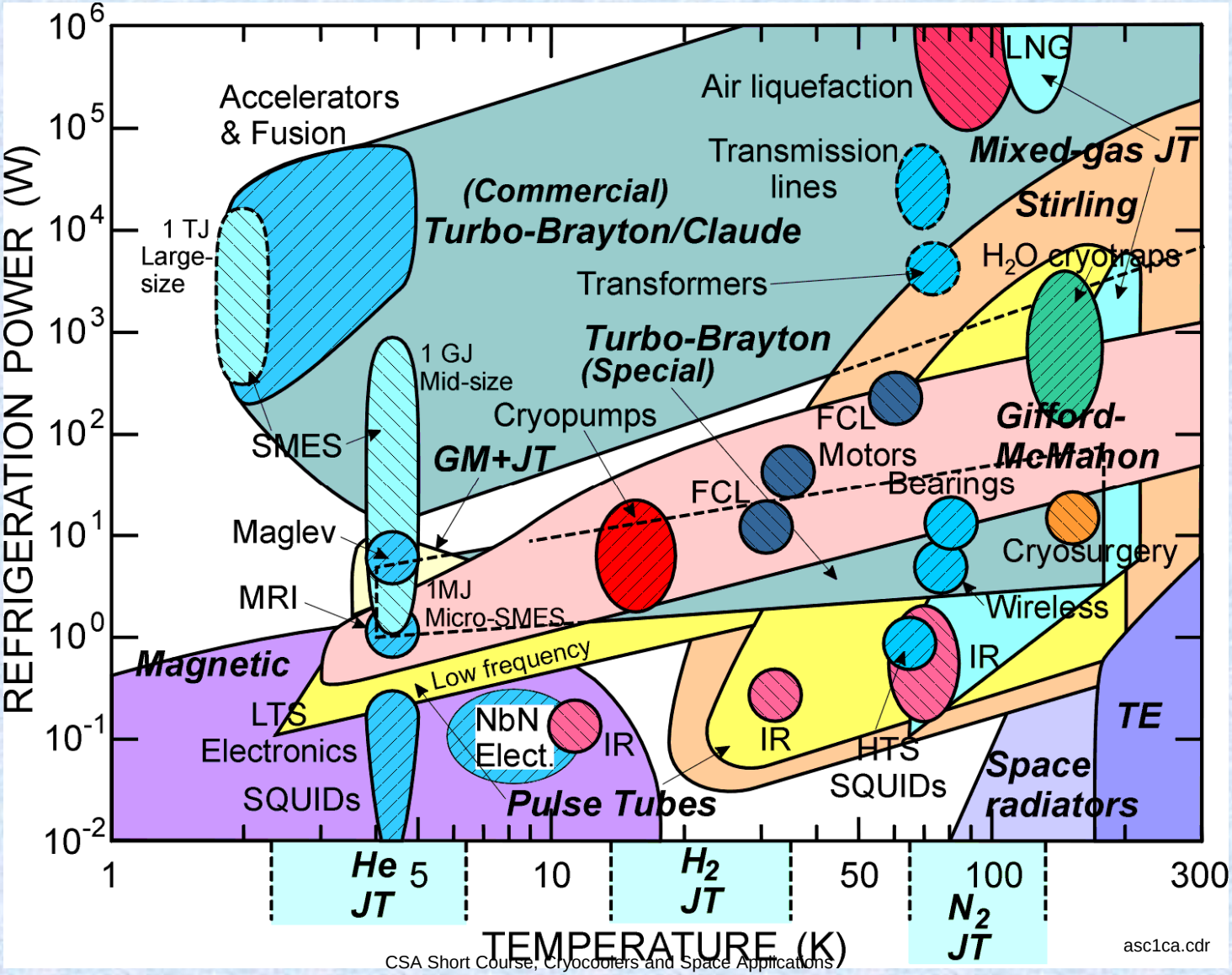
Recuperative Cryocoolers
(Steady Flow)



Regenerative Cryocoolers
(Oscillating Flow)



Cryocooler Applications and Operating Regions

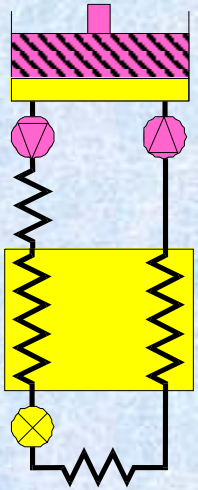


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- Heat exchangers

Joule-Thomson (JT) Cycle
(Throttle Cycle)



ADVANTAGES

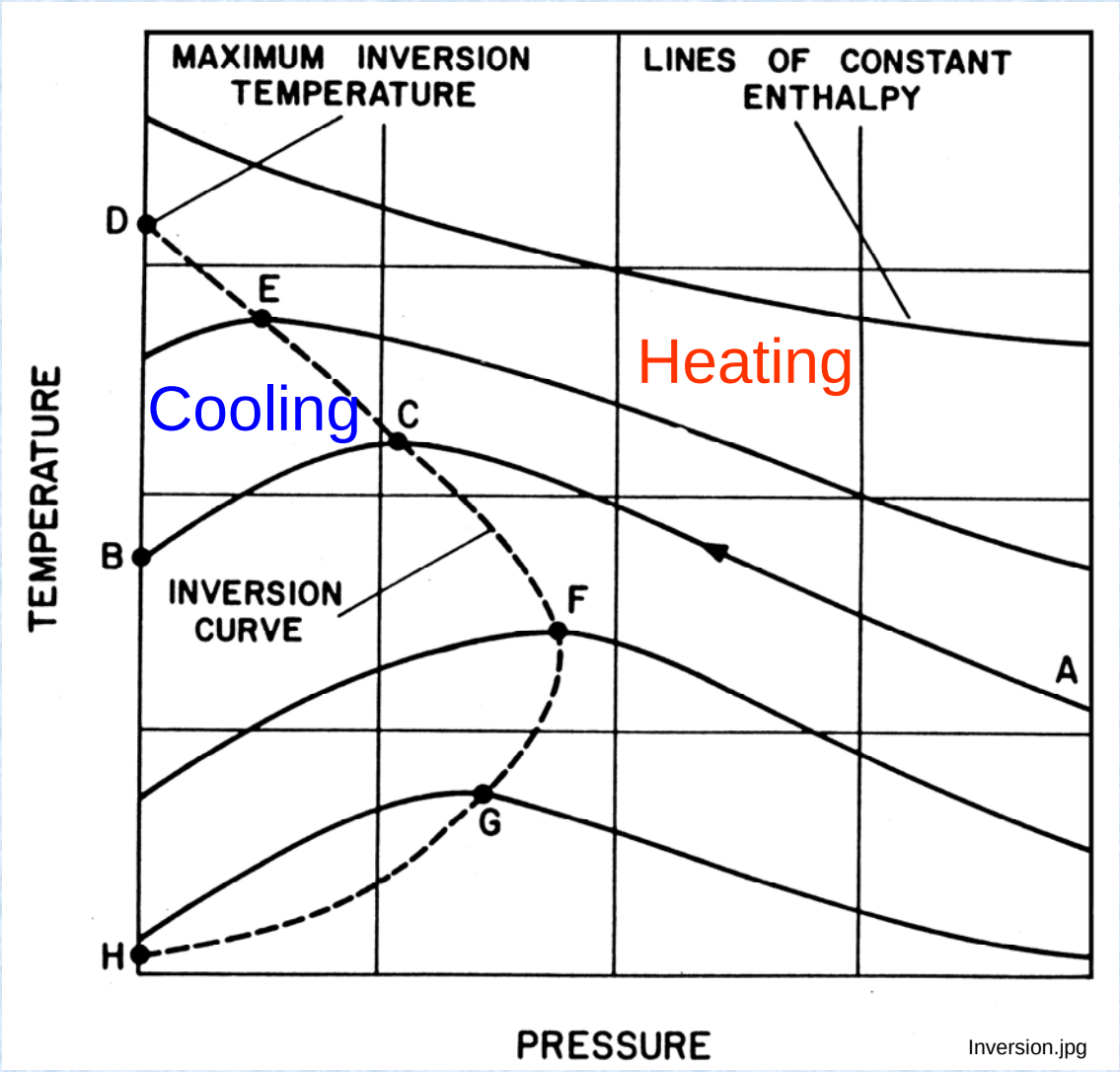
- No cold moving parts
- Steady flow (no vibration)
- Transport cold long distance
- Cold end can be miniaturized

DISADVANTAGES

- Relies on real-gas behavior
- Requires high pressures 200:1 (compr. wear)
  - 25:1 with mixed gases (T > 70 K)
- Small orifice susceptible to clogging
- Low efficiency below 90 K

lcec14c.cdr

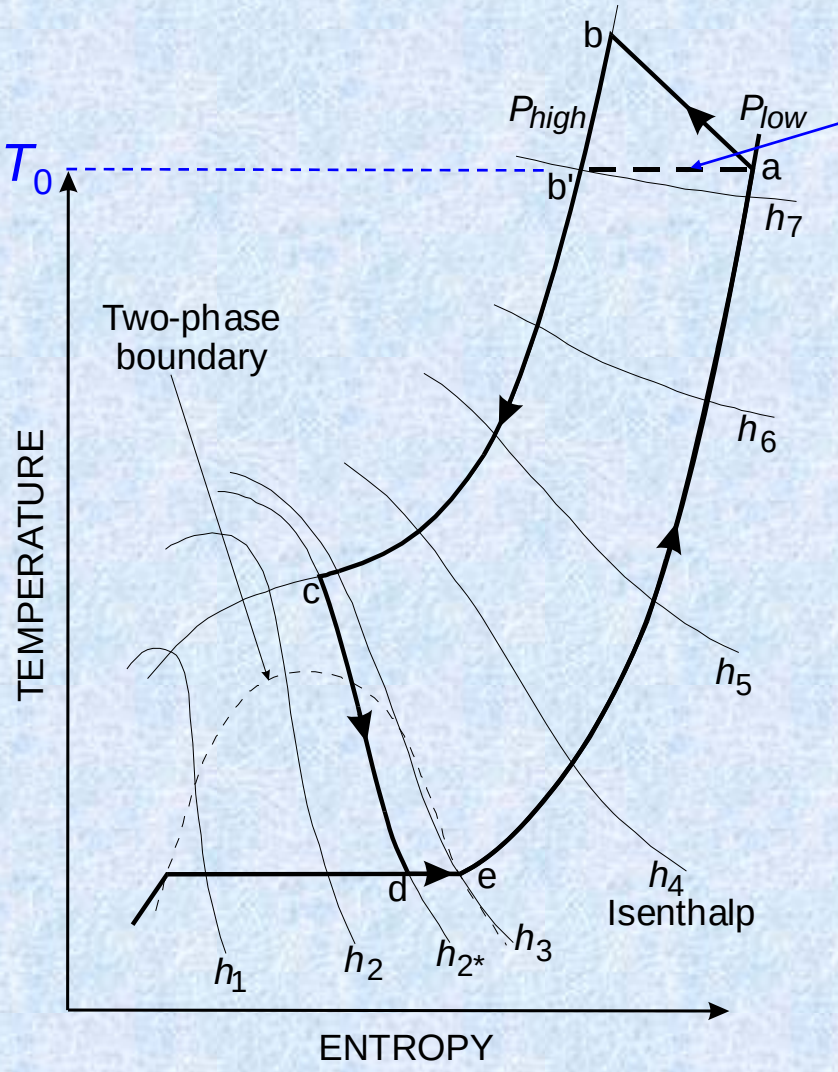
Inversion Curve



Inversion.jpg

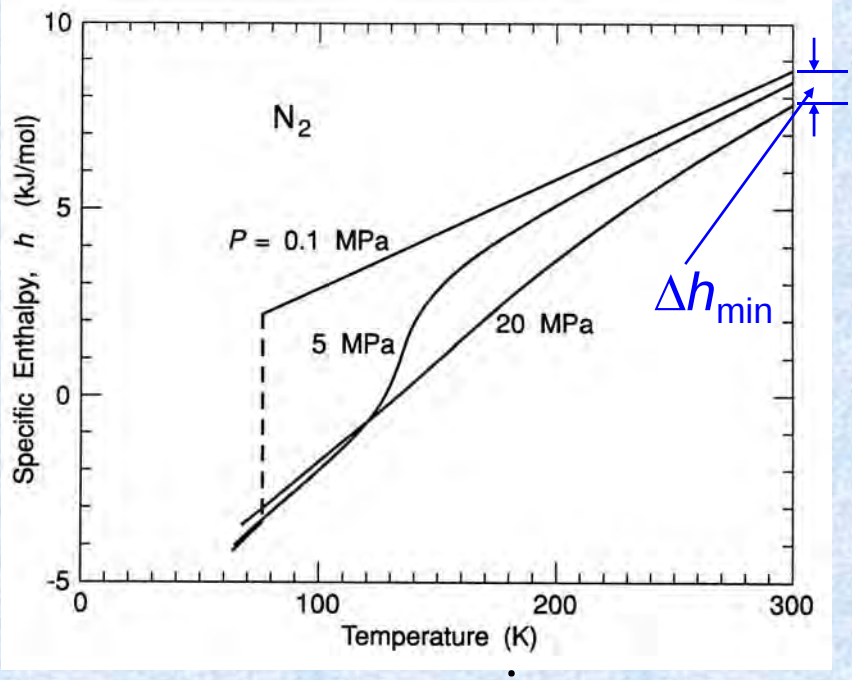
Gas	Max. Inversion Temperature (K)
Ammonia	1994
Carbon dioxide	1500
Methane	939
Oxygen	761
Argon	794
Nitrogen	621
Air	603
----------	300 K
Neon	250
Hydrogen	205
Helium-4	40

Analysis of the JT Cycle



Minimum power input (Reversible):

$$\dot{W} = \dot{m}\Delta g_0 = \dot{m}\Delta(h - T_0 s)$$



Refrigeration power: $\dot{Q}_c = \dot{m}\Delta h_{\min}$

Ideal COP: $\dfrac{\dot{Q}_c}{\dot{W}} = \dfrac{\Delta h_{\min}}{\Delta g_0}$

Ideal Efficiency of JT Cycle

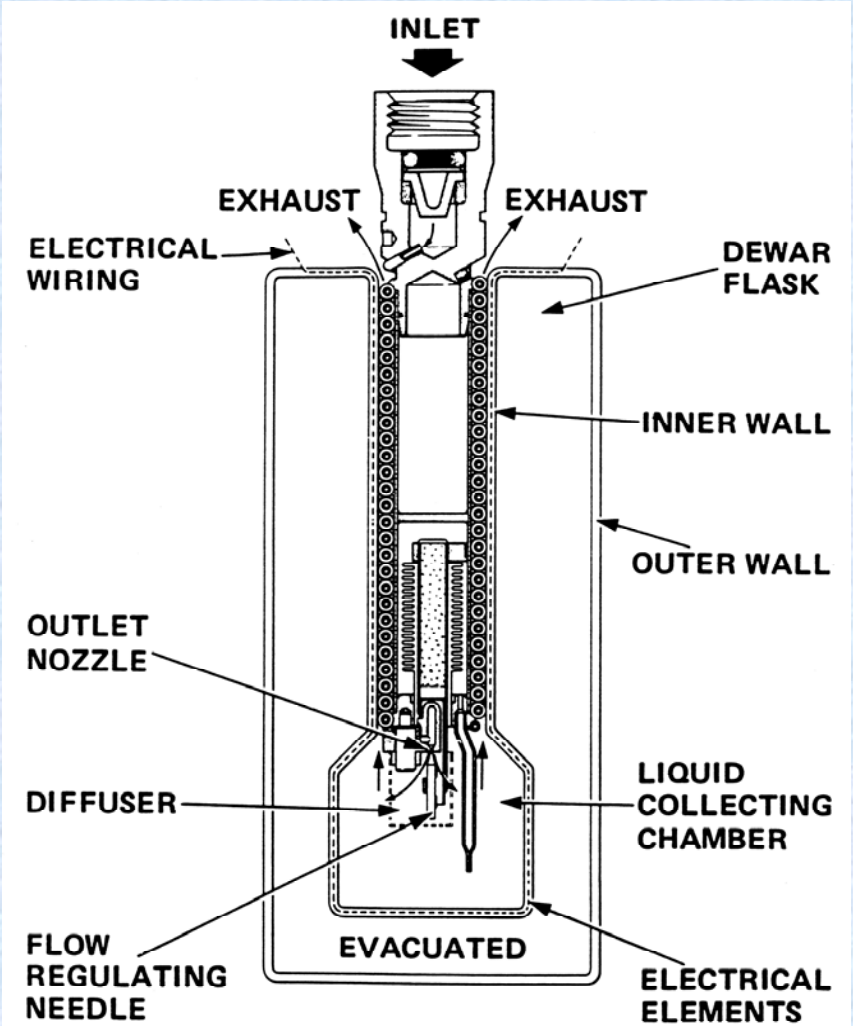
Pure Fluids

P_L = 0.1013 MPa = 1 atm

Gas	P_H (MPa)	T_c (K)	COP_{ideal}	η %Carnot
N_2	5	77	0.0296	8.5
N_2	20	77	0.0682	19.6
Ar	5	87	0.0381	9.3
Ar	20	87	0.1014	24.7
O_2	5	90	0.0404	9.4
O_2	20	90	0.1074	25.0

Miniature Joule-Thomson Cryostat

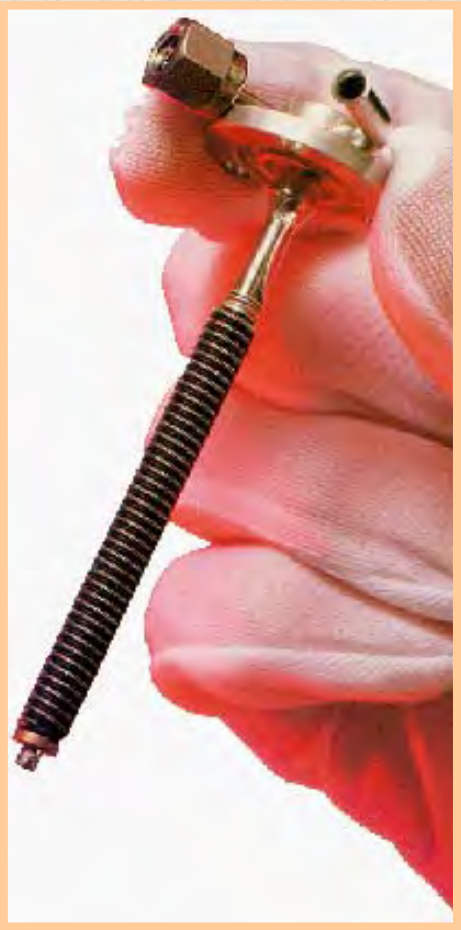
- Open cycle
  - Gas vents
  - Gas supply from cylinder
  - Fast cooldown



Hymatic Engineering Co. Ltd JT cryostat.jpg

Miniature Joule-Thomson Cryocoolers



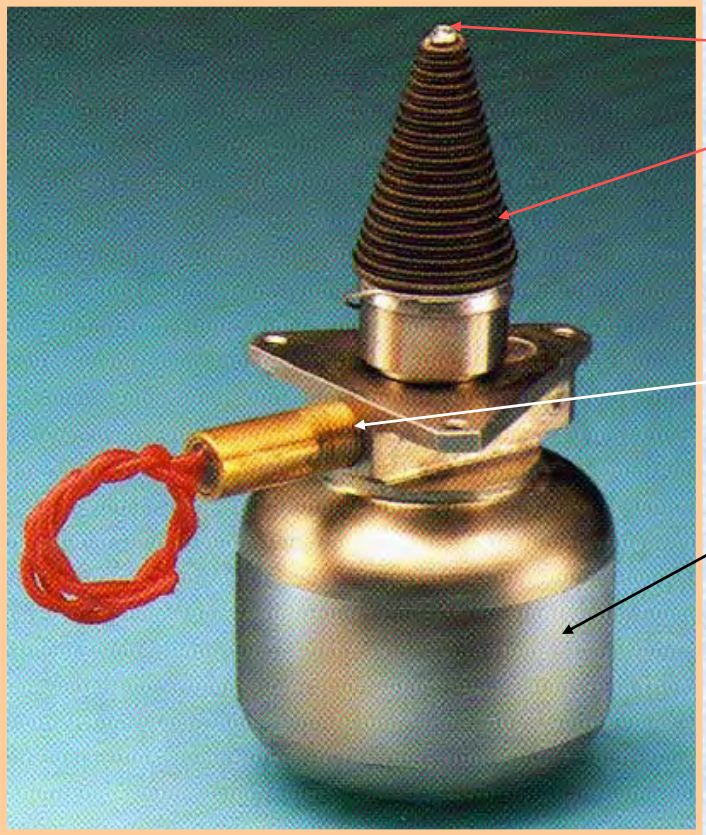


- Used for direct cooling of infrared sensors
- First direct coupling of cryocooler to device (late 1950s)

Air Products and APD Cryogenics

miniaturejt.cdr

Joule-Thomson Cryocooler
(Open Cycle)



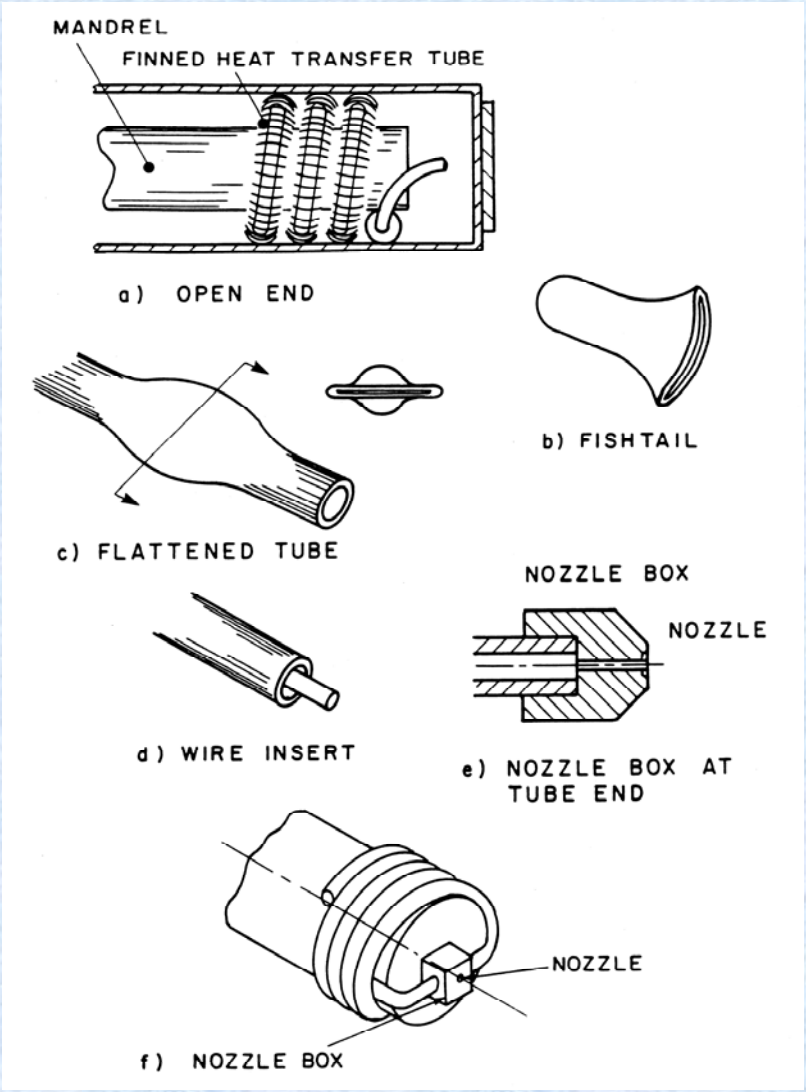
Requires delivery of gas from separate compressor

- For cooling infrared sensors
- Missile guidance
- 3 s cool down

Carleton

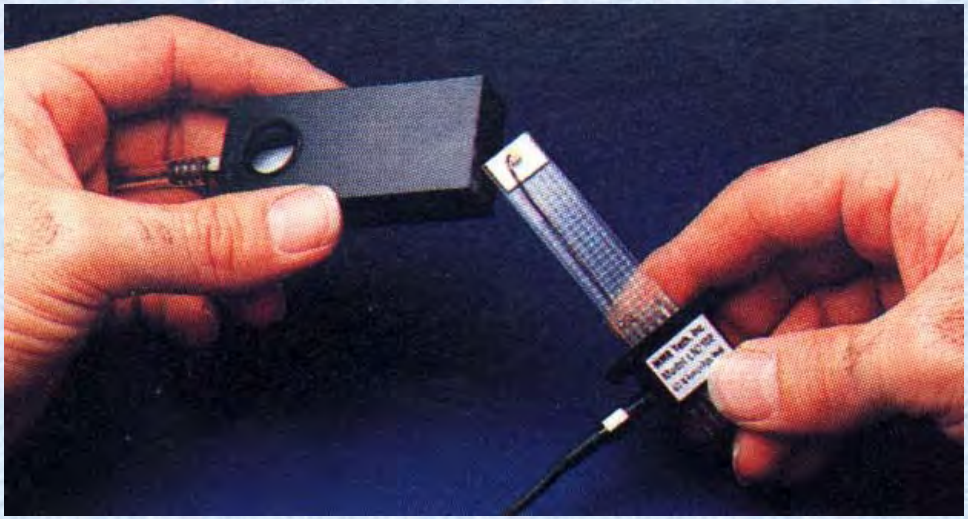
openjt.cdr

Types of Fixed JT Nozzles

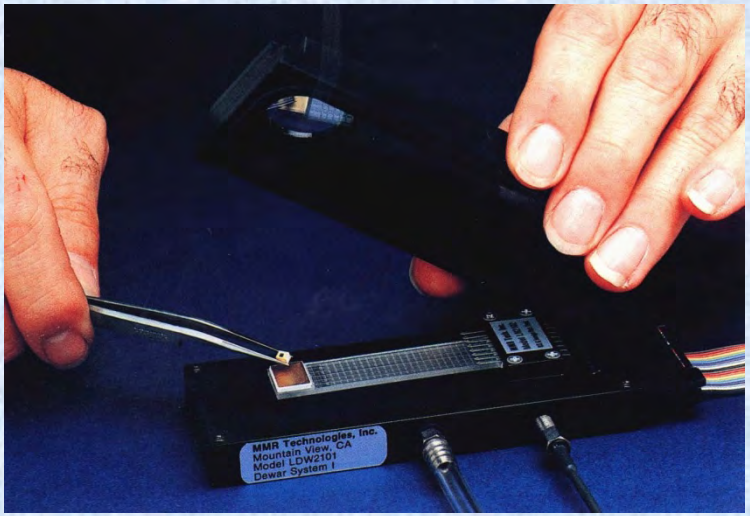


Miniature JT Cryocoolers in Early 1980s

Etched glass heat exchangers



MMRJT2.jpg

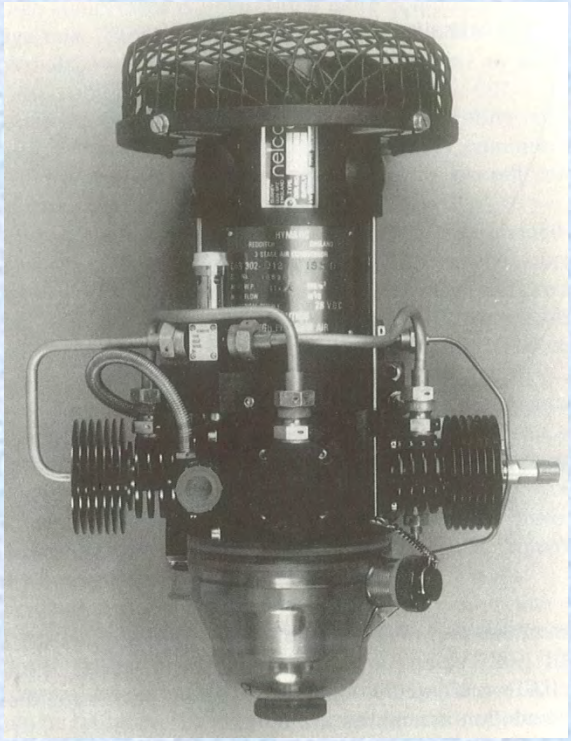


MMRJT1.jpg

- Open cycle
- Etched glass flow channels
- Laboratory experiments
- MMR
- N_2 gas from cylinders

Joule-Thomson Compressors

High pressure to about 15 MPa (150 bar) in three stages



Oil-lubricated; Hymatic (1960s)



- Oil-free; RIX Microboost (modern)
- ½ hp (370 W) input
- 1.4 SLPM (3.4% isothermal efficiency)

Sorption Compressors

- ***PHYSISORPTION***

  Best with high boiling point gases

  Microporous carbon (charcoal) works best

  Uses heat source

- ***CHEMISORPTION***

  Good for oxygen and hydrogen

  Reached 10 K in periodic mode (BETSCE)

  Uses high temperature heat source

Icec18bw.cdr

165 K Microscale JT Cooler

Ethylene Refrigerant

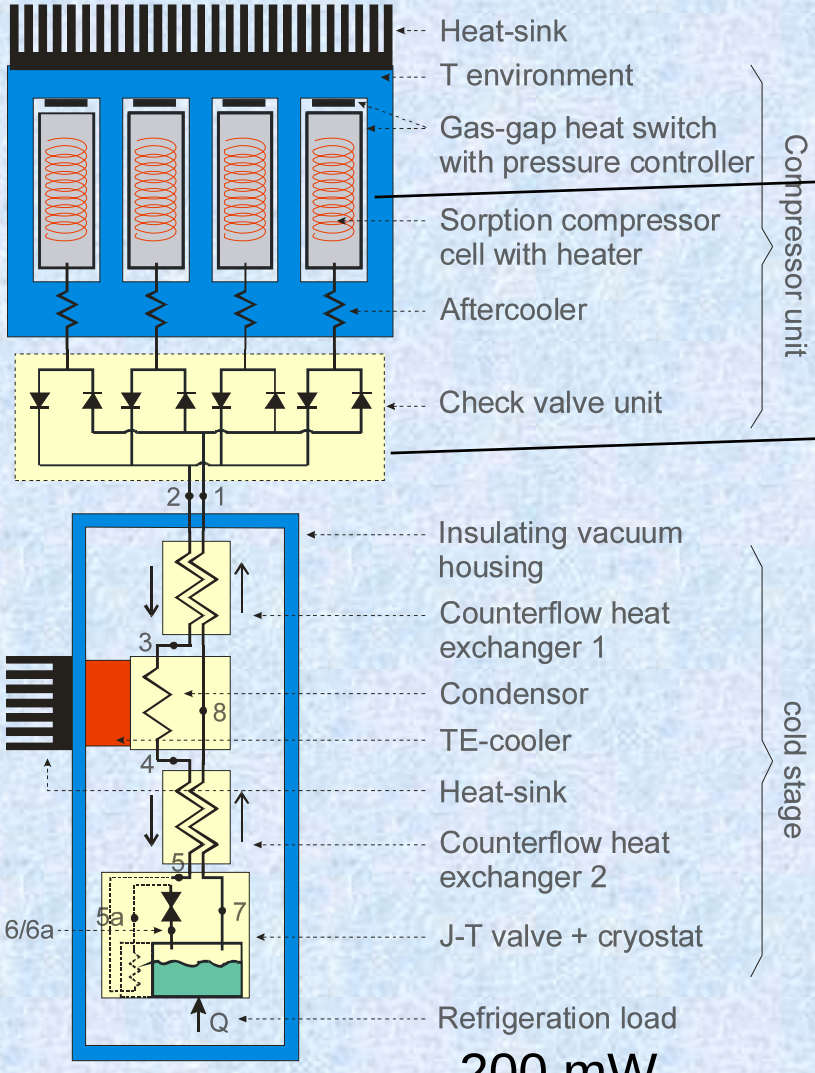


Fig 1.cdr

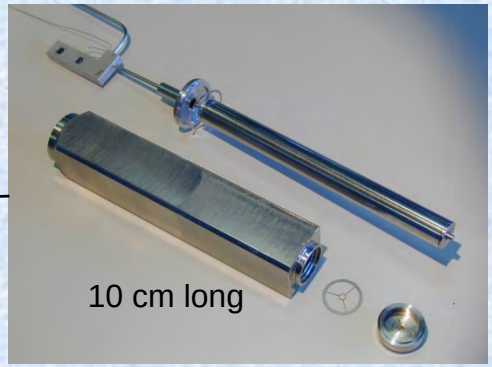
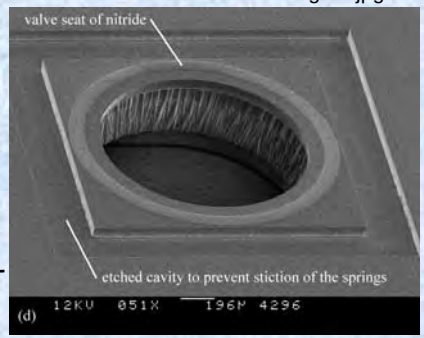


Fig 3b.jpg

Sorption Compressor

Fig 5d.jpg



MEMS Valves

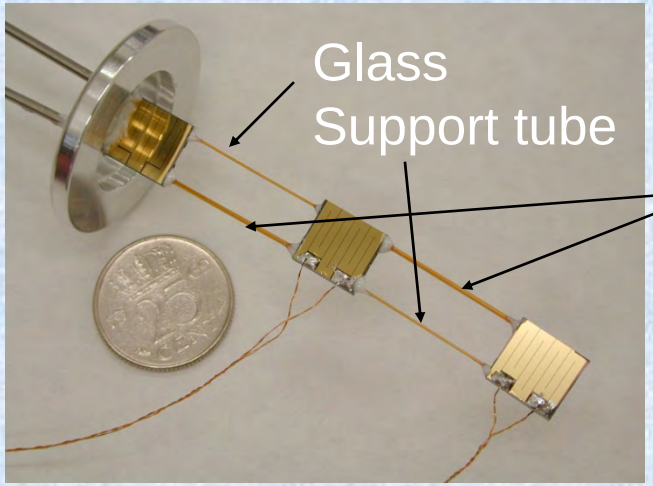


Fig 9b.jpg

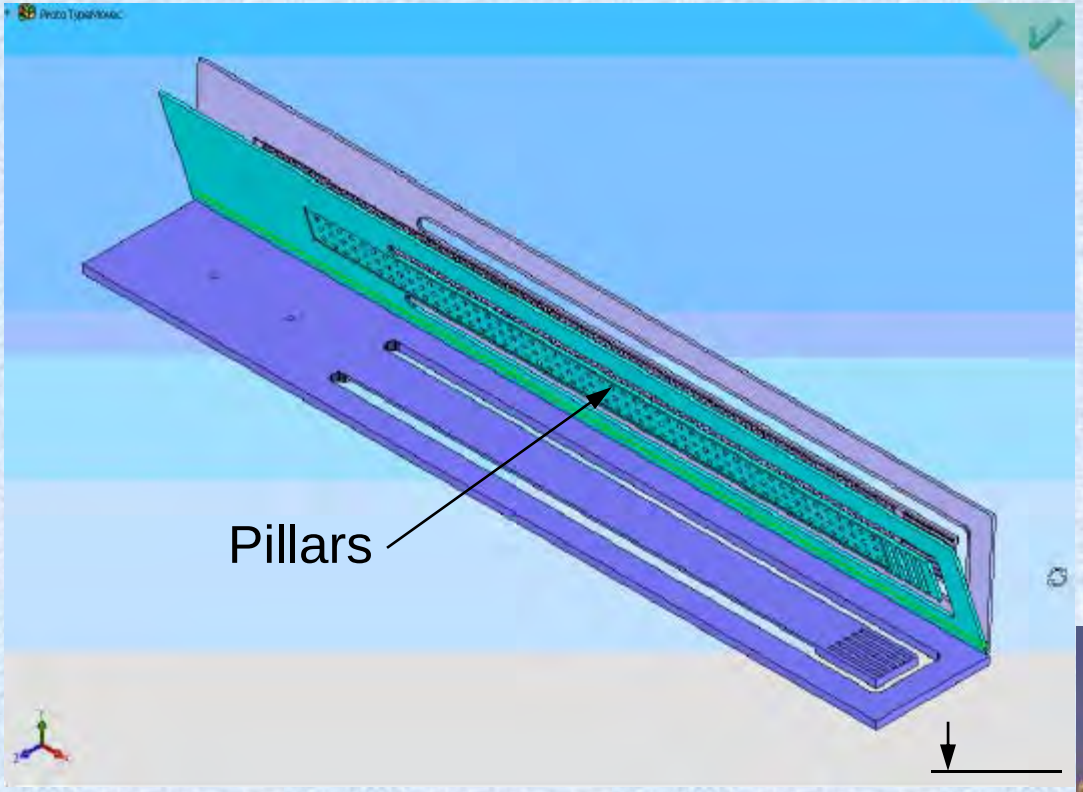
Coaxial Glass Tube HX

ID/OD:
0.1/0.36
0.53/0.67

Courtesy: J. F. Burger, Univ. Twente
ICC11 (2000), pp. 551-560

200 mW

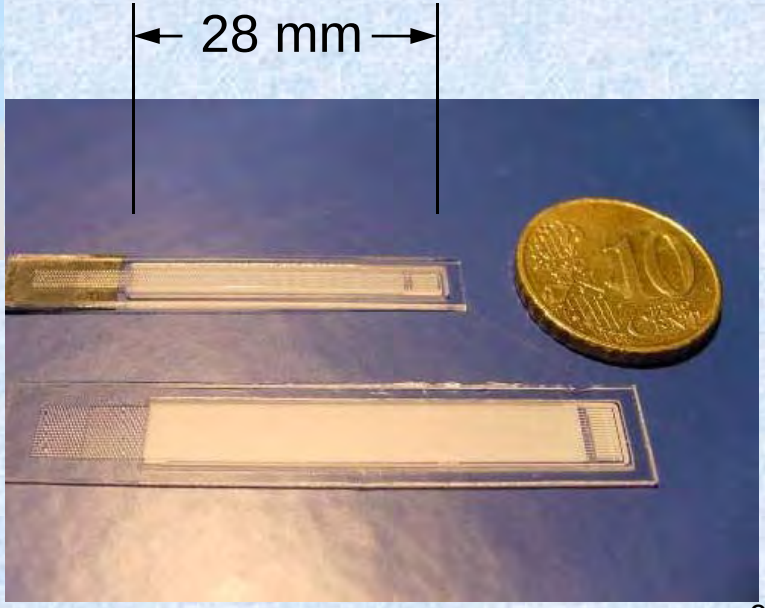
Sorption Microcryocooler (JT System)



Glass Heat Exchanger

Specs

- Gas: N_2 36 μmol/s
- P_H = 80 bar
- P_L = 6 bar
- T_c = 96 K
- Q_{net} = 10 mW

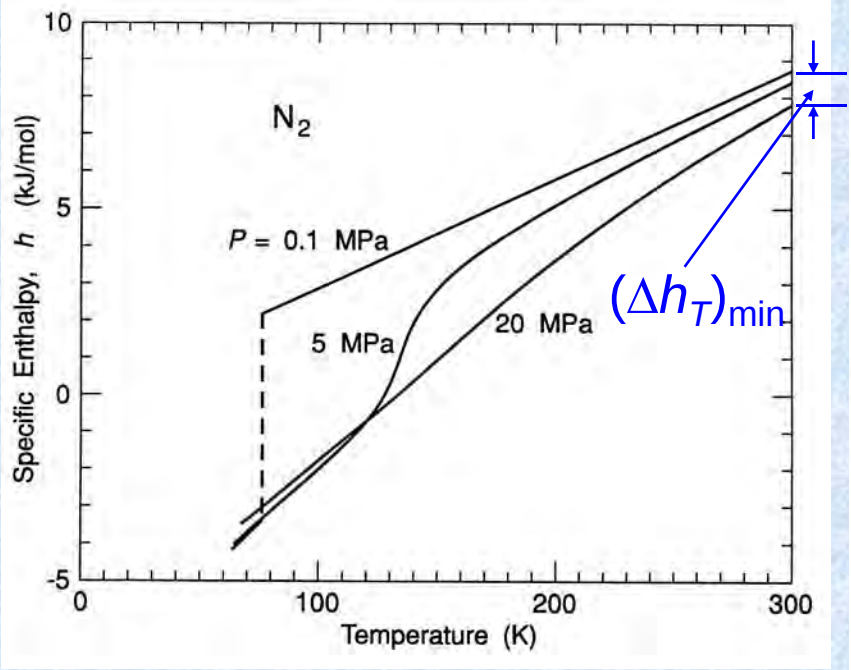


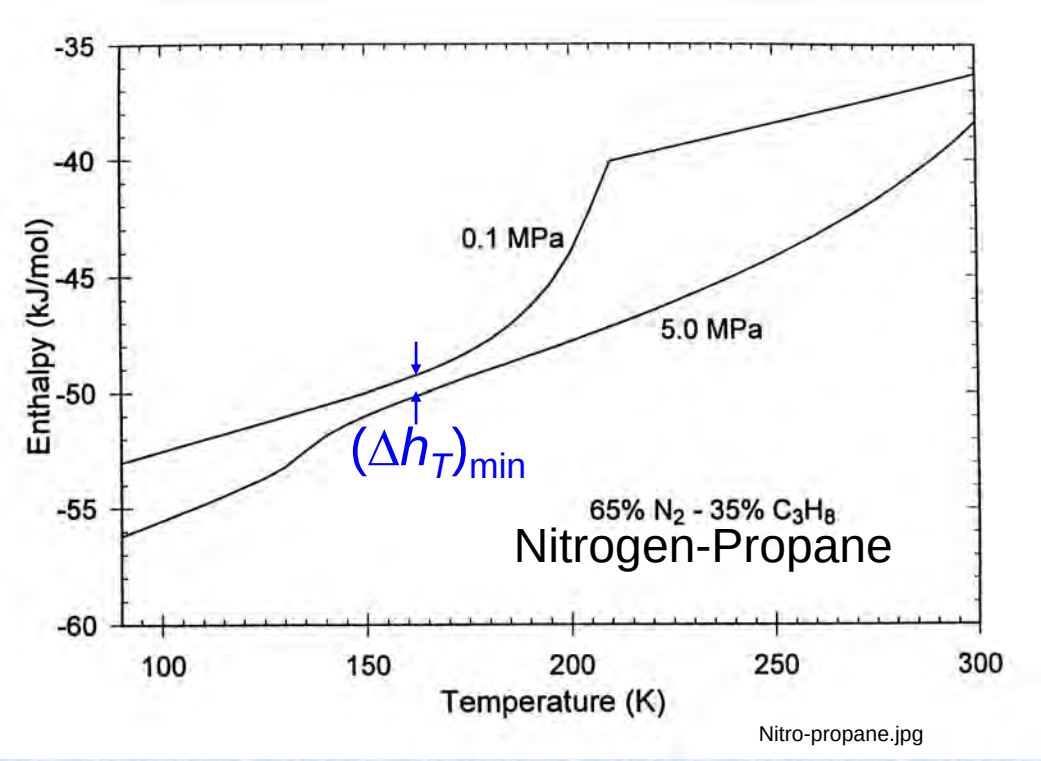
Univ. of Twente

Courtesy: Marcel Ter Brake (5/22/2006)

Enthalpy of Mixed Refrigerants

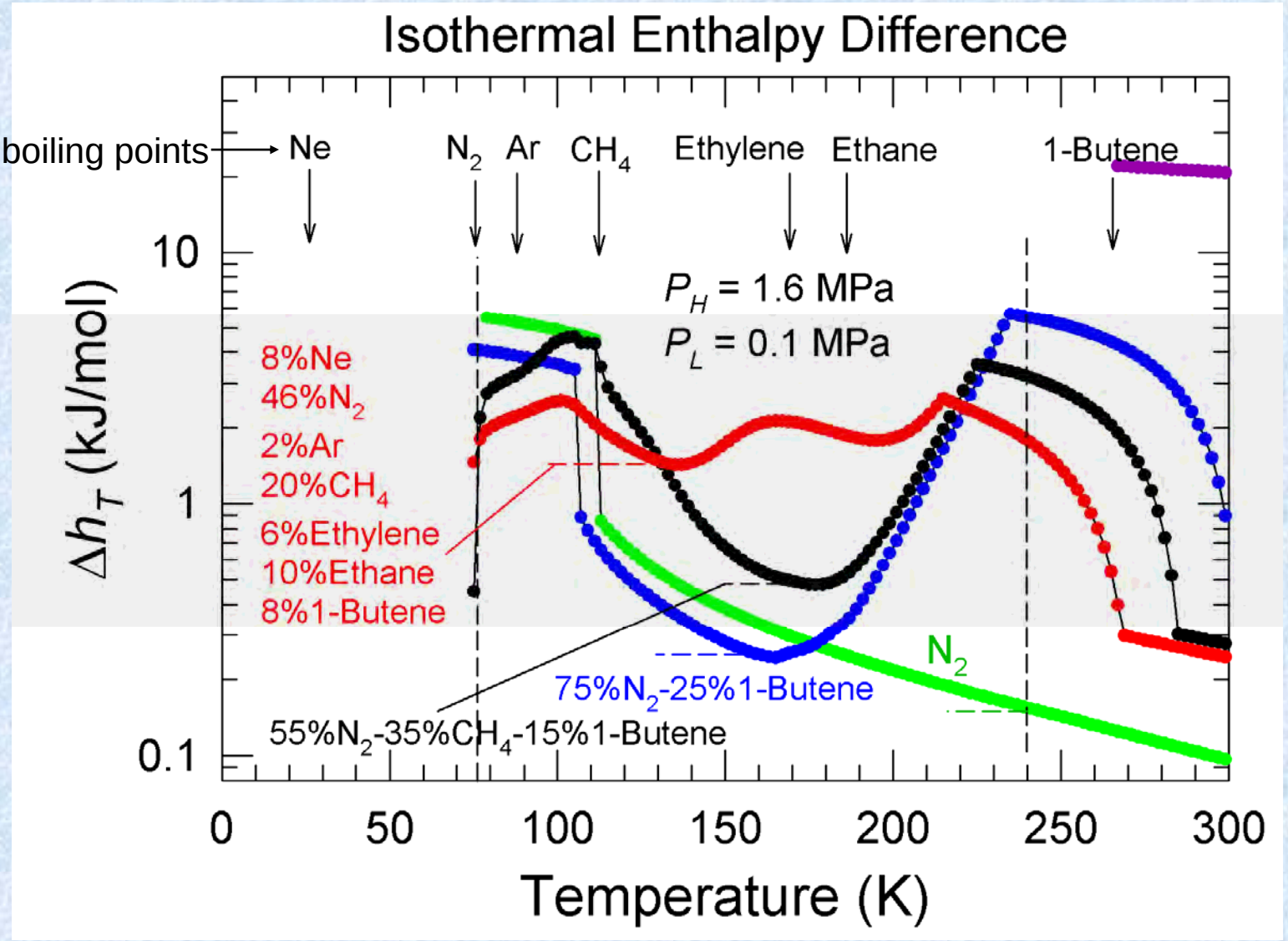
Refrigeration power: $\dot{Q}_c = \dot{m}(\Delta h_T)_{\min}$





Nitro-propane.jpg

Mixed Refrigerants Comparison



Comparison of Refrigerants in JT Cycle

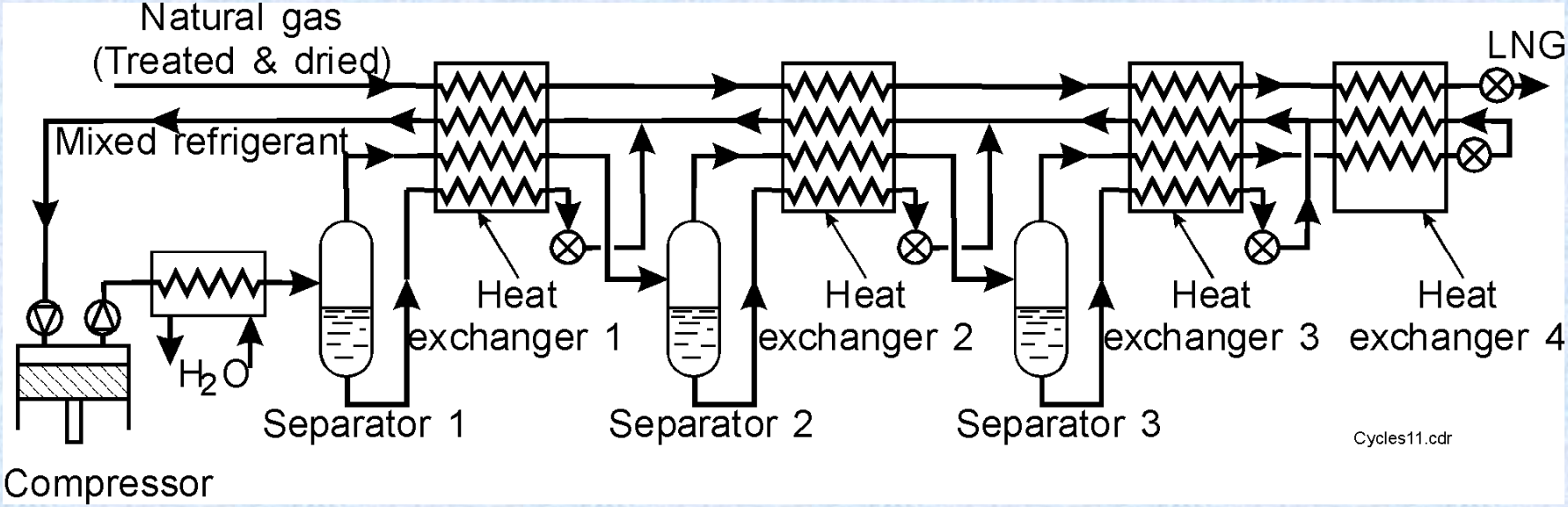
P_L = 0.10 MPa = 1 bar; T_c = 75 K, T_h = 240 K

Compressor at 300 K

Gas	P_H (bar)	P_H (MPa)	$(\Delta h_T)_{min}$ (kJ/mol)	COP_{ideal}	η %Carnot
N_2	50	5	0.468	0.0480	14.4
N_2	25	2.5	0.232	0.0289	8.7
N_2	16	1.6	0.146	0.0211	6.3
Mixture 7-comp	16	1.6	1.576	0.228	68.4

Mixed-Refrigerant Cascade (MRC) Cycle

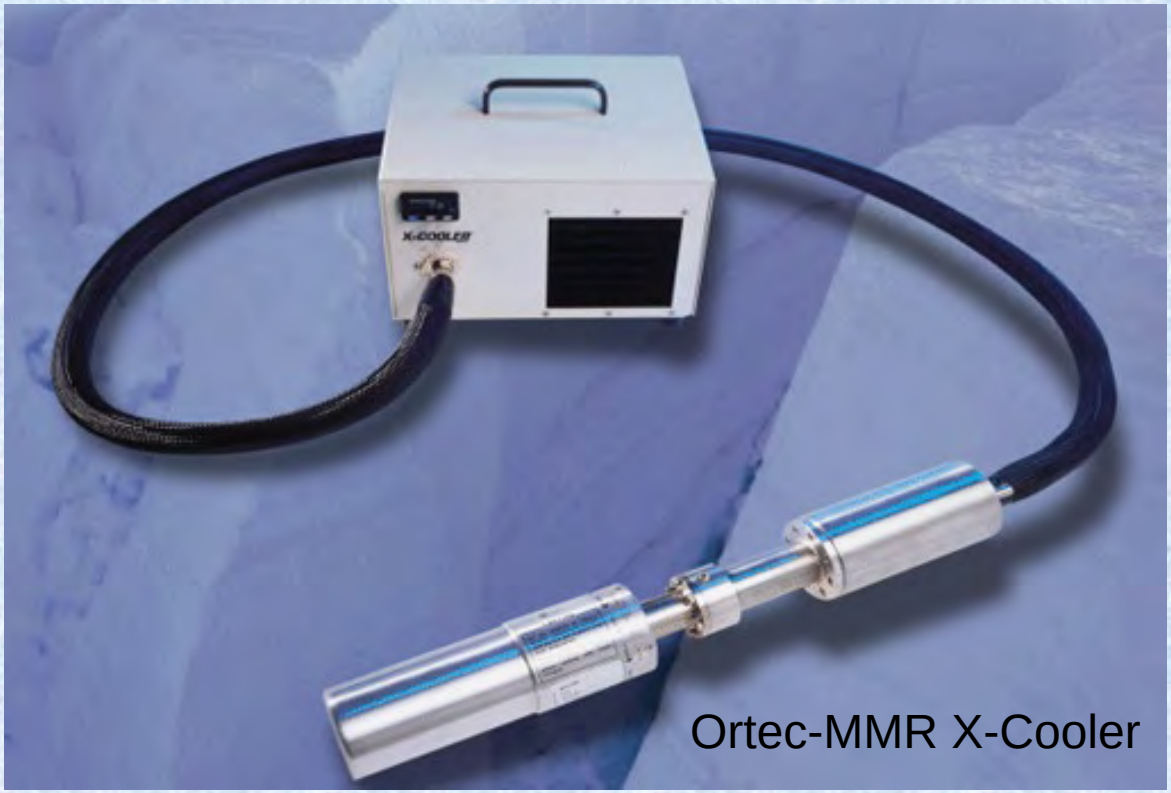
Use of Phase Separators



Often used for liquefaction of natural gas

Sometimes called Kleemenko Cycle

High Purity Germanium γ-Ray Detector



Ortec-MMR X-Cooler

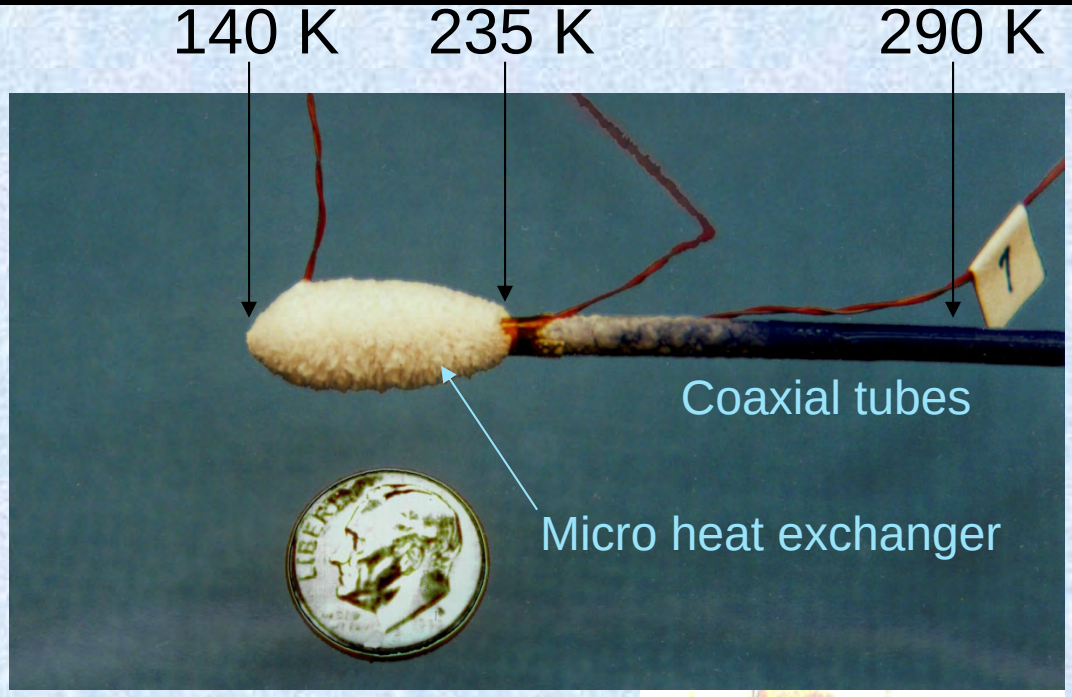
Courtesy: MMR

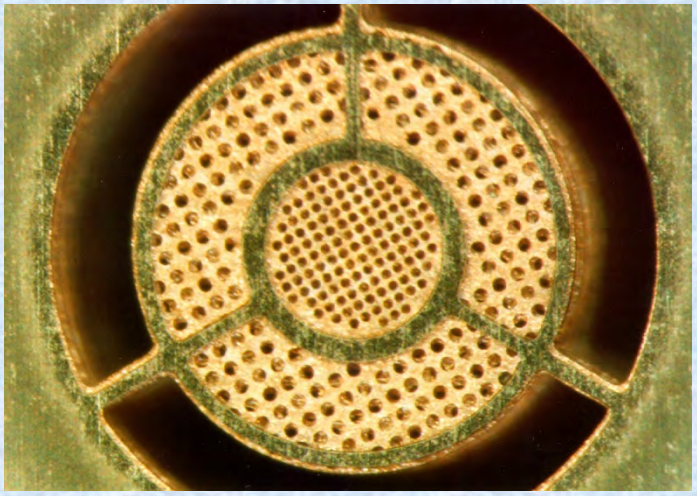
- Kleemenko mixed-refrigerant cryocooler
- 5 W @ 80 K; 300 W input power
- >900 in field use (2006)
- >7 yr continuous operation
- 11.4 kg

Gamma-ray ortec-mmr xcooler-crop.jpg

With phase separator

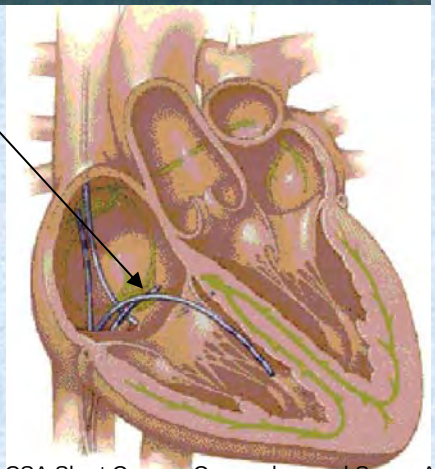
Mixed-Refrigerant Cryogenic Catheter
Early 1990s NIST/CryoGen/CryoCor Development







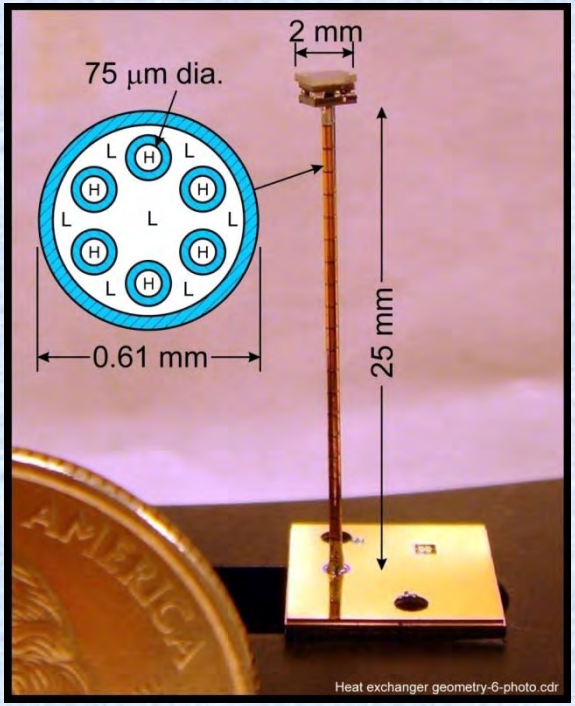
- 3.0 mm dia. catheter
- Heart arrhythmia
- 85 K in air
- Distributed cooling
- HTS electronics



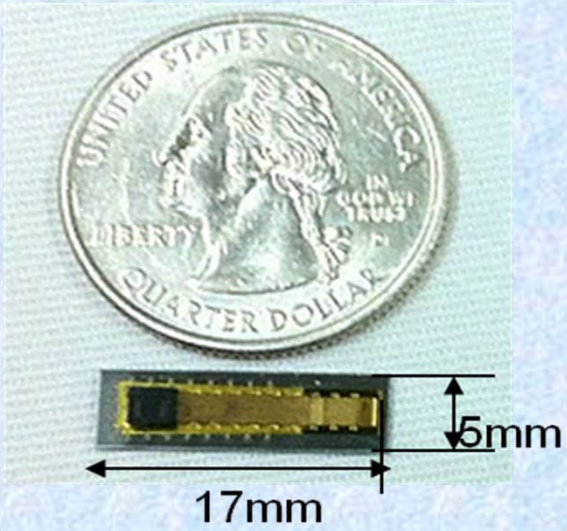
- Micro heat exchanger
- 2.5 mm diameter

DARPA Microcryocooler Program

- University of Colorado/NIST
  - Cold head: (1) Glass capillaries, (2) Planar polyimide
  - Refrigerant: Mixed; T_L = 180 K, 10 mW
  - Compressor: Miniature Stirling with valves; P_r = 6:1

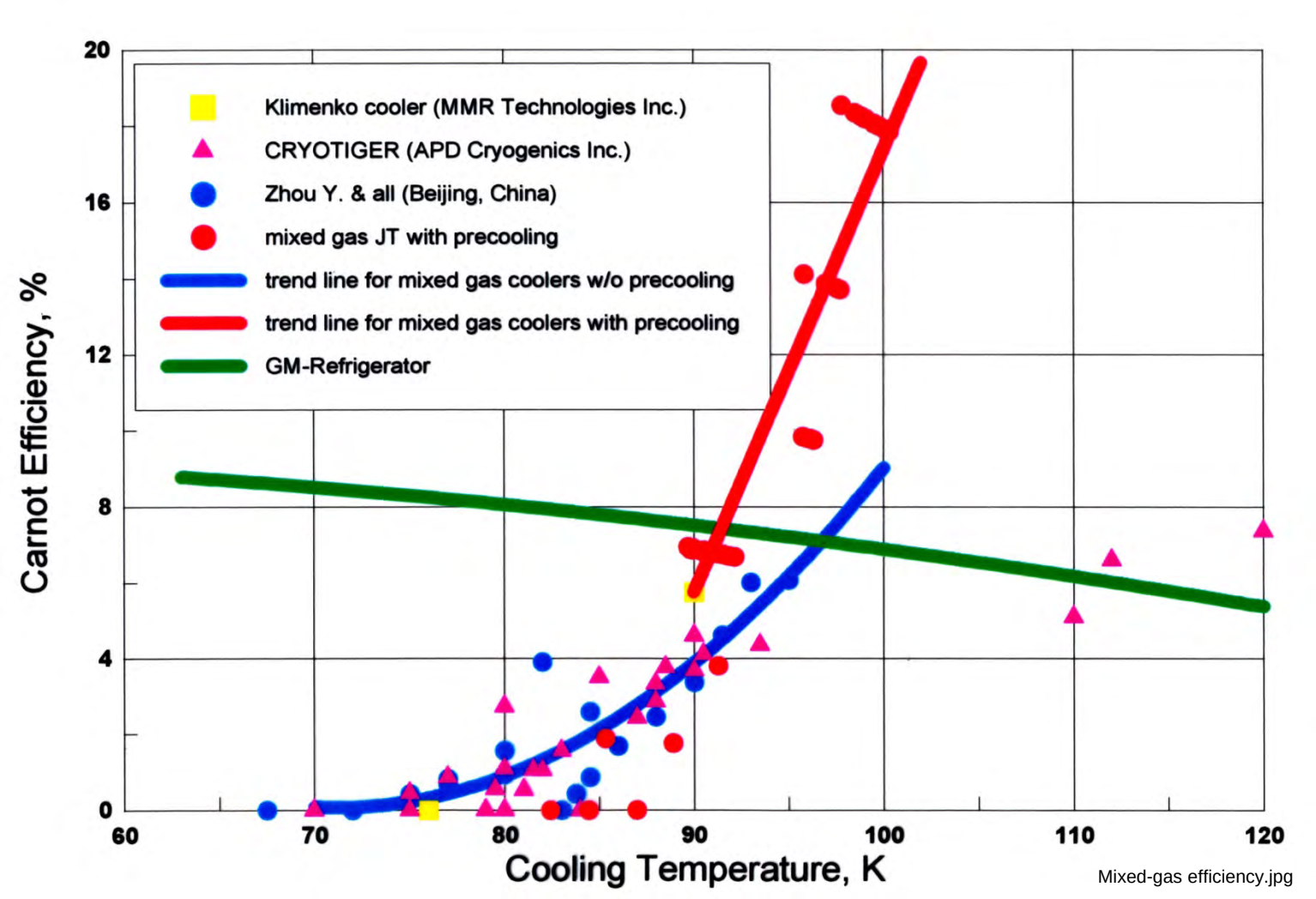






Wafer-level fabrication

Efficiencies of Mixed-Refrigerant JT



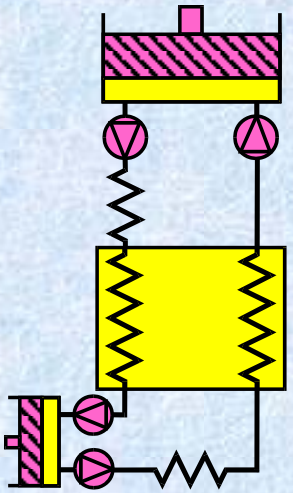
Courtesy, H. Quack

Outline

Session 1

- Information sources
- Definitions
- History
- Applications and problems
- Thermodynamics
- Cryocooler types and operating principles
- Joule-Thomson (JT) cryocoolers
- Brayton cryocoolers
- Claude systems
- Heat exchangers

Brayton Cycle



ADVANTAGES

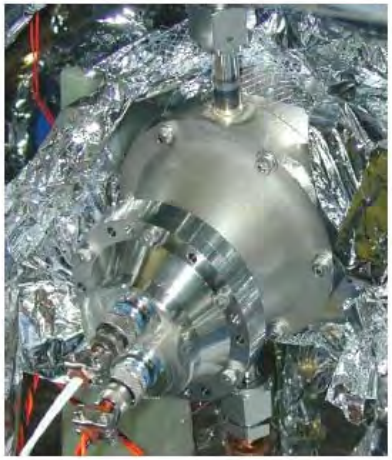
- Steady flow (low vibration, turbo-expander)
- Long lifetime (gas bearings, turbo system)
- Transport cold long distance
- Good efficiency except in small sizes

DISADVANTAGES

- Difficult to miniaturize
- Requires large heat exchanger
- Expensive to fabricate

lcec19c.cdr

Brayton Cycle: Key Components





Compressors



Recuperators



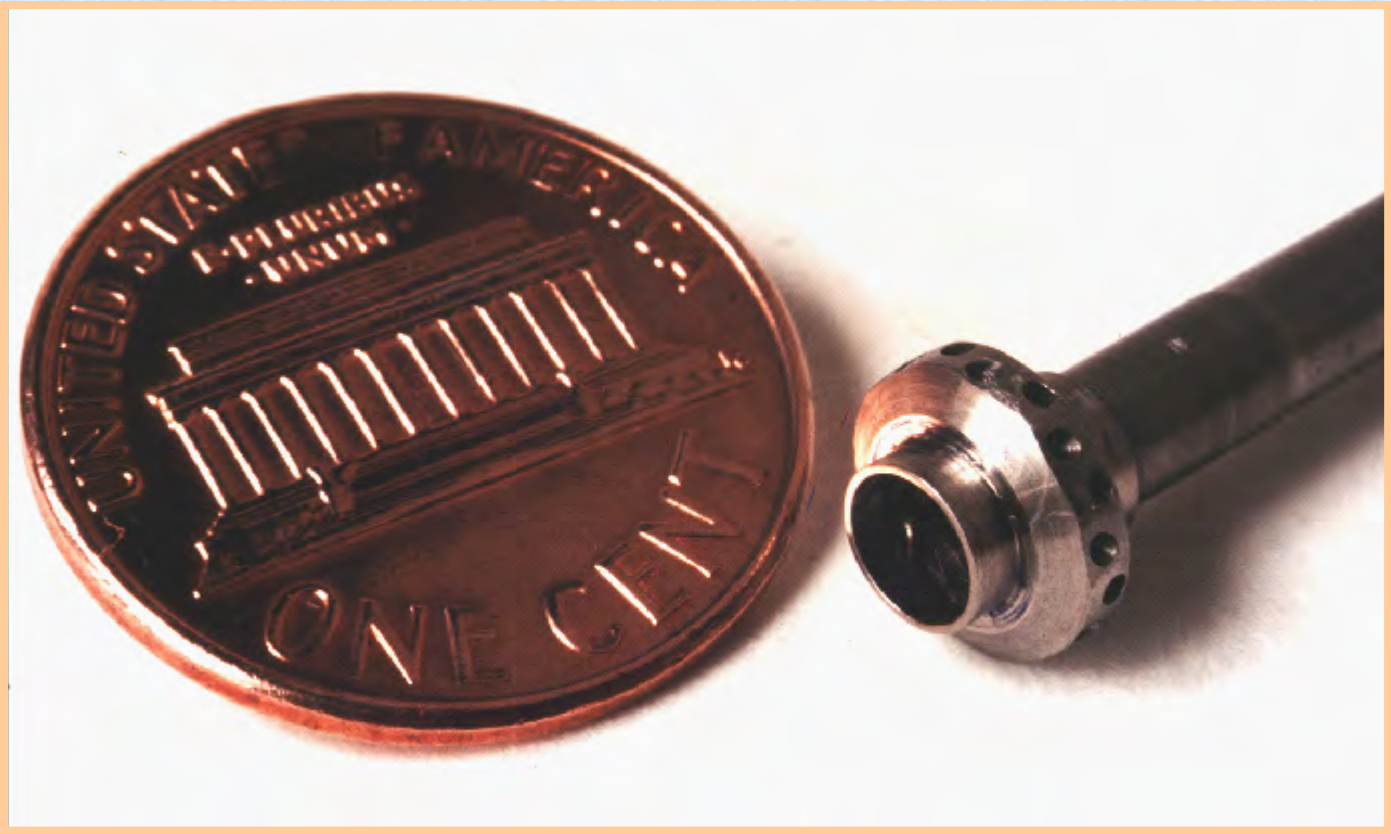


Turbines

(courtesy: Mark Zagarola, Creare)

Copyright © 2006, Creare Inc.
An unpublished work. All rights reserved.

Miniature Turboexpander
(Turboalternator)



Creare

Mini-turbine.cdr

- Cryogenic alternator drive
- 4000 rev/s
- Tilt-pad gas bearings
- 2 W @ 65 K, 100 W input

NICMOS Brayton Cryocooler

- Hubble Space Telescope
- Creare
- March 2002 installation
- 8.0% of Carnot



NICMOS2.cdr

- 7.7 W @70 K
- 315 W input

Outline

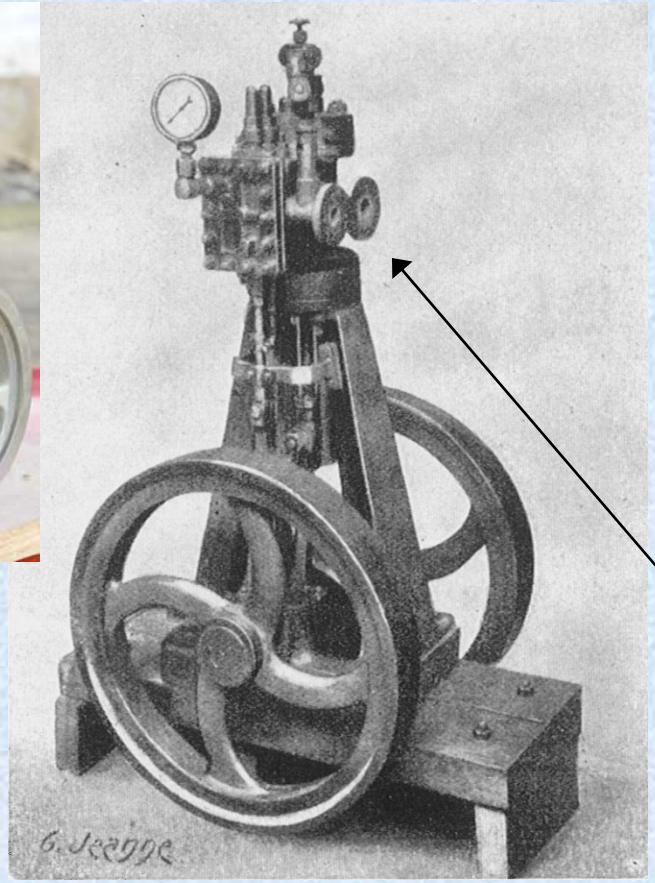
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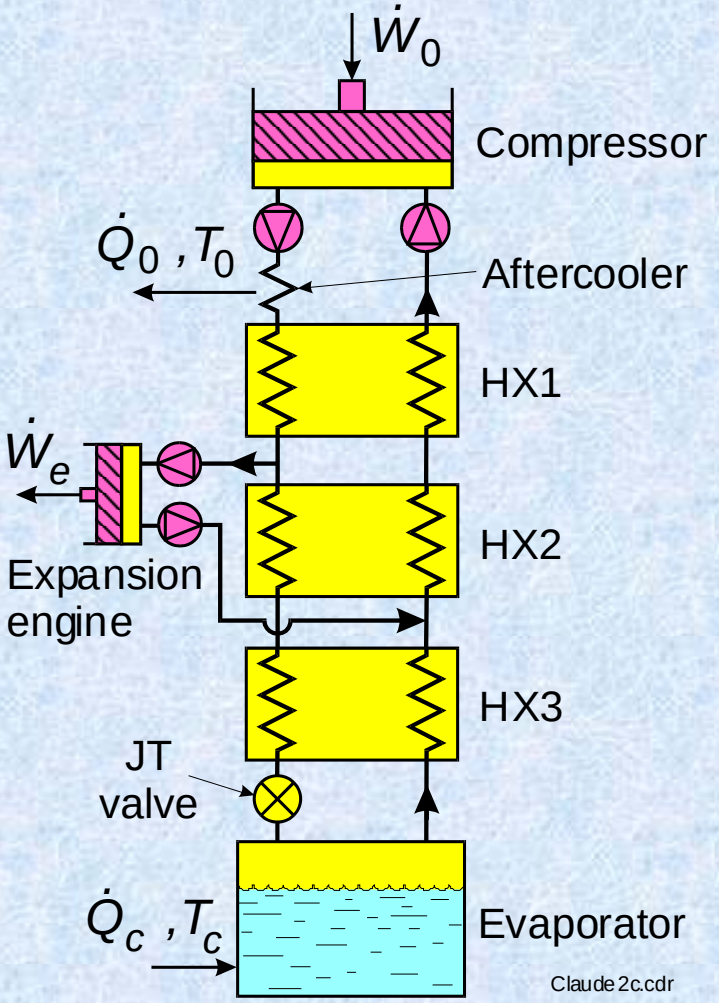
Claude Cycle



Steam engine

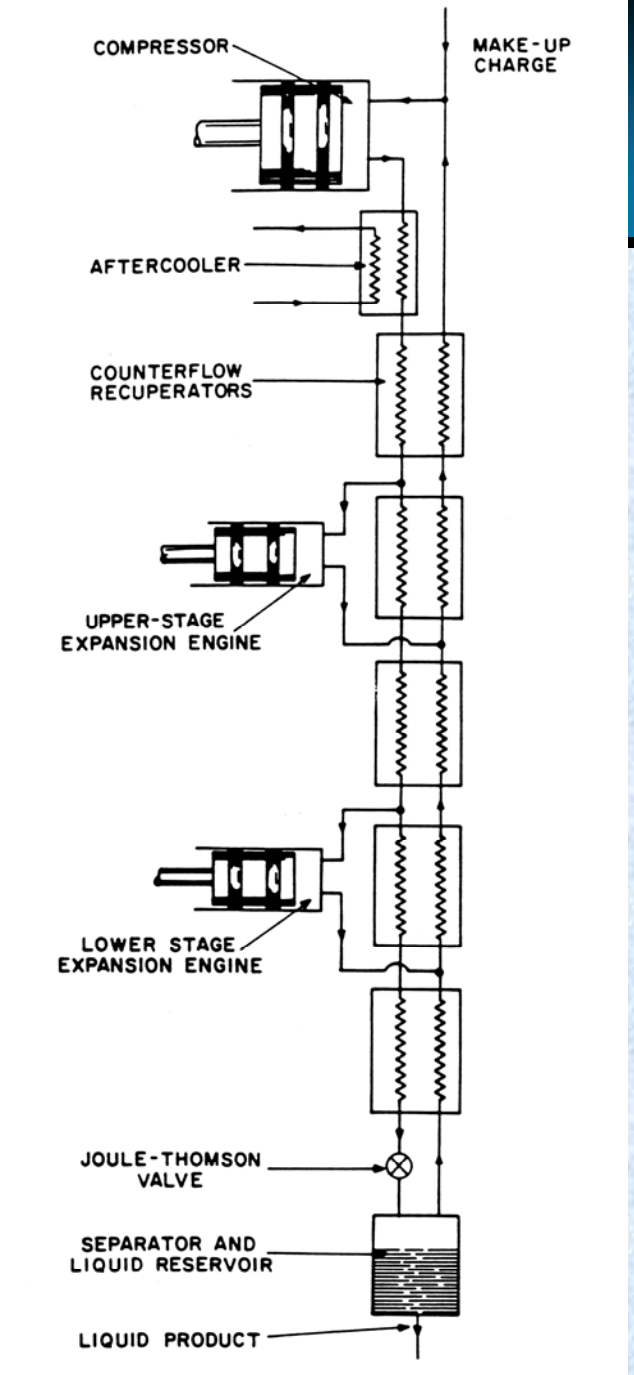


Claude Expansion Engine
Air liquefaction (1902) 2.5 MPa



G. Claude, *Liquid Air, Oxygen, and Nitrogen* (1913)

Claude Cycle



Collins-claude.jpg

Collins Helium Liquefier



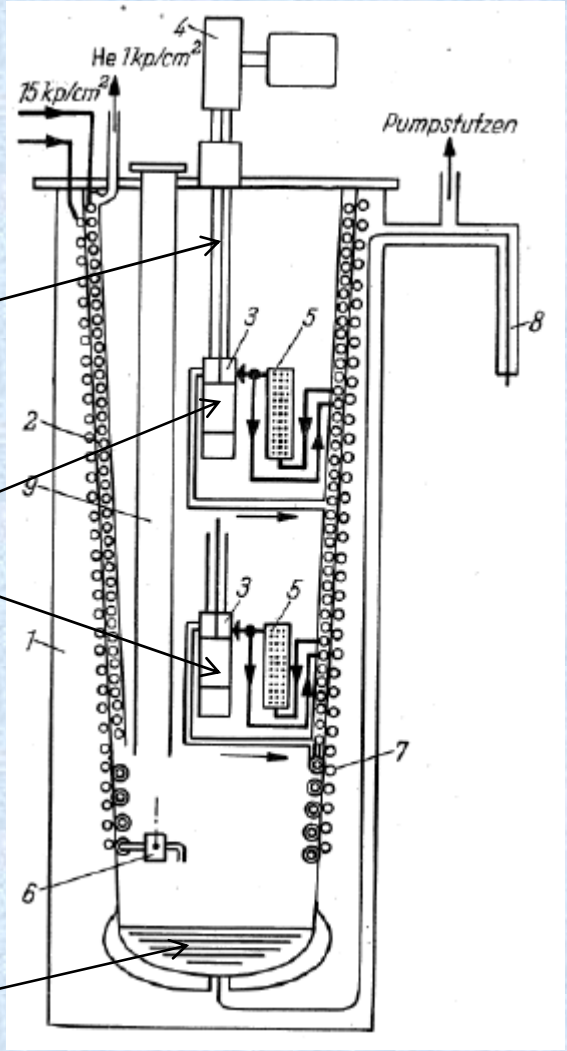
Collins He liquefier-Adl1.jpg

Collins Helium Liquefier

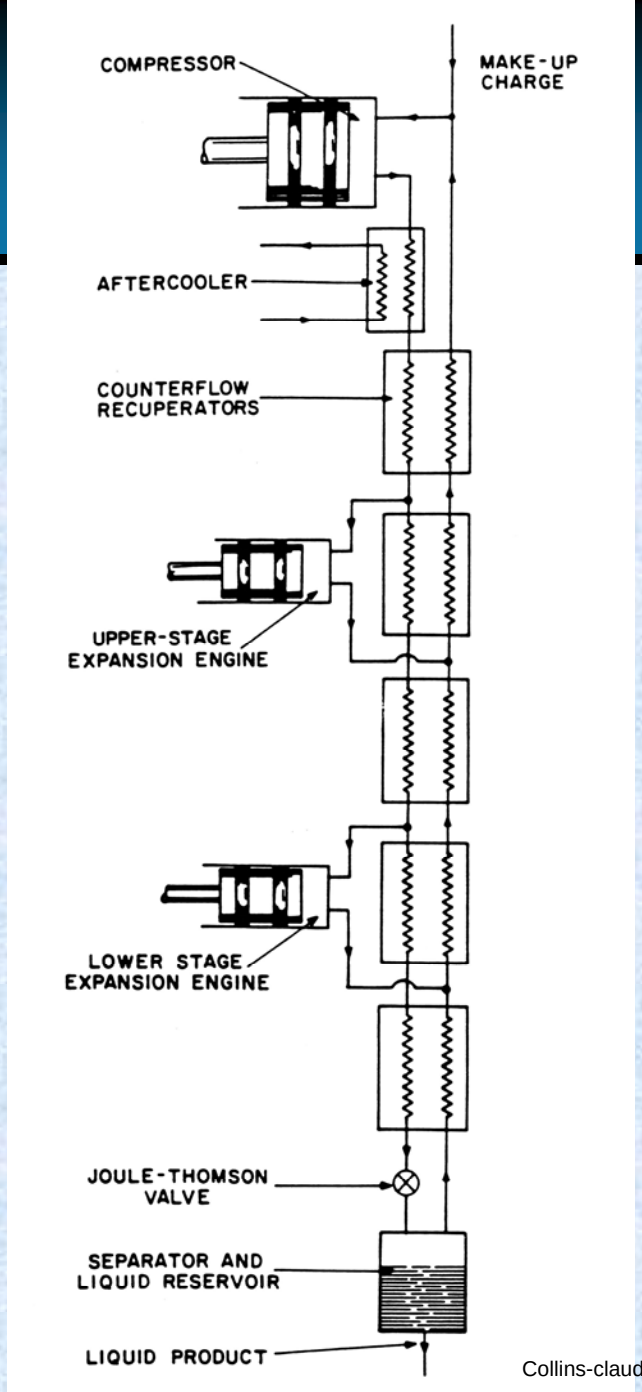
Expander tension rods

Expanders in He vapor

Liquid He



Claude Cycle



Collins-claude.jpg

Modern Helium Liquefier



Helium liquefier-Linde-1410 liquefier.jpg

Courtesy; Linde Corp.

8 L/hr

Gas-Bearing Expansion Turbine



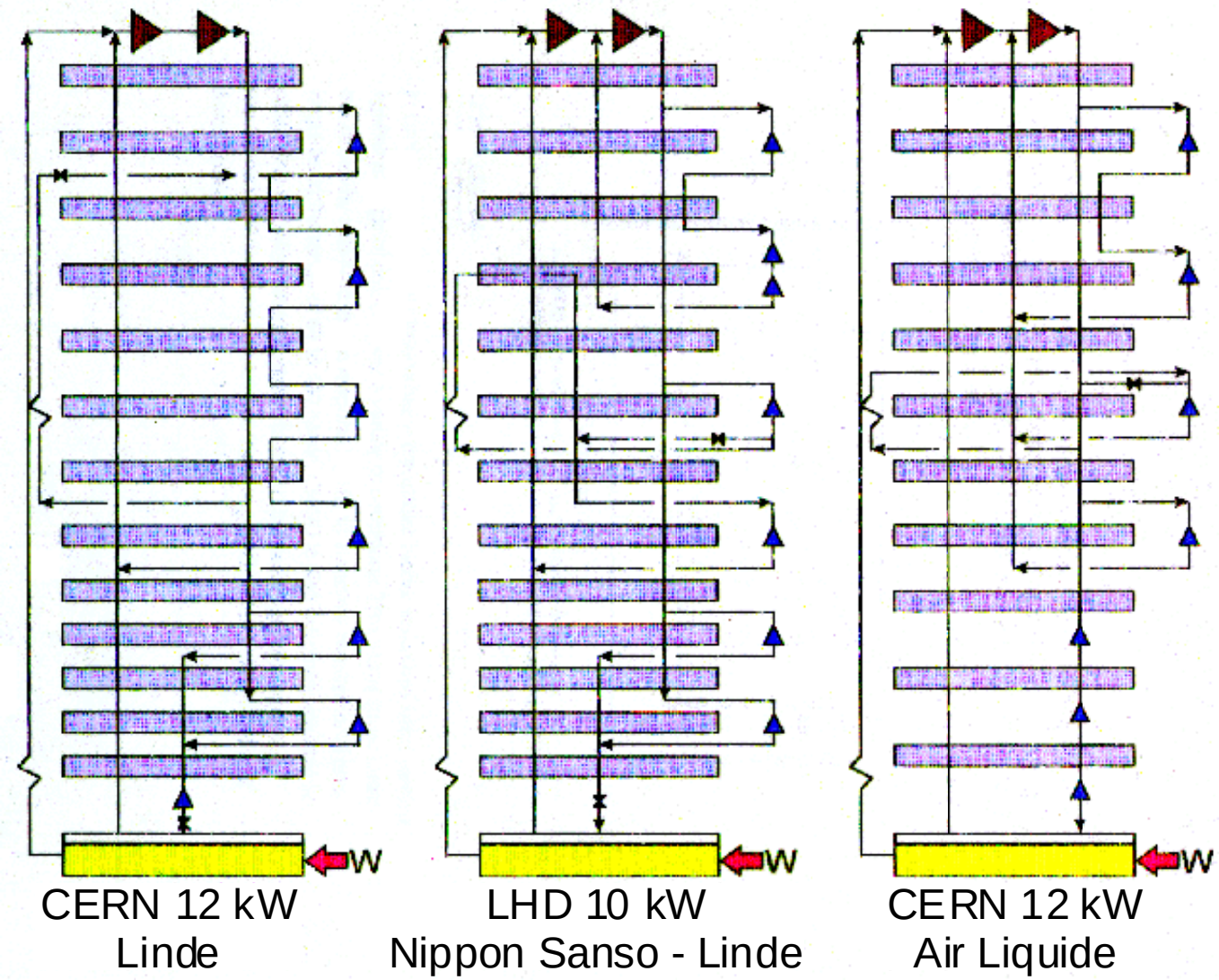
Isentropic expansion
Efficiency 65 % in 1990s
75 % to 80 % today



Speeds to 5400 Hz

Often used as wet expander in place of JT valve

Large Helium Liquefiers (Claude Cycle)



CERN 12 kW
Linde

LHD 10 kW
Nippon Sanso - Linde

CERN 12 kW
Air Liquide

Helium Liquefaction Plant



Linde

heliumplant.cdr

- HERA (DESY, Hamburg)
- 3 x 10,600 W @ 4.2 K

Recent Advances

Recuperative Systems

- JOULE-THOMSON (Low efficiency-low cost)
  a. Mixed refrigerants
  b. Sorption compressors
- BRAYTON/CLAUDE (High efficiency-high cost)
  a. Miniaturization
  b. Wet expanders
  c. Cold compressors
  d. High effectiveness heat exchangers

asc6c.cdr

Outline

Session 1

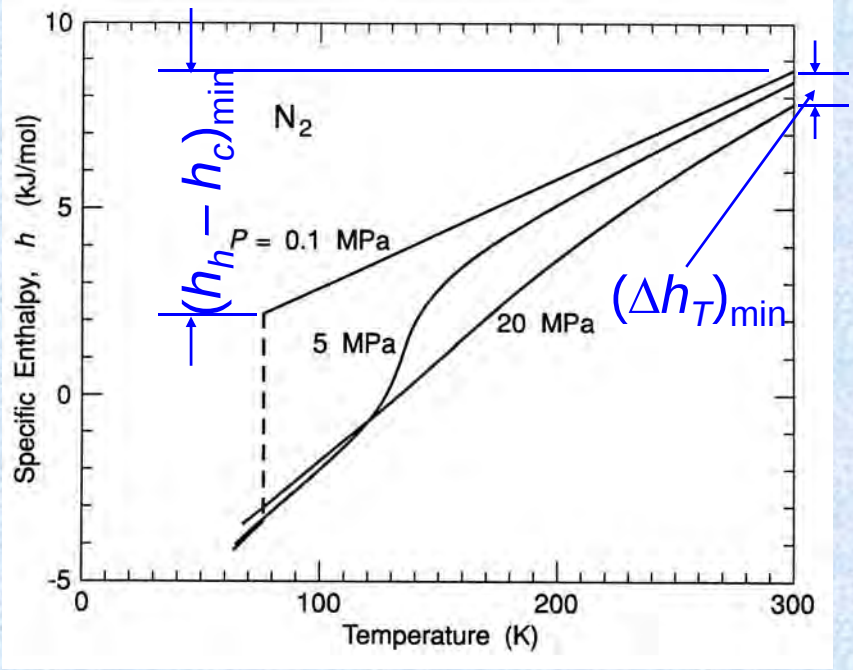
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Heat Exchanger Thermal Loss

$$\dot{Q}_r = \dot{n}(\Delta h_T)_{\min} = \dot{Q}_{net} + \boxed{\dot{Q}_{hx}} + \dot{Q}_{cond} + \dot{Q}_{\Delta P} + \dot{Q}_{rad}$$

Heat exchanger ineffectiveness:

$$\lambda = 1 - \varepsilon = \frac{\dot{Q}_{hx}}{\dot{n}(h_h - h_c)_{\min}}$$



Maximum ineffectiveness:

$$\dot{Q}_{hx} = \dot{Q}_r$$

$$\lambda_{\max} = \frac{(\Delta h_T)_{\min}}{(h_h - h_c)_{\min}}$$

$$\lambda_{\max} = \frac{2q_r}{(2/\pi)(h_h - h_c)_{\min}}$$ regenerative

Maximum Ineffectiveness

Examples: T_c = 80 K, T_h = 300 K

Cycle	Gas	P_L (MPa)	P_r	q_r (J/mol)	λ_{max}
JT	N_2	0.1	25	145	0.022
JT	N_2	0.1	150	744	0.115
JT	MR	0.1	25	850	0.070
Isoth.	He, Ne	0.1	2	461	0.100
Regen.	He	2.5	1.3	43	0.030

Maximum Ineffectiveness (Helium)

Examples: T_c = 4.2 K

Cycle	T_h (K)	P_L (MPa)	P_r	q_r (J/mol)	λ_{max}
JT	18	0.1	20	39.82	0.129
JT	20	0.1	20	32.34	0.092
JT	25	0.1	20	18.49	0.041
Isoth.	20	0.1	20	51.17	0.146
Isoth.	18	0.1	4	9.17	0.026

Outline

Session 2

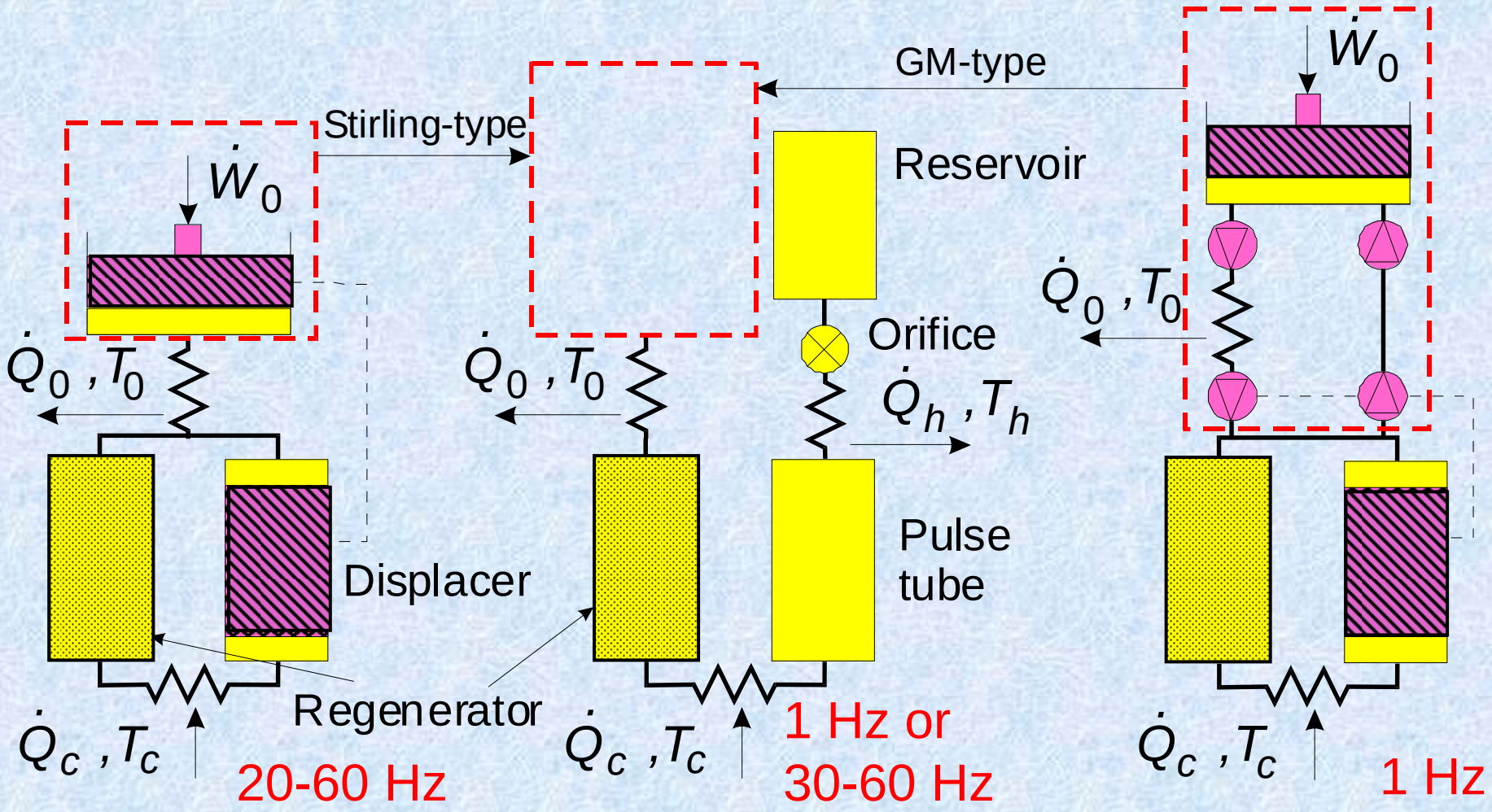
- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

Outline

Session 2

- Regenerative cycles
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- Gifford-McMahon cryocoolers
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Regenerative Cryocoolers
(Oscillating Flow)

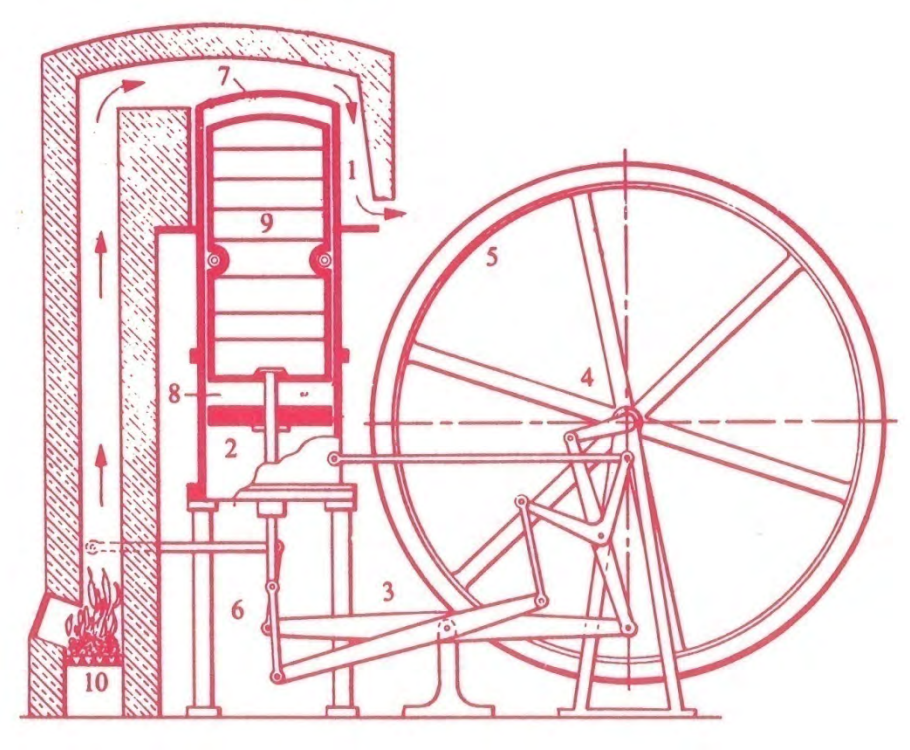


Outline

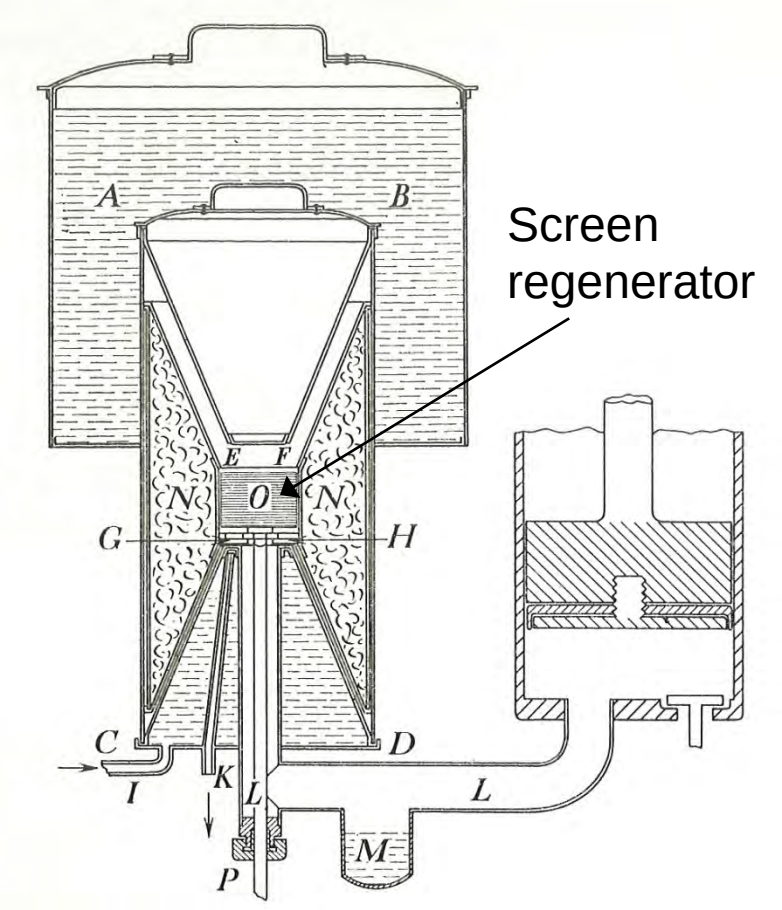
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Stirling Engine/Cooler



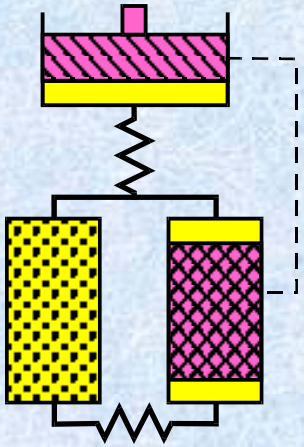
Stirling Engine (1816)
Piston-displacer
Economizer (regenerator)



Kirk Ice Machine (1861)
Stirling Cooler

A.C. Kirk, *Min. Proc. Istn. Civ. Engrs.* **37**, 244 (1874)

Stirling Cryocooler



ADVANTAGES

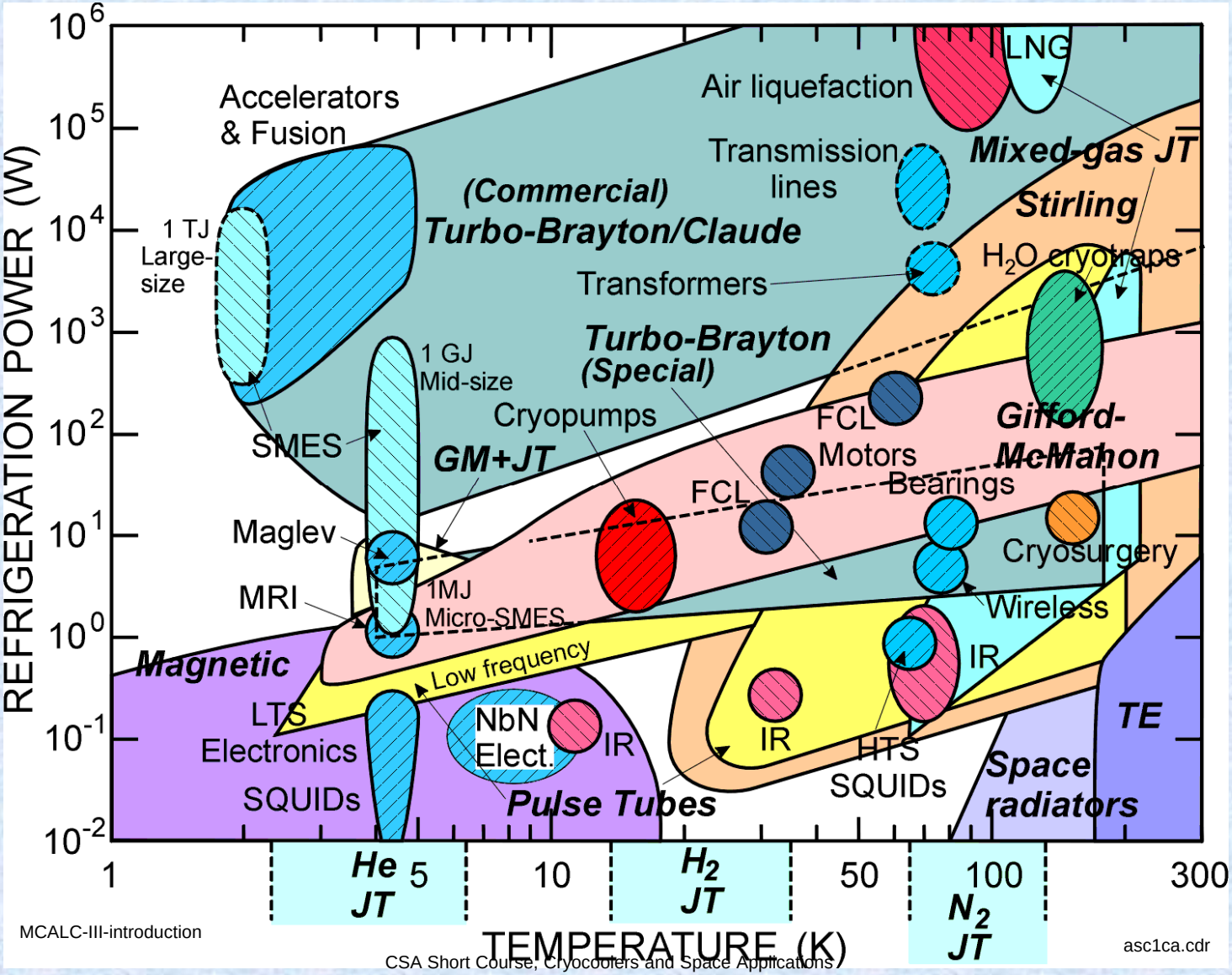
- High efficiency
- Moderate cost
- Small size and weight
- Over 140,000 made to date

DISADVANTAGES

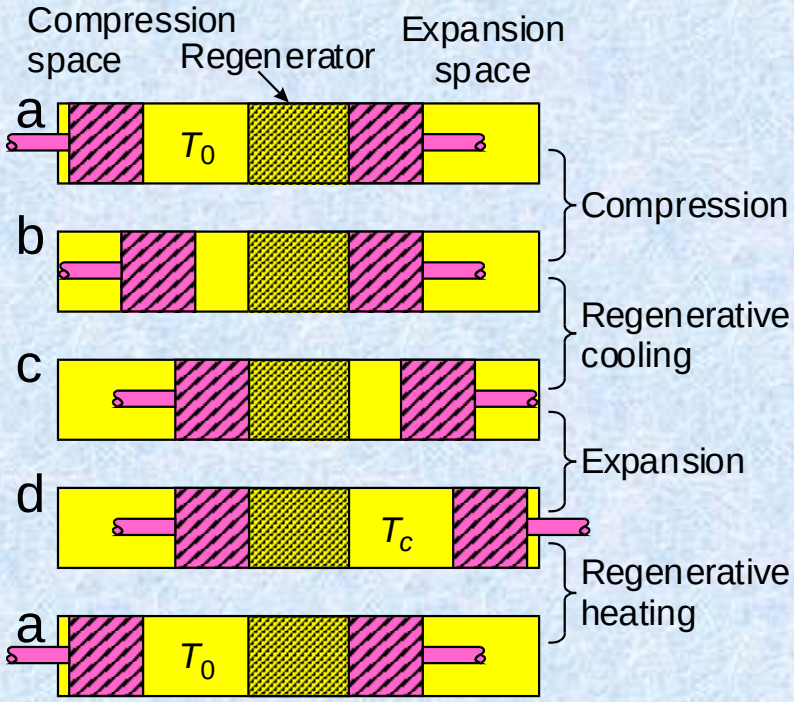
- Dry or no lubrication
- Intrinsic vibration from displacer
- Long lifetime expensive (3-10 yrs)

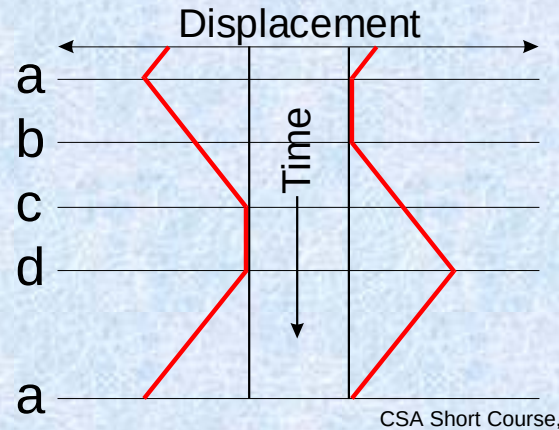
lcec21c.cdr

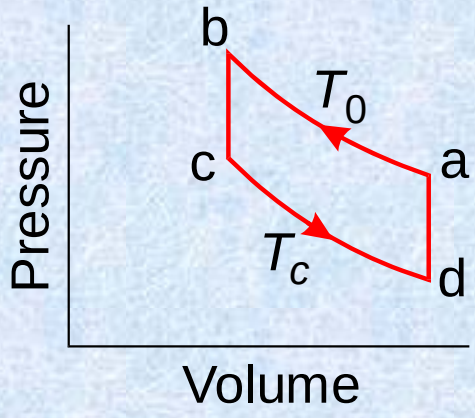
Cryocooler Applications and Operating Regions

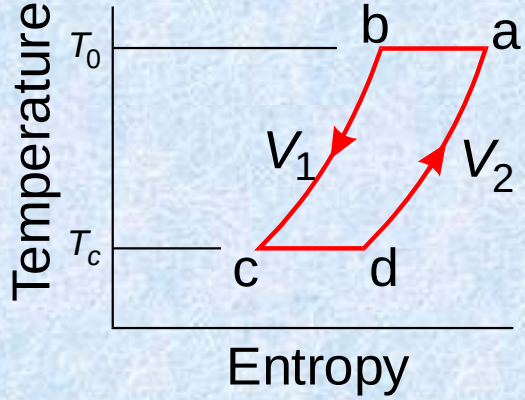


Ideal Stirling Cycle





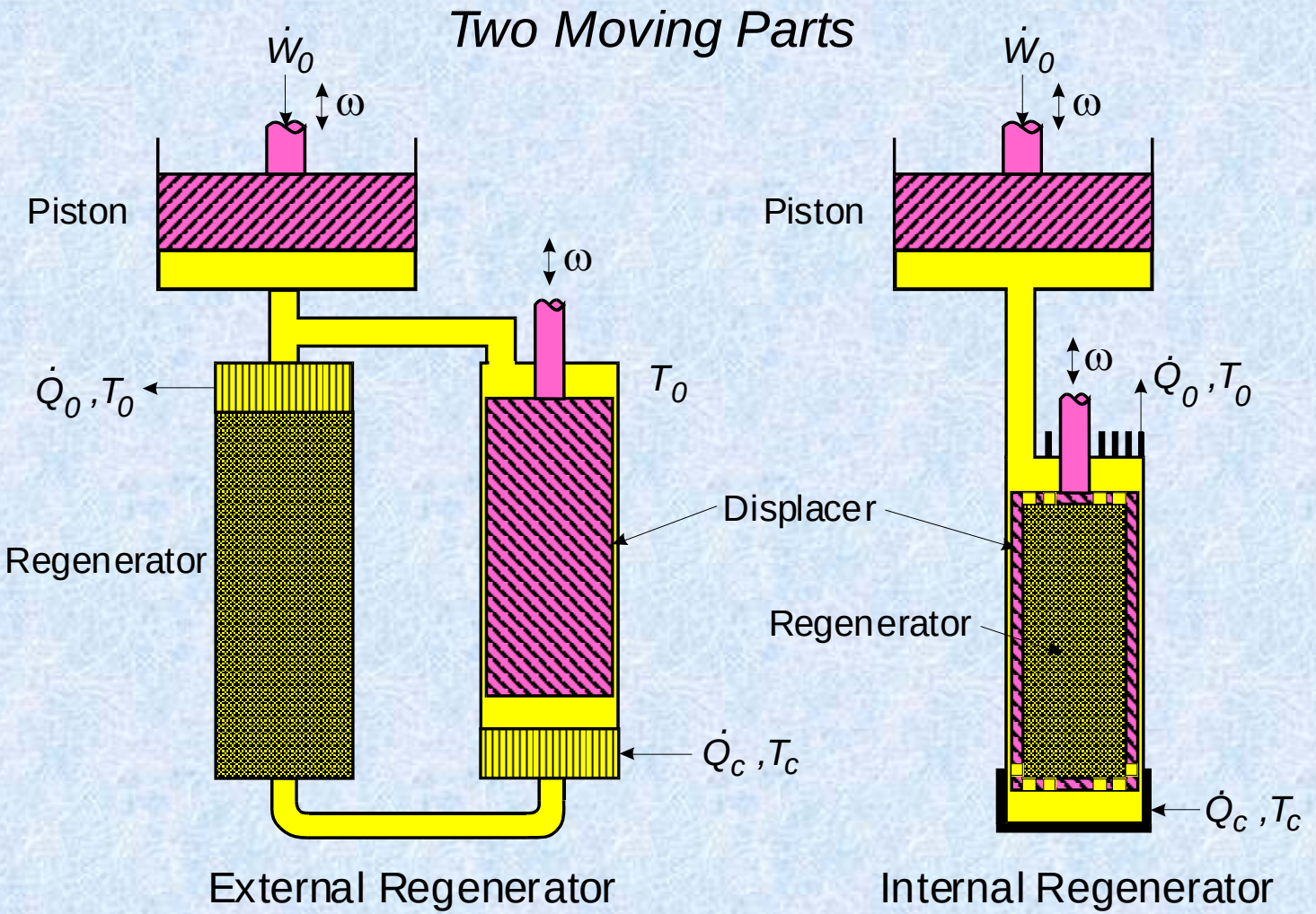




Working fluid: Ideal gas (helium)

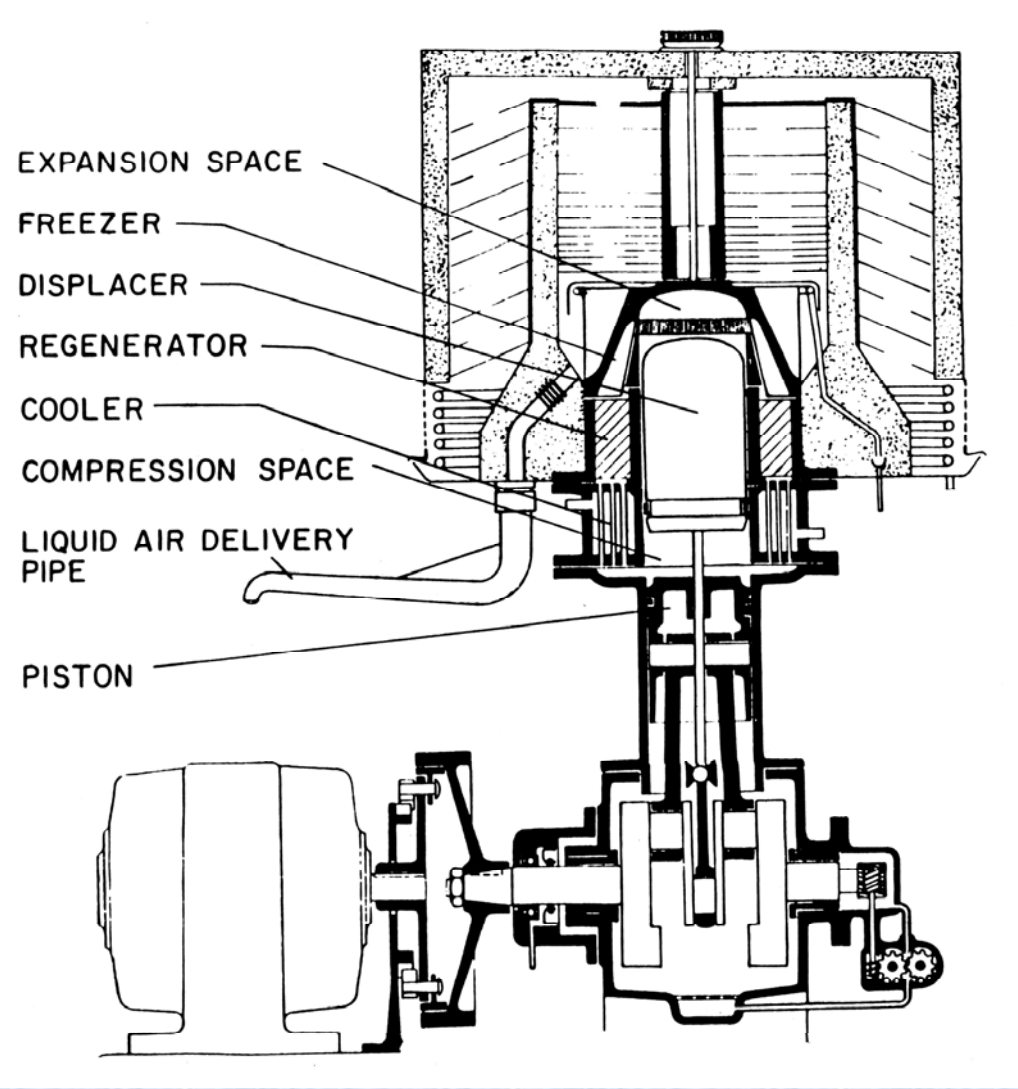
Stirling ideal 1c.cdr

Stirling Cryocooler
(Piston-Displacer Arrangement)

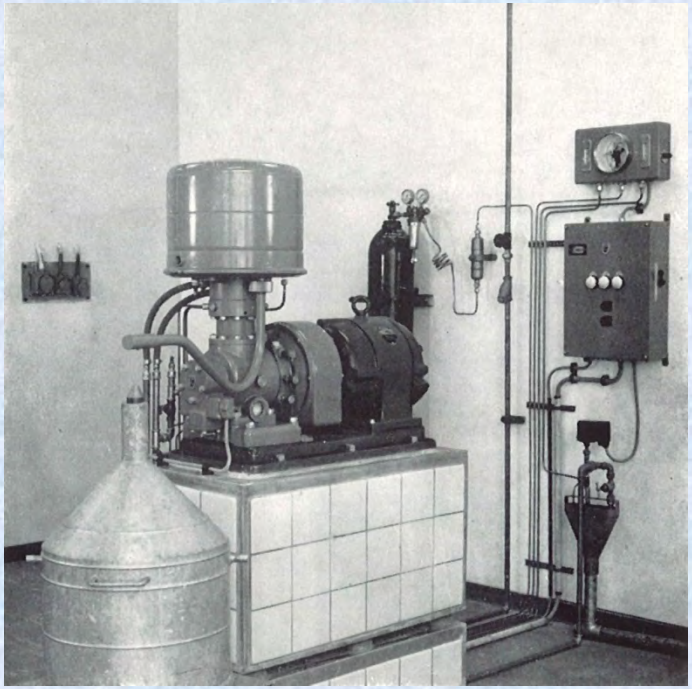


stirling3c.cdr

Early Philips Stirling Air Liquefier



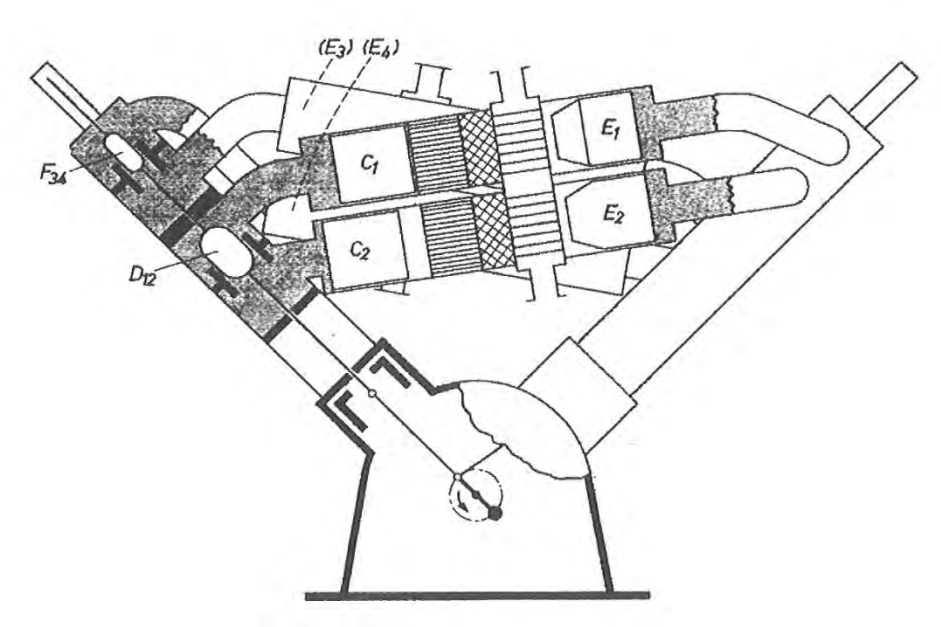
- First developed in 1953

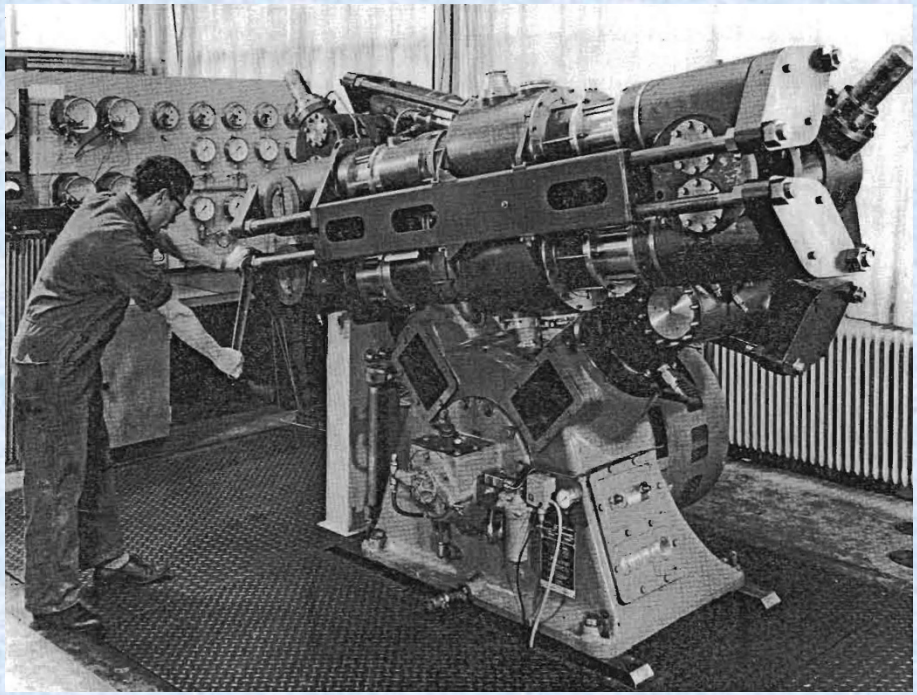


Courtesy: G. Walker, *Cryocoolers*, Plenum (1983)

Largest Stirling Cryocooler
Most Efficient Cryocooler

Philips Stirling Model C (1965)





- 20 kW at 70 K
- 41 % Carnot
- 4 cylinder (25 Hz)
- Hydraulically driven

A.A. Dros, “An industrial gas refrigerating machine with hydraulic drive,” *Philips technical review* **26**, (1965), pp. 297-308.

Modern Stirling Air Liquefier



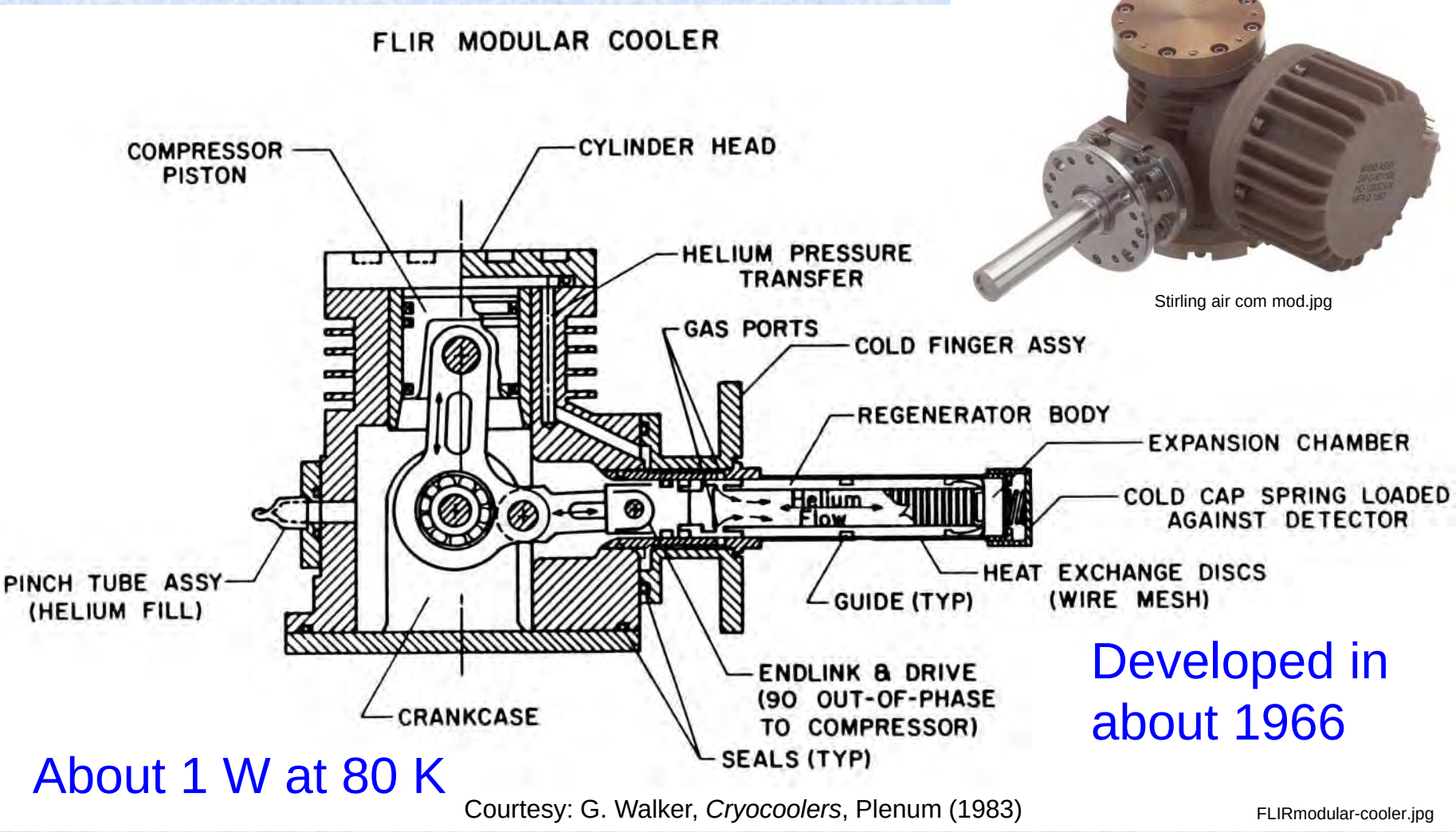
StirLIN_1_compact.jpg

- 700 W @ 80 K
- 500 W @ 65 K
- 12 kW input
- 15% Carnot @ 80 K

Common-Module Integral Stirling Cooler



Stirling air com mod.jpg



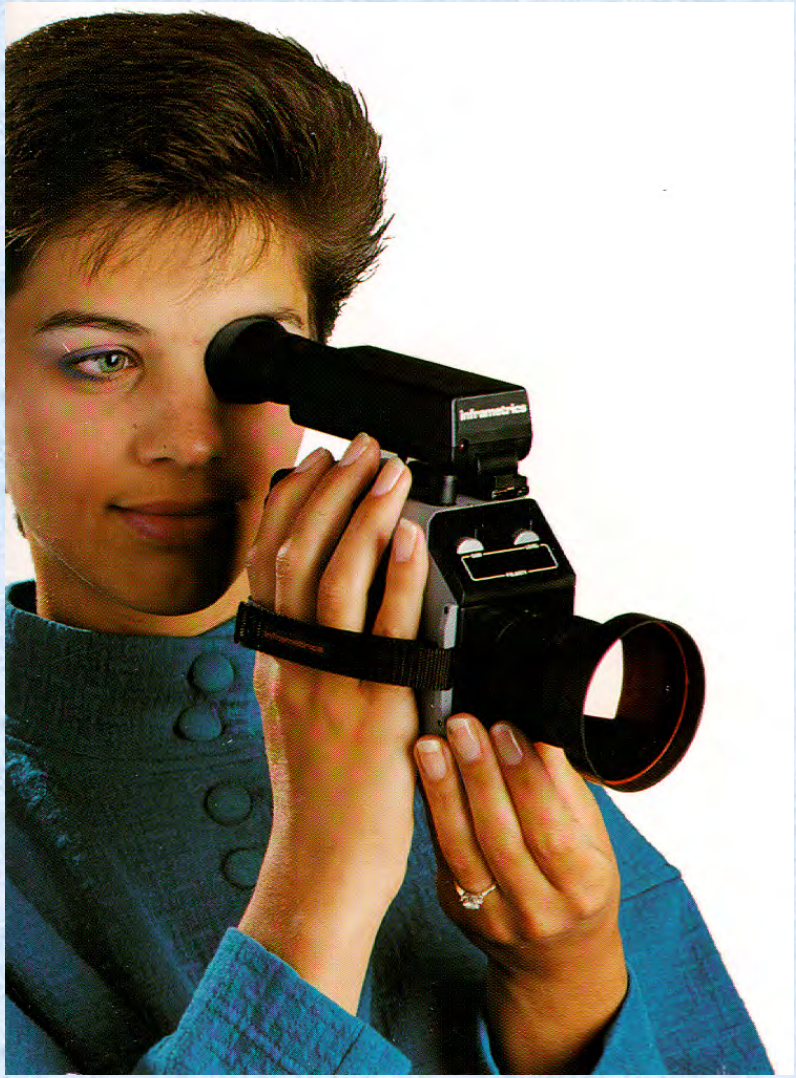
About 1 W at 80 K

Developed in about 1966

Courtesy: G. Walker, *Cryocoolers*, Plenum (1983)

FLIRmodular-cooler.jpg

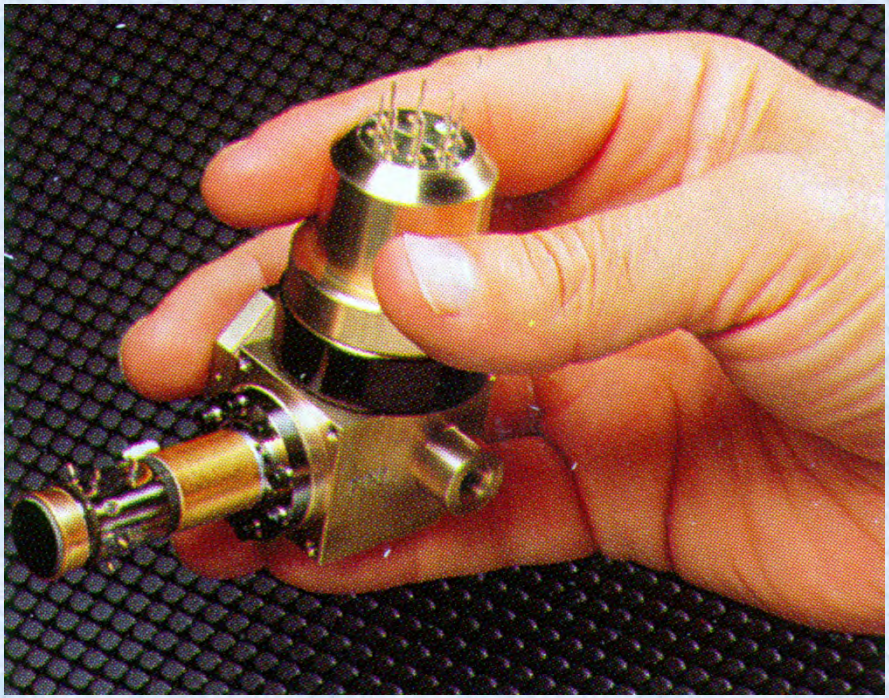
Infrared Camcorder



Inframetrics/FLIR Systems

Inframetric-01.jpg

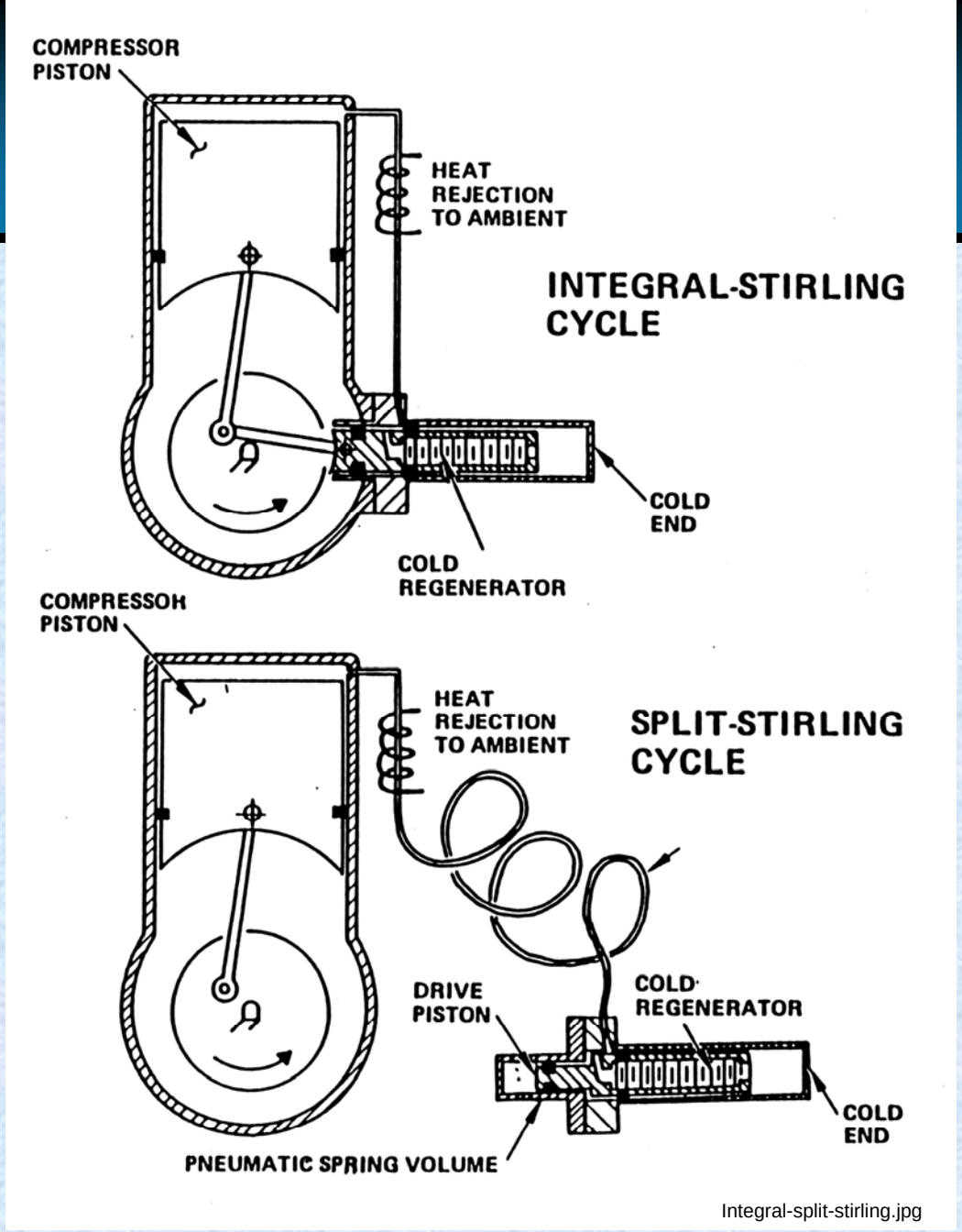
With Stirling Microcooler



Inframetrics-02.jpg

- 0.15 W @ 80 K
- 3 W input power
- 14% of Carnot

Integral vs Split Configuration

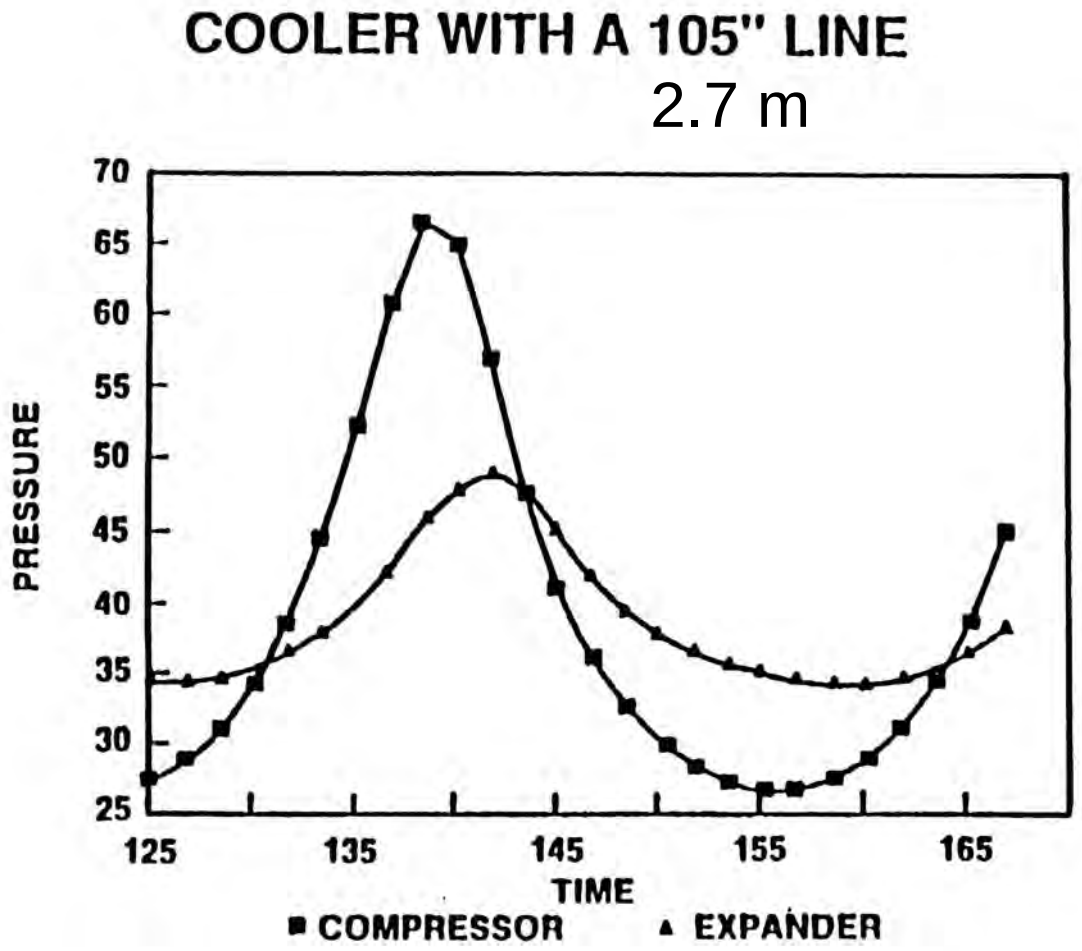


Integral-split-stirling.jpg

Courtesy: Hughes Aircraft (W. Gully)

- Displacer driven by mechanical linkage
- Displacer driven pneumatically by oscillating pressure

Effect of Long Transfer Line

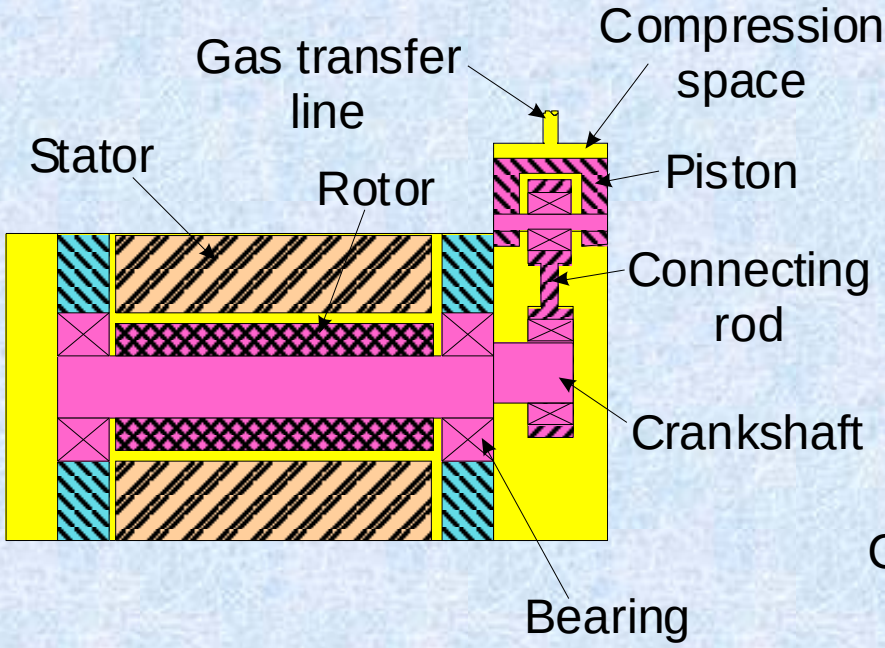


Courtesy: Hughes Aircraft (W. Gully)

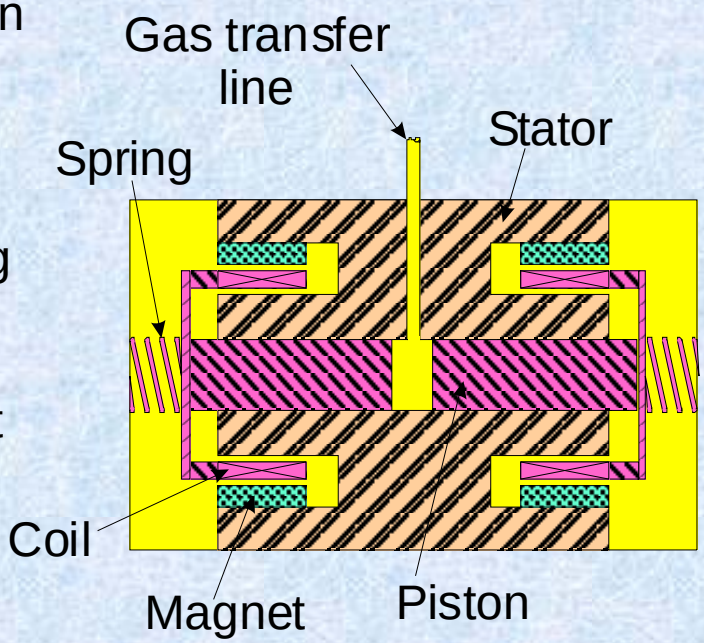
105-line.jpg

Rotary vs. Linear Drive
(Comparison)

Rotary Reciprocating Compressor



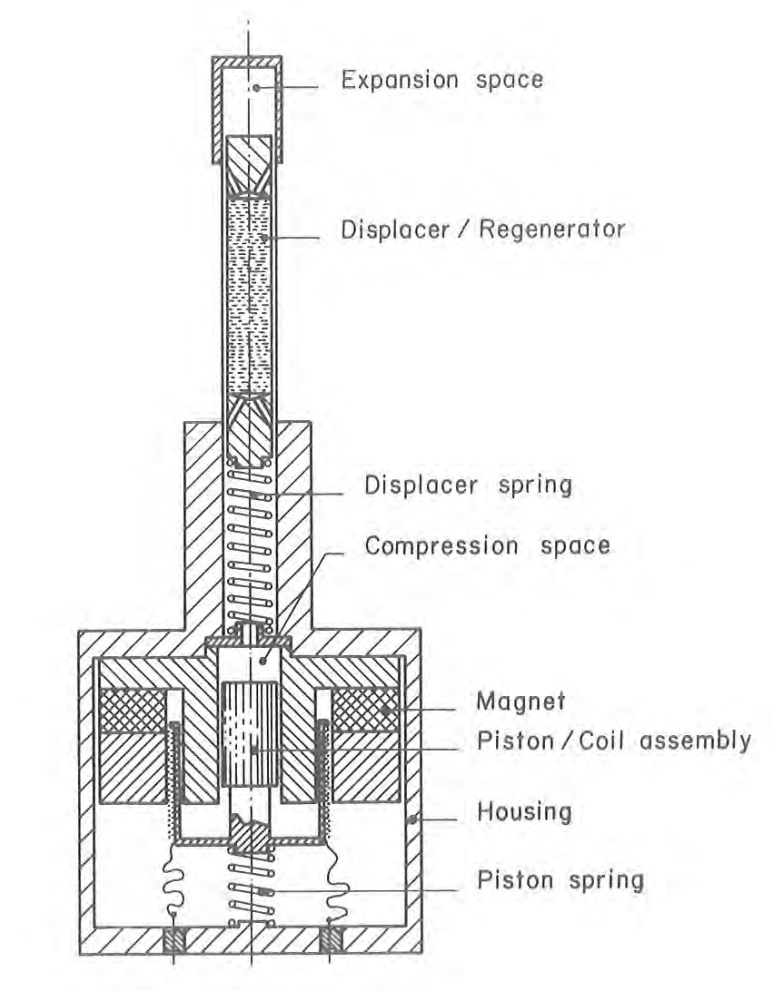
Linear Drive Compressor



- Side loads eliminated
- Bearings eliminated
- Balanced by design

comp lin rot.cdr

First Linear Stirling Cryocooler (1978)



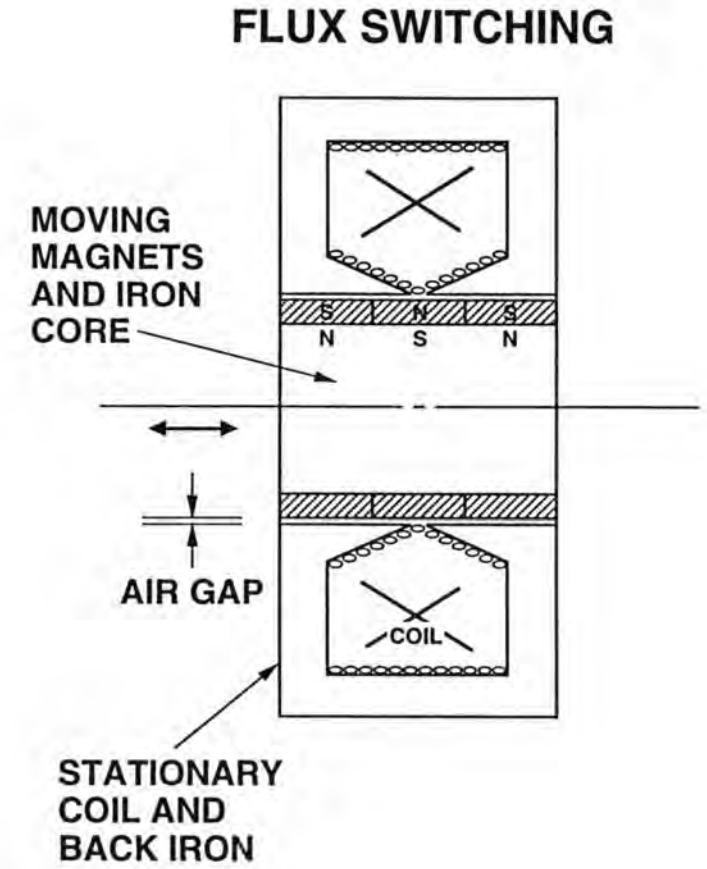
Philips mc 80 Linear Stirling Cryocooler

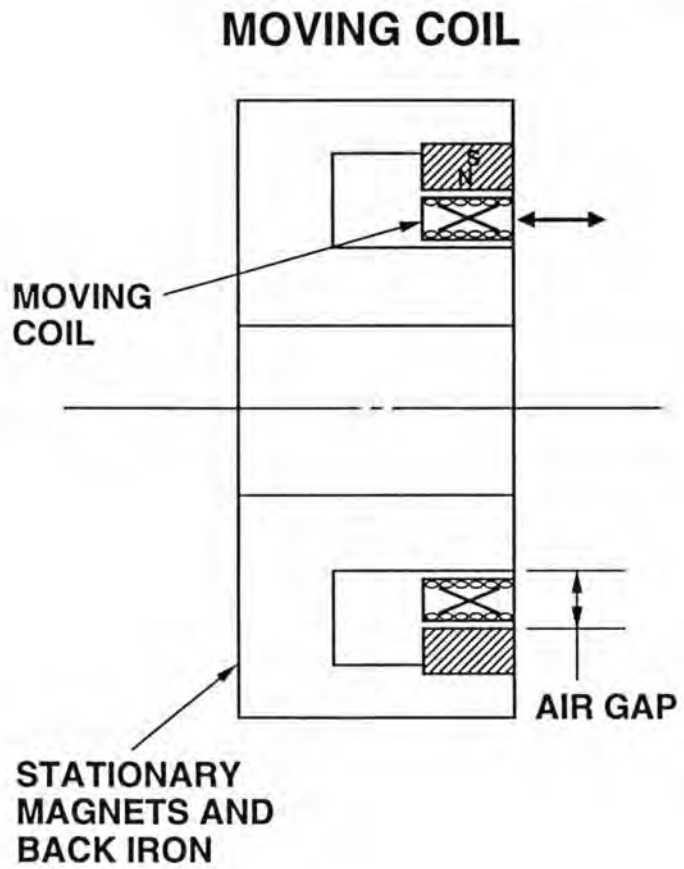


Courtesy: G. Walker, Miniature Refrigerators for Cryogenic Sensors and Cold Electronics (1989)

G. J. Haarhuis, "The mc 80 – a magnetically driven Stirling refrigerator," *Cryogenics*, vol. 18 (1978), pp. 656-658

Types of Linear Motors

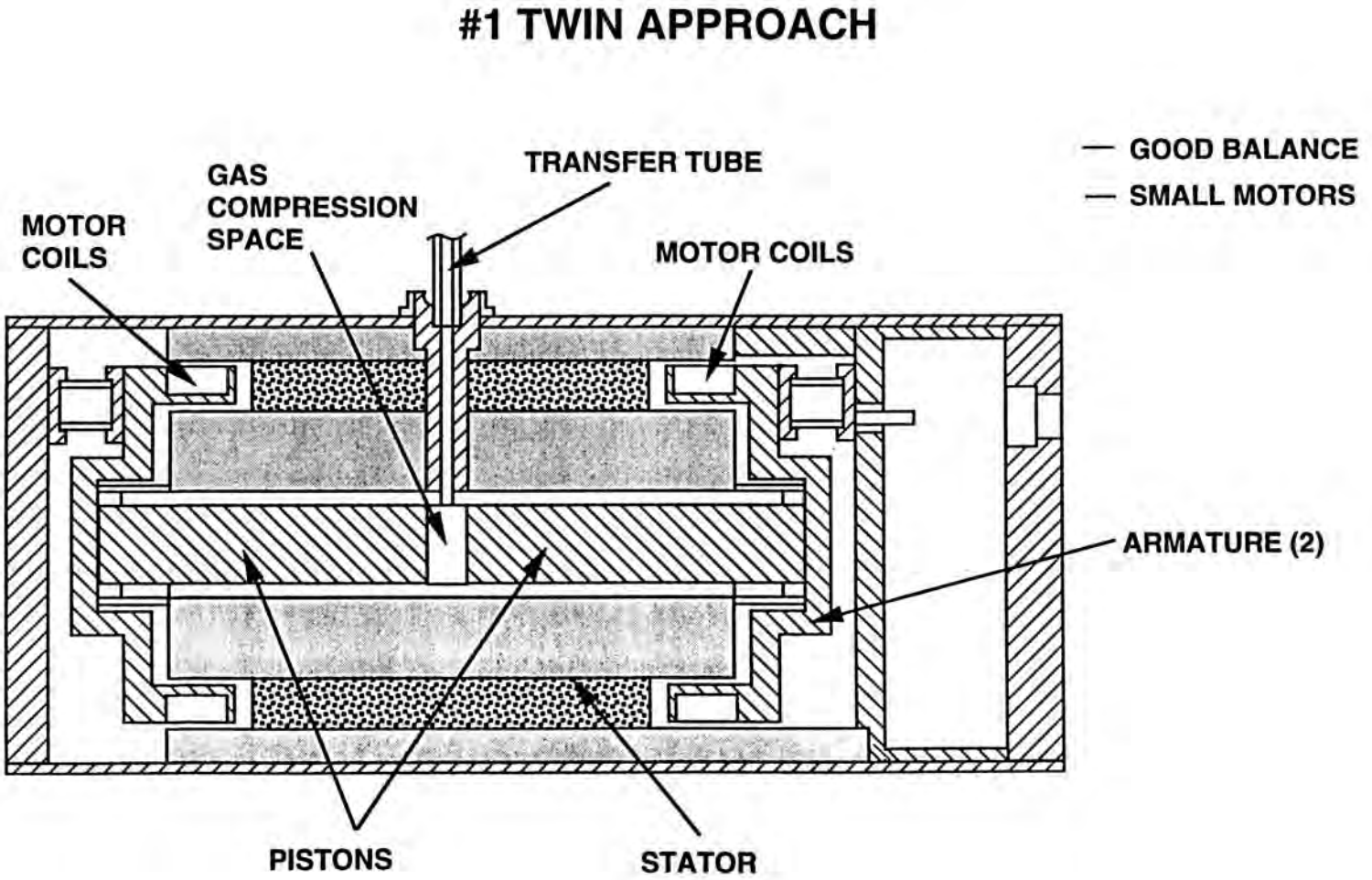




Courtesy: Hughes Aircraft (W. Gully)

Linear motors.jpg

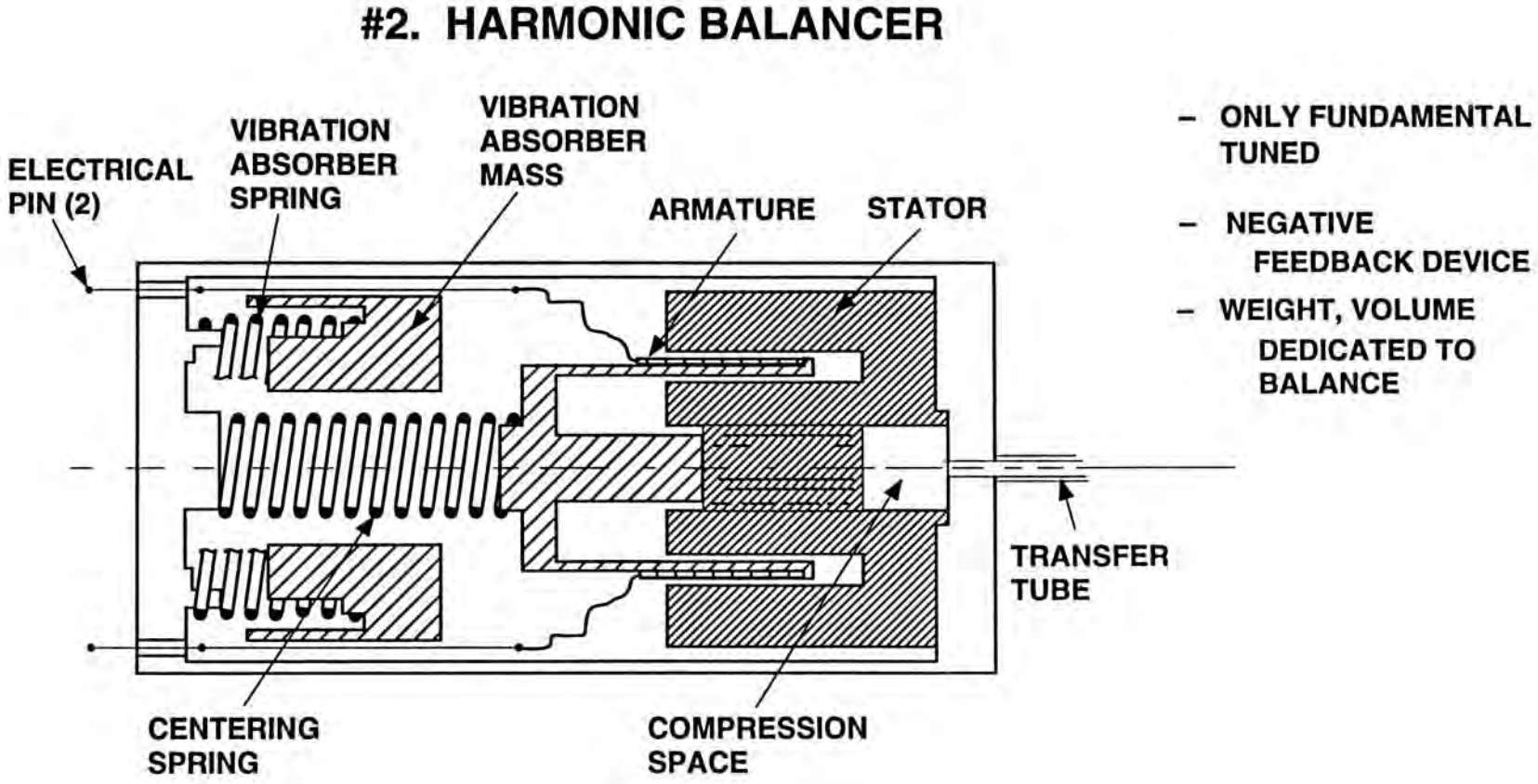
Linear Compressor Vibration Damping



Courtesy: Hughes Aircraft (W. Gully)

Twin approach.jpg

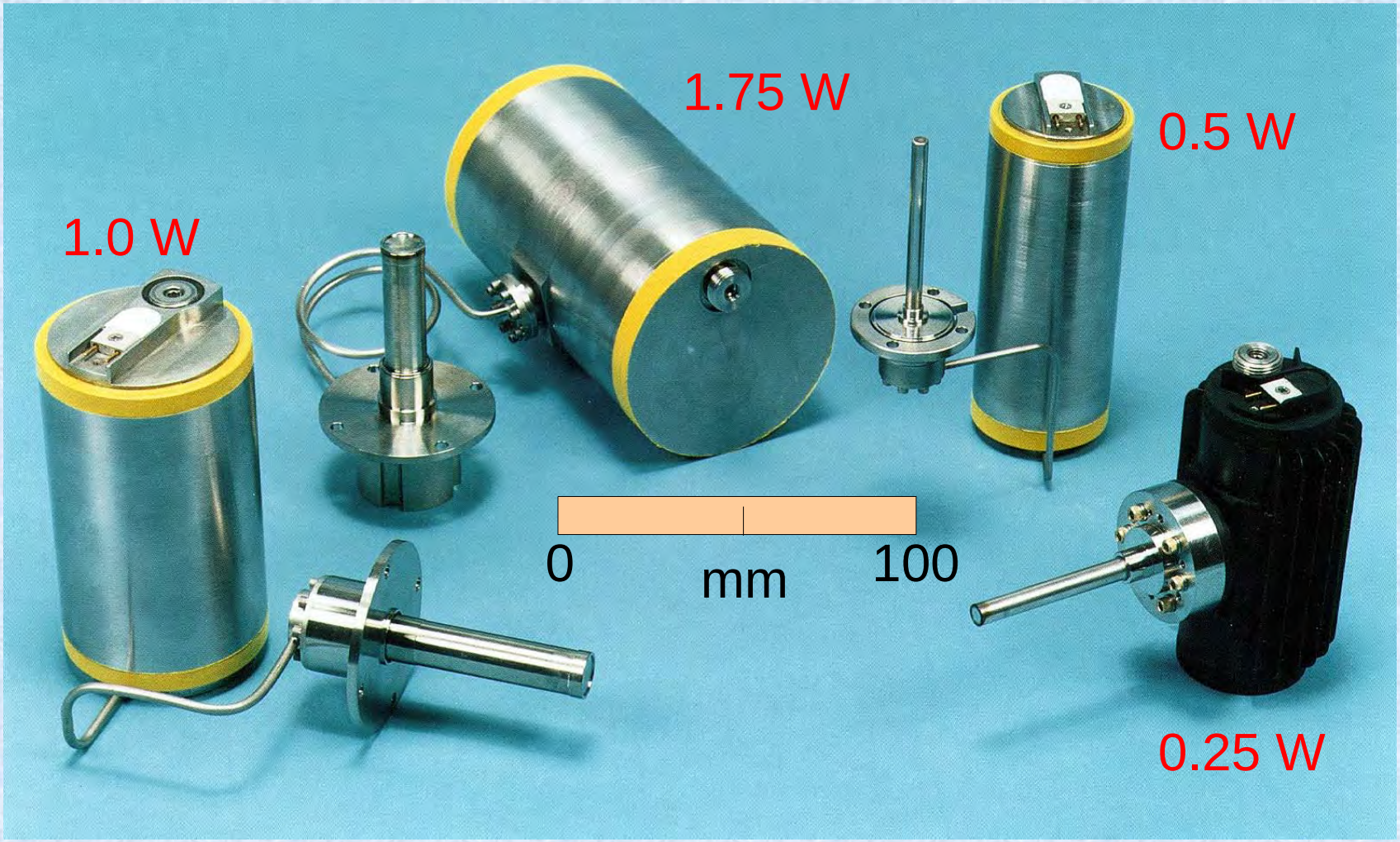
Linear Compressor Vibration Damping II



Courtesy: Hughes Aircraft (W. Gully)

Harmonic Balancer.jpg

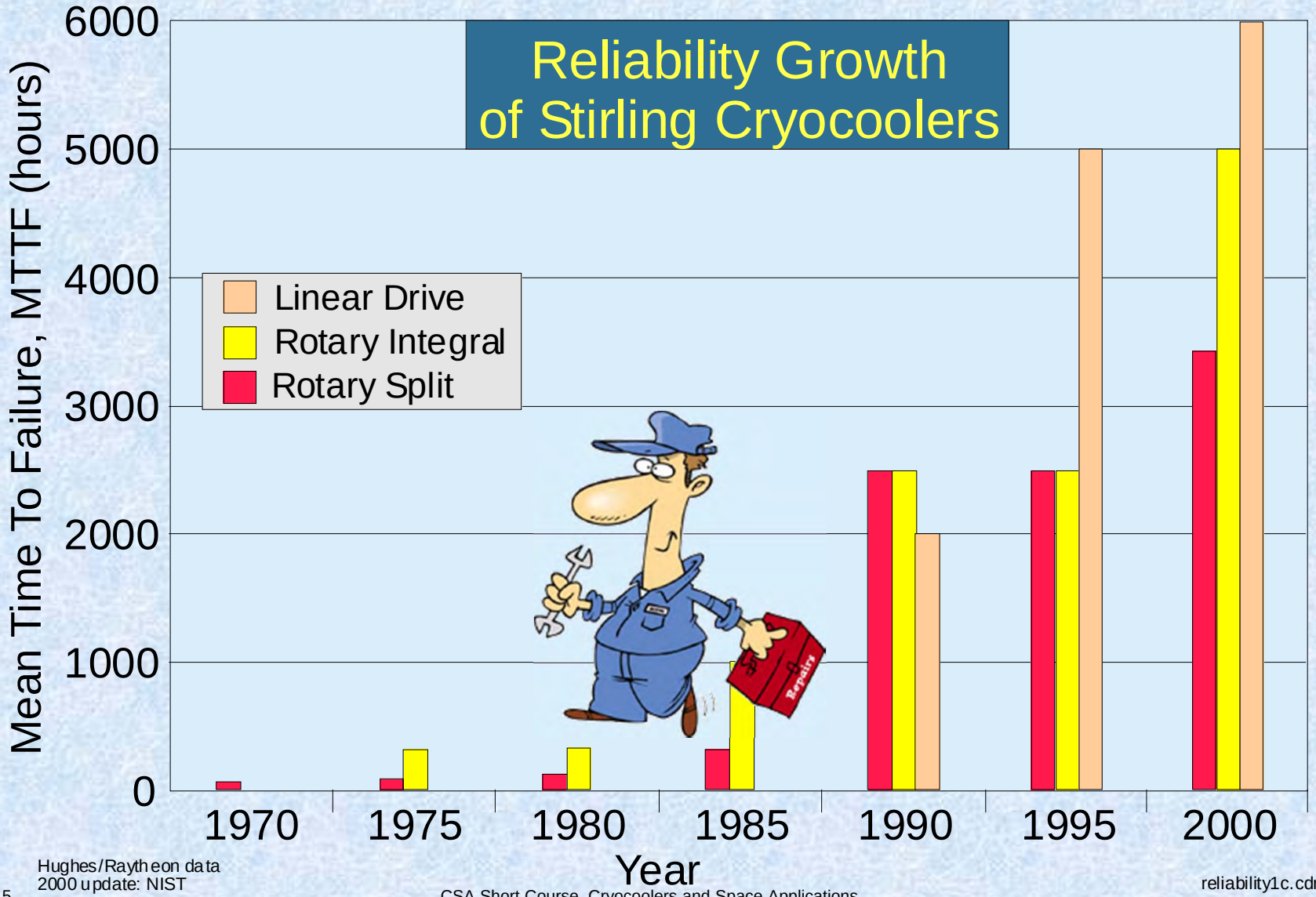
Stirling Cryocoolers (Linear Drive)
80 K



DRS Infrared Technologies

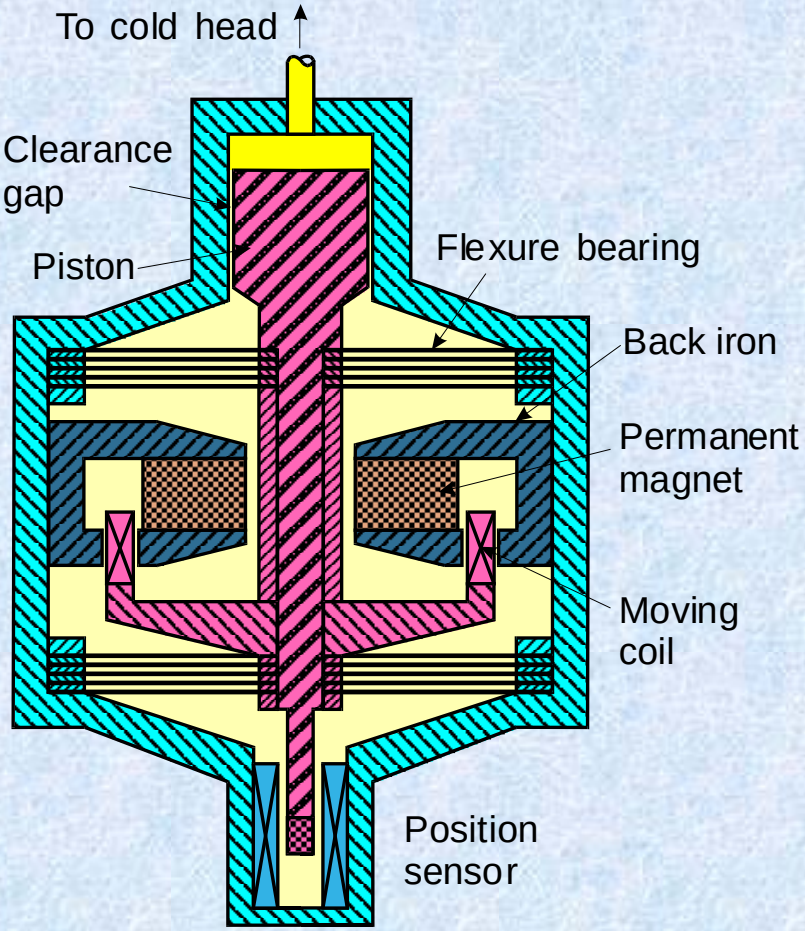
Stirlinglin1.cdr TI—01.jpg

Reliability of Stirling Cryocoolers

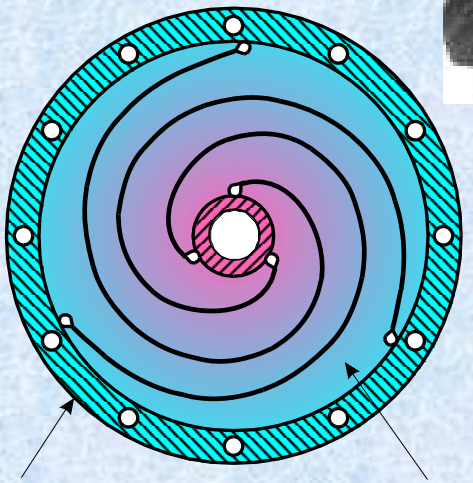


Hughes/Raytheon data
2000 update: NIST

Flexure-Bearing Linear Compressor



a **Linear compressor**

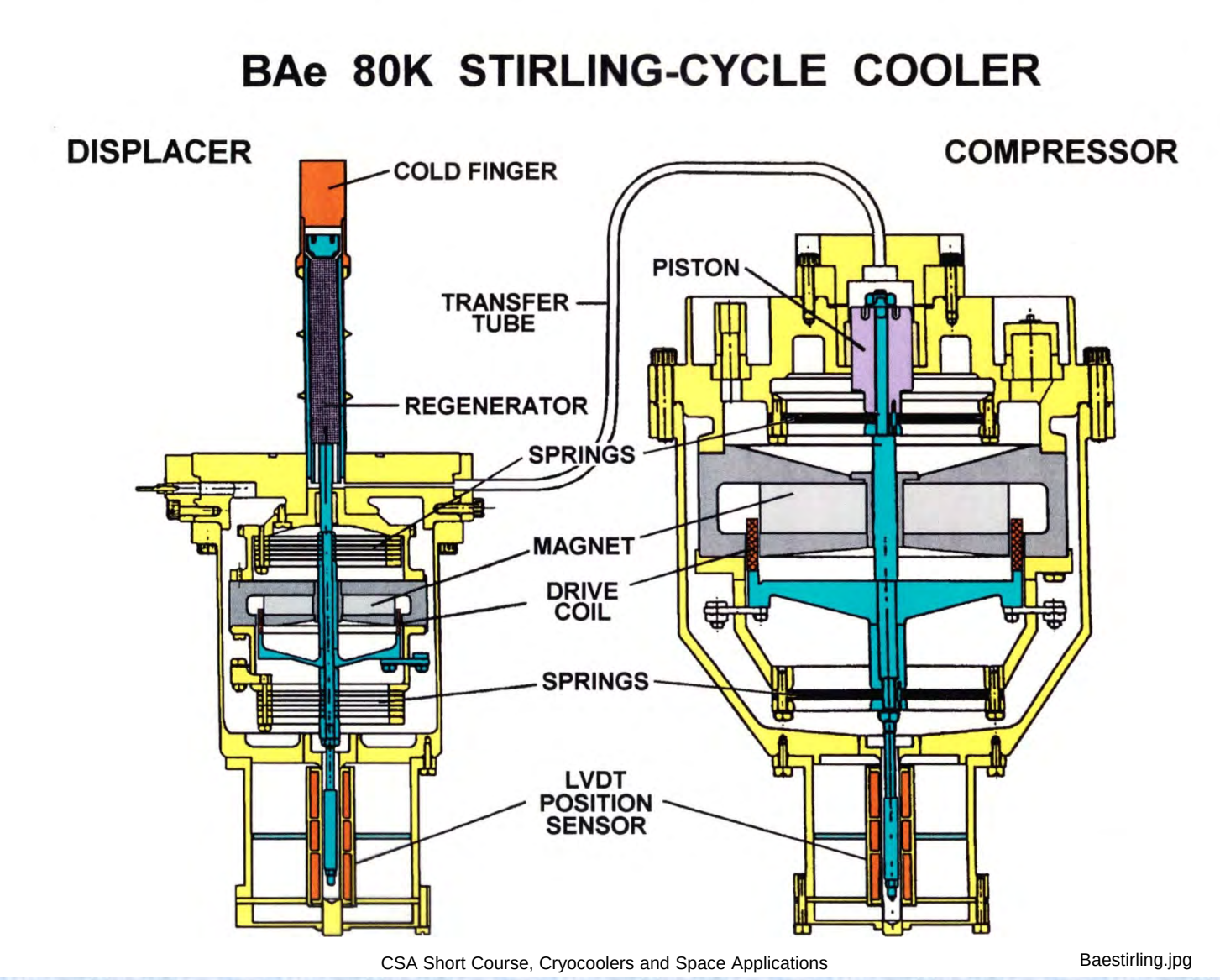


b **Spiral flexure bearing**



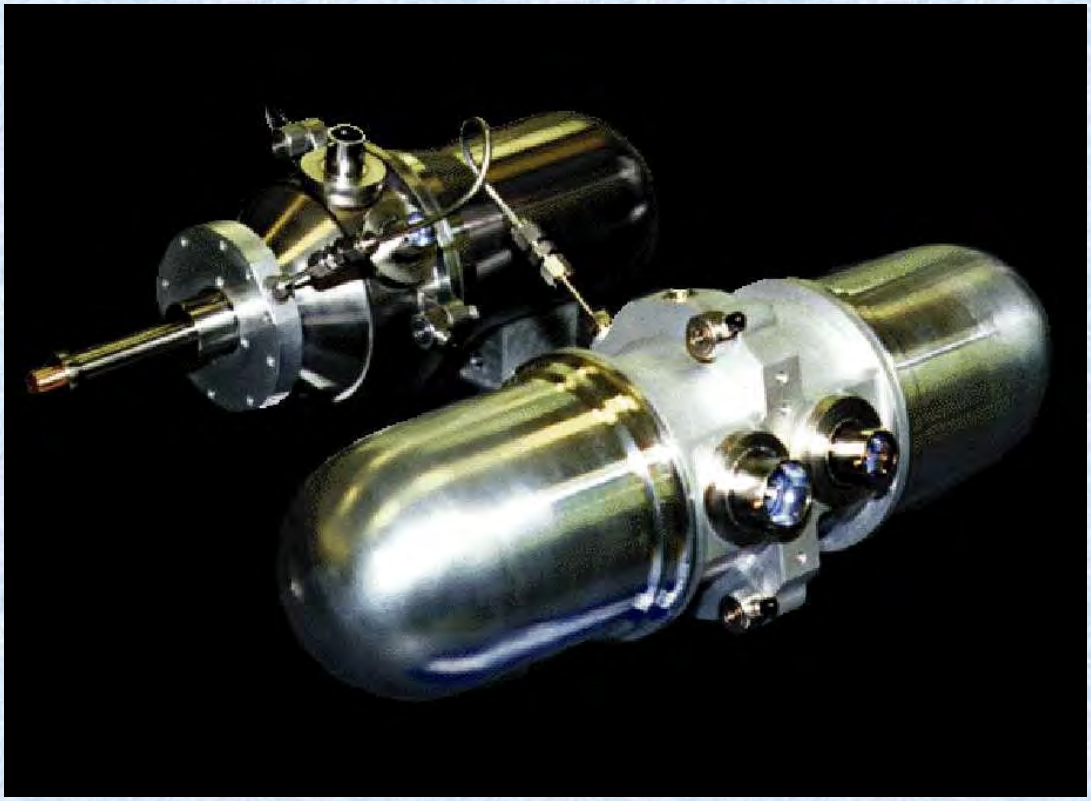
Oxford3ca.cdr

Flexure-Bearing Stirling Cryocooler



Baestirling.jpg

Protoflight Space Cryocooler (PSC)



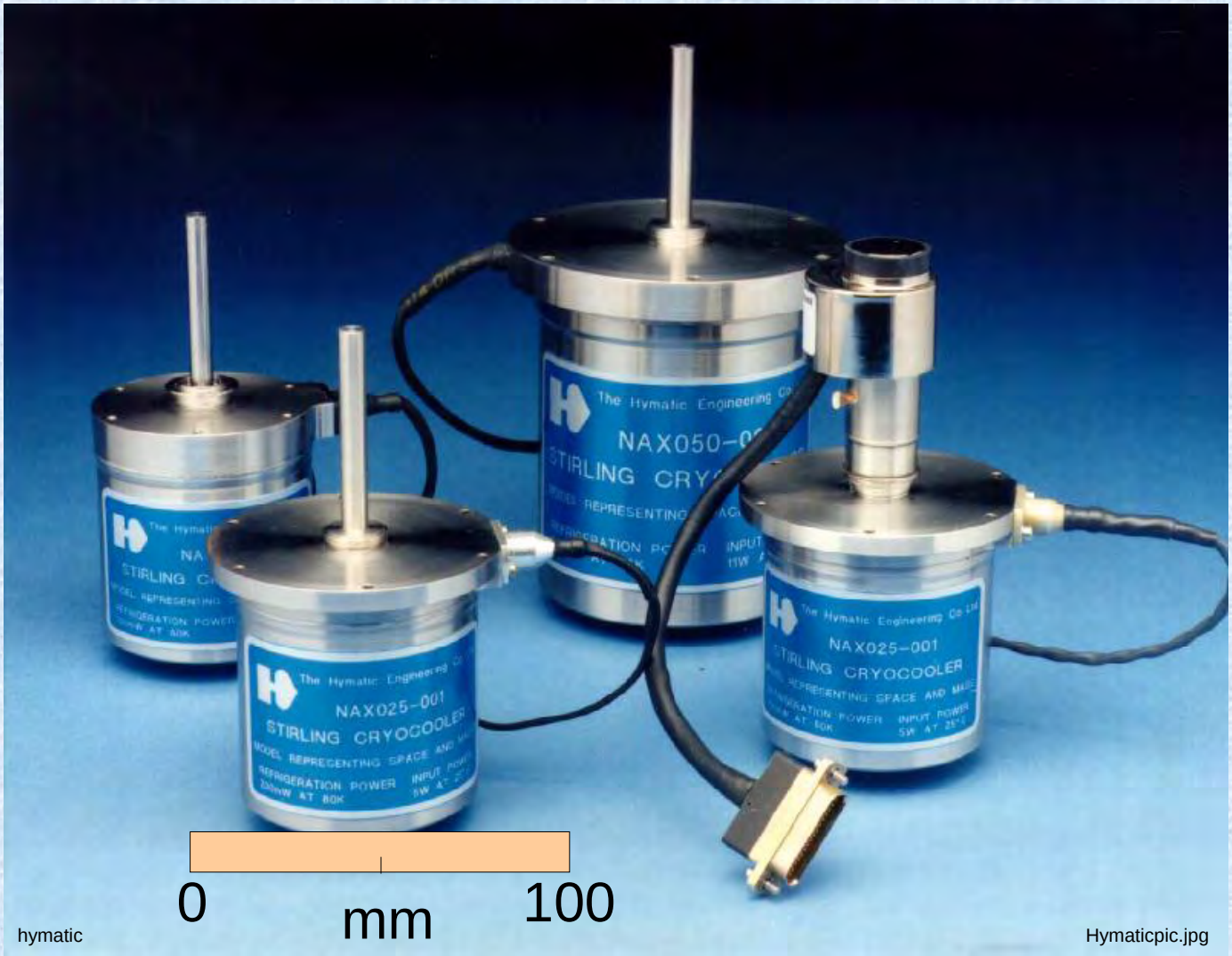
- Flexure bearing Stirling
- 9.0% of Carnot @ 35 K

Hughes/Ratheon

- 3 W @ 60 K, 87 W input
- 1.2 W @ 35 K, 100 W input
- Raytheon

psc1.cdr

Commercial Flexure-Bearing Stirling Coolers

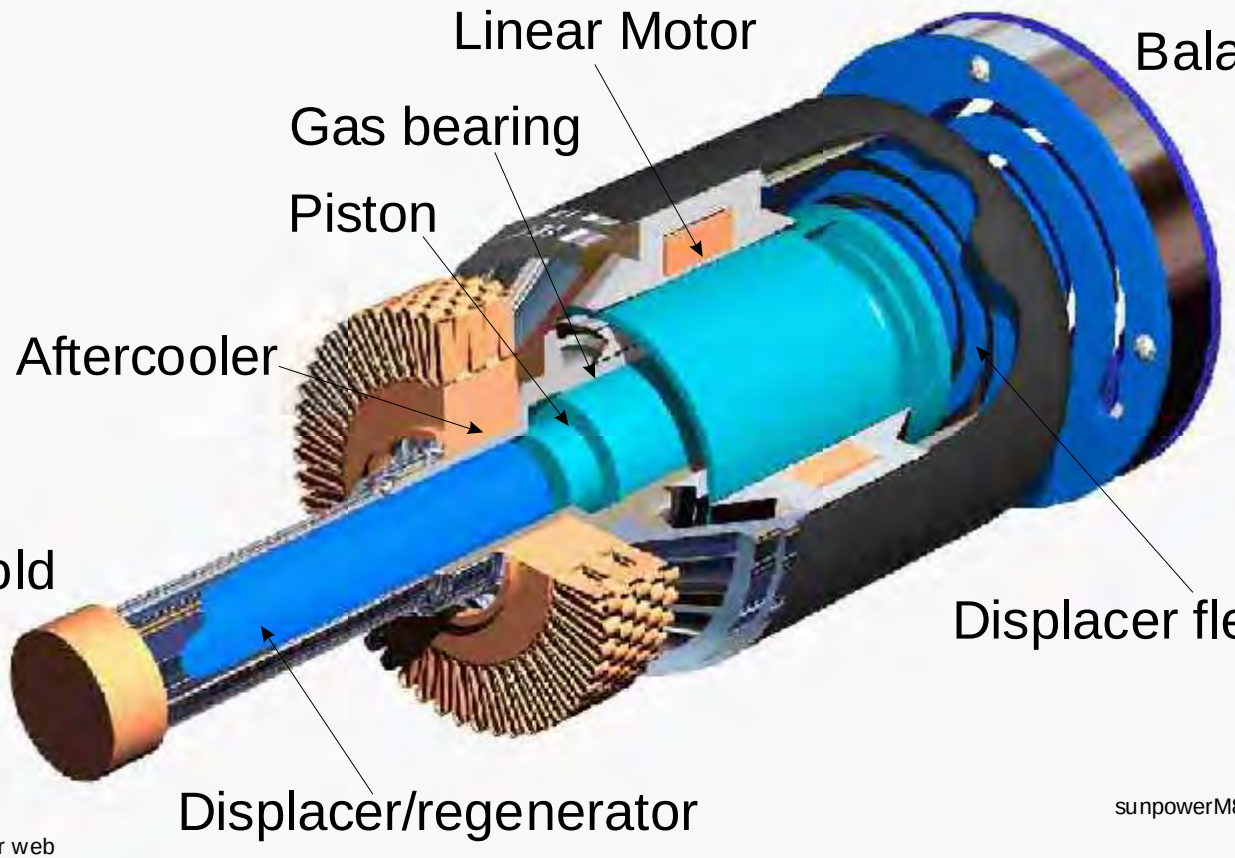


hymatic

Hymaticpic.jpg

Gas-Bearing Stirling Cryocooler

Cutaway View



Gas-Bearing Stirling Cryocooler



sunpowerM87.jpg

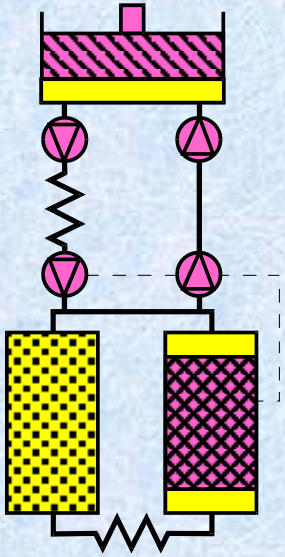
- 7.5 W @ 77 K
- 150 W input
- 15% of Carnot
- 3 - 5 yr life
- Sunpower

Outline

Session 2

- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

Gifford-McMahon Cryocoolers



ADVANTAGES

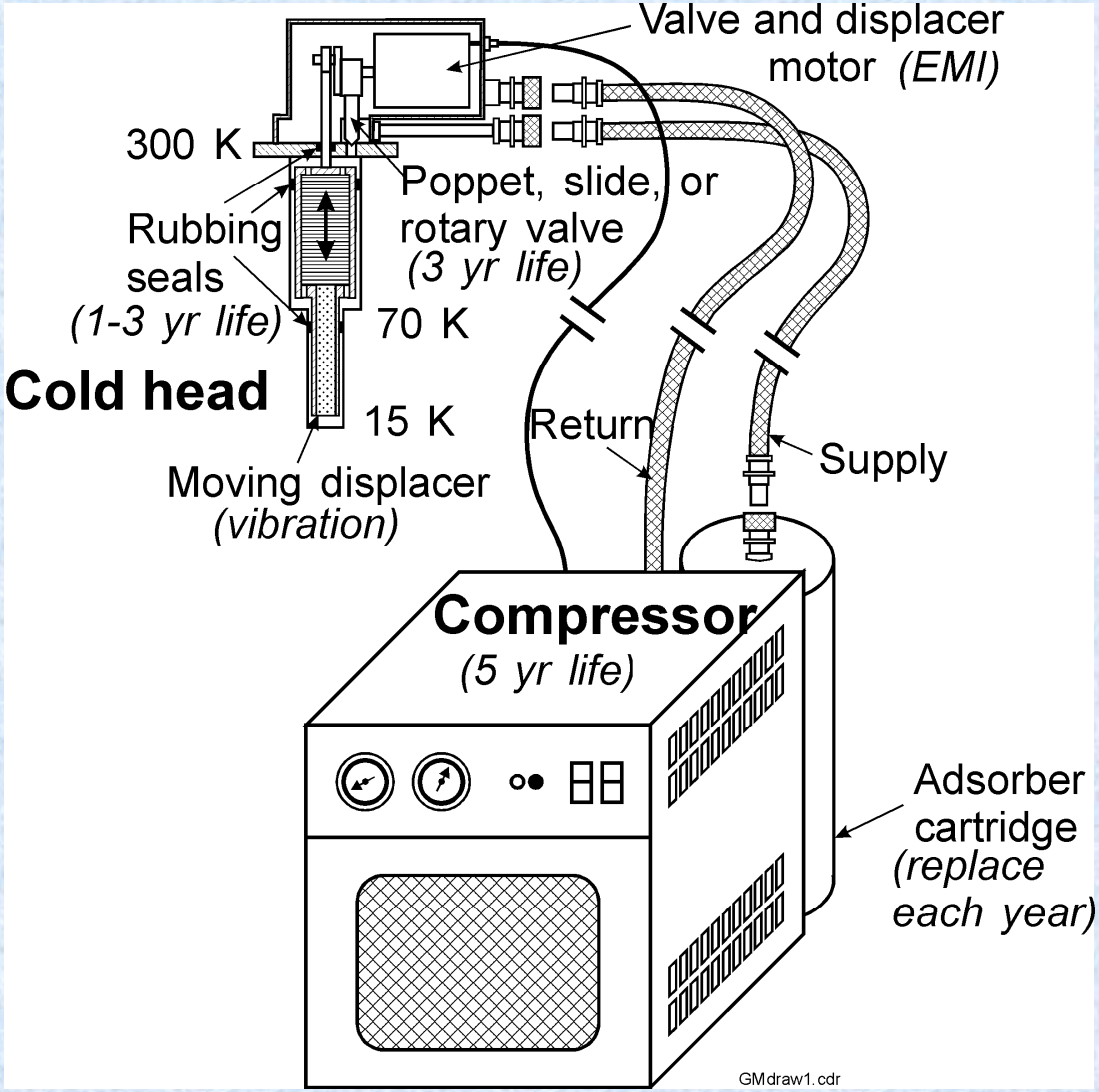
- High reliability (1-3 yrs)
- Moderate cost
- Good service
- Over 20,000/yr made

DISADVANTAGES

- Large and heavy
- Intrinsic vibration from displacer
- Low efficiency

lcec28c.cdr

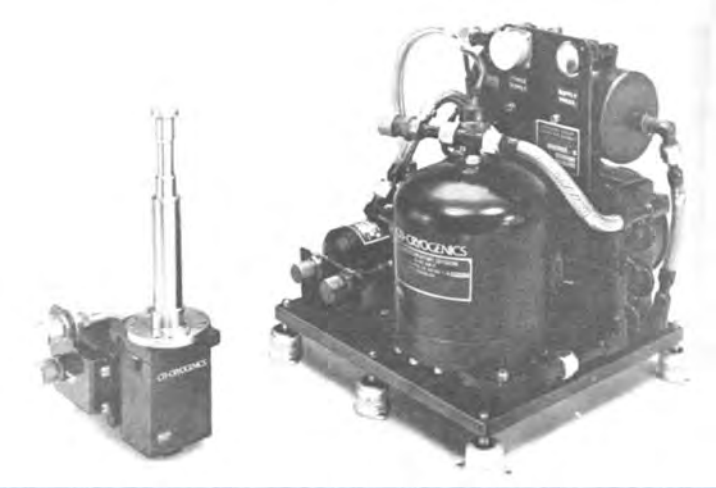
Gifford-McMahon (GM) Cryocooler System



- Used for cryopumps
- Used for MRI shield cooling or zero boiloff

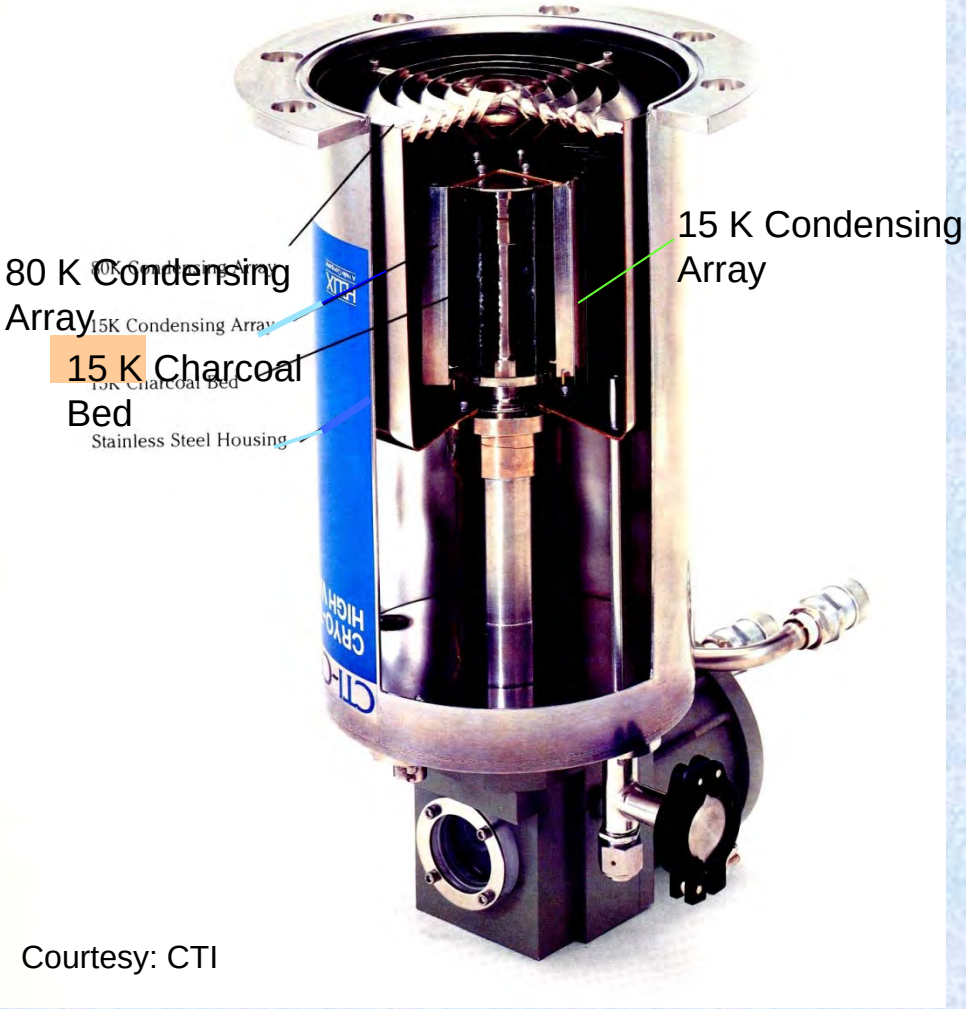
GM Cryocooler Early History (1960)

Gifford-McMahon (GM) Cryocooler Invented 1960



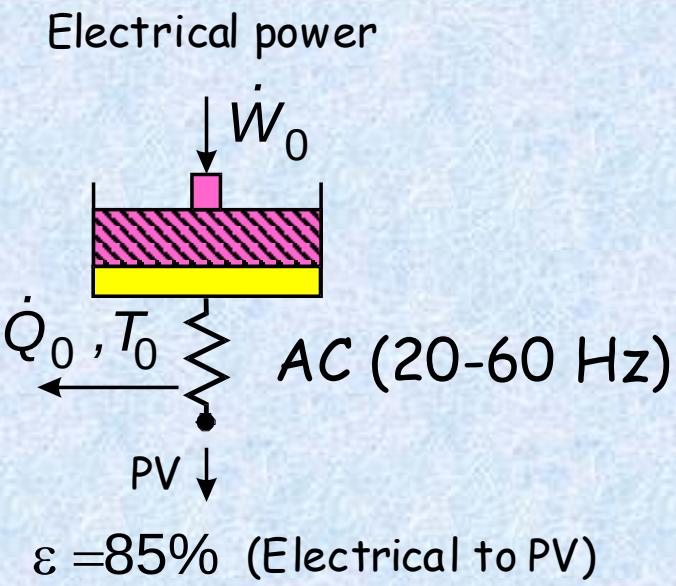
- 1970s 2-stage GM
- 1 W at 26 K
- CTI model 0120

1983 Cryopump with GM cryocooler

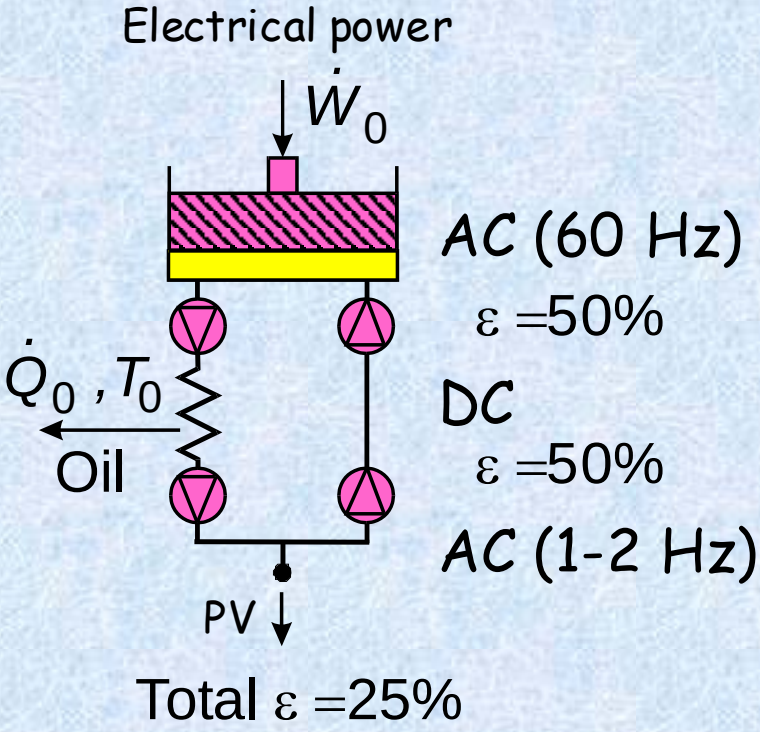


Courtesy: CTI

Efficiency of Pressure Oscillators



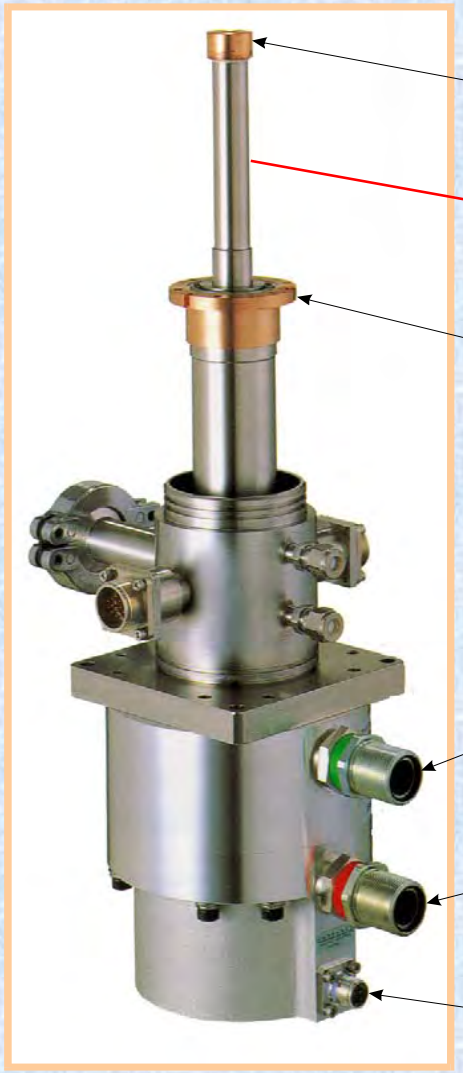
(a) Valveless compressor
(Stirling type)



(b) Valved compressor
(Gifford-McMahon type)

2-Stage Gifford-McMahon Cryocooler

Cold Head

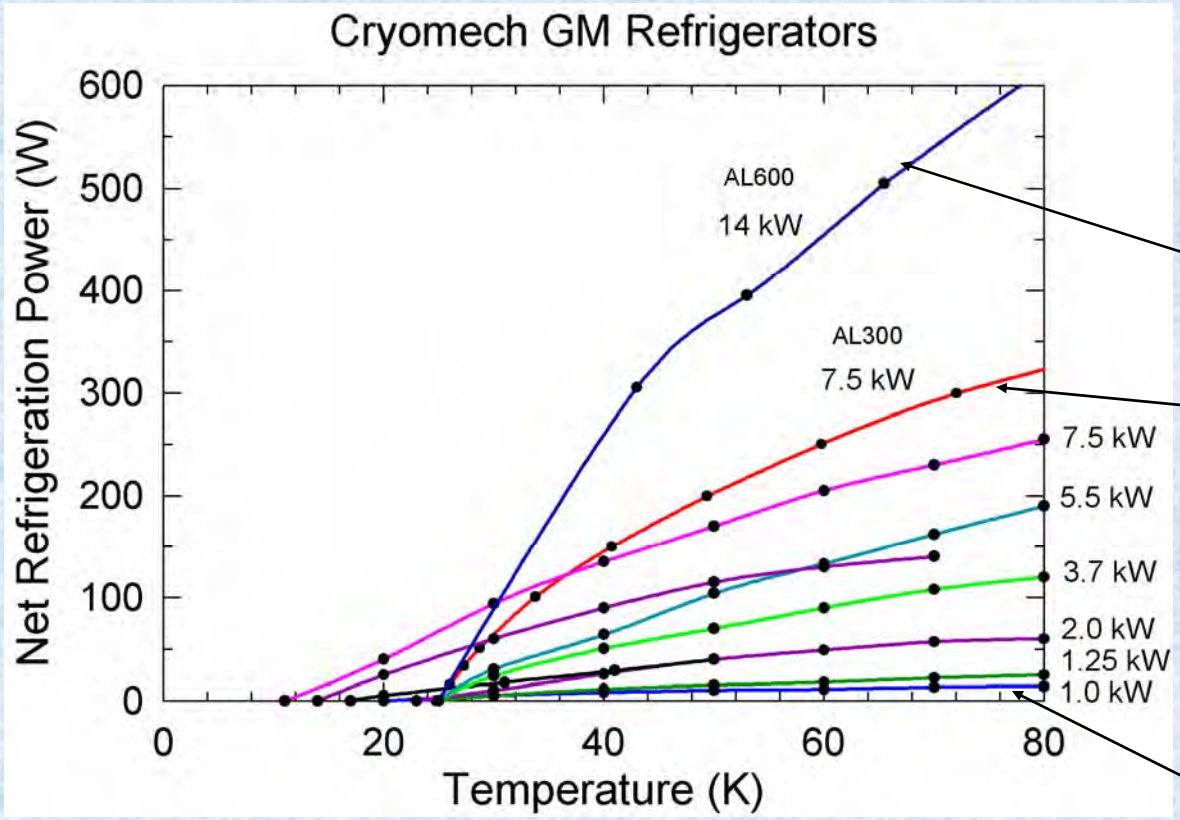


GM2stage.cdr



Range of Commercial GM Refrigerators

One-Stage Coolers







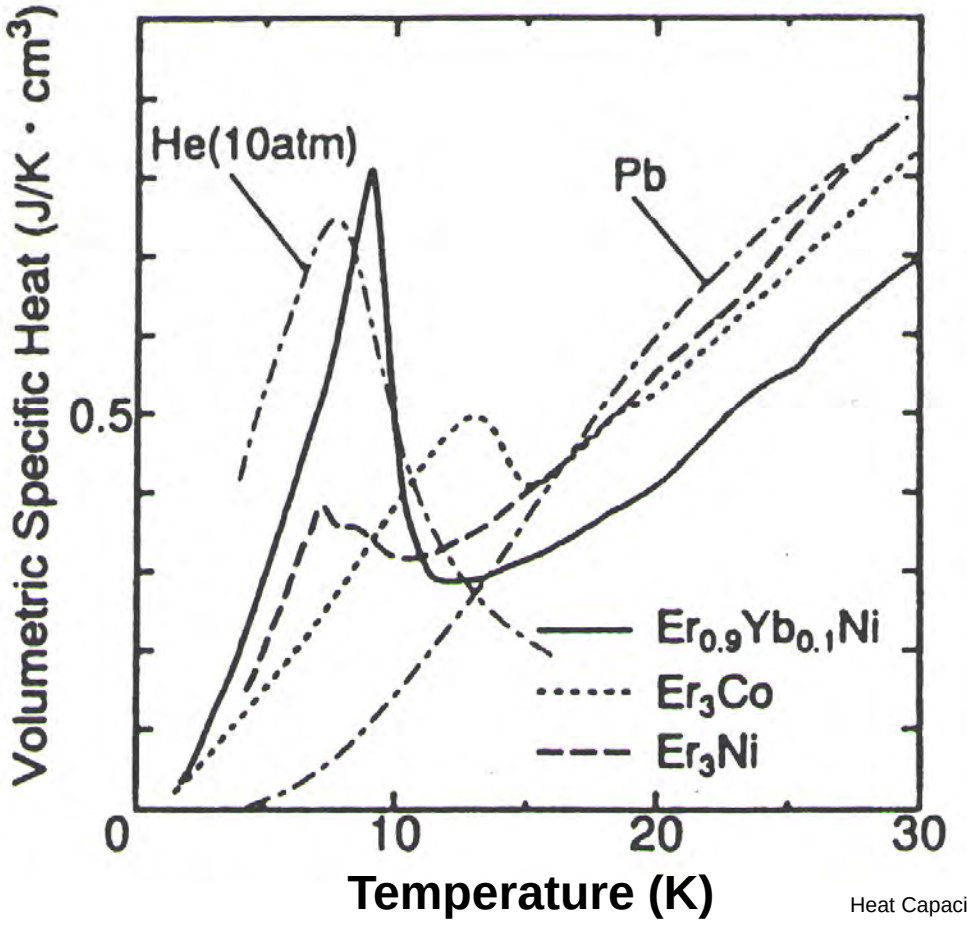


Outline

Session 2

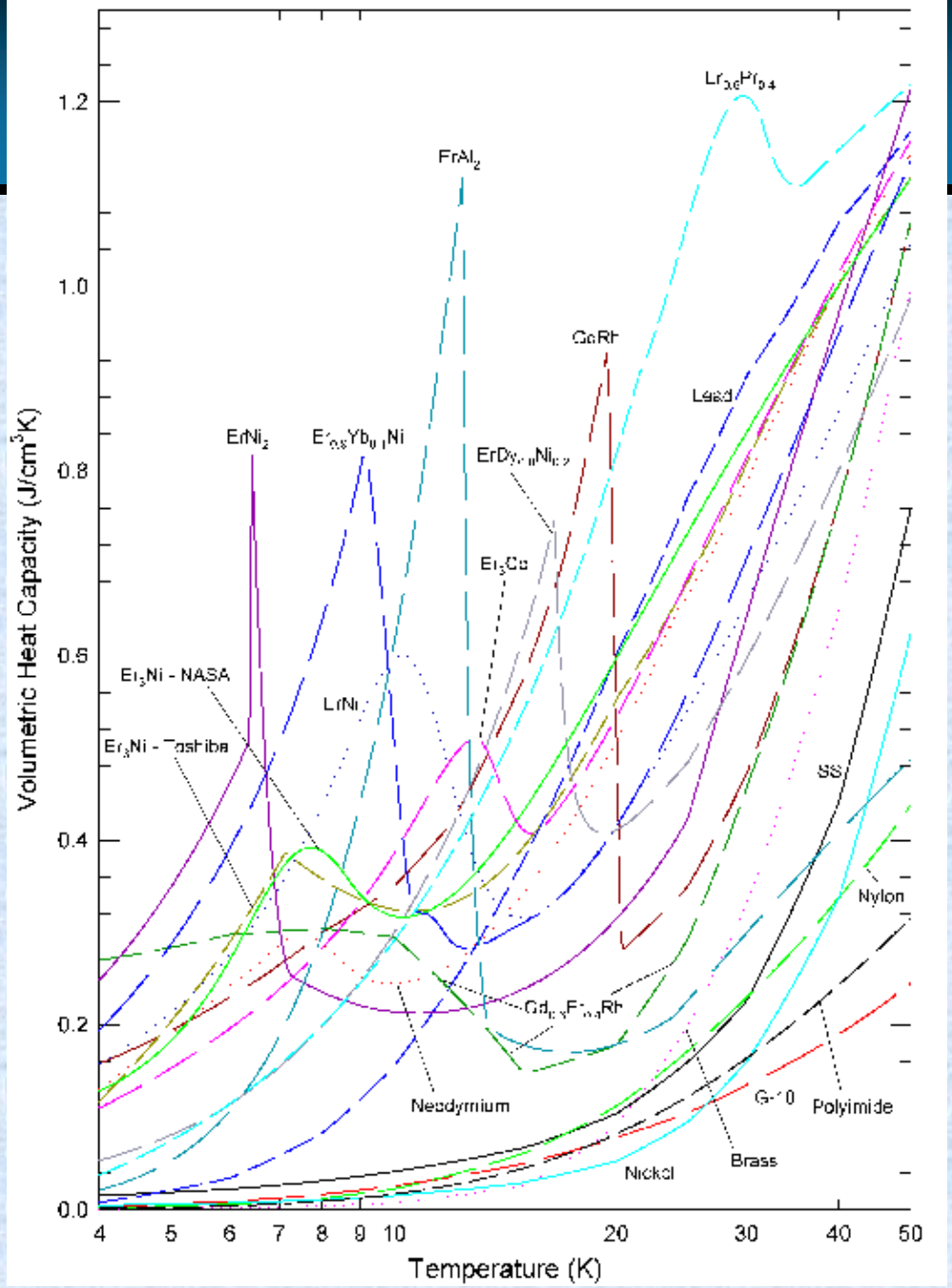
- Regenerative cycles
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Volumetric Heat Capacity



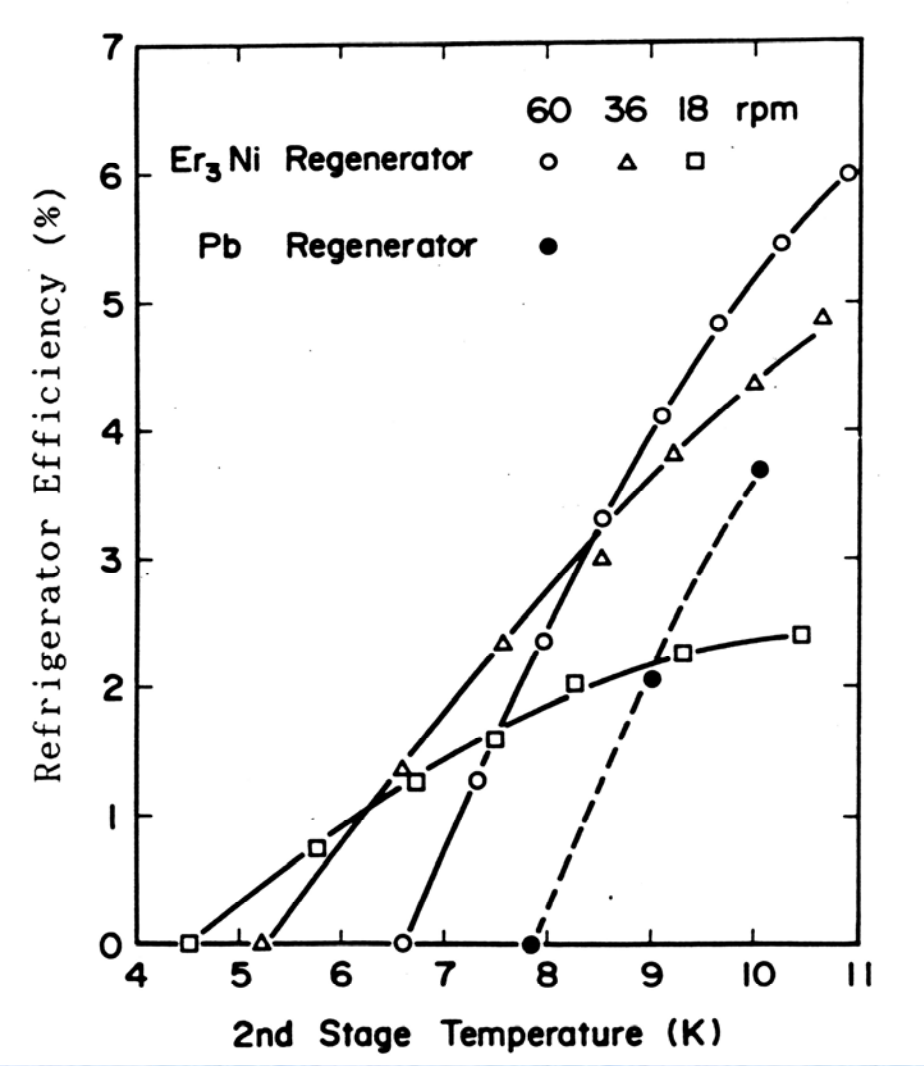
Heat Capacity03.tif

Materials in REGEN3.3



REGENheatcap1.cdr

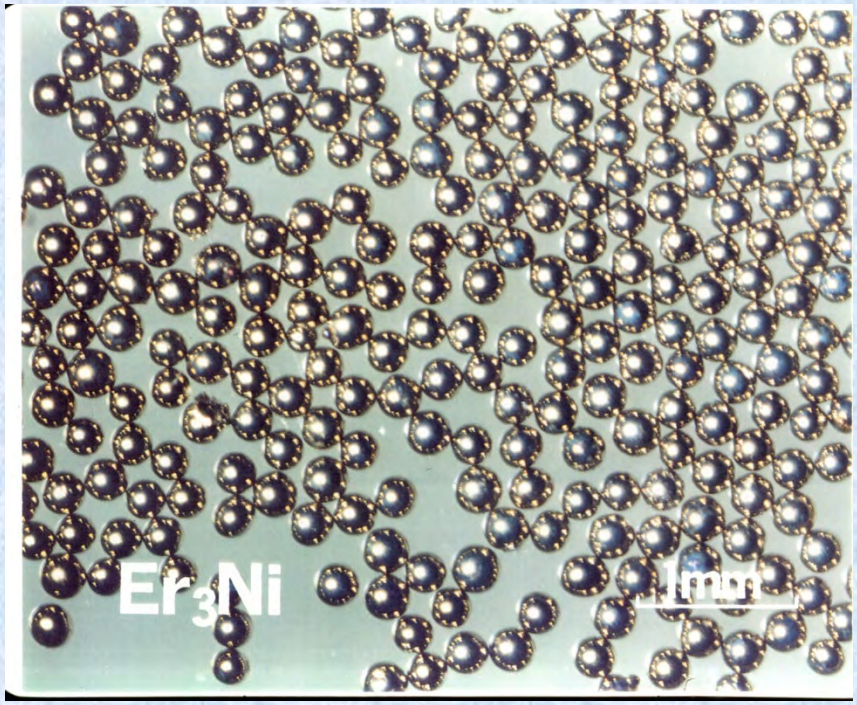
First 4 K GM Cryocooler Performance



From: Kuriyama *et al.* (1989)

Gifford McMahon.jpg

Regenerator Materials



Sphere porosity = 0.38

Irregular particle porosity > 0.50

- Er_3Ni spheres developed by Toshiba in 1989
- Kuriyama, *et al.* reached 4.5 K in 1989
- Ushered in development of 4 K GM and PT cryocoolers

GM Cryocoolers for MRI Magnets

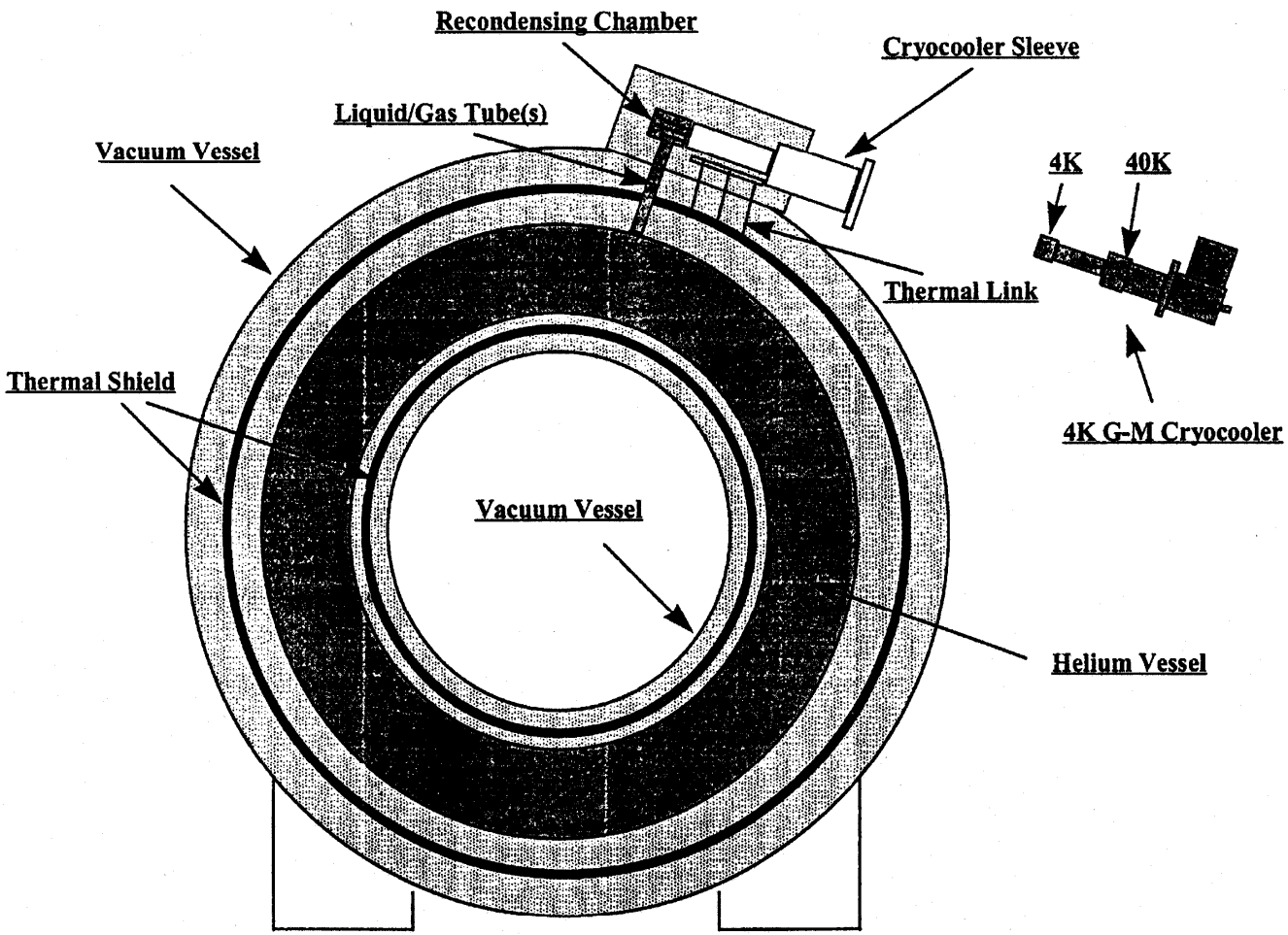


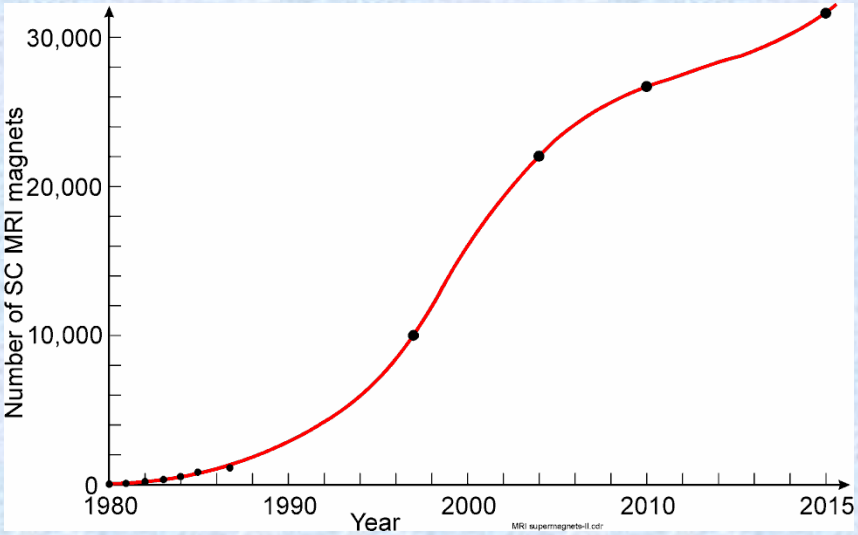
Figure 6. Schematic of the K4 zero-boil-off magnet and cryostat.

Zero boil-off

4 K GM

Magnetic Resonance Imaging (MRI)

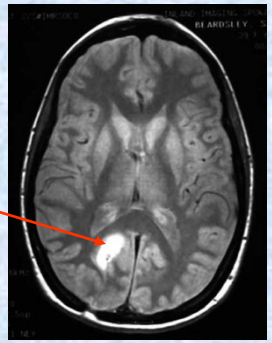




Cumulative number of MRI superconducting magnets sold

- 1.5 T Superconducting magnets
- 1 W at 4 K
- Non-magnetic regenerators
- >40,000 4 K GM cryocoolers since 1995

Tumor

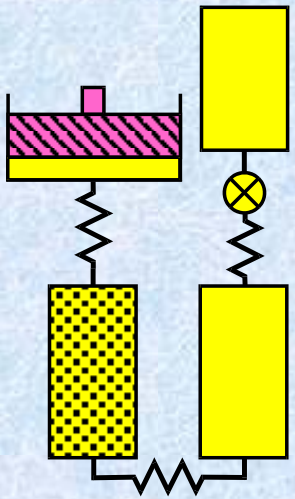


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Pulse Tube Cryocoolers



lcec23cb.cdr

ADVANTAGES

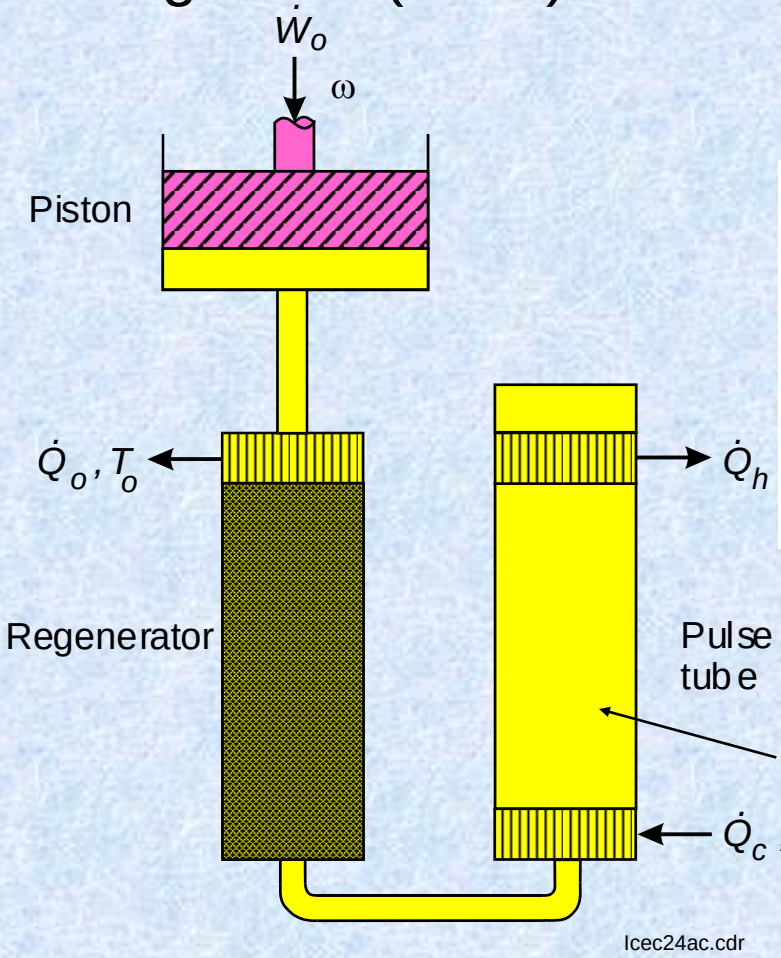
- High efficiency (15-20% Carnot at 80 K)
- No cold moving parts
  - Higher reliability
  - Lower vibration and EMI
  - Lower cost

DISADVANTAGES

- Short history (OPTR since 1984)
- Gravity-induced convective instability
- Lower limit to size for efficient pulse tube

Early Pulse Tube Refrigerators

Basic Pulse Tube Refrigerator (1967)

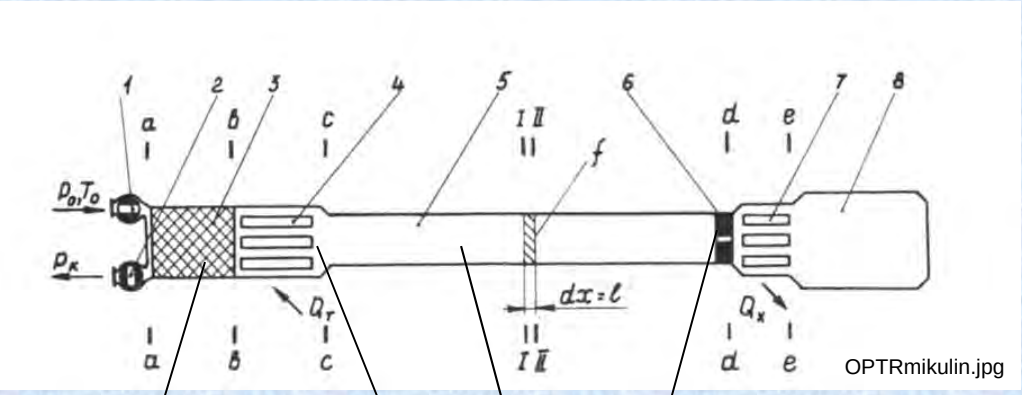


Icec24ac.cdr

Gifford and Longsworth

Orifice Pulse Tube Refrigerator (GM type) (1983)

Mikulin, *et al.* (1983)



Regenerator

105 K

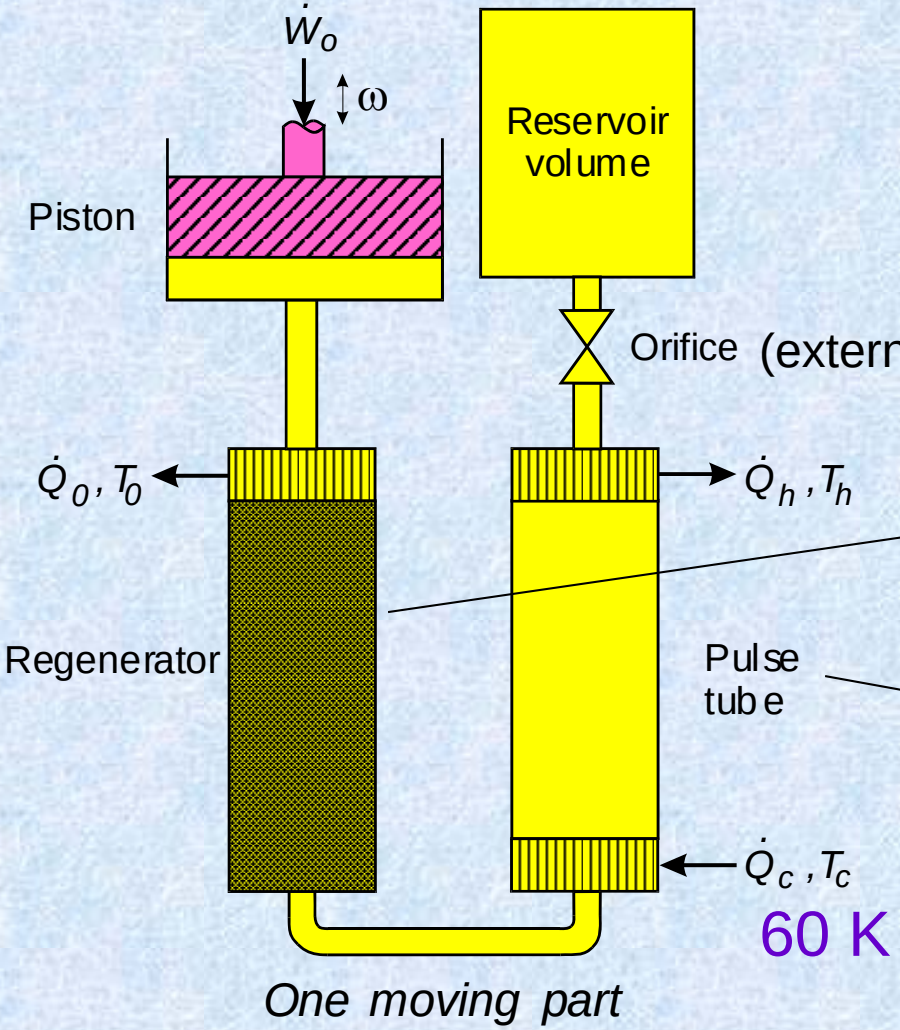
Pulse tube

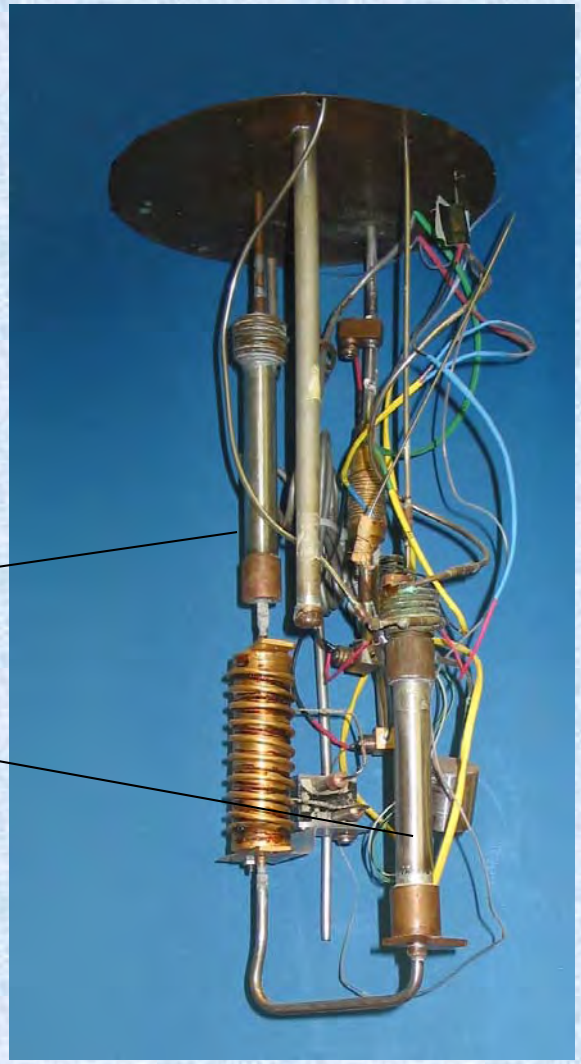
Orifice (Internal)

Requires irreversible heat transfer between gas and tube

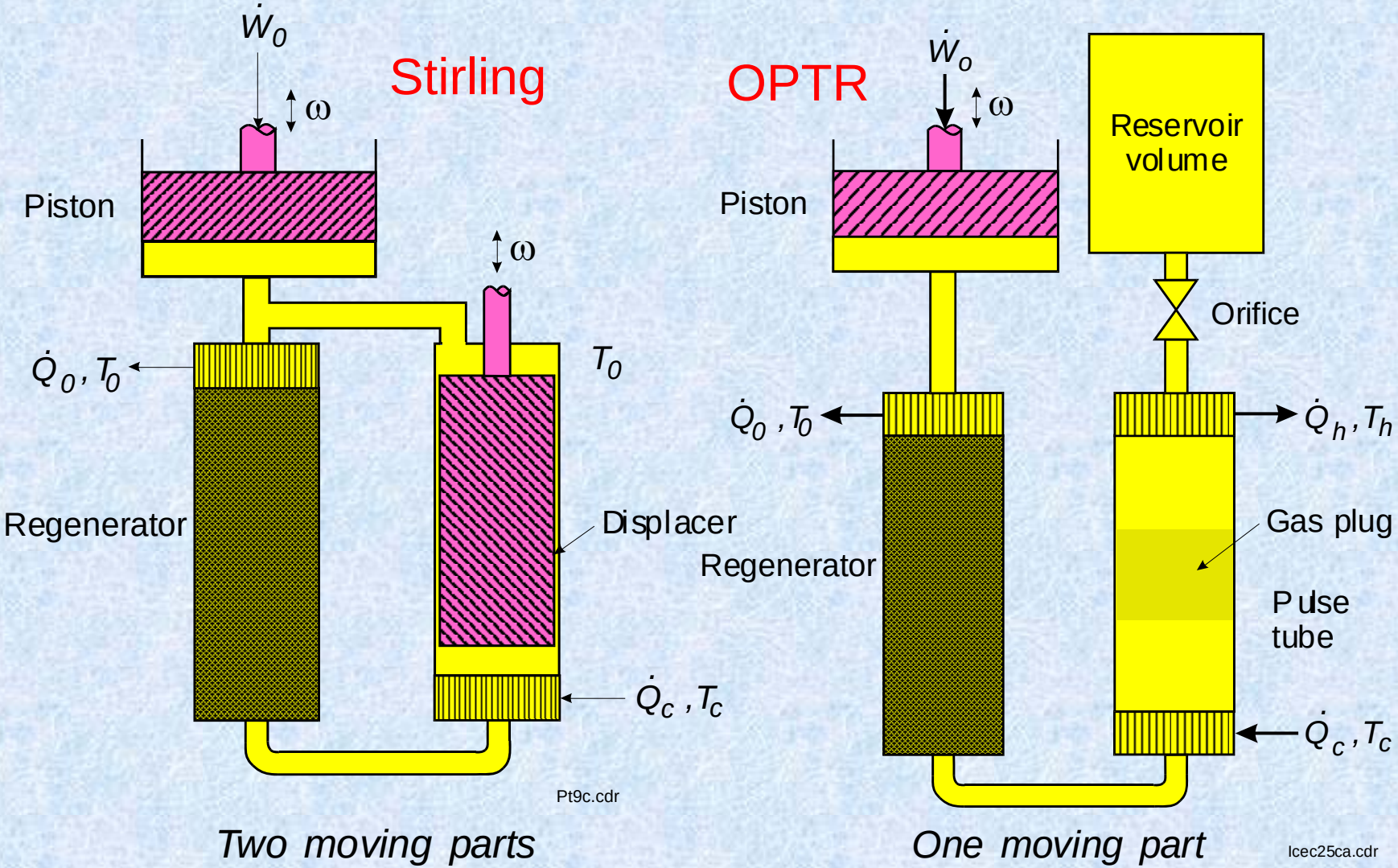
NIST Orifice Pulse Tube Refrigerator

Stirling type 1985

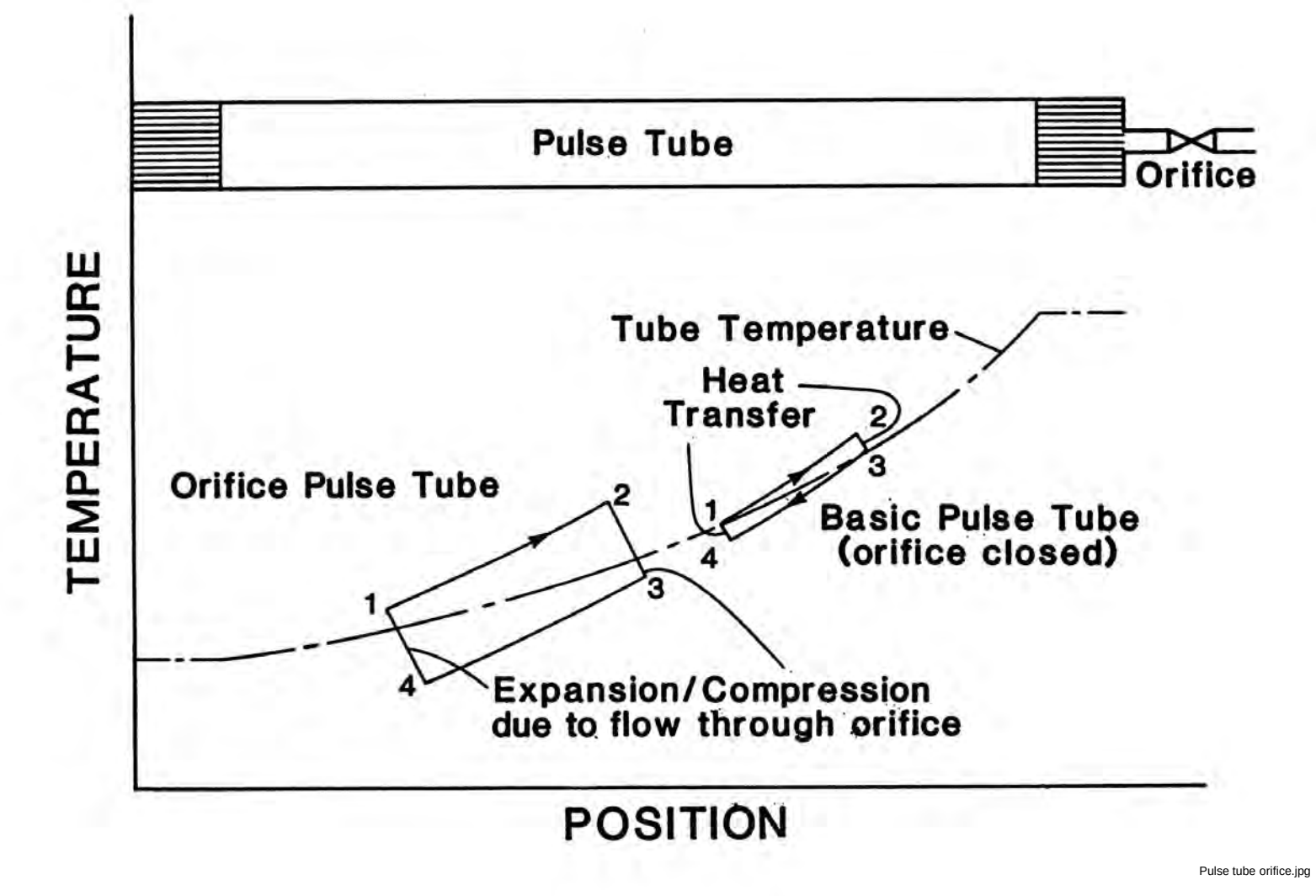




Comparison of Stirling and Orifice Pulse Tube Refrigerators (OPTR)

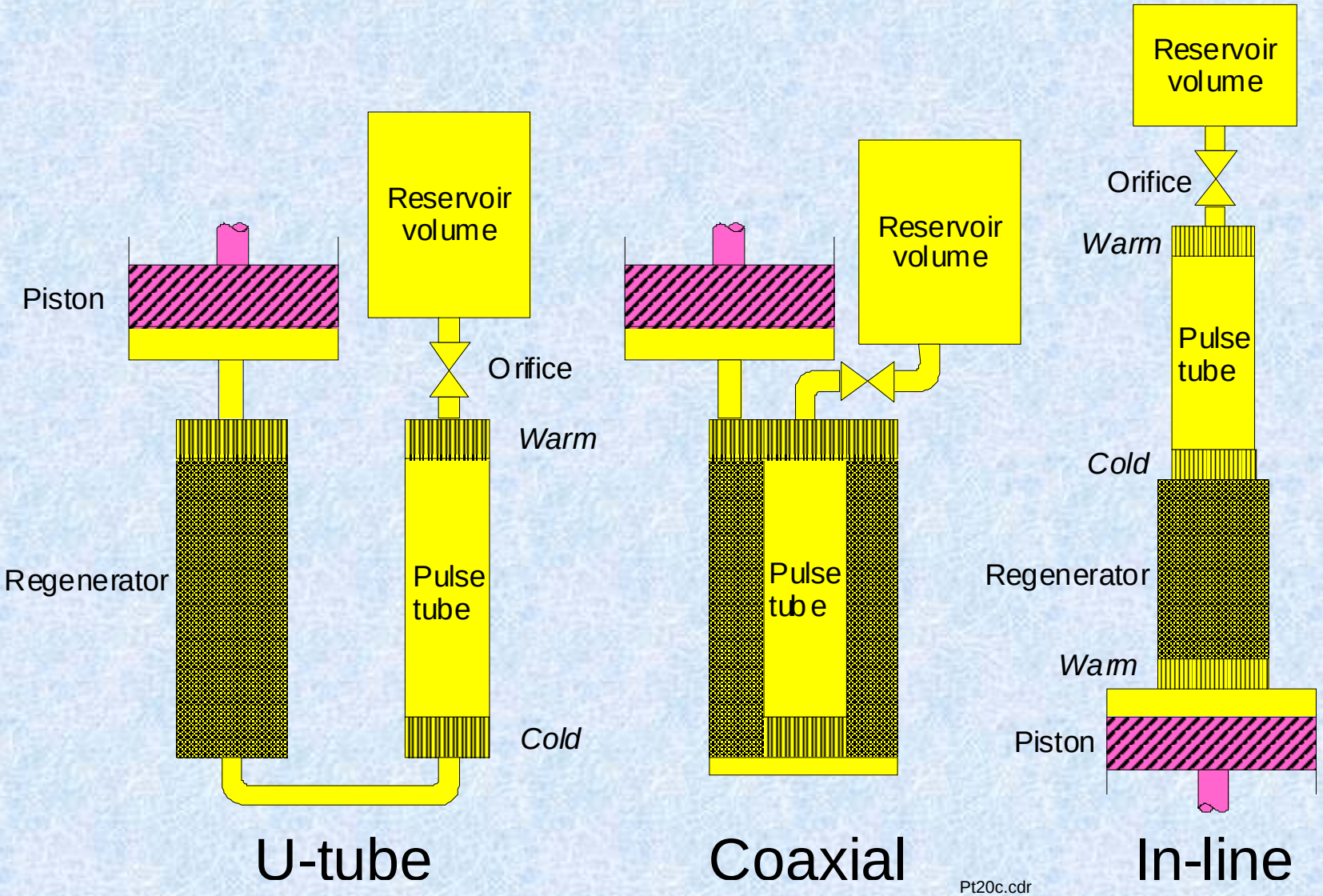


Comparison of Orifice and Basic Pulse Tubes



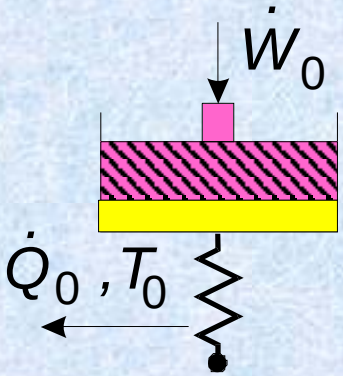
Pulse tube orifice.jpg

Pulse Tube Geometry

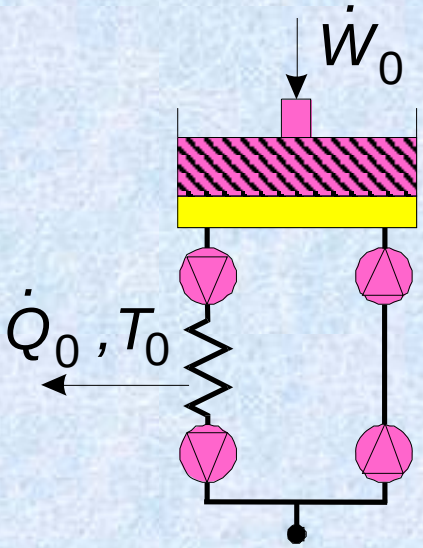


Pt20c.cdr

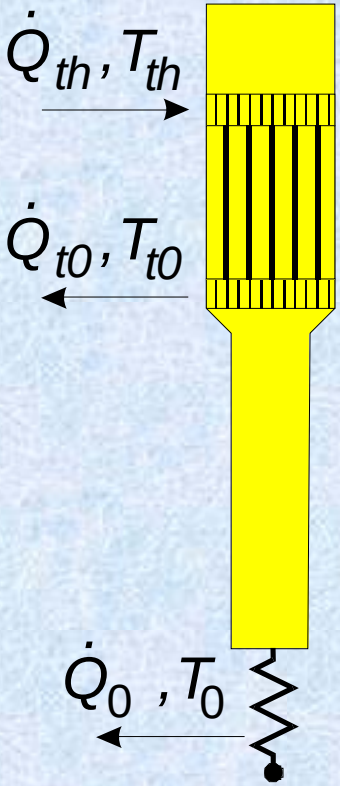
Types of Pressure Oscillators



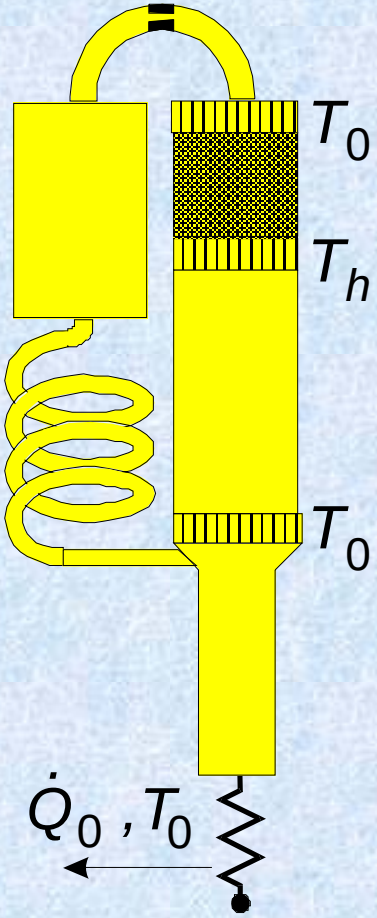
(a) Valveless compressor
(Stirling type)



(b) Valved compressor
(Gifford-McMahon type)



(c) TAD
ThermoAcoustic driver



(d) TASHE
ThermoAcoustic Stirling Heat Engine

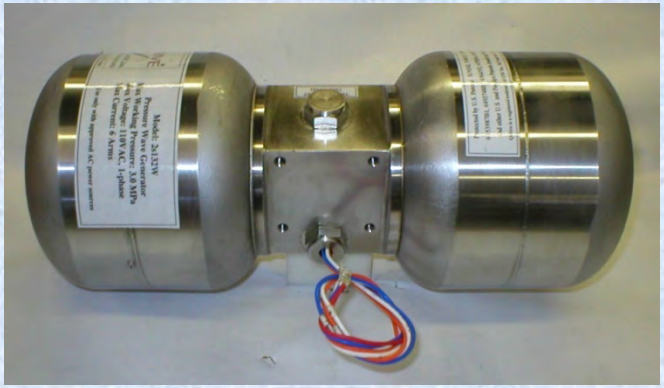
Csa6c.cdr

Pressure Oscillators (Stirling-Type)

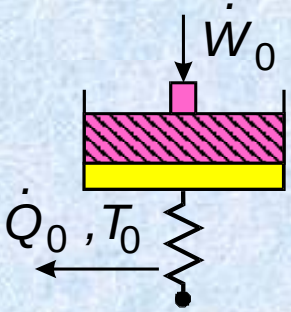


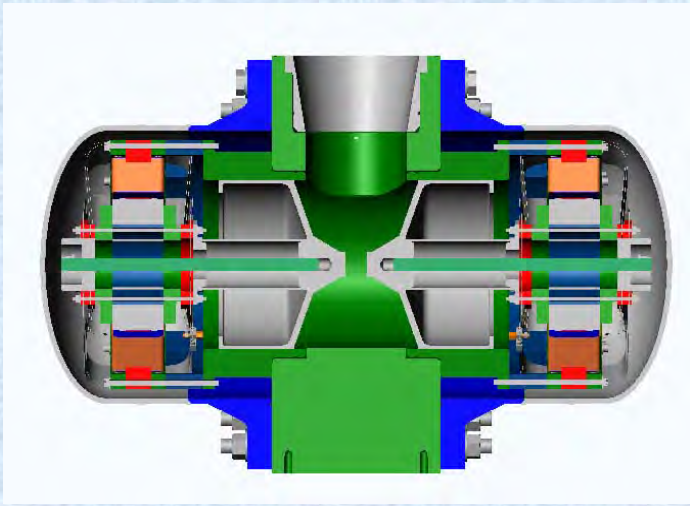


CFIC 210 W 213 mm
8.5 kg



CFIC 500 W 351 mm
11.4 kg



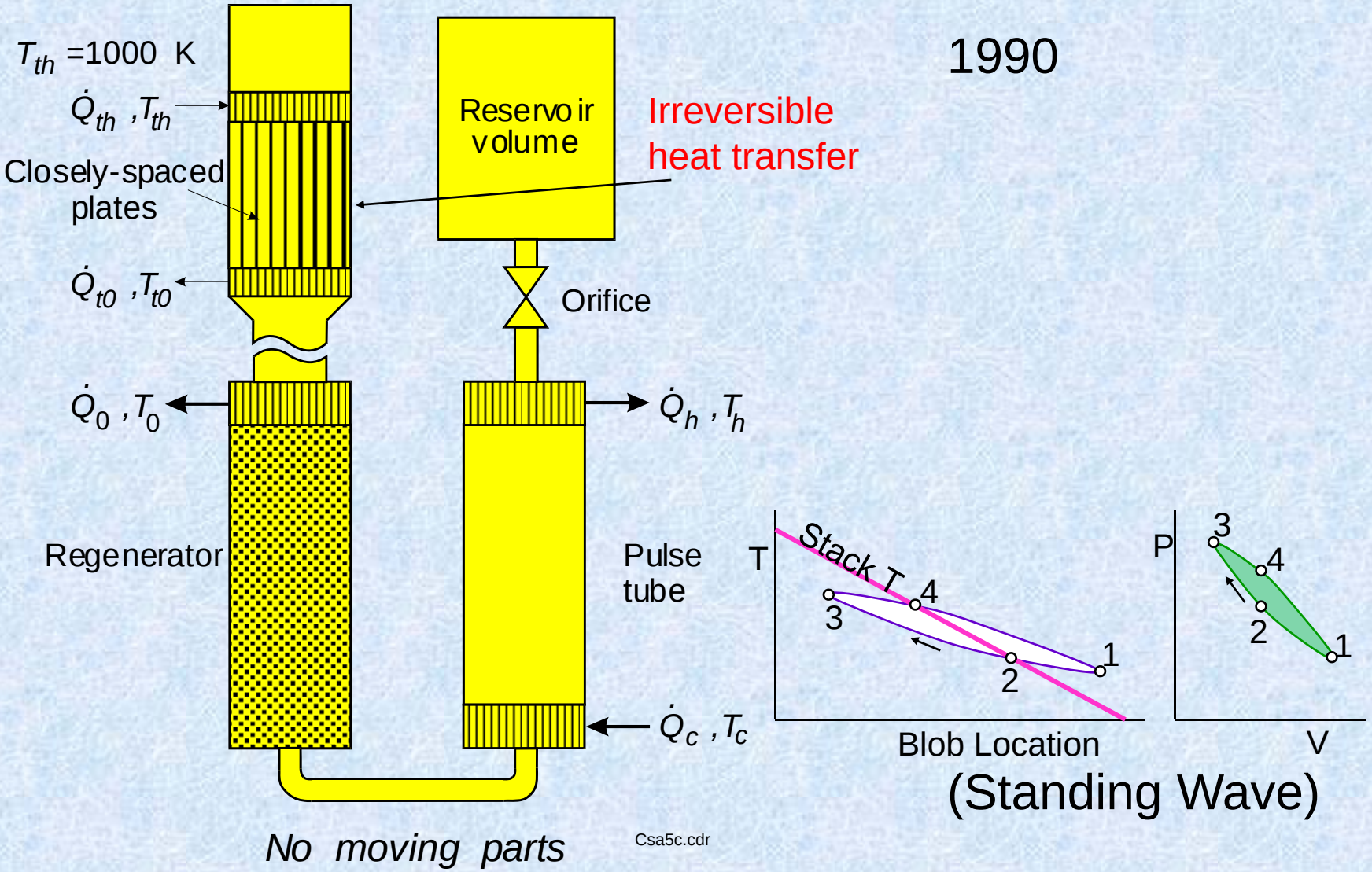




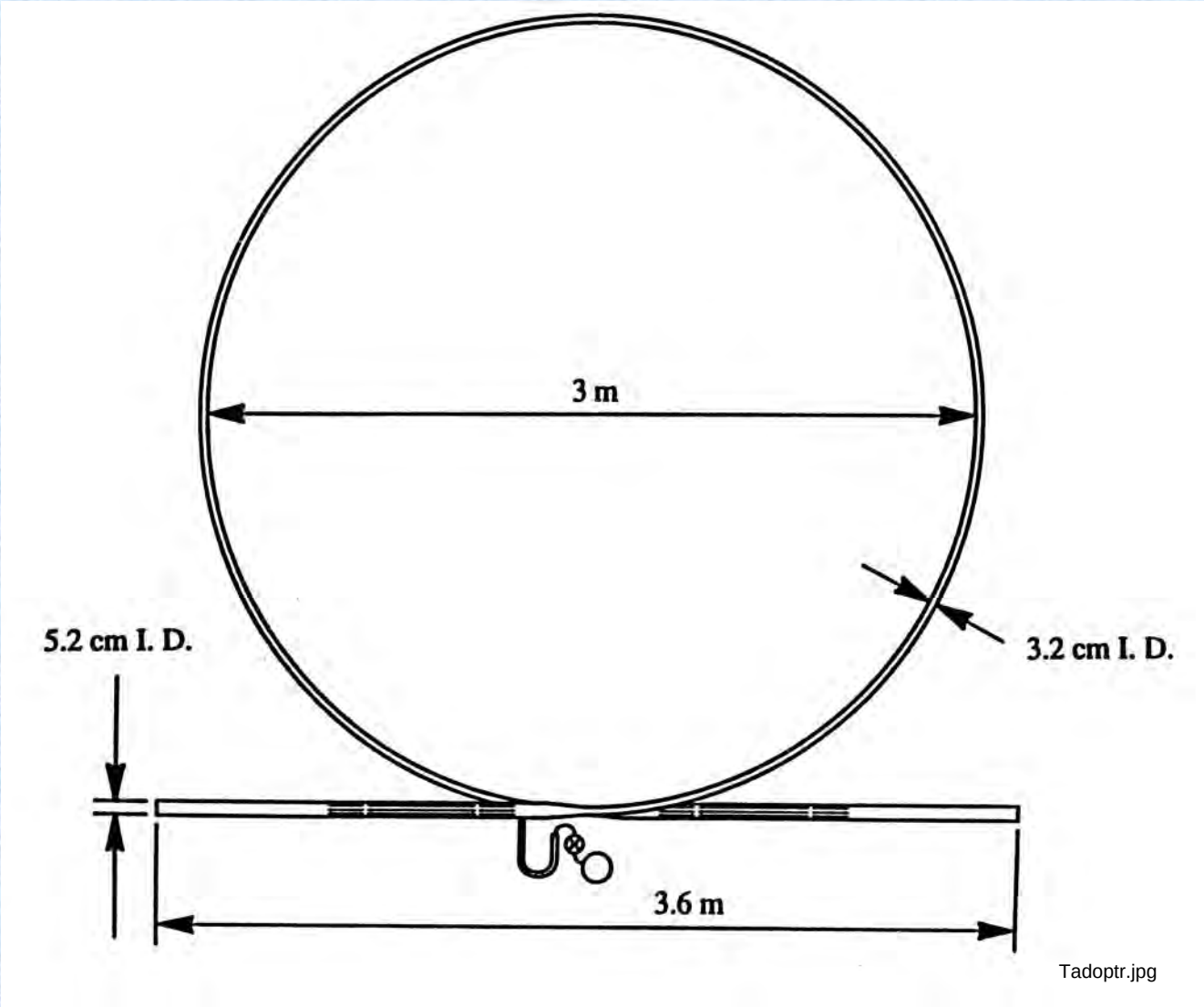
CFIC
20 kW
954 mm
413 kg

www.cficinc.com

Thermoacoustically Driven Orifice Pulse Tube Refrigerator (TADOPTR)

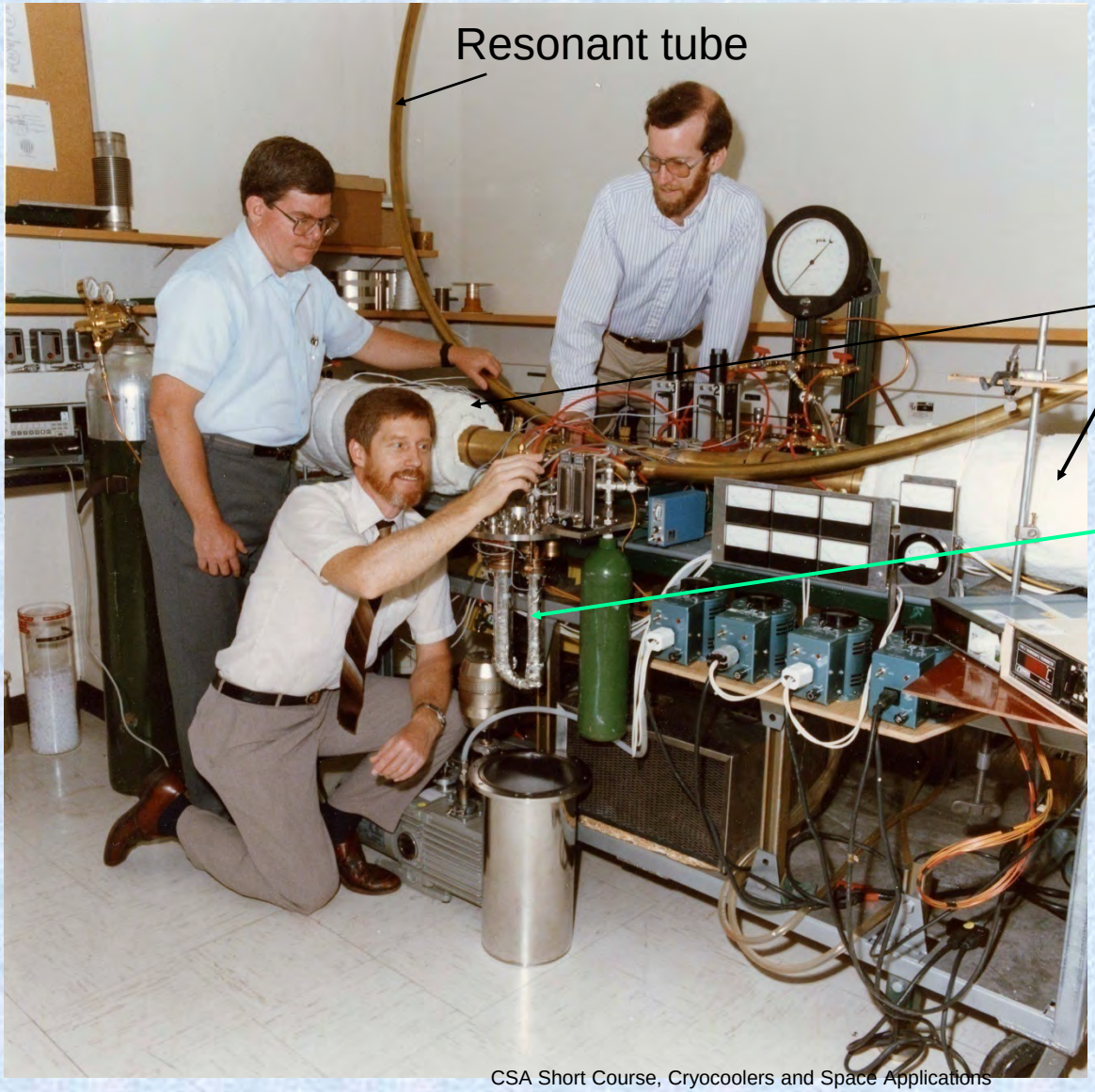


NIST/LANL TADOPTR



Tadoptr.jpg

First TADOPTR: NIST/LANL



First cryocooler with no moving parts (1990)

- T_{min} = 90 K
- f = 28 Hz
- Inaudible

TADOPTRorig.jpg

TADOPTR Natural Gas Liquefier

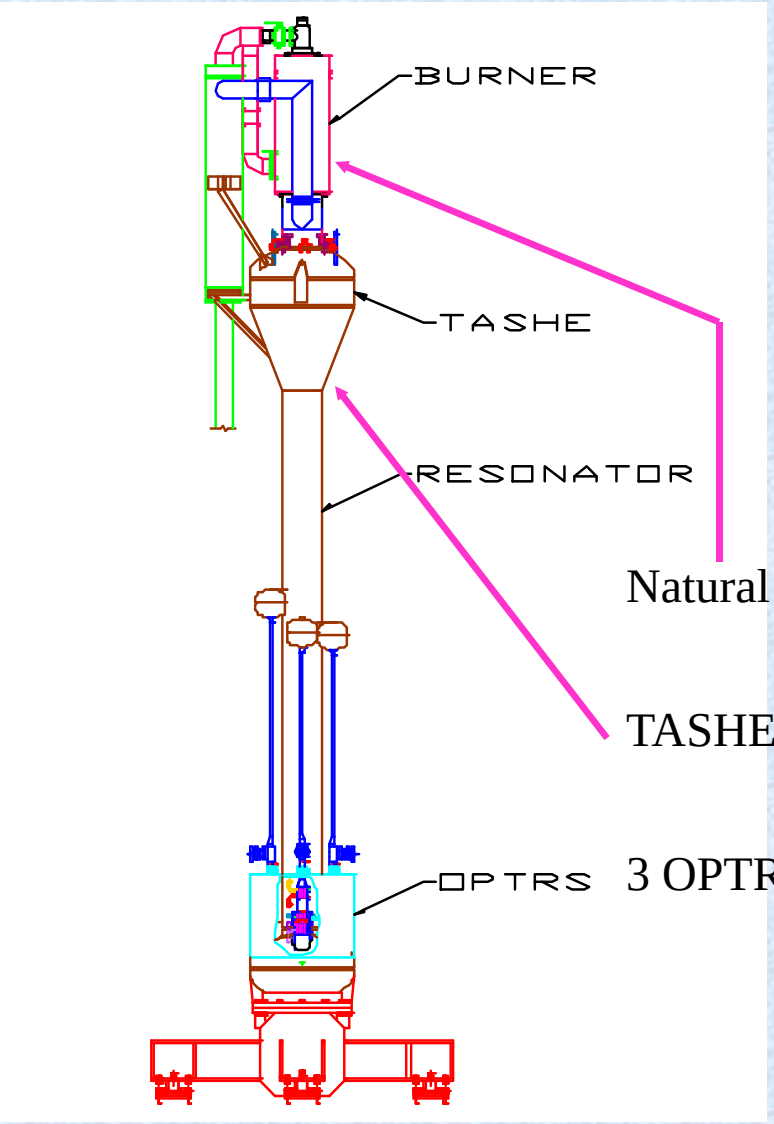


TADOPTR3c.jpg

- 2000 W at 120 K
- 600 L/day (150 gal/day) LNG
- Praxair/LANL/NIST

TASHEOPTR





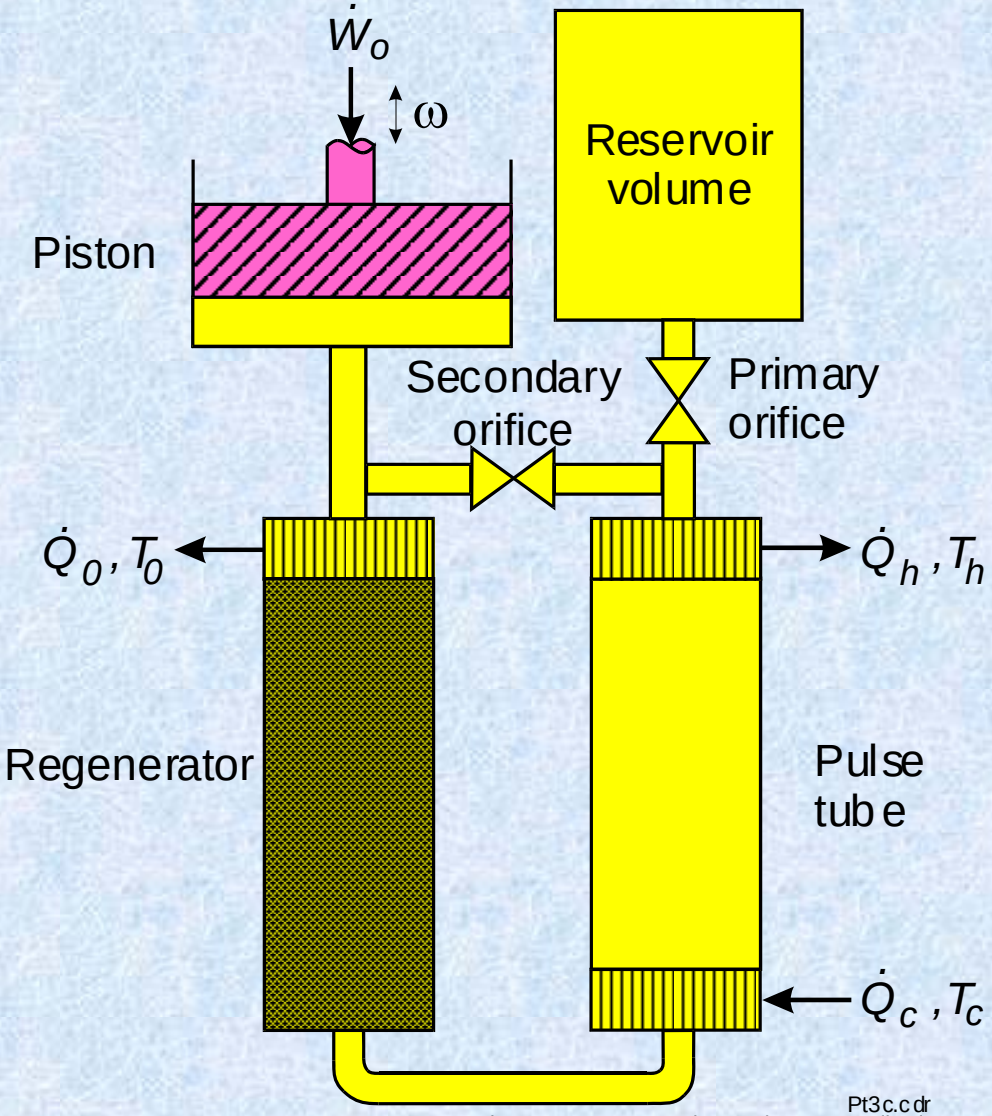
Praxair/LANL (2000)

Natural gas burner - 150 kW heat @ 700 °C

TASHE - 60 kW acoustic power @ 50 °C

3 OPTRs - 8 kW cooling power @ -150 °C

Double Inlet Pulse Tube Refrigerator



Zhu, Wu, and Chen, Xi'an Jiaotong Univ., *Cryogenics* **30**, 514 (1990)

Single	Double
173 K ⟶	132 K
52 K ⟶	43 K

Outline

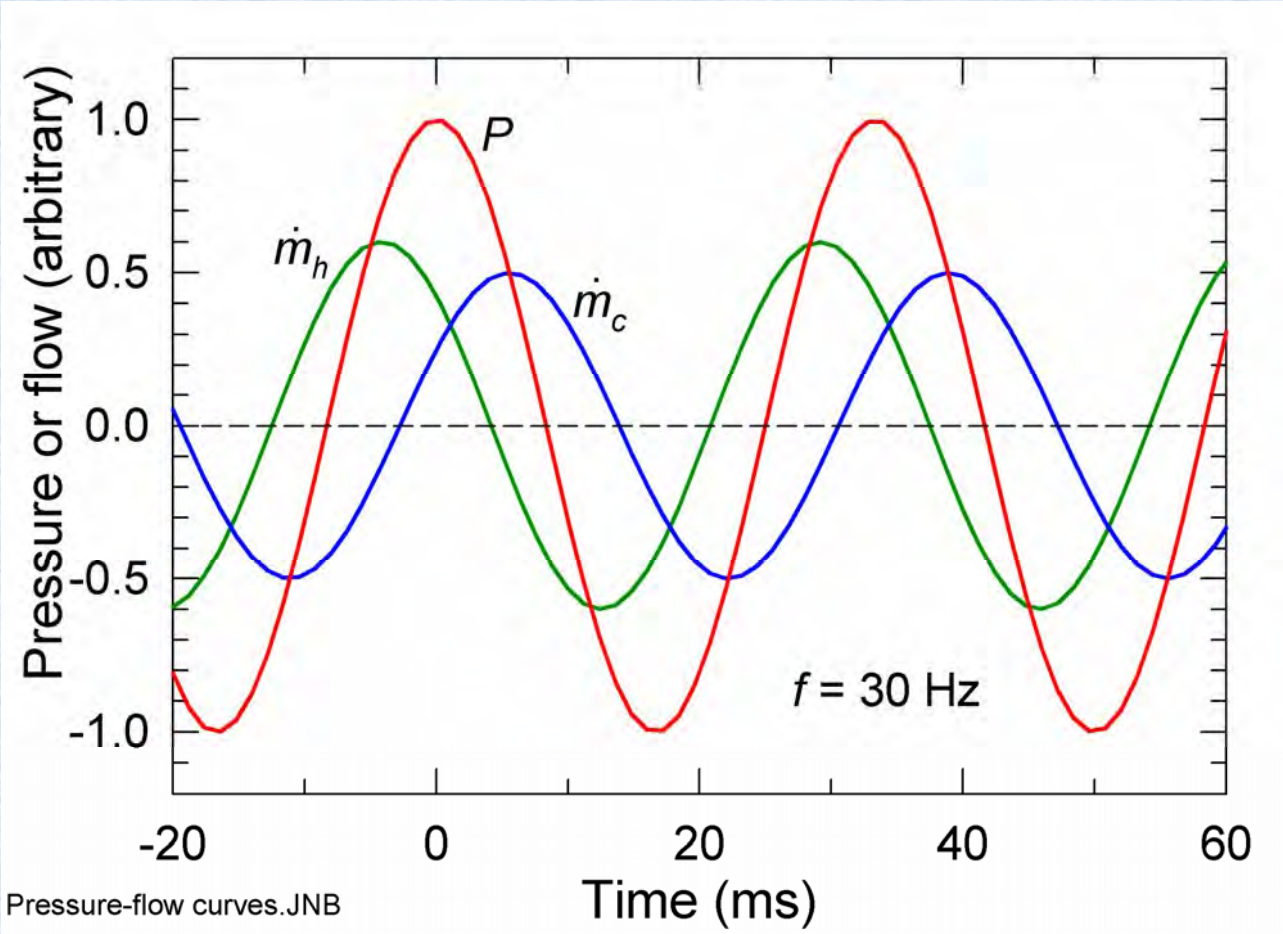
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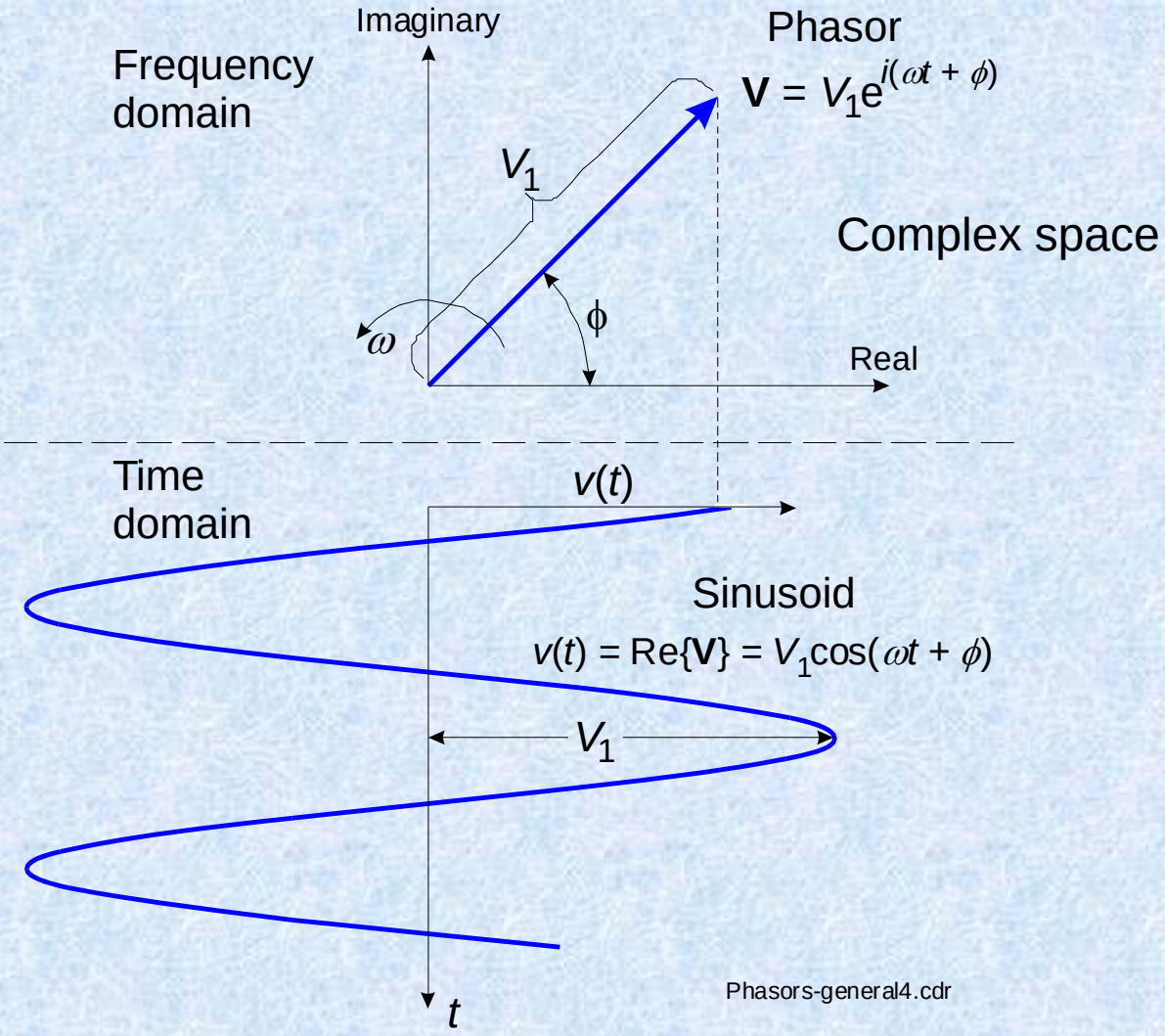
Pressure and Flow in Regenerative Cryocoolers

Time domain

- Confusing with many curves
- Difficult to visualize algebra



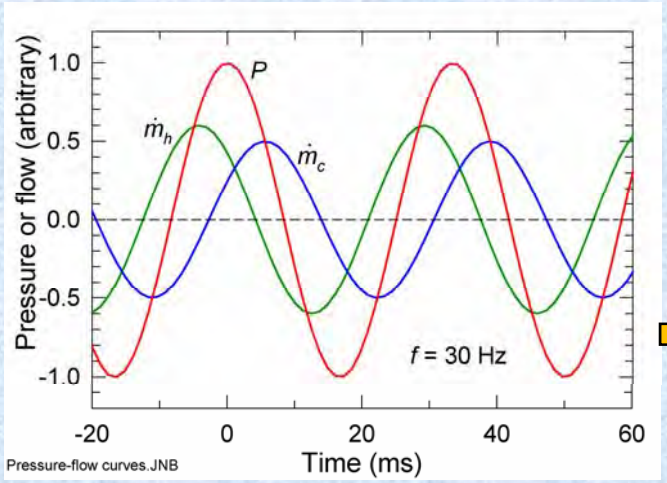
Phasors and Sinusoids



Sinusoid to Phasor Transformation

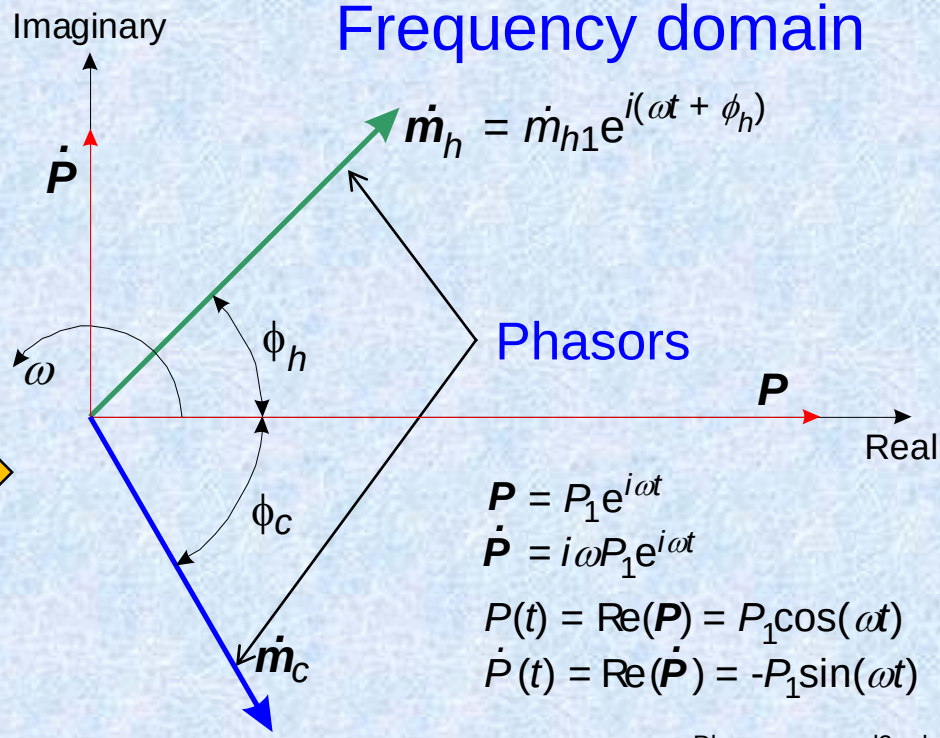
Sinusoidal pressure and flow curves

Time domain



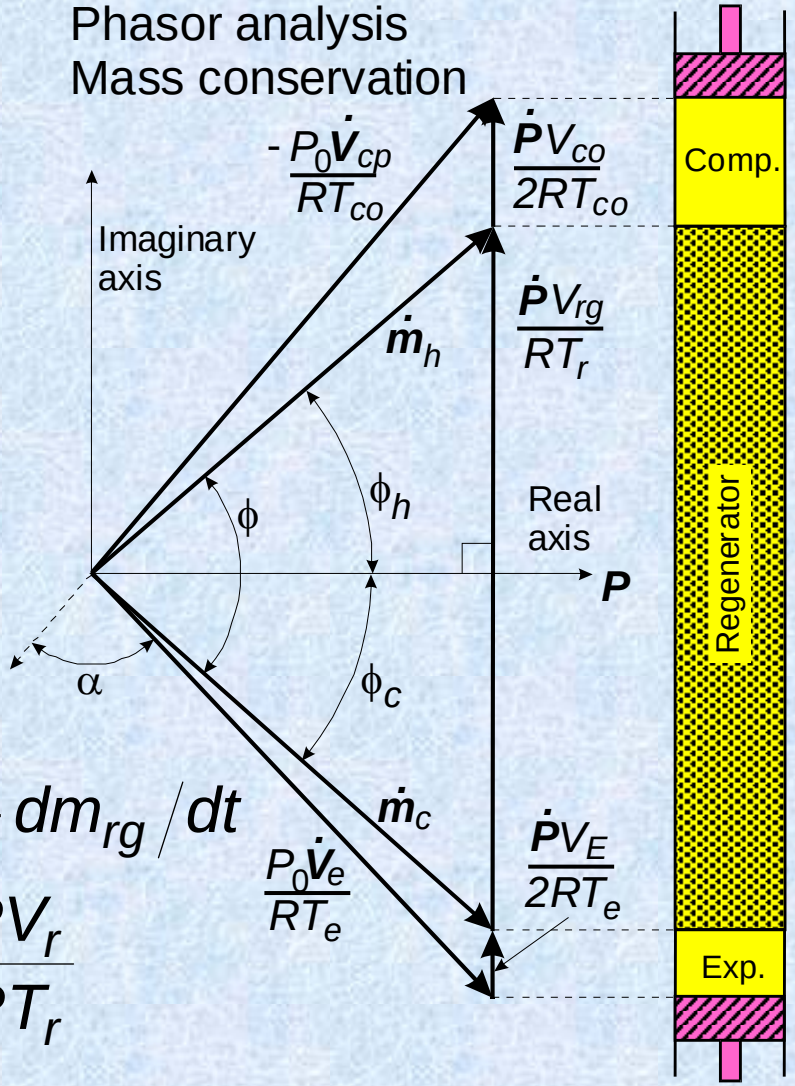
- Sinusoids
- Confusing with many curves
- Difficult to visualize algebra
- Math: differential equations

Frequency domain



- Phasors (t = 0, complex space)
- Amplitude and phase easily visualized
- Math: algebra, real and imaginary parts

Phasor Analysis for Mass Conservation



$$\dot{m}_h = \dot{m}_c + dm_{rg}/dt$$

$$\frac{dm_{rg}}{dt} = \frac{\dot{P}V_r}{RT_r}$$

STIRLING REFRIGERATOR

Acoustic Power Flow

$$\langle \dot{W} \rangle = \langle P\dot{V} \rangle = \tfrac{1}{2}|P||\dot{V}|\cos\theta$$

$$= \tfrac{1}{2}RT\frac{|P|}{P_0}|\dot{m}|\cos\theta$$

$$\propto |\dot{m}|\cos\theta$$

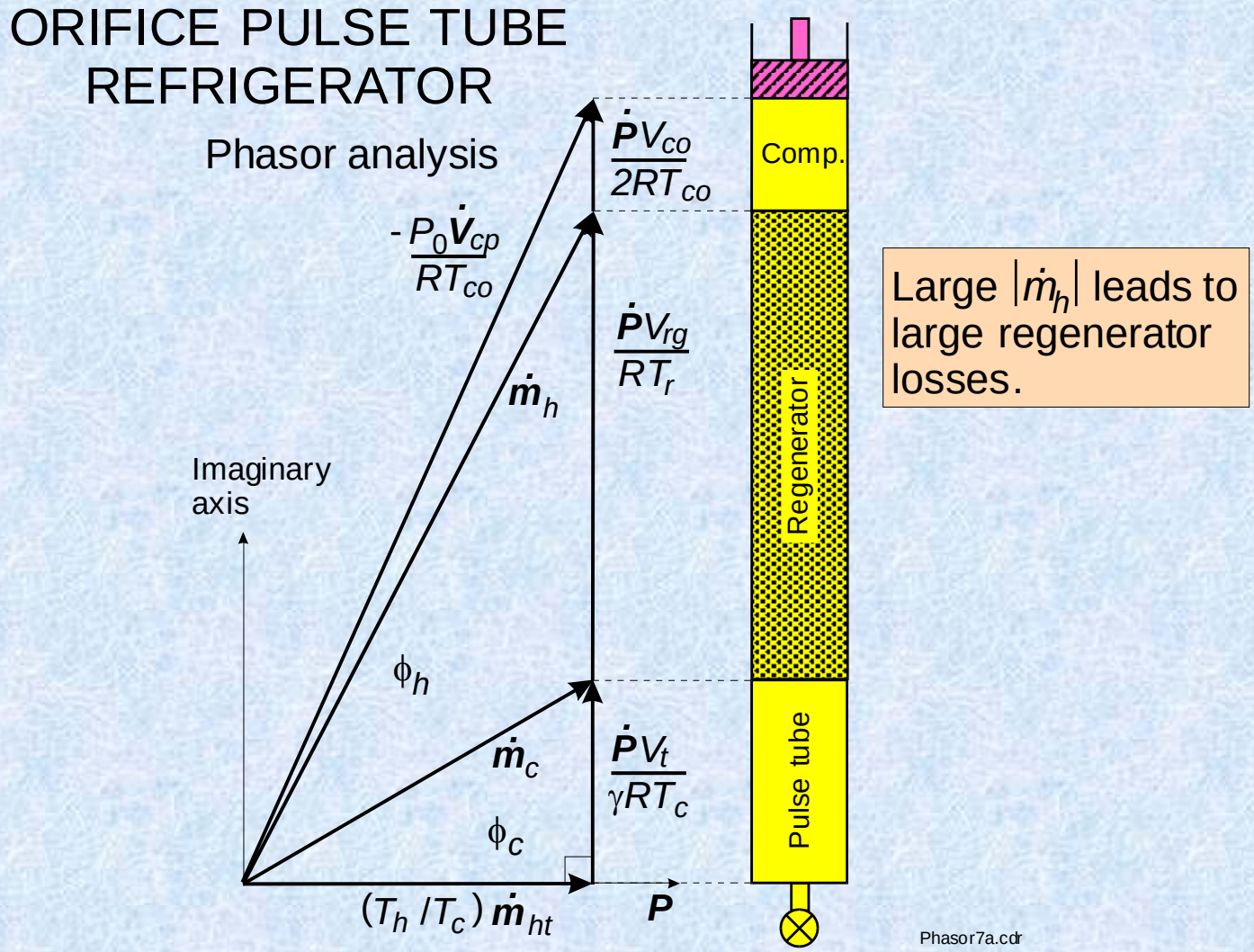
θ: phase between P and flow

Regenerator Losses

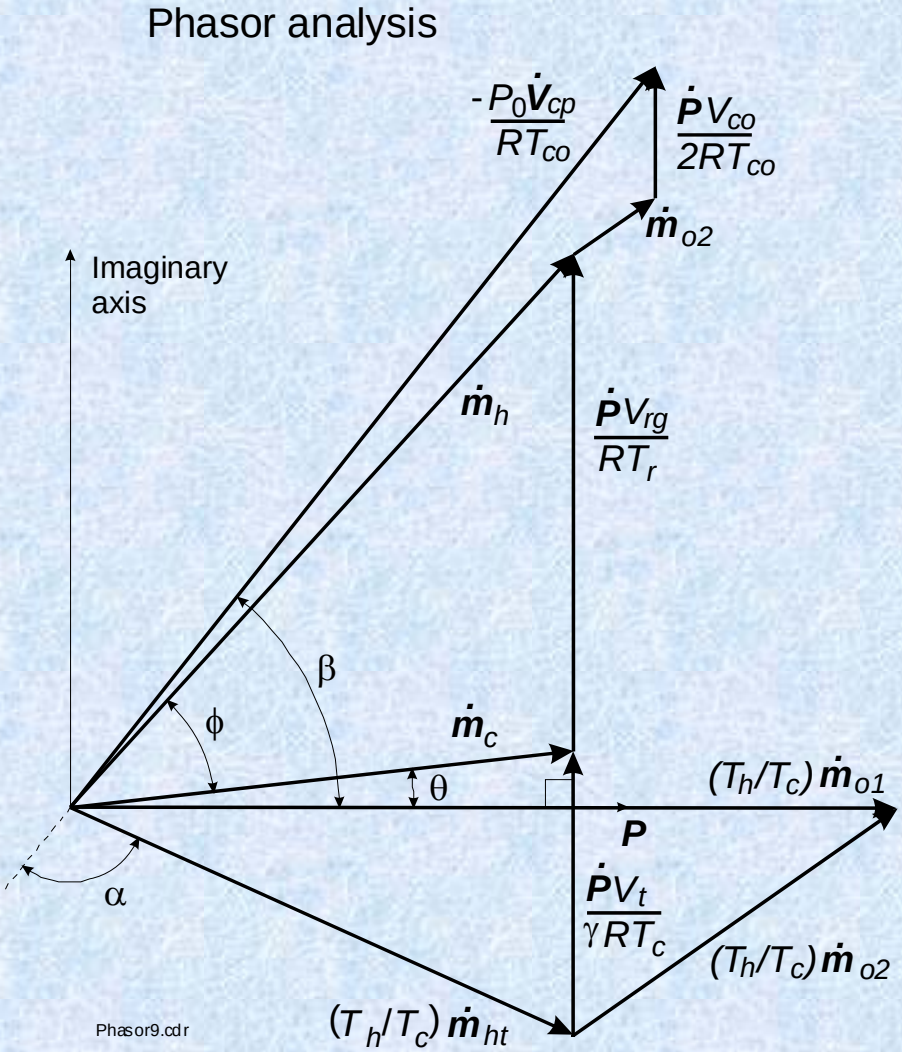
$$\propto |\dot{m}|$$

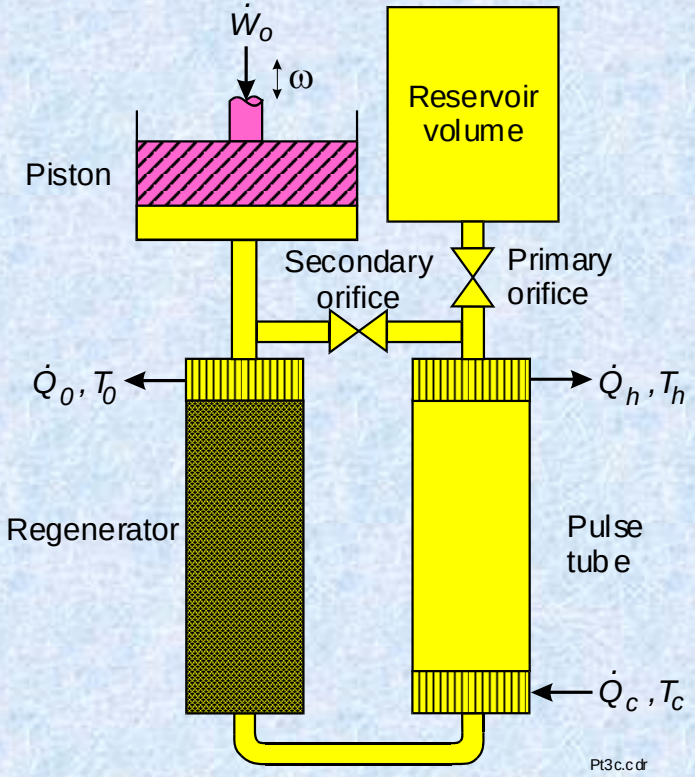
Phasor6b.cdr

Phasor Analysis in Pulse Tube Cryocooler

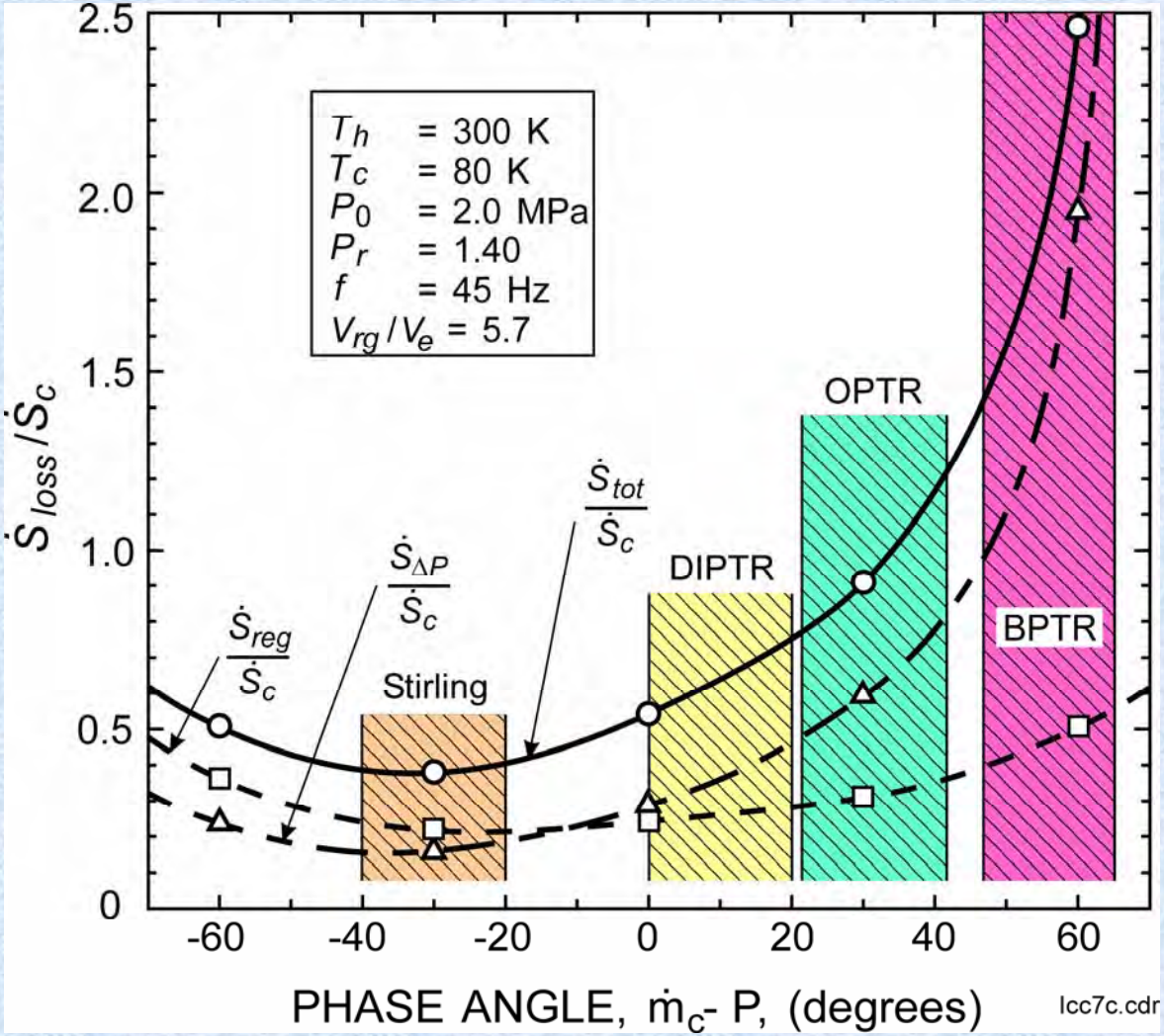


Phasor Analysis: Double Inlet





Regenerator Losses



Inertance Effects

Conservation of momentum equation:

$$-\frac{\partial \boldsymbol{P}}{\partial x} = \frac{f_r |\dot{m}| \dot{\boldsymbol{m}}}{2 r_h \rho A_g^2} + \frac{\partial}{\partial t}\left(\frac{\dot{\boldsymbol{m}}}{A_g}\right) + \frac{\partial}{\partial x}\left\{\frac{1}{\rho}\left(\frac{\dot{\boldsymbol{m}}}{A_g}\right)^2\right\}$$

pressure gradient | **flow friction** | **inertial effect** | **momentum divergence**

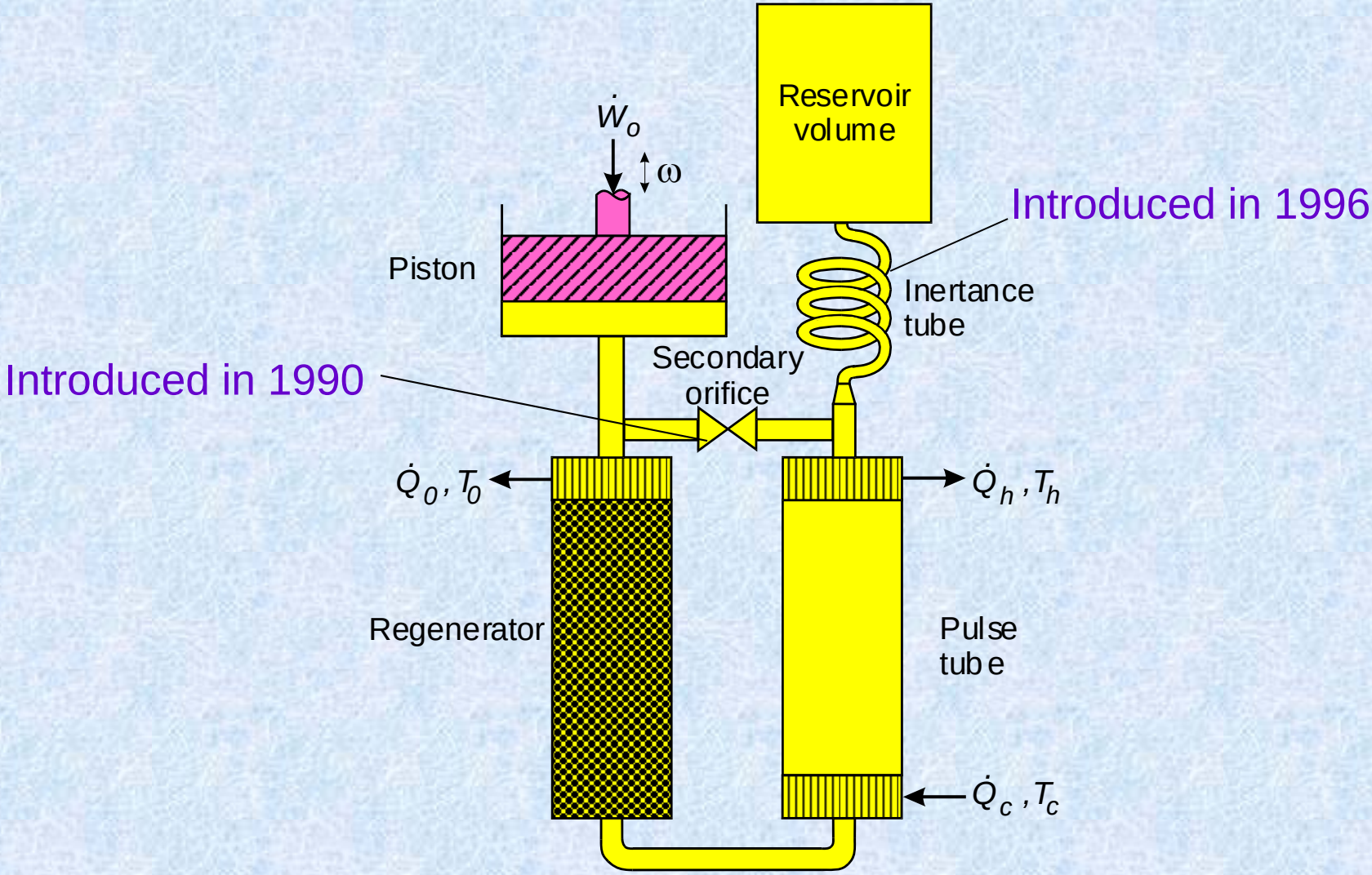
Conservation of mass equation:

$$\frac{\partial}{\partial x}\left(\frac{\dot{\boldsymbol{m}}}{A_g}\right) = -\frac{\partial \rho}{\partial t} = -\frac{\dot{\boldsymbol{P}}}{RT}$$

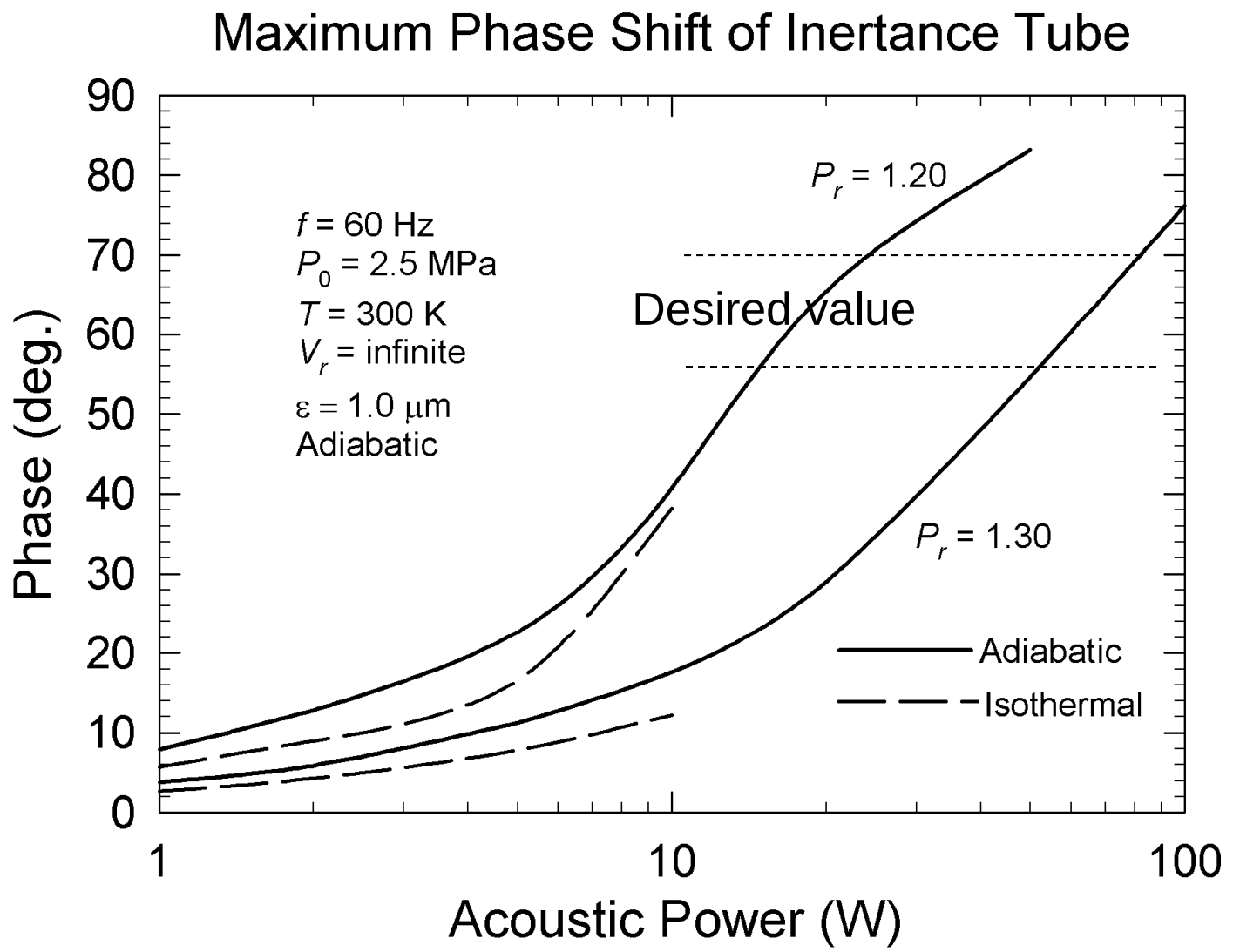
← Compliance effect

mass divergence | **mass change**

Double Inlet and Inertance Tube



Transmission Line Model



Outline

Session 2

- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

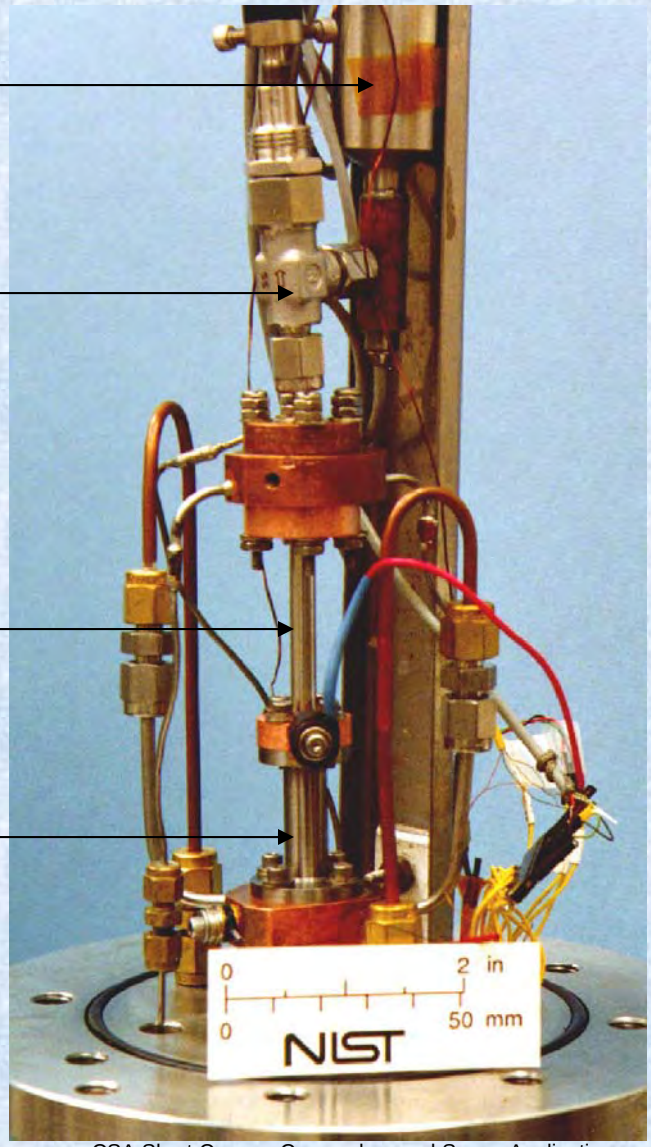
NIST Mini Pulse Tube Experimental System

Reservoir

Needle Valve (orifice)

Pulse Tube

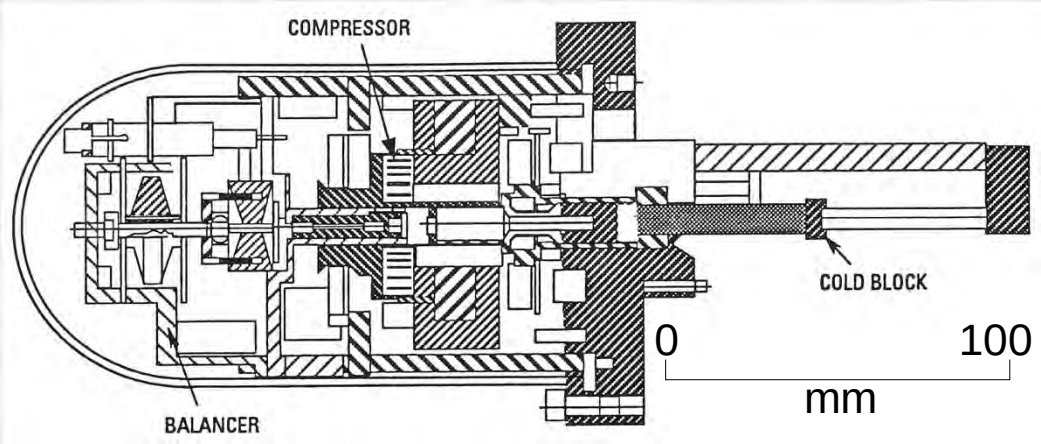
Regenerator



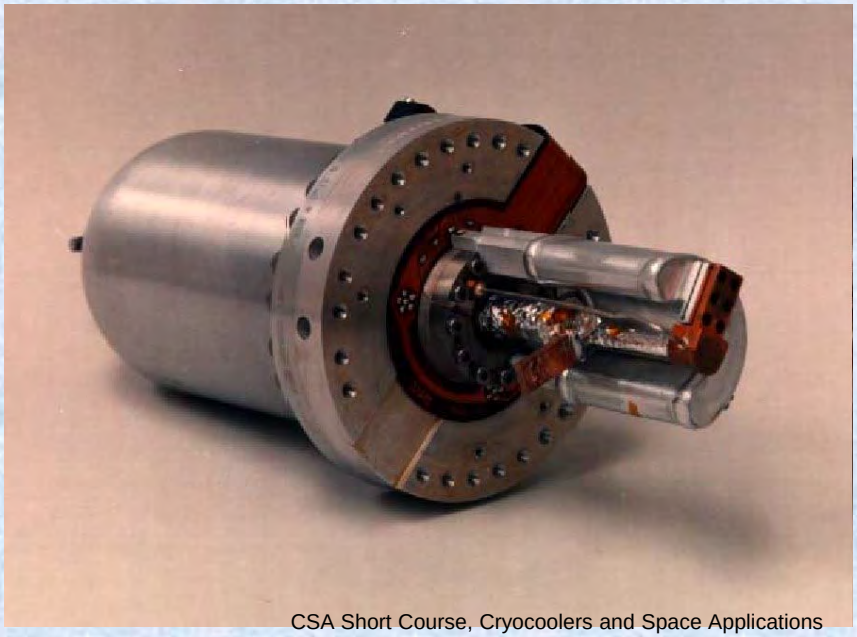
Inline geometry

- 3.4 W @ 80 K
- 4 cc compressor

Miniature Pulse Tube Cryocooler for Space Applications



Inline geometry



- 0.5 W at 80 K
- 17 W input power
- Space applications

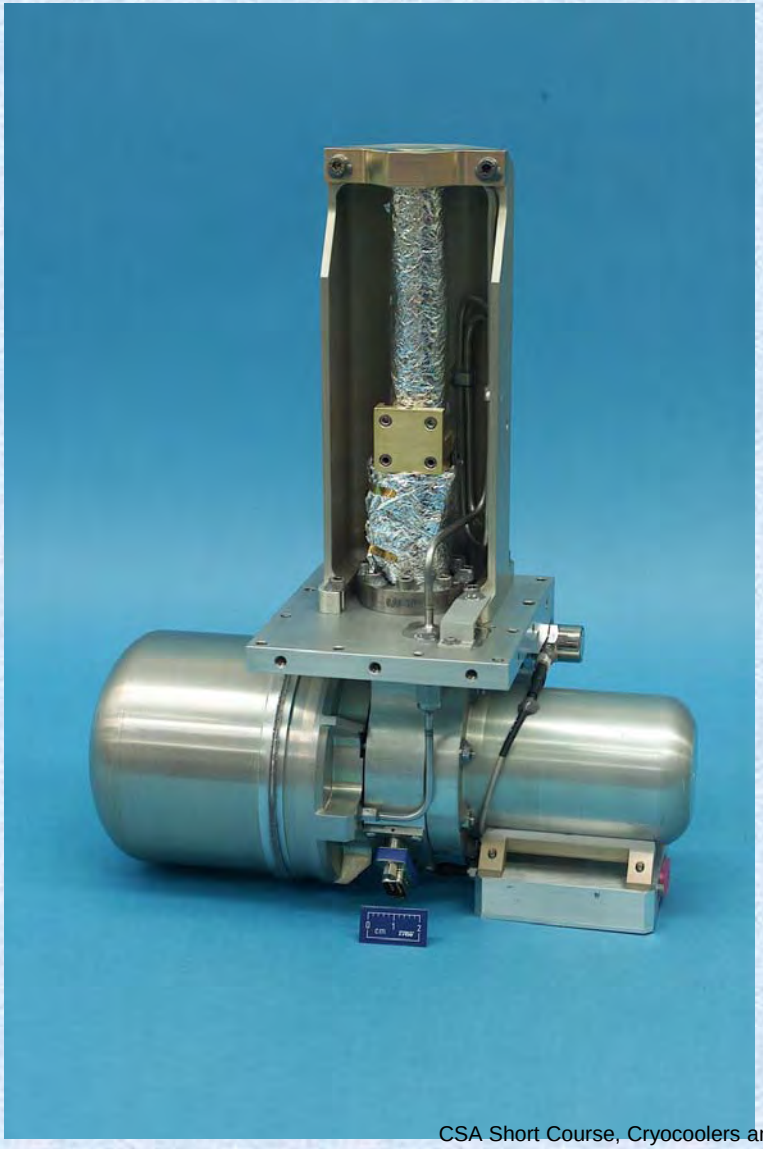
Mini Pulse Tube Cryocooler for Space Applications



U-tube geometry

- 0.62 W at 80 K
- 20 W input power
- Space applications
- 1.3 kg

High Efficiency Cooler (HEC) (Pulse Tube)



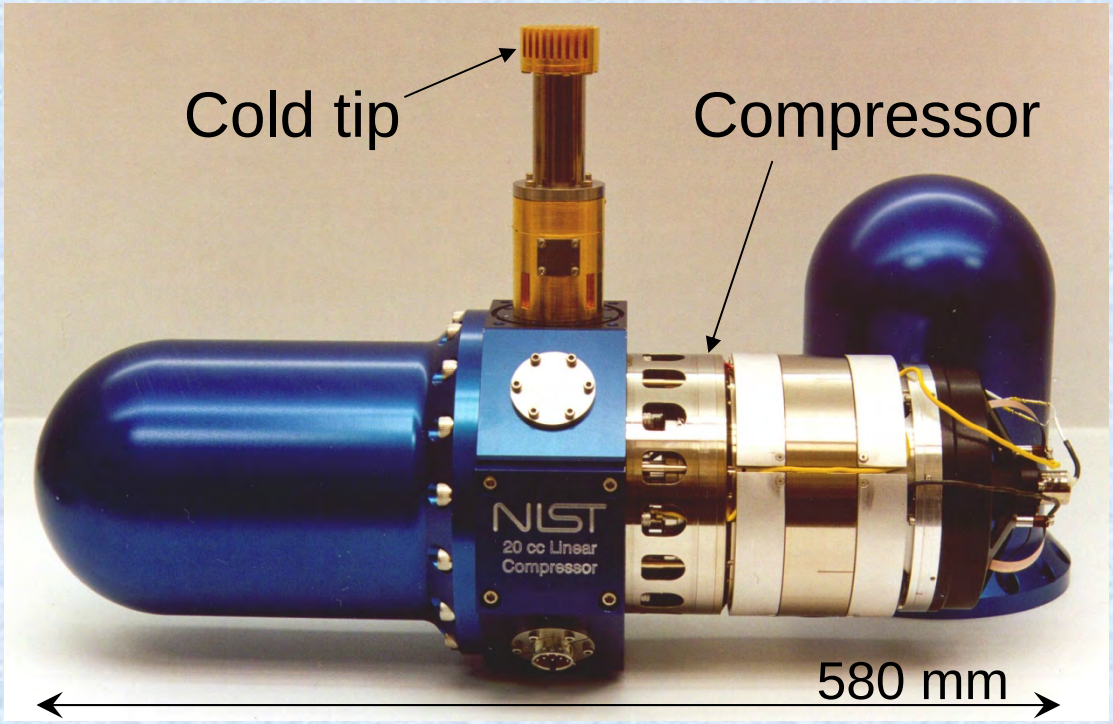
Inline geometry

- 7.3 W at 80 K
- 107 W comp. input power
- 19% Carnot @ 80 K
- Space applications
- 4.3 kg

TRW-HEC011.jpg

Mars Oxygen Liquefier Prototype

- Transport of O_2 to Mars too heavy
- Mars atmosphere 95% CO_2
- $CO_2 \rightarrow O_2$ chemically
- Liquefy gaseous O_2
- 2007 flight goal
- Funded by NASA



Coaxial geometry

19 W @ 90 K

- NIST laboratory pulse tube liquefier
- Used NIST modeling programs
- Uses flexure bearings and inertance tube
- Extremely efficient: 20% Carnot (PV power)

Iwatani First Commercial Pulse Tube

GM-type

U-tube geometry

1993

Originally 5 W @ 77 K

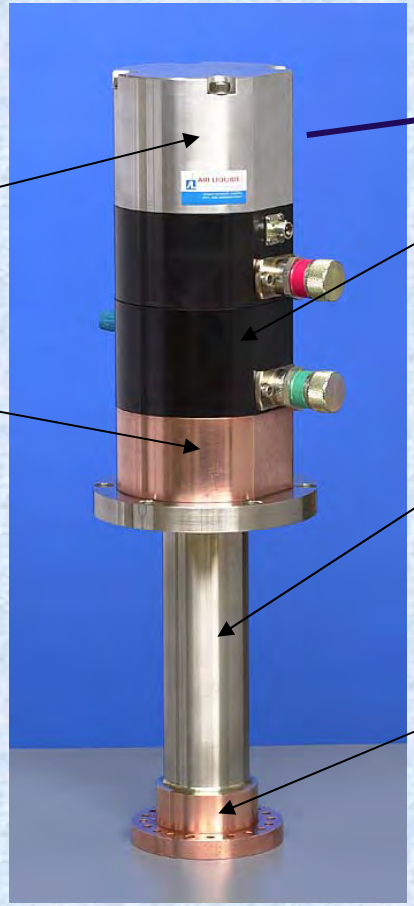
Later 15 W @ 77 K



Large GM-Type Pulse Tube Cryocooler

Large capacity low frequency coaxial pulse tube cold head (left)

Final integration on a 40 kVA HTS transformer demonstrator (right)





Courtesy: T. Trollier
Air Liquide

100W @ 80 K / 5 kW

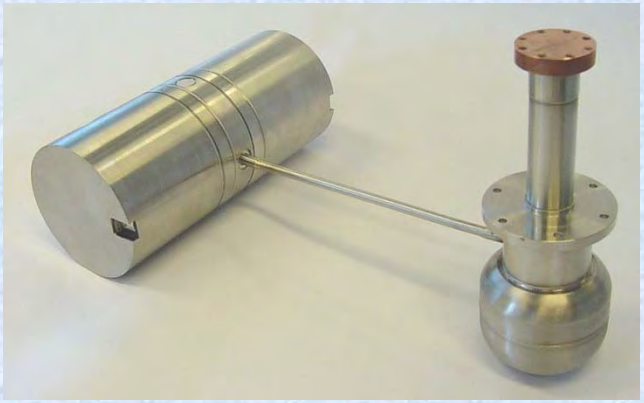
Small Commercial Pulse Tube Cryocoolers

Stirling-Type

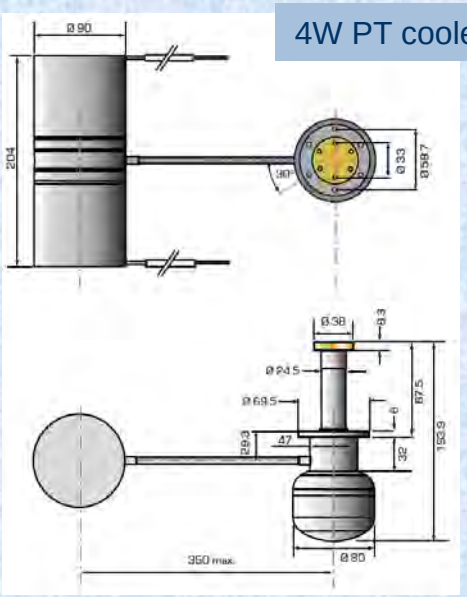
- **Coaxial shaped pulse tubes commercially available (licence from CEA/SBT)**

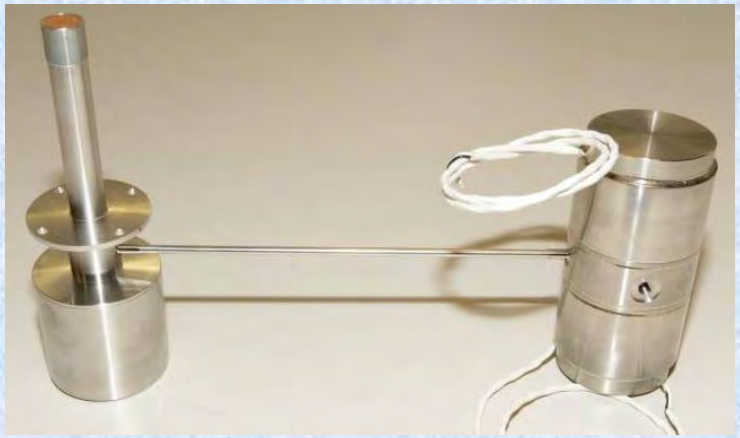


4W @ 80 K / 150 W input

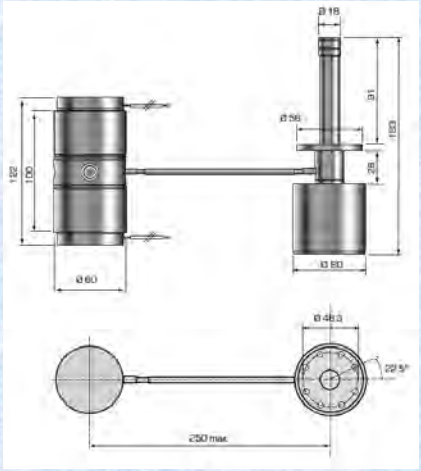


4W PT cooler





1W @ 80 K / 60 W input



1W PT cooler

THALES

Courtesy: T. Trollier
Air Liquide

Large Commercial Pulse Tube Cryocoolers

Stirling-Type



8 W @77 K
250 W input

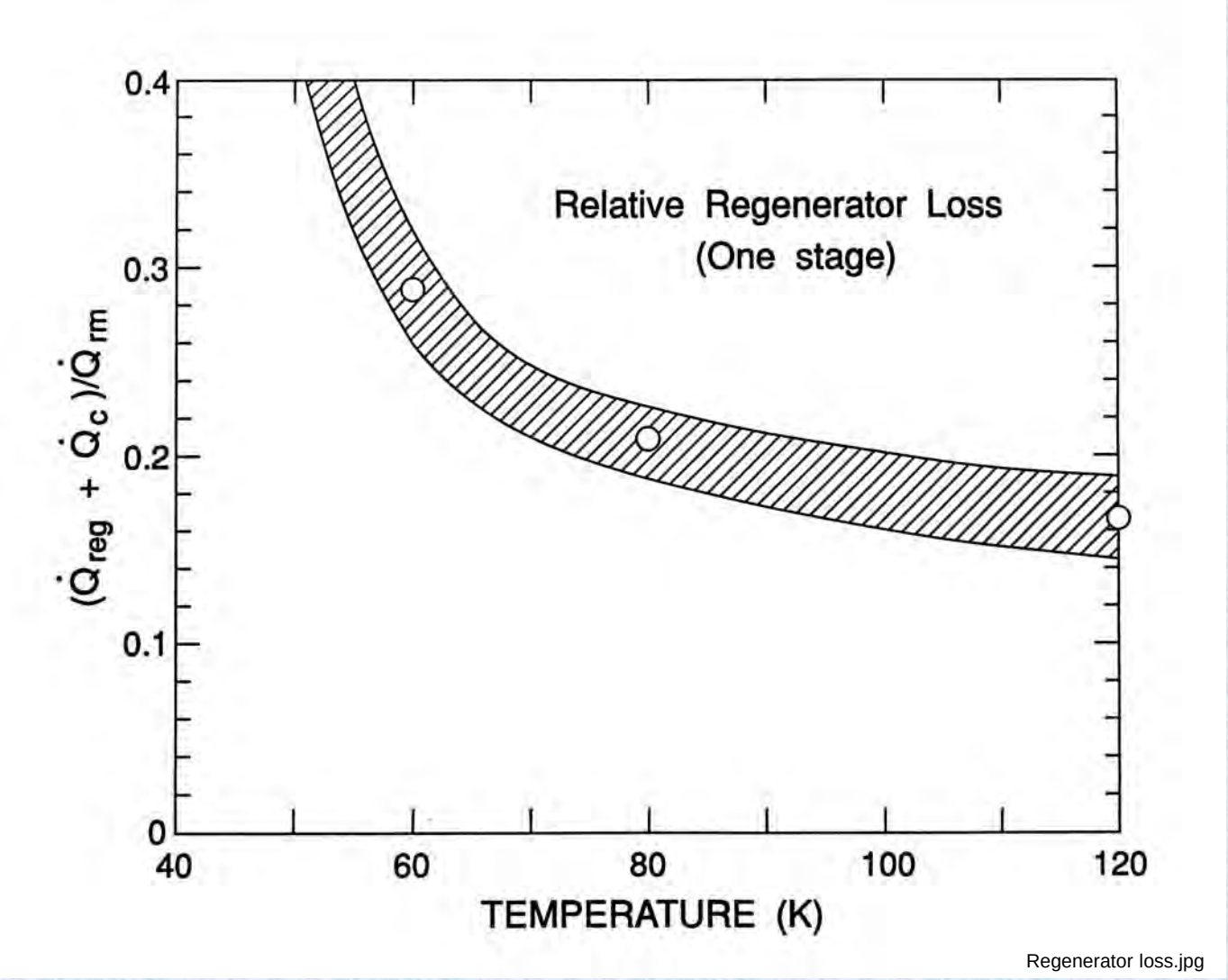


1 kW @ 77 K
21 kw input



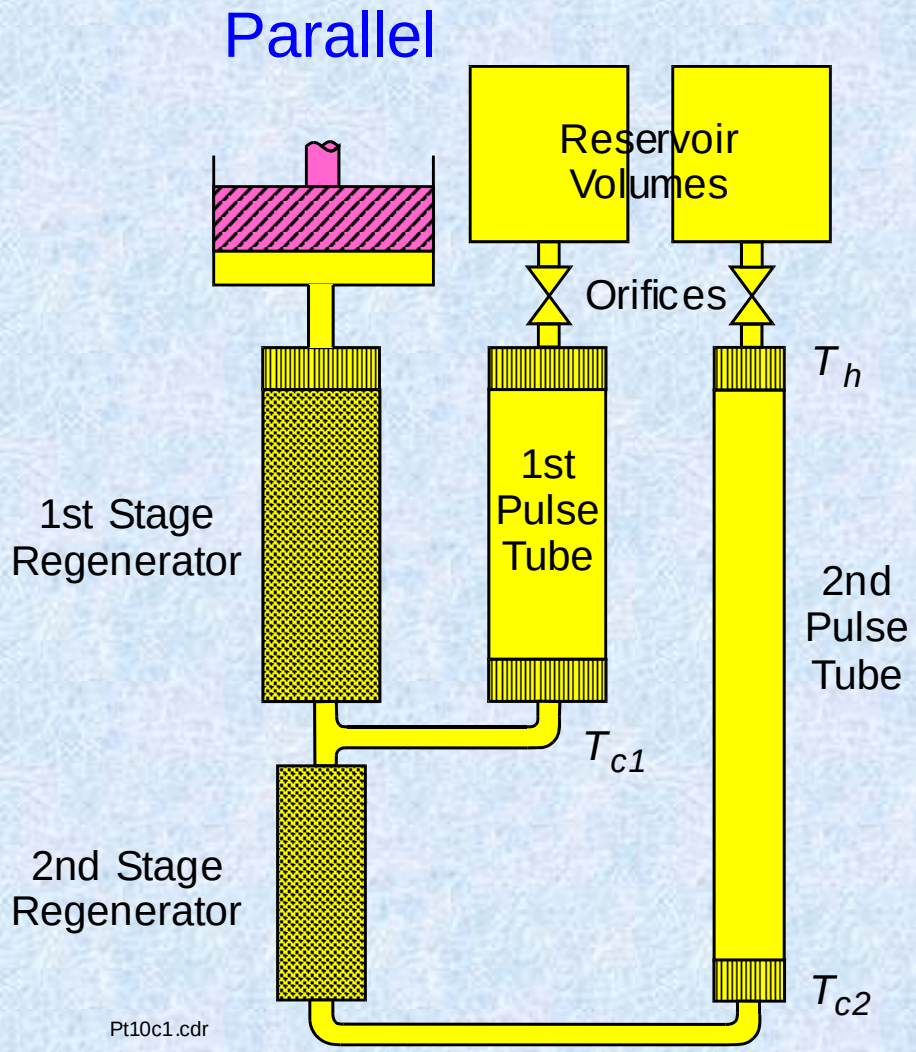
20 W @ 77 K
500 W input

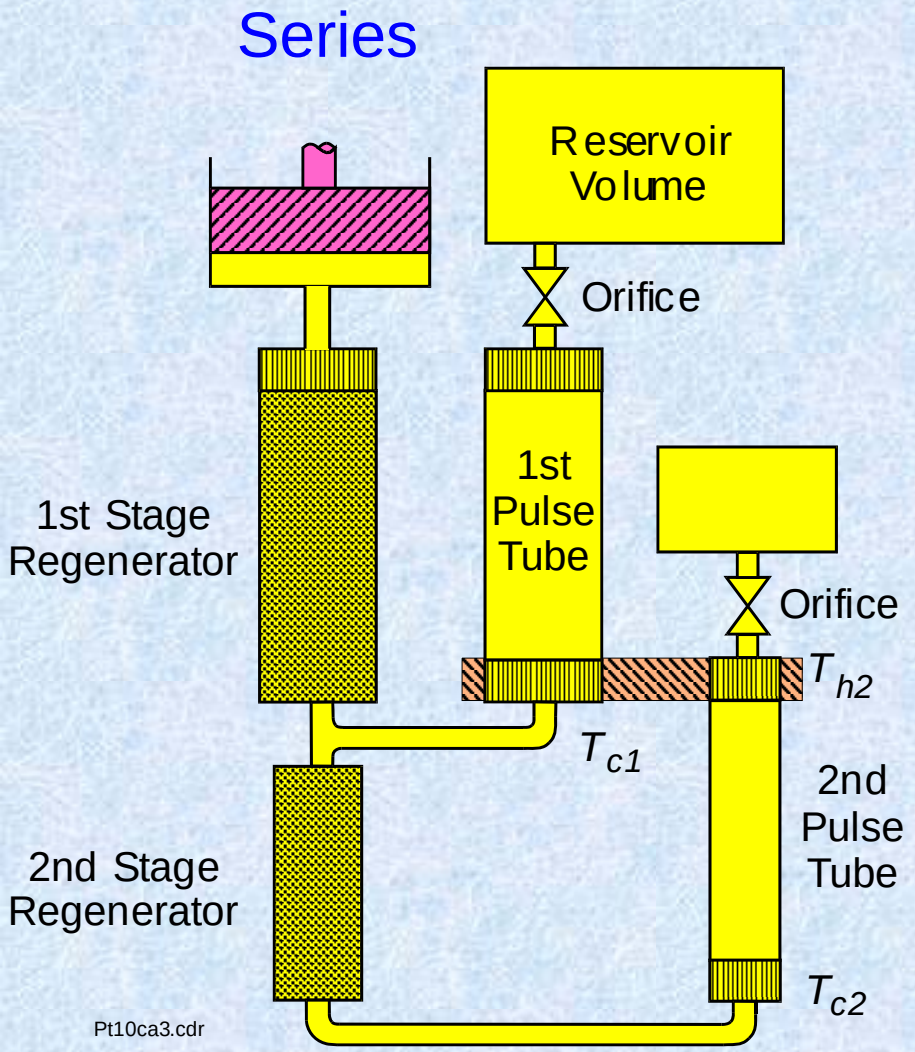
Regenerator Losses vs. Cold Temperature



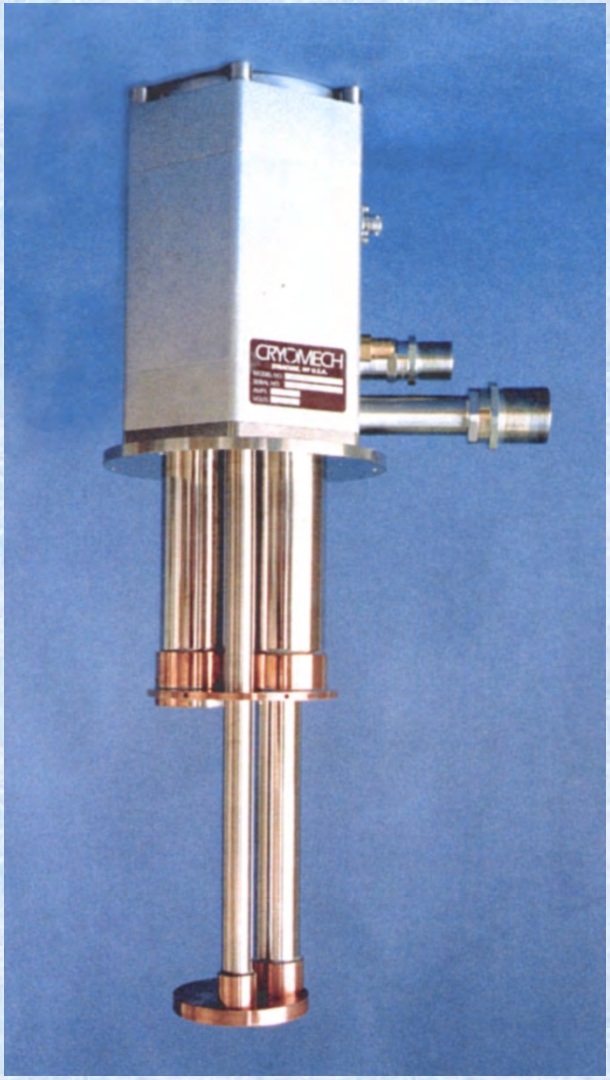
Regenerator loss.jpg

Two-Stage Pulse Tube Cryocooler





Commercial 4 K Pulse Tube Cryocooler



Cryomech pt405

Cryomechpt405.cdr

U-tube geometry
Parallel arrangement

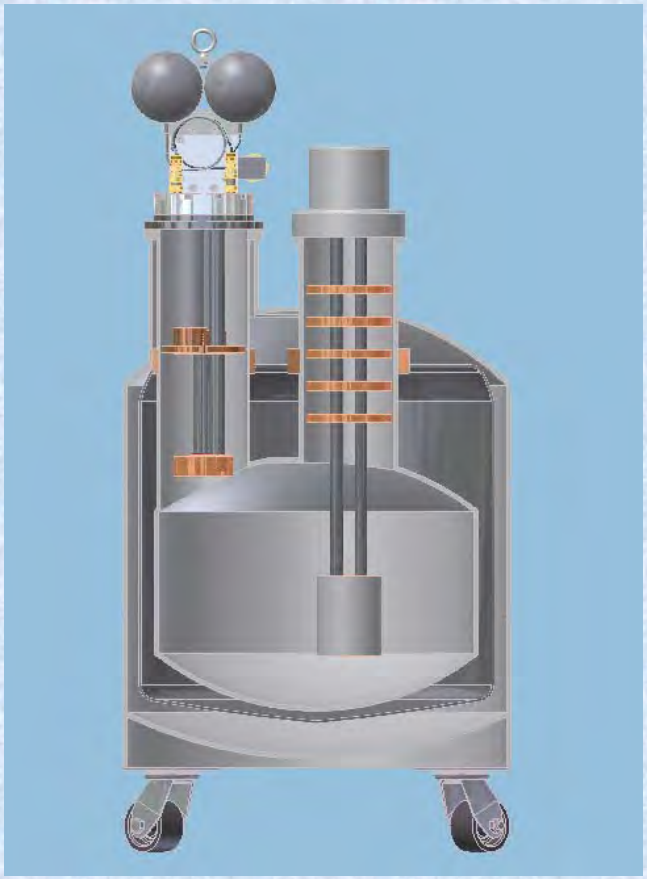
- 0.5 W at 4 K
- 30 W at 65 K
- 5.5 kW input power
- 20,000 hr maintenance
- 1 Hz frequency (GM type)

Pulse Tube Helium Liquefier

12-18 liters per day of LHe

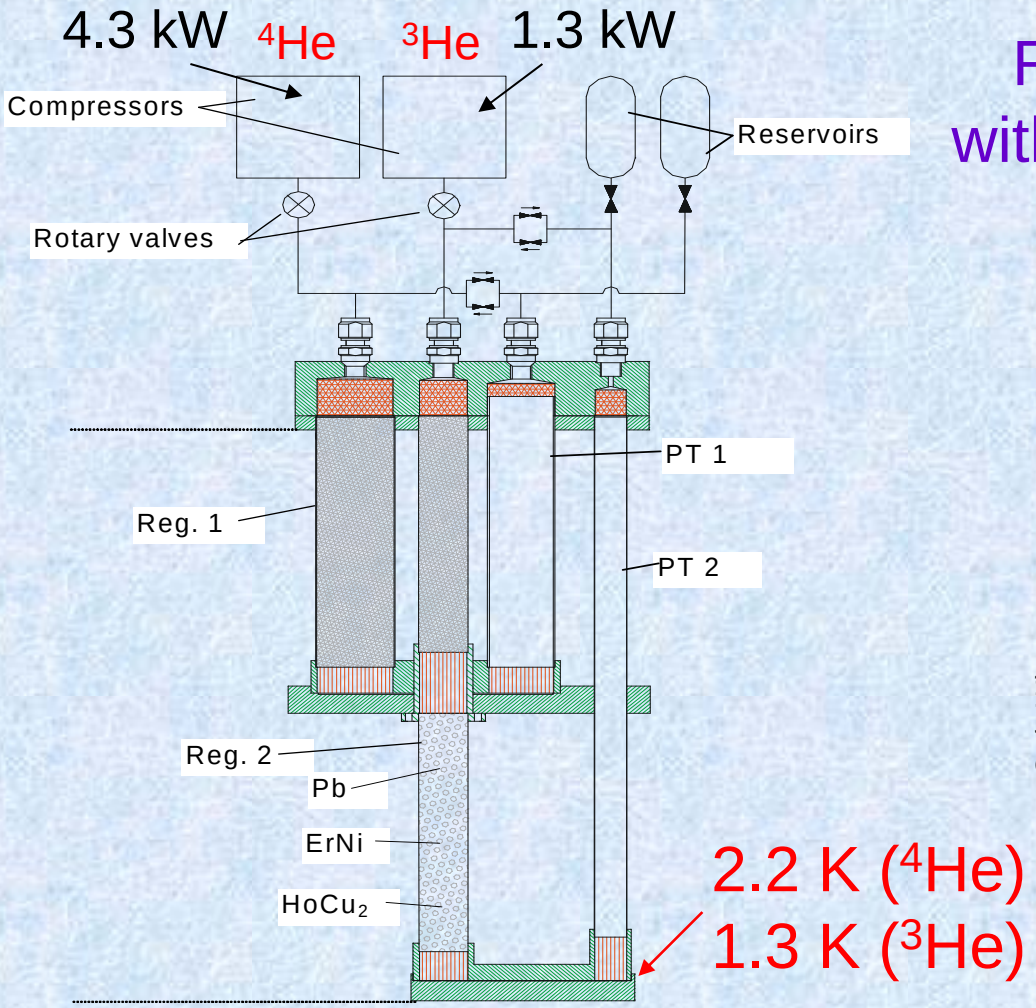
Helium recondensing





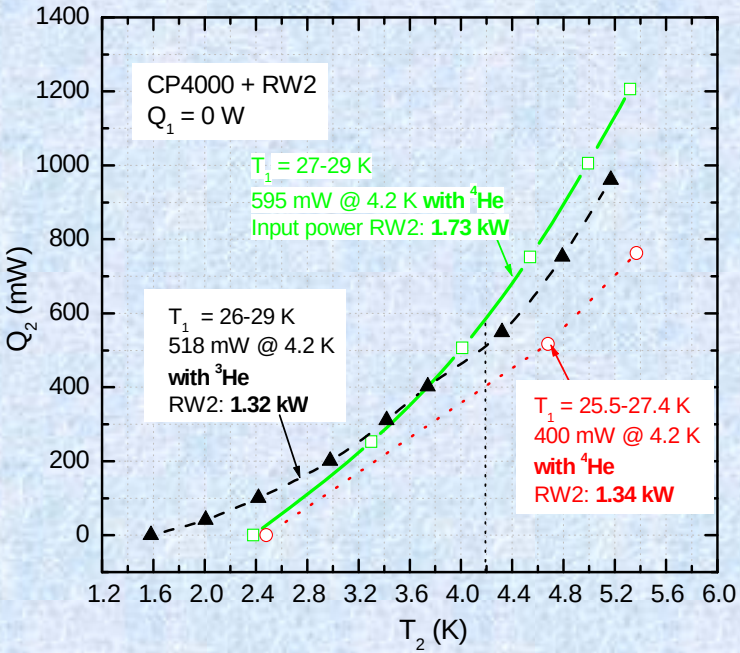
Photos courtesy of Cryomech

^{3}He Pulse Tube Cryocooler



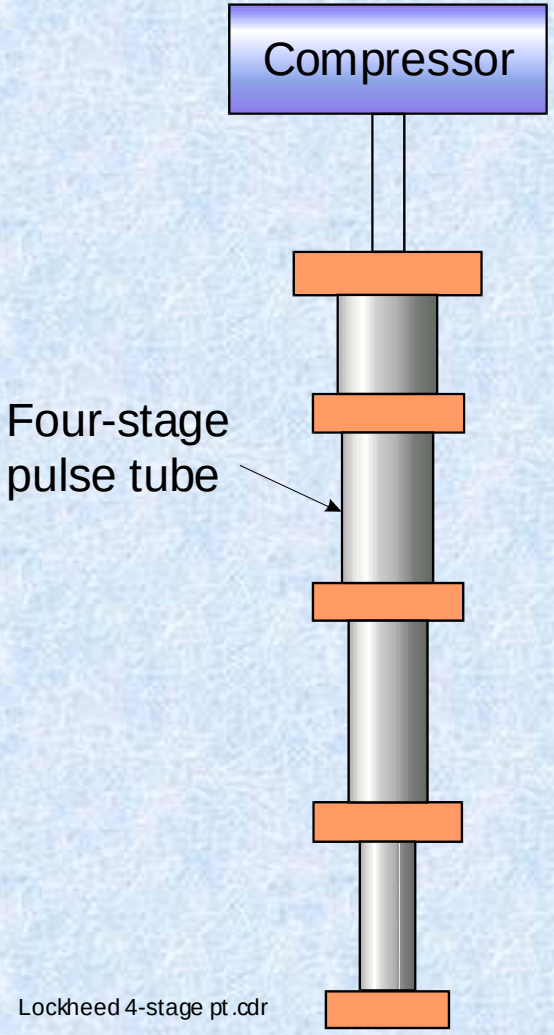
Record low temperature with gas regenerative cycles

Univ. of Giessen 2003

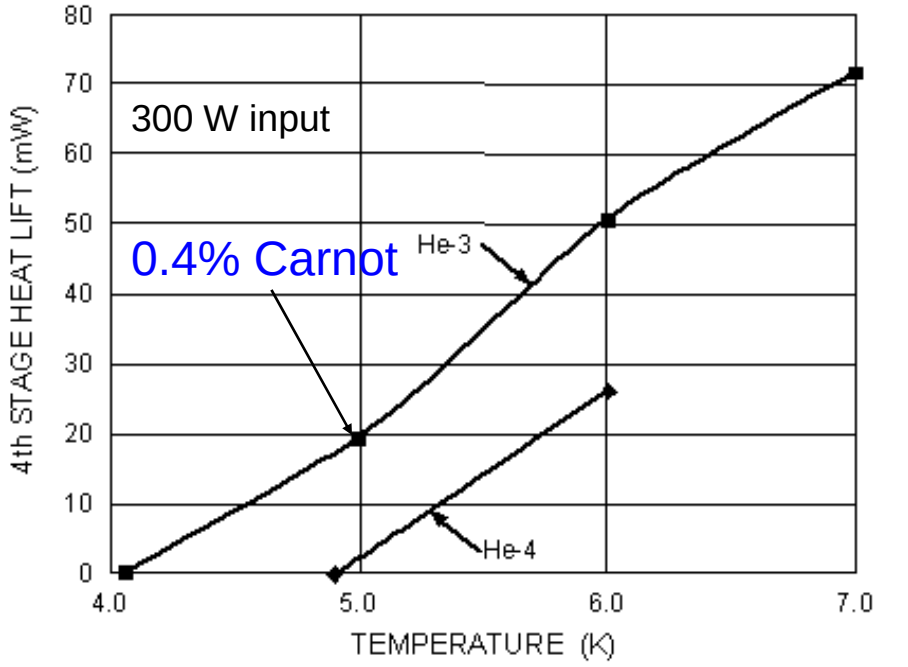


Courtesy: G. Thummes, Univ. of Giessen

4 K Stirling-Type Pulse Tube



- Lockheed Martin development
- 3.8 K achieved
- Four stages
- 31 Hz drive frequency
- 300 W input power



T. Nast, *et al.*, *Cryocoolers 14* (2007), pp. 33-40

Outline

Session 2

- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

Regenerative Cryocooler Models

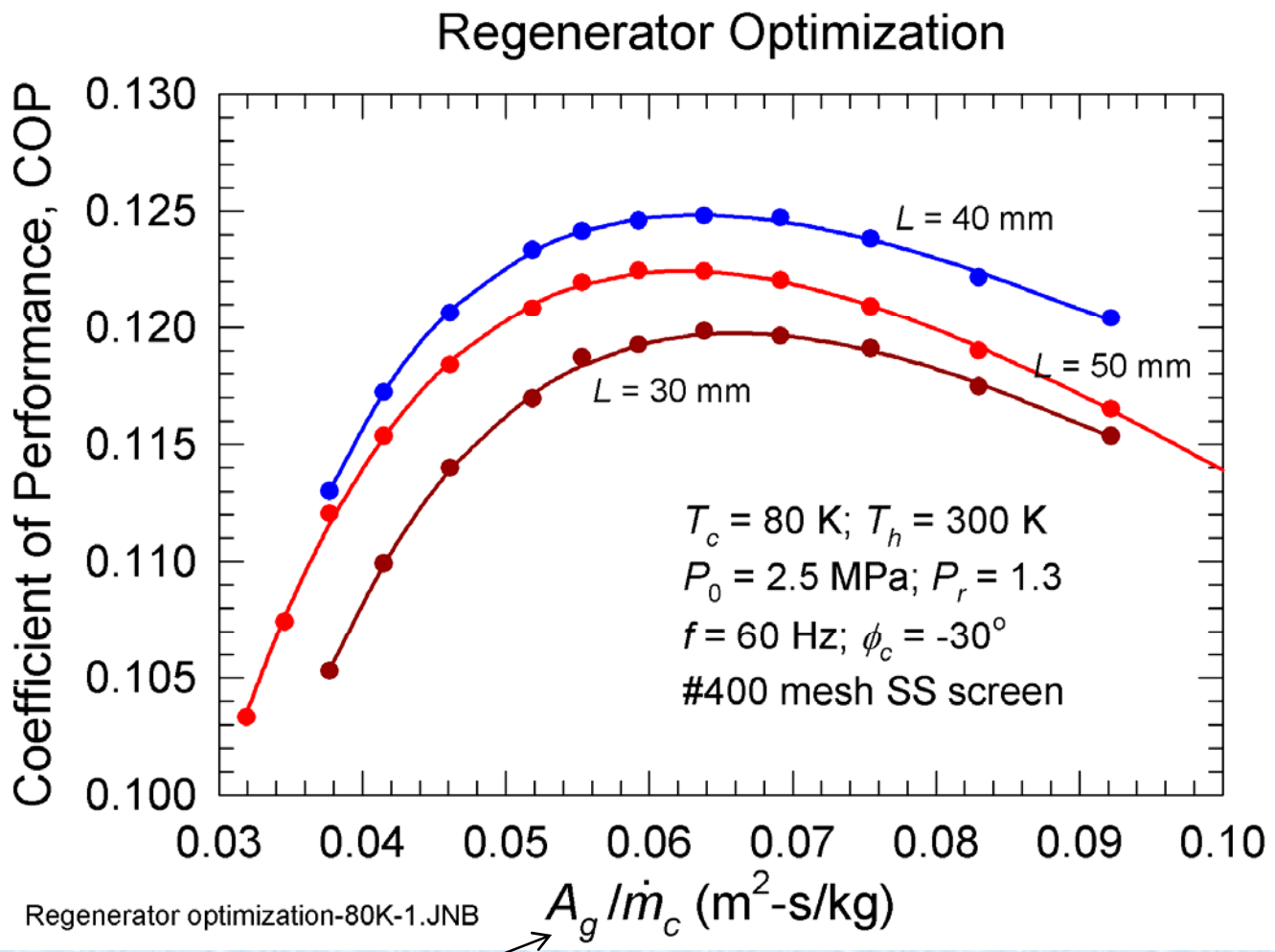
- ANALYTICAL MODELS
  - Shows effect of various parameters
  - Harmonic approximation
  - Approximate results
- NUMERICAL MODELS (1-dimensional)
  - Uses finite difference equations
  - Oscillating boundary conditions
  - REGEN3.3, DeltaE, Sage
  - Turbulence and jetting in pulse tube not modeled
- NUMERICAL MODELS (2 and 3 dimensions)
  - Computational Fluid Dynamic (CFD) models
  - Fluent
  - Model pulse tube and transitions

REGEN3.3 Model (NIST)

- REGENERATOR NUMERICAL MODEL
  - Finite difference model for energy, mass, and momentum
  - Sinusoidal mass flow
- VARIABLE BOUNDARY CONDITIONS
- MATERIALS
  - Helium-4, helium-3, and ideal gas
  - 30 matrix materials (can layer combinations)
- 5 GEOMETRIES
  - Steady-state flow correlations (Kays and London)

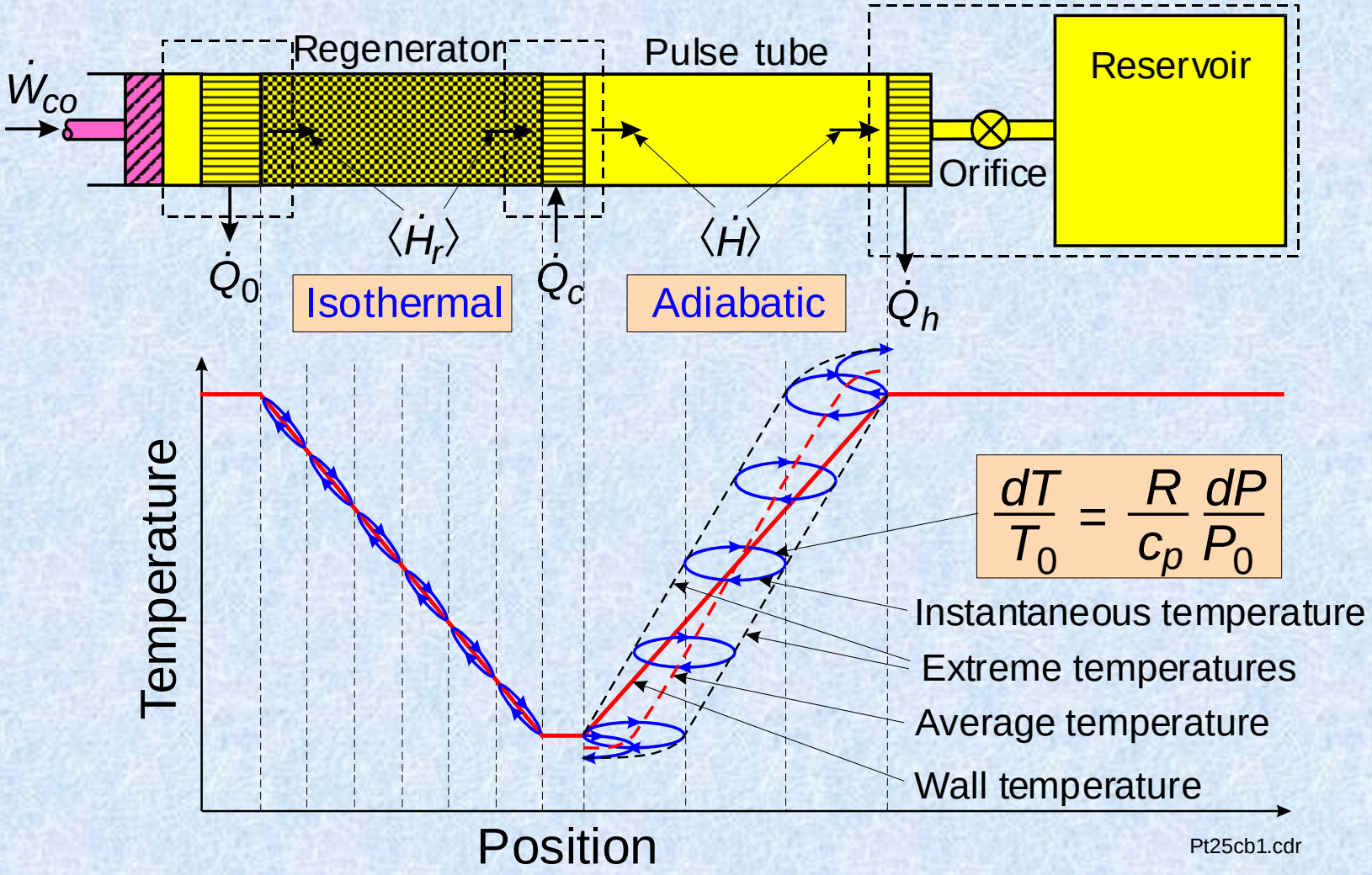
Available free from NIST at: (http://www.cryogenics.nist.gov)

Regenerator Optimization
(Use of REGEN3.3)

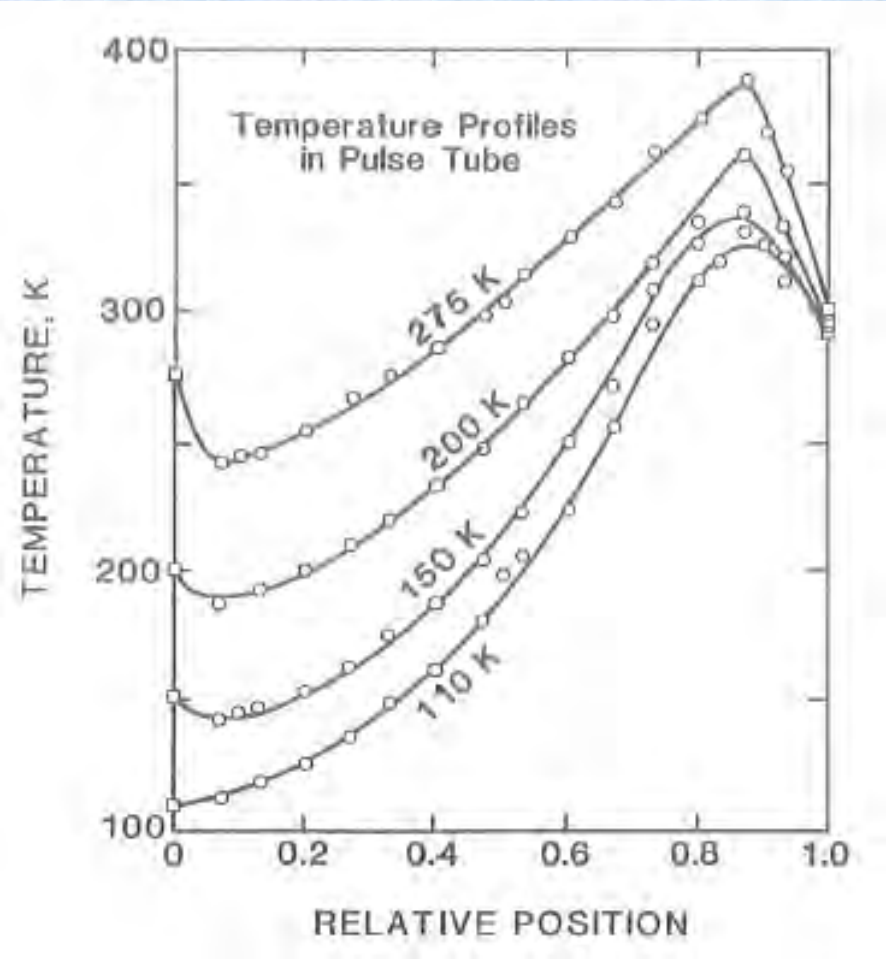


Independent of cooling power

Pulse Tube Cryocooler Analysis

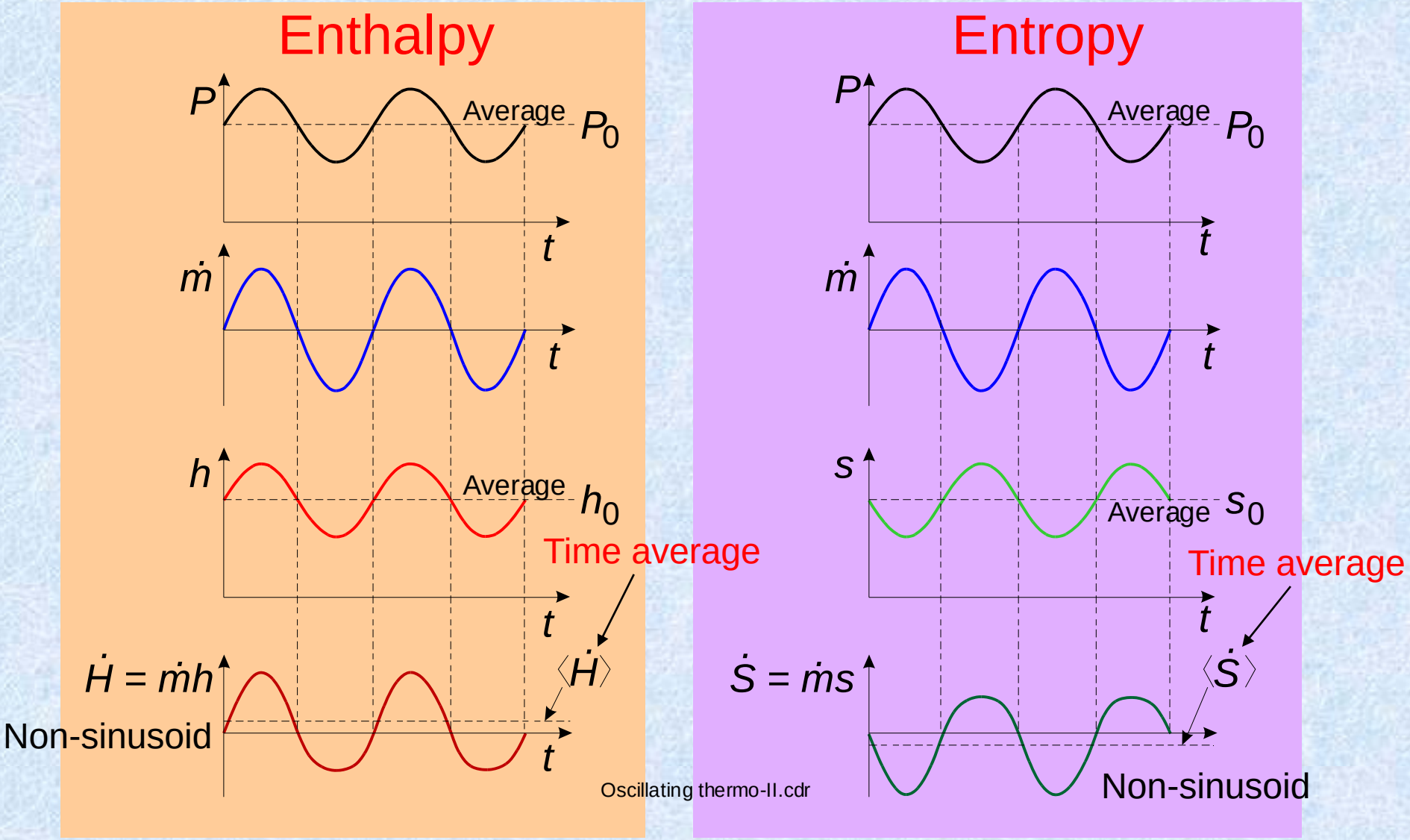


Average Pulse Tube Temperatures



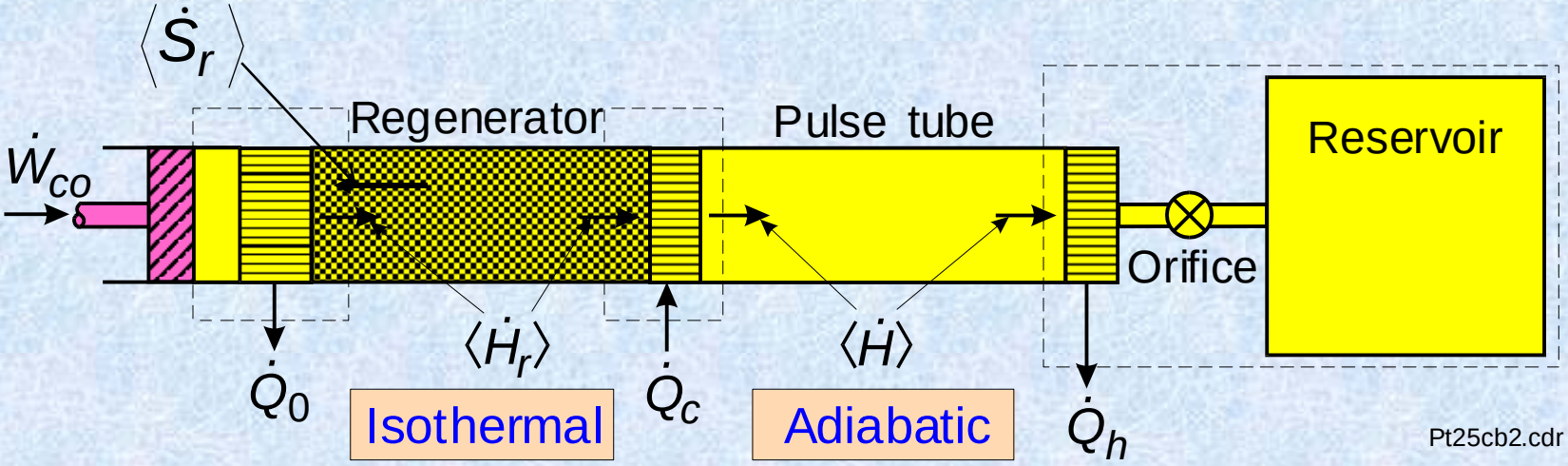
Storch and Radebaugh, Adv. Cry. Eng. **33** (1988) pp. 851-859

Oscillating Flow Parameters



Oscillating thermo-II.cdr

First and Second Law Analysis for Oscillating Systems



1st Law for an open system:

$$\dot{Q} = \dot{W} + \dot{m}(h_e - h_i) + \frac{d}{dt}(mu)$$

2nd Law for an open system:

$$\frac{\dot{Q}}{T} + \dot{S}_{irr} = \dot{m}(s_e - s_i) + \frac{d}{dt}(ms)$$

For oscillating systems:

$$\langle \dot{H} \rangle = (1/\tau)\int_0^\tau \dot{m}h\,dt$$

$$\langle \dot{m} \rangle = 0$$

For ideal gas: $h = c_p T$

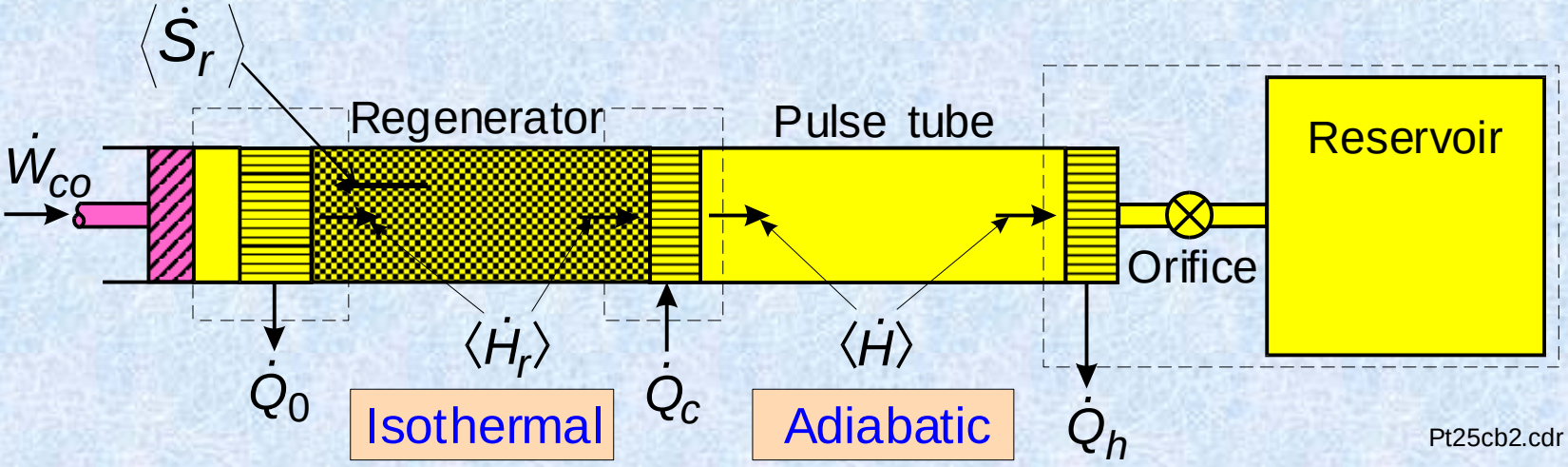
For oscillating systems:

$$\langle \dot{S} \rangle = (1/\tau)\int_0^\tau \dot{m}s\,dt$$

$\langle \dot{S} \rangle = 0$ for adiabatic & rev.

$\langle \dot{H} \rangle = 0$ for isothermal

Heat and Energy Flows

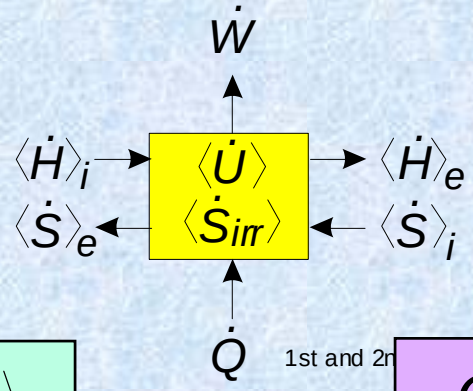


$$\dot{Q}_0 = \dot{W}_{co} - \langle \dot{H} \rangle_r = T_0 \langle \dot{S} \rangle_r \qquad \dot{Q}_c = \langle \dot{H} \rangle - \langle \dot{H} \rangle_r = T_c \langle \dot{S} \rangle_r$$

$$\dot{Q}_h = \langle \dot{H} \rangle$$

What happens to $\dot{Q}_c$? Where does it go?

Gibbs Free Energy and Acoustic Power



1st and 2n

1st Law:

$$\dot{Q} = \dot{W} + \langle \dot{H} \rangle_e - \langle \dot{H} \rangle_i$$

Equilibrium

2nd Law:

$$\frac{\dot{Q}}{T} + \langle \dot{S}_{irr} \rangle = \langle \dot{S} \rangle_e - \langle \dot{S} \rangle_i$$

$$\dot{W} = \left[\langle \dot{H} \rangle_i - T \langle \dot{S} \rangle_i \right] - \left[\langle \dot{H} \rangle_e - T \langle \dot{S} \rangle_e \right] - T \langle \dot{S}_{irr} \rangle$$

$$\left[\langle \dot{H} \rangle - T \langle \dot{S} \rangle \right] \equiv \langle \dot{G} \rangle$$ Gibbs free energy

$$\dot{W}_{rev} = \langle \dot{G} \rangle_i - \langle \dot{G} \rangle_e$$ Reversible, isothermal work at temperature T

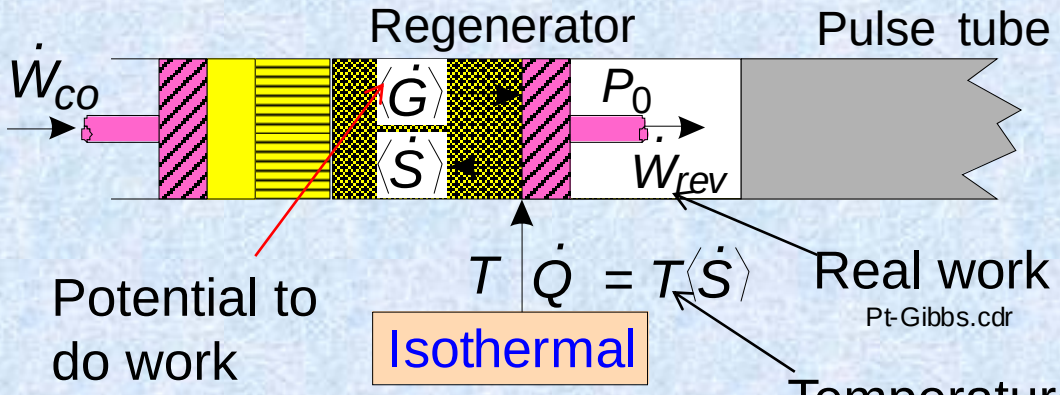
$$\langle \dot{G} \rangle_e = 0$$ For exit at ground state (mean *P* and *T*)

$$\langle \dot{W} \rangle_{\max} = \langle \dot{G} \rangle \equiv \langle P\dot{V} \rangle$$

Acoustic power or *PV* power flow, not actual power

Potential to do reversible work referred to ground state

Meaning of Gibbs Free Energy Flow



Other interchangeable terms:

Acoustic power
PV power flow
Work flow
$\langle P\dot{V} \rangle$

General Thermodynamic Relation

$$\left\langle P\dot{V} \right\rangle = \left\langle \dot{H} \right\rangle - T \left\langle \dot{S} \right\rangle$$

Potential
Maximum Power
(Acoustic Power)
Gibbs Free Energy

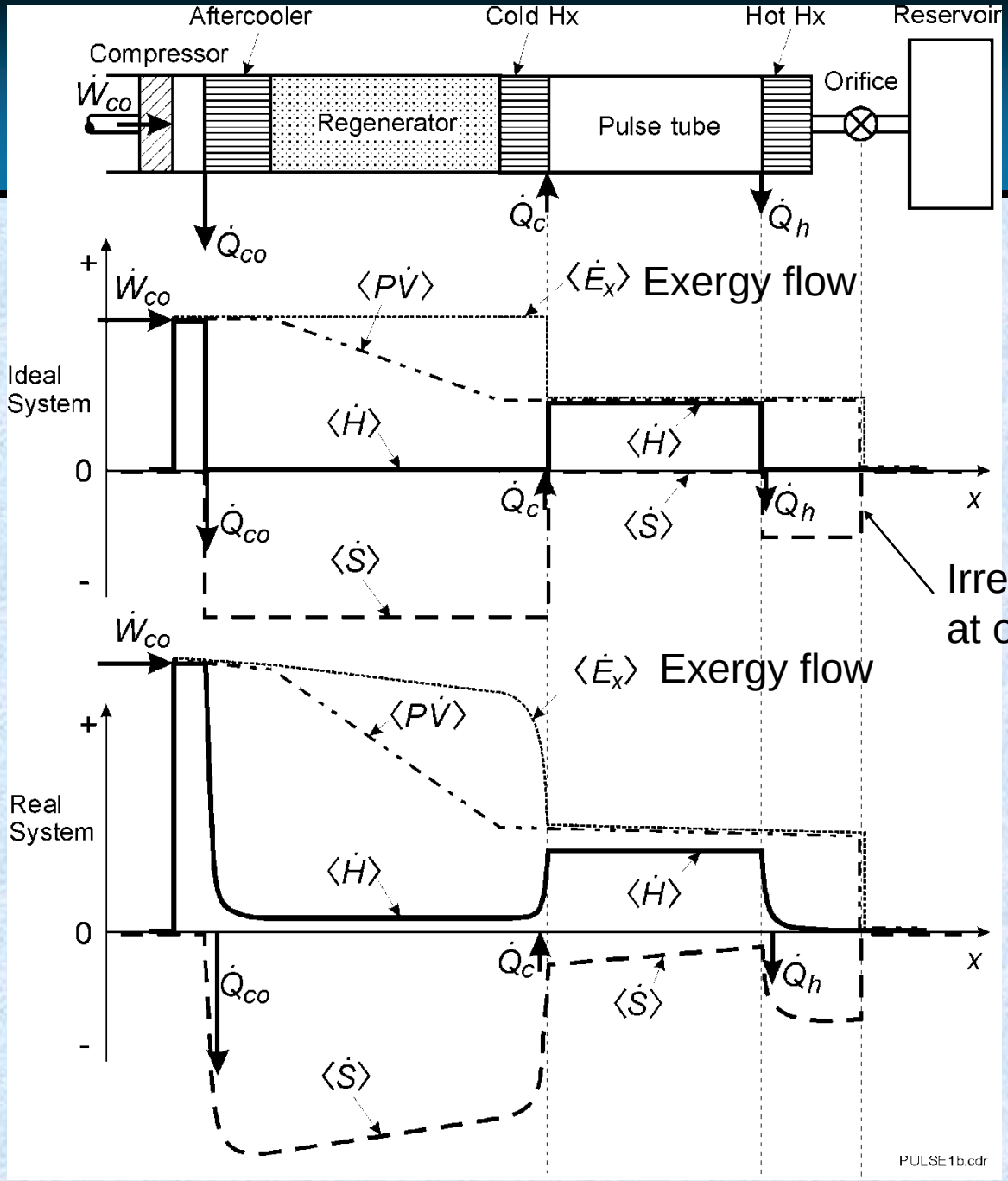
Not actual power

Enthalpy Flow

Entropy Flow

Isothermal Temperature

Energy Flows



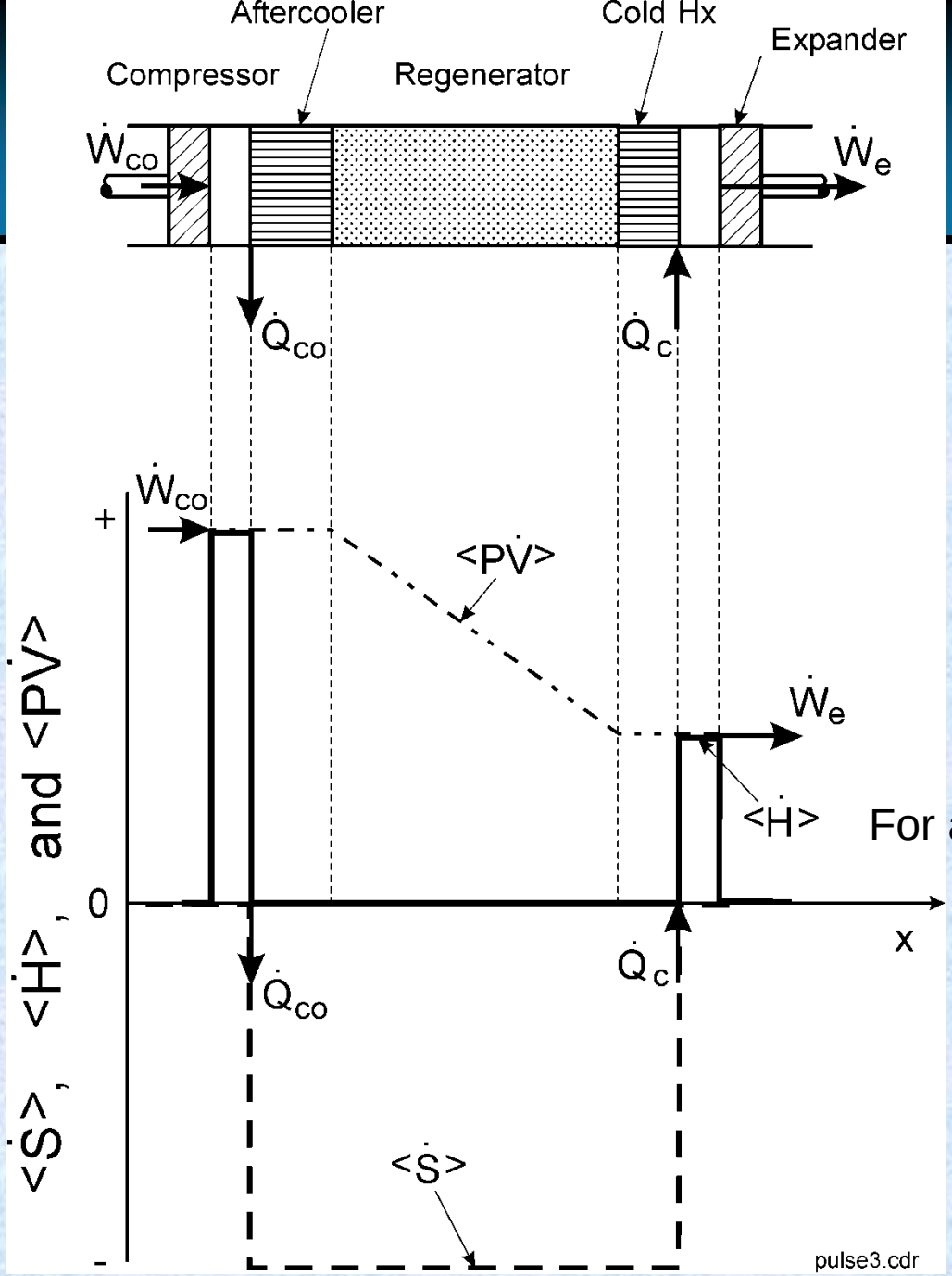
$$\langle \dot{E}x \rangle = \langle \dot{H} \rangle - T_0 \langle \dot{S} \rangle$$

$$\langle P\dot{V} \rangle = \langle \dot{G} \rangle = \langle \dot{H} \rangle - T \langle \dot{S} \rangle$$

Ideal OPTR

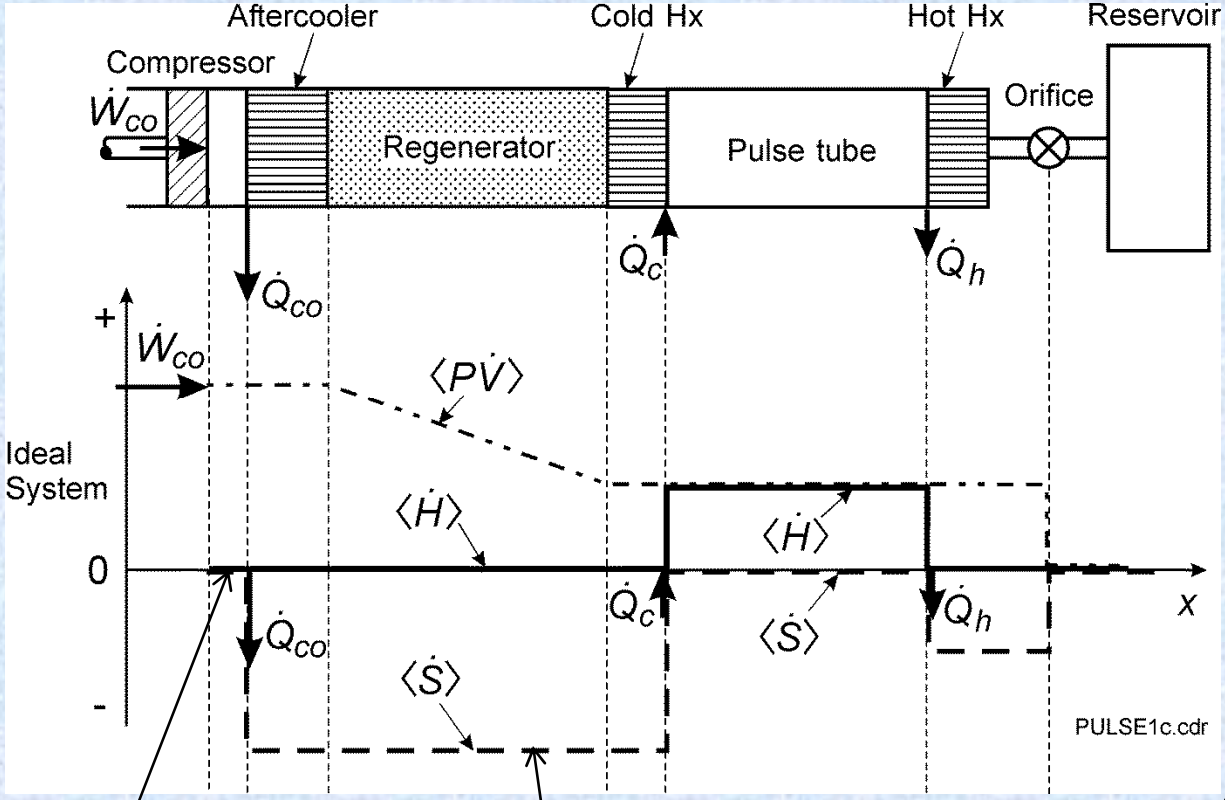
Real OPTR (losses)

Stirling Cryocooler Energy Flows



For adiabatic process

Ideal Pulse Tube COP



Isothermal

$\langle \dot{S} \rangle_r$

$$\mathrm{COP} = \frac{\dot{Q}_c}{\dot{W}_{co}}$$

$$\dot{Q}_c = T_c \langle \dot{S} \rangle_r$$

$$\dot{Q}_0 = T_h \langle \dot{S} \rangle_r$$

$$\dot{W}_{co} = \dot{Q}_0$$

$$\mathrm{COP} = \frac{T_c \langle \dot{S} \rangle_r}{T_h \langle \dot{S} \rangle_r} = \frac{T_c}{T_h}$$

Ideal Orifice Pulse Tube Efficiency

Coefficient of Performance, *COP*:

$$COP_{pt} \equiv \frac{\dot{Q}_c}{\dot{W}_{co}} = \frac{T_c}{T_0}$$

Stirling and Carnot

$$COP_s = \frac{T_c}{T_0 - T_c}$$

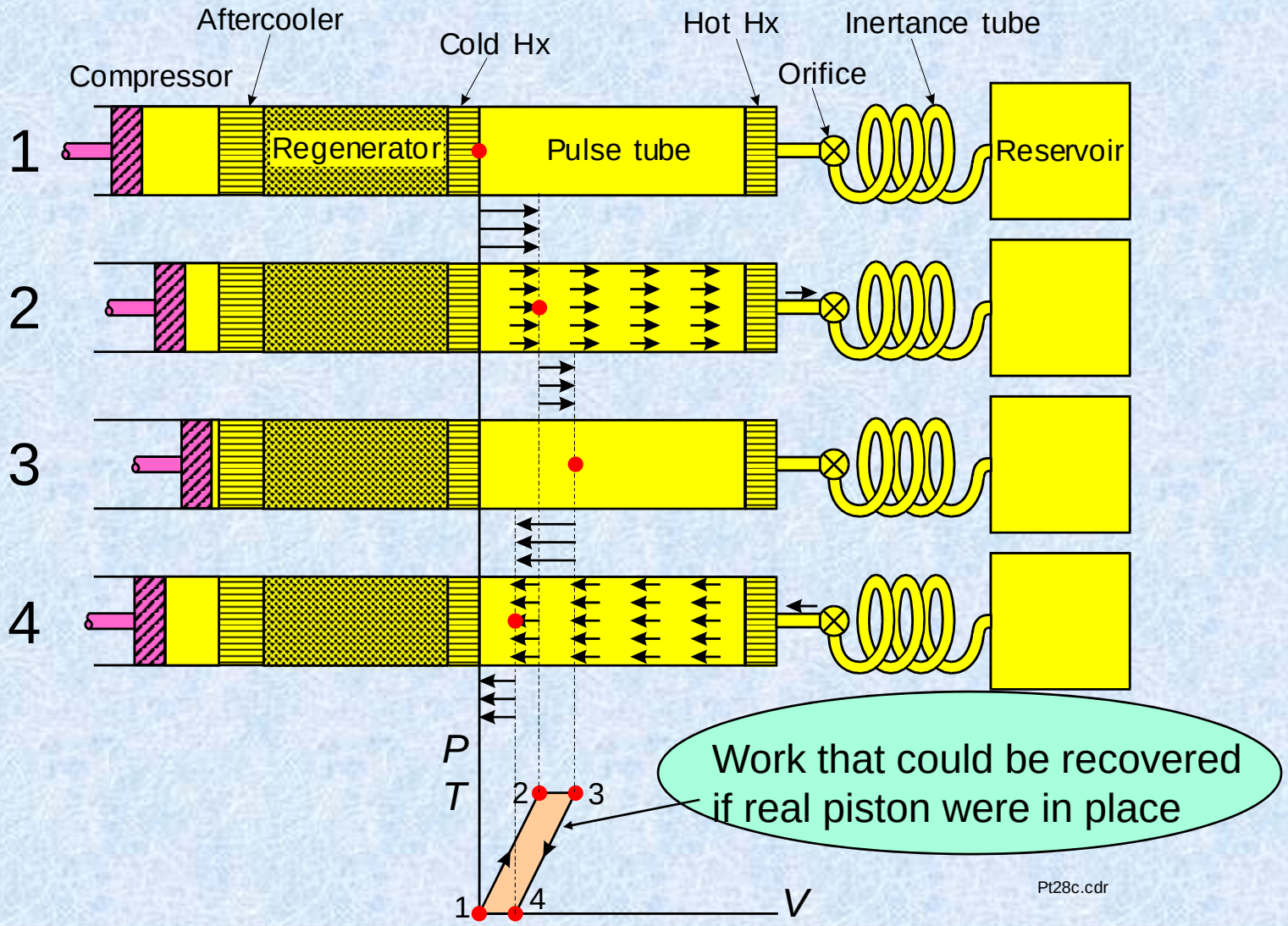
Recovered expansion work

$$\eta_{pt} = \frac{COP_{pt}}{COP_s} = \frac{T_0 - T_c}{T_0}$$

For T_0 = 300 K

T_c :	75 K	250 K
η_{pt} :	76%	17%

P-V Diagram Swept by Virtual Piston



Pulse Tube Effectiveness

Inside the pulse tube component we have

$$\langle P\dot{V} \rangle = \langle \dot{H} \rangle - T \langle \dot{S} \rangle$$

For real pulse tubes with losses

$$\langle \dot{S} \rangle < 0 \qquad \text{(Flow toward compressor)}$$

Gross refrigeration power:

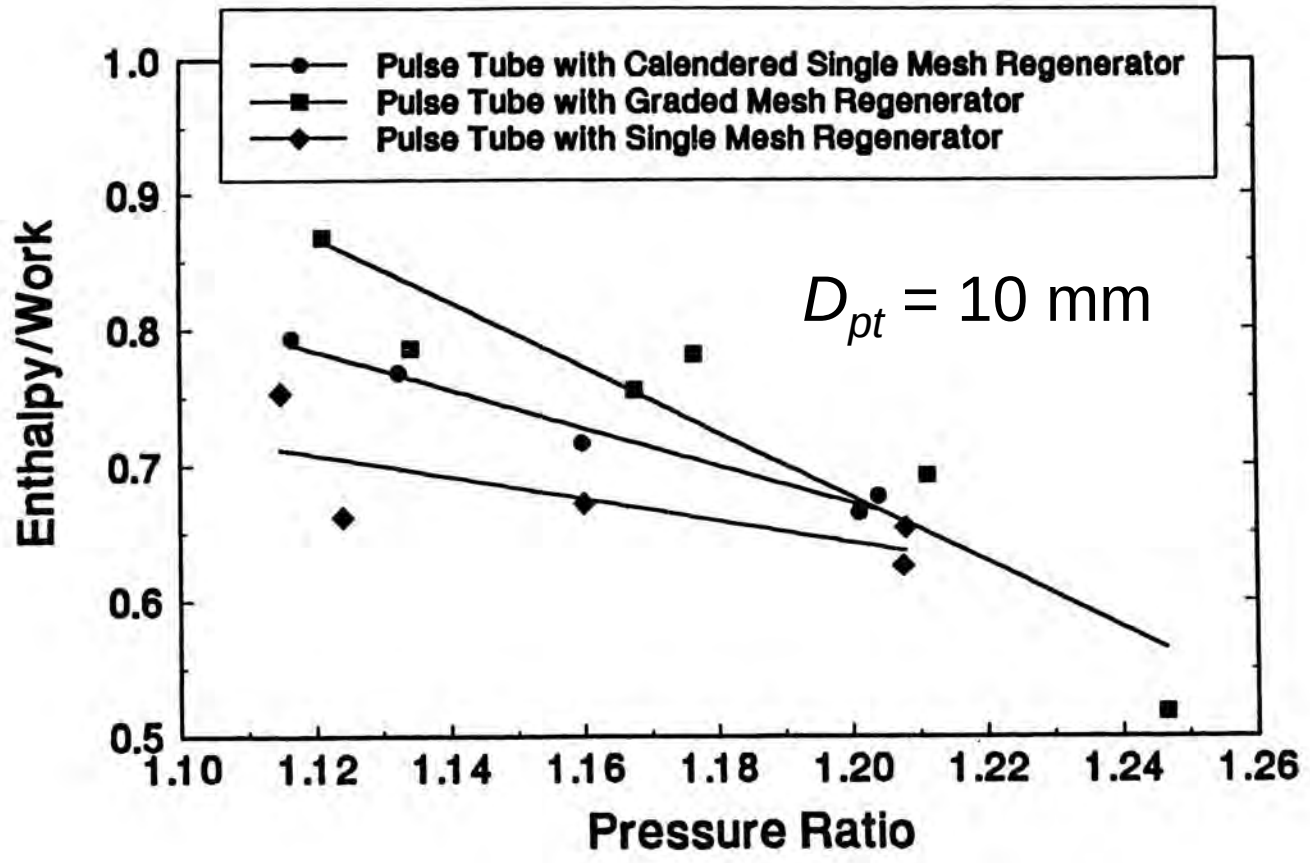
$$\langle \dot{Q}_c \rangle + \langle \dot{H} \rangle_r = \langle \dot{H} \rangle = \langle P\dot{V} \rangle + T \langle \dot{S} \rangle$$

Pulse tube effectiveness (or figure of merit)

$$\varepsilon_{pt} = \frac{\langle \dot{H} \rangle}{\langle P\dot{V} \rangle} = 1 + \frac{T \langle \dot{S} \rangle}{\langle P\dot{V} \rangle}$$

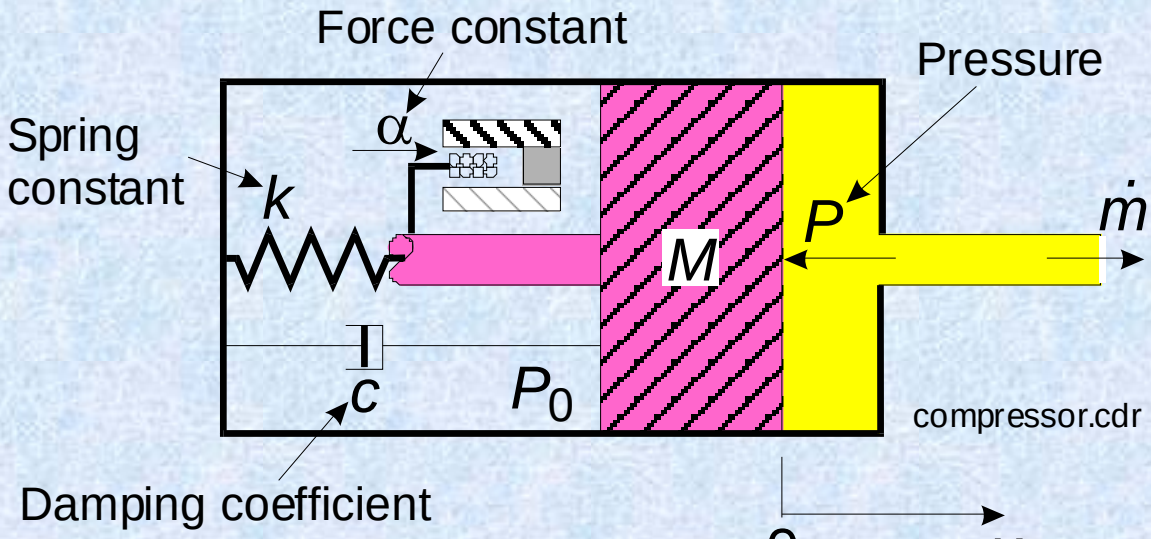
Measured Pulse Tube Effectiveness

Rawlins, Radebaugh, Bradley, Timmerhaus, Adv. Cry. Eng. **39**, 1449 (1994)



Pulse tube refrigerators.jpg

Compressor Analysis



$$F_t = M\ddot{x}$$

$$\alpha I - kx - P_d A - c\dot{x} = M\ddot{x}$$

$$\alpha I = kx + P_d A + c\dot{x} + M\ddot{x} \qquad P_d = P - P_0$$

$$\alpha I_1 e^{i\theta_m} = kx_1 + P_{d1} A e^{i\theta_P} + ic\omega x_1 - M\omega^2 x_1$$

$$x = x_1 e^{i\omega t}$$

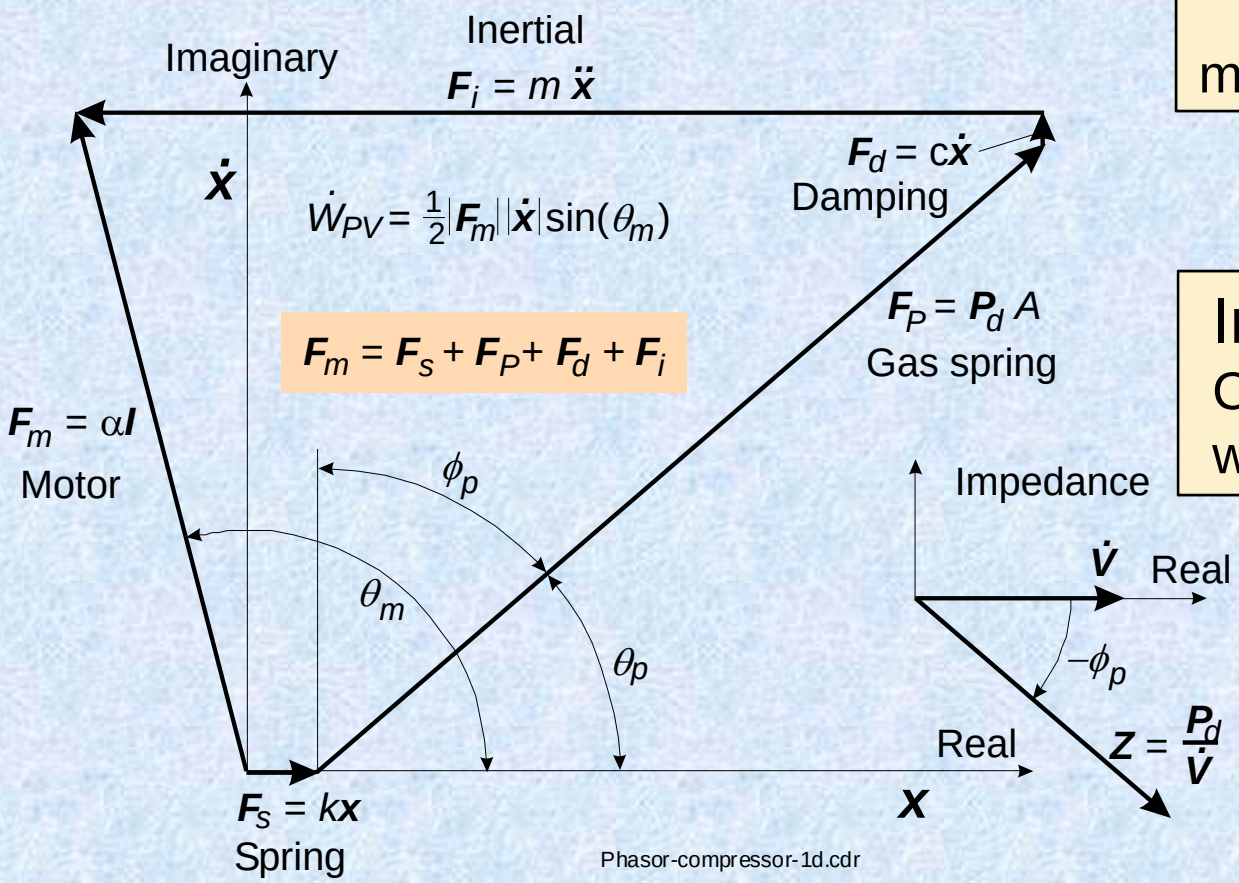
$$\dot{x} = i\omega x_1 e^{i\omega t}$$

$$\ddot{x} = -\omega^2 x_1 e^{i\omega t}$$

$e^{i\omega t}$ components cancel

Solve real and imaginary parts for I_1 and θ_m

Linear Compressor Phasor Analysis
(Force Balance)



Resonance:

$\theta_m = 90°$

minimum *I* for fixed power

Impedance matched:

Compressor in resonance with coldhead impedance

Compressor Characterization

- Parameters necessary for impedance matching

  Maximum stroke amplitude (m): x_1
  Spring constant (N/m): k
  Piston diameter (m): $A = \pi D^2/4$
  Damping coefficient (kg/s): c
  Moving mass (kg): M

  F_s
  F_P
  F_d
  F_i

  Natural freq.

  $$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{M}}$$

  Need $f > f_0$ for resonance

- Other important parameters:

  Motor force constant (N/A): α
  Coil resistance (Ω): R
  Maximum current (A): I_{max}
  Motor inductance (H): L_m

  I
  I^2R
  Heating limit
  V_L

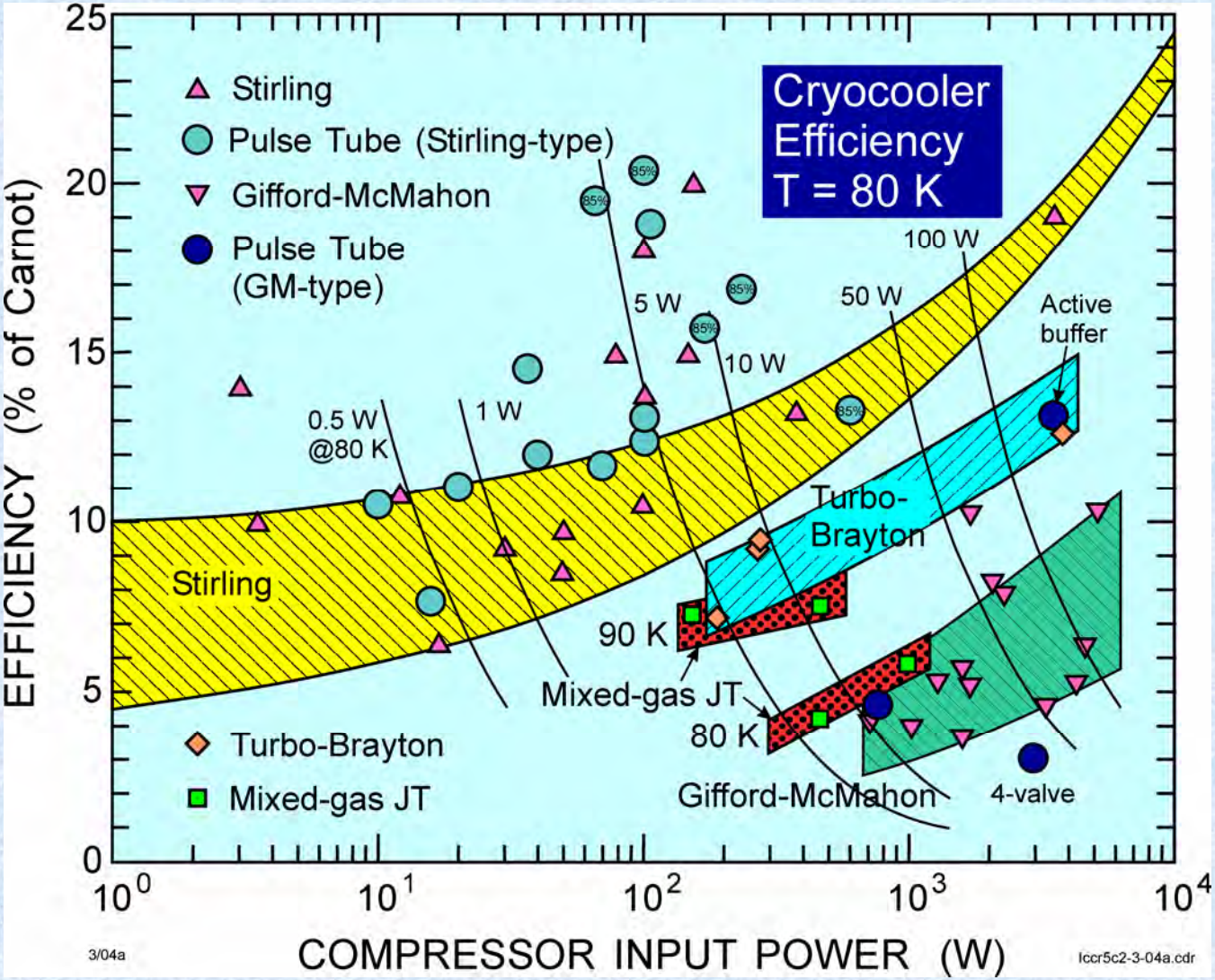
Typical: 20 Hz < f_0 < 40 Hz

Outline

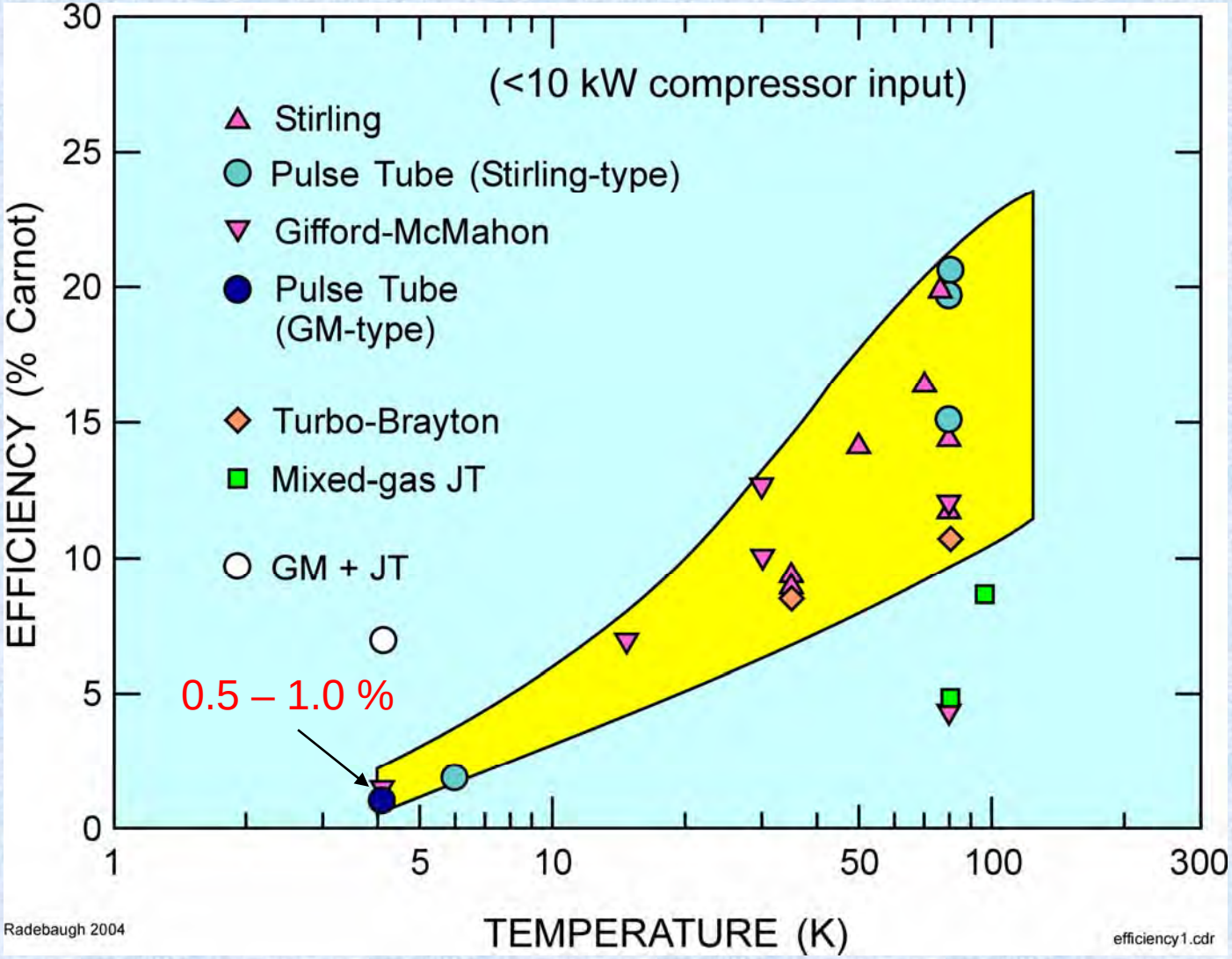
Session 2

- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

Cryocooler Efficiency Comparisons



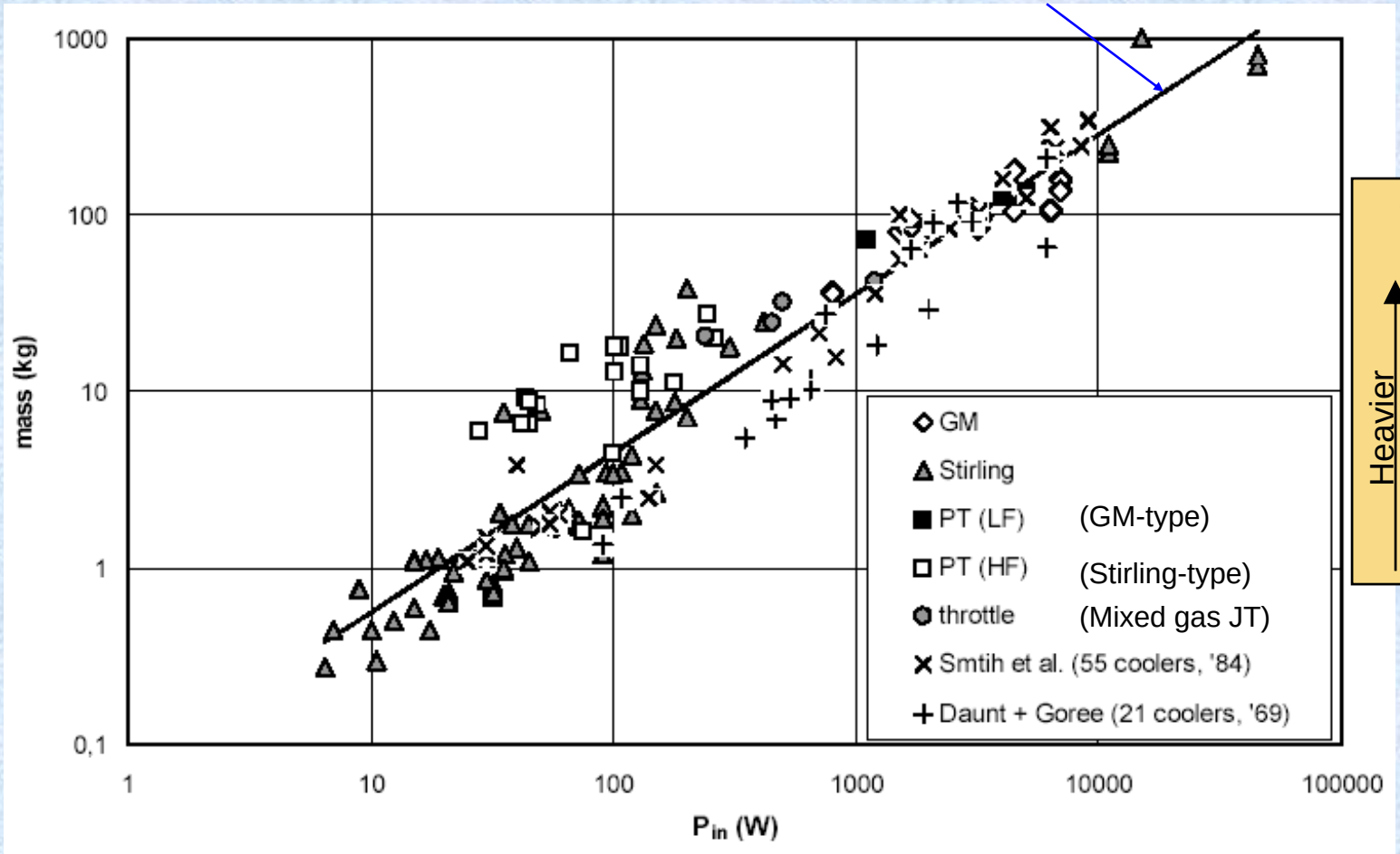
Low T Efficiency Comparisons



Size and Weight Comparisons

Density = 0.8 kg/L

Mass (kg) = 0.0204 x P_{in} (W)



General Rank

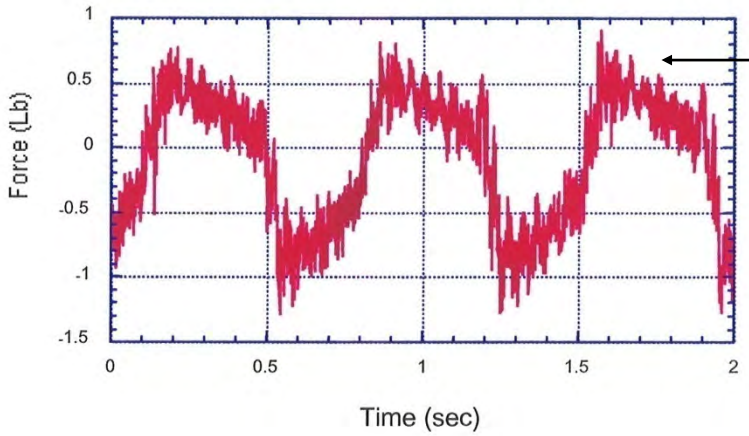
Heavier

- PT (LF)
- GM
- JT Mixed gas)
- Turbo Brayton
- PT (HF)
- Stirling

For fixed cooling power

From: H. J. M. ter Brake and G. F. M. Wiegerinck, *Cryogenics*, vol. 42, (2002) pp. 705-718

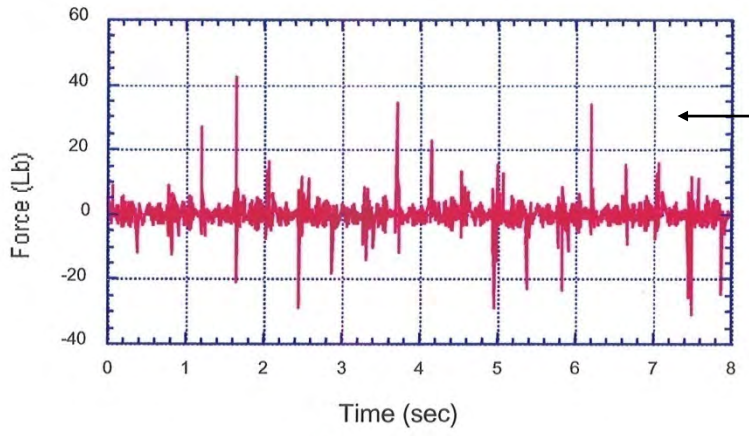
Vibration Comparison



0.7 lb (3.1 N)

About 10 μm displacement along z-axis

4 K **(a) Pulse Tube** PT405



30 lb (133 N)

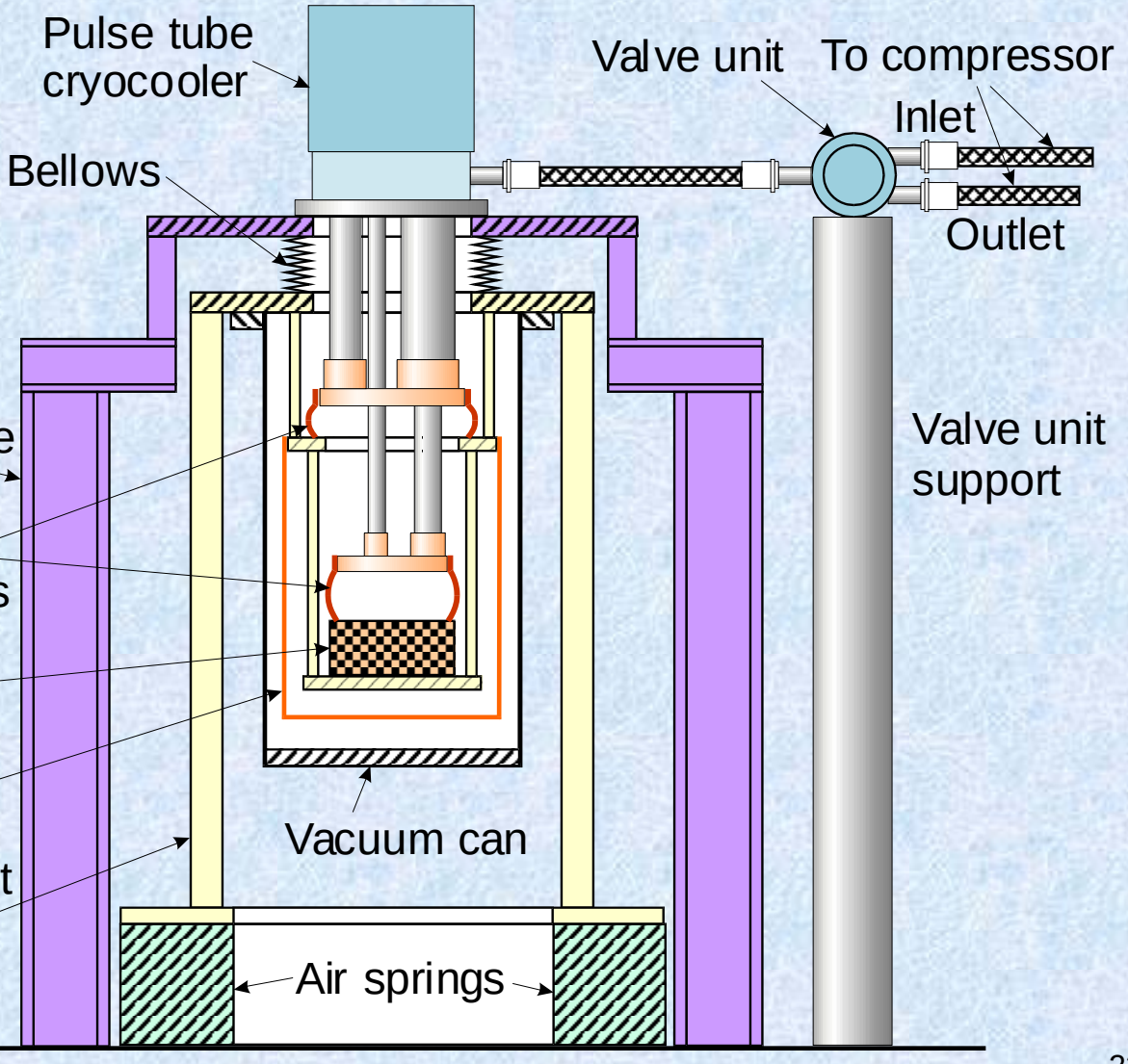
4 K **(b) GM**

Courtesy: Cryomech

Ultra-Low Vibration System

For laser interferometer in gravitational wave detection

From: Ikushima *et al*. *Cryogenics* **48** (2008) 406-412

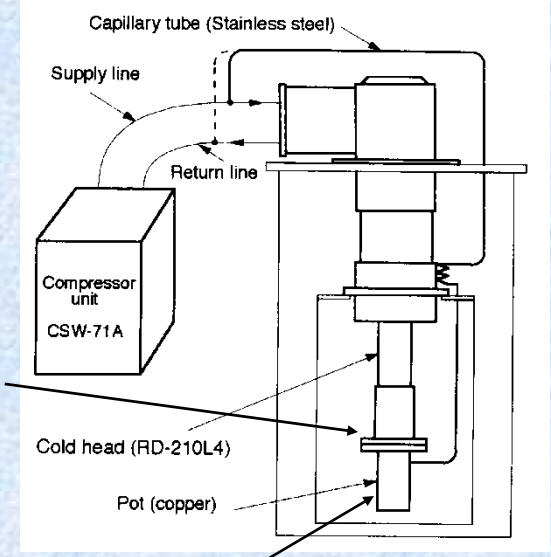


Vibration (p-p)
Cryocooler: 8 μm
Sample: 0.2 μm
(z axis) 0.02 μm

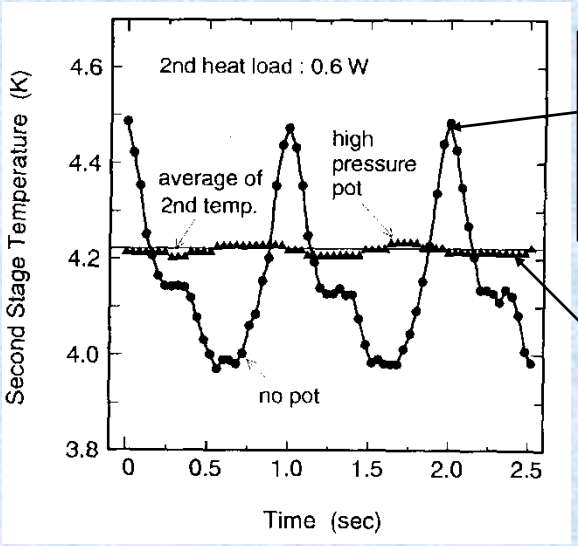
Vibration damping-II.cdr

Temperature Oscillation at Low Frequency

4 K GM Cryocooler (1 Hz)

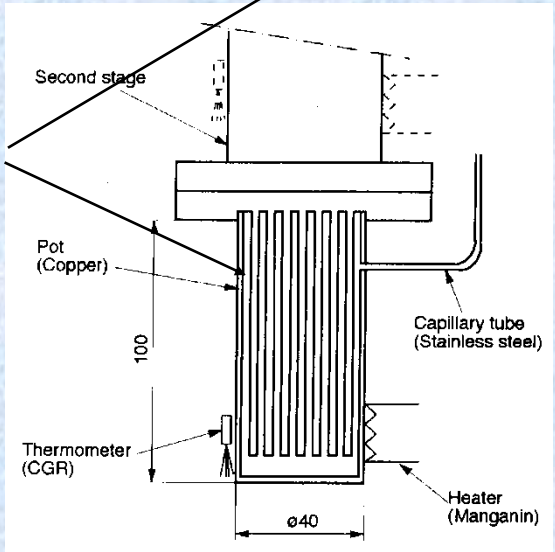


0.6 W @4 K

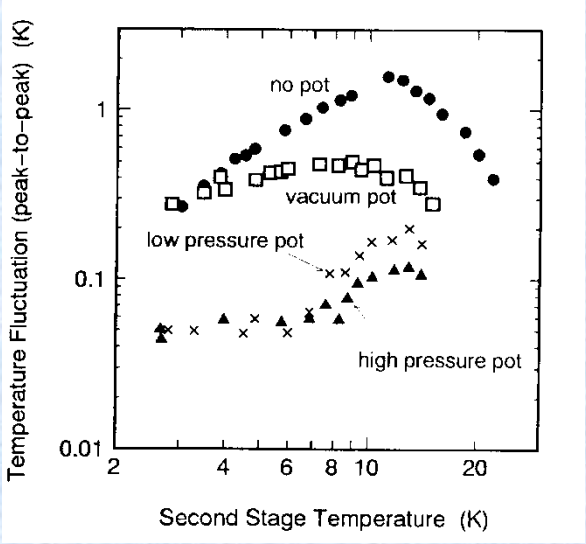


Without pot:
Peak-peak ΔT = 0.53 K $\propto$ 1/*f*

With pot:
Peak-peak ΔT = 0.054 K

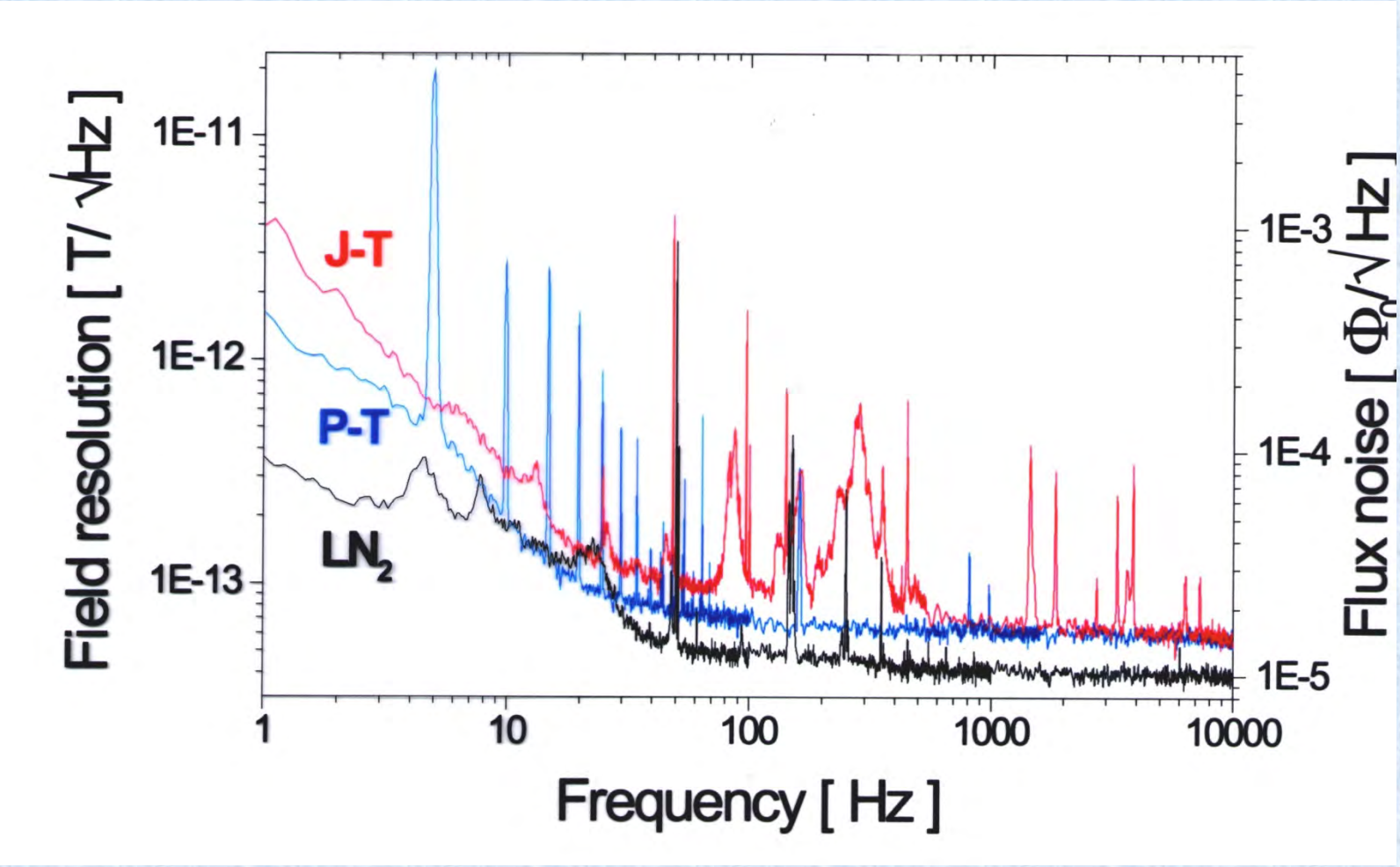


He^4 pot



Source:
R. Li, *et al.*, *Cryocoolers 9* pp. 765-771 (1997)

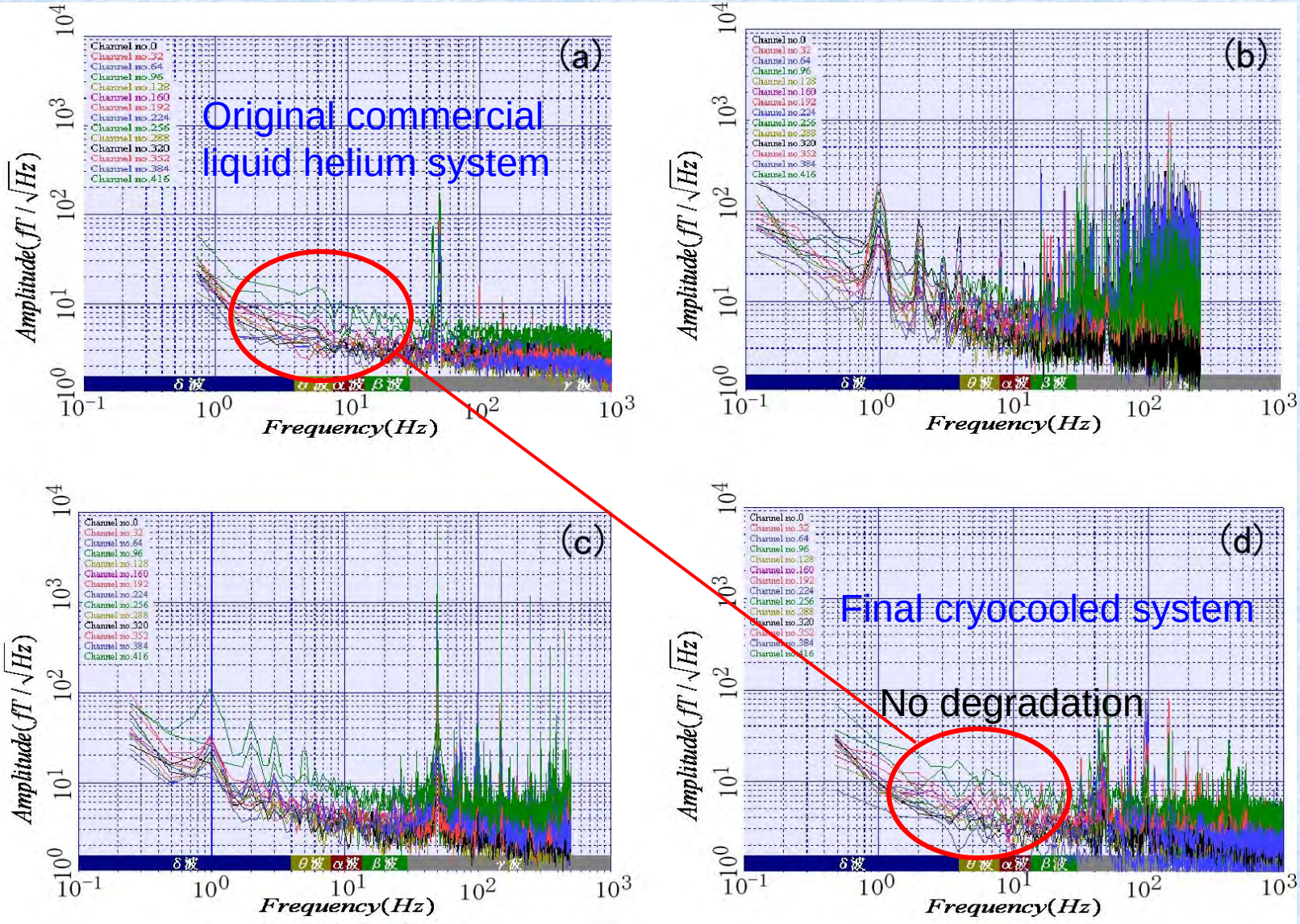
SQUID Noise With JT, PT, and LN_2 Cooling



From R. Hohmann et al., 1998 ASC, Vol. 9, p. 3688

SQUID Noise in Cryocooled MEG System

(From T. Takeda, *et al.*, Cryogenics **49** (2009) pp144-150)



Outline

Session 2

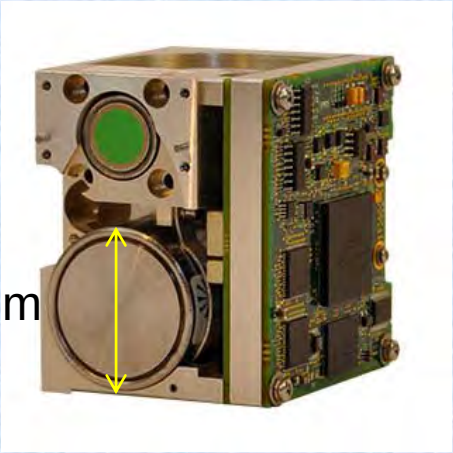
- Regenerative cycles
- Stirling cryocoolers
- Gifford-McMahon cryocoolers
- 4 K regenerators
- Pulse tube cryocoolers
- Phasor analysis
- Pulse tube cryocooler examples
- Modeling and thermodynamics
- Comparisons
- New research areas

New Research Areas

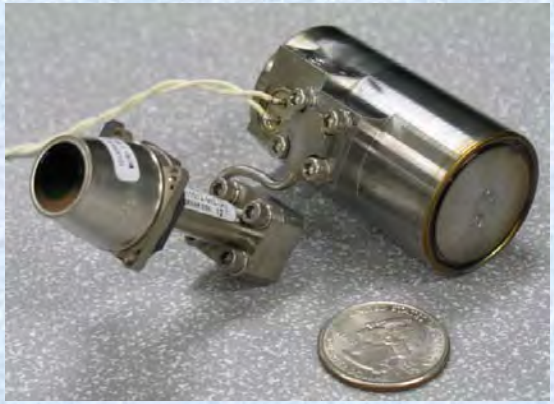
- Higher temperatures (150 K to 200 K)
  - Miniaturization (<2 W input power)
- Higher frequencies
  - Increase *f* > 100 Hz; smaller sizes
- Warm displacers
- CFD modeling of pulse tube and transitions
- Large pulse tube cryocoolers
  - Higher ε; reduce temperature inhomogeneities
- 4 K cryocoolers
  - Increase efficiencies above 1% of Carnot
- Microcryocoolers (JT)
  - Refrigeration powers of 1-50 mW at 80-150 K

Miniature Coolers for 150 K (HOT FPA)

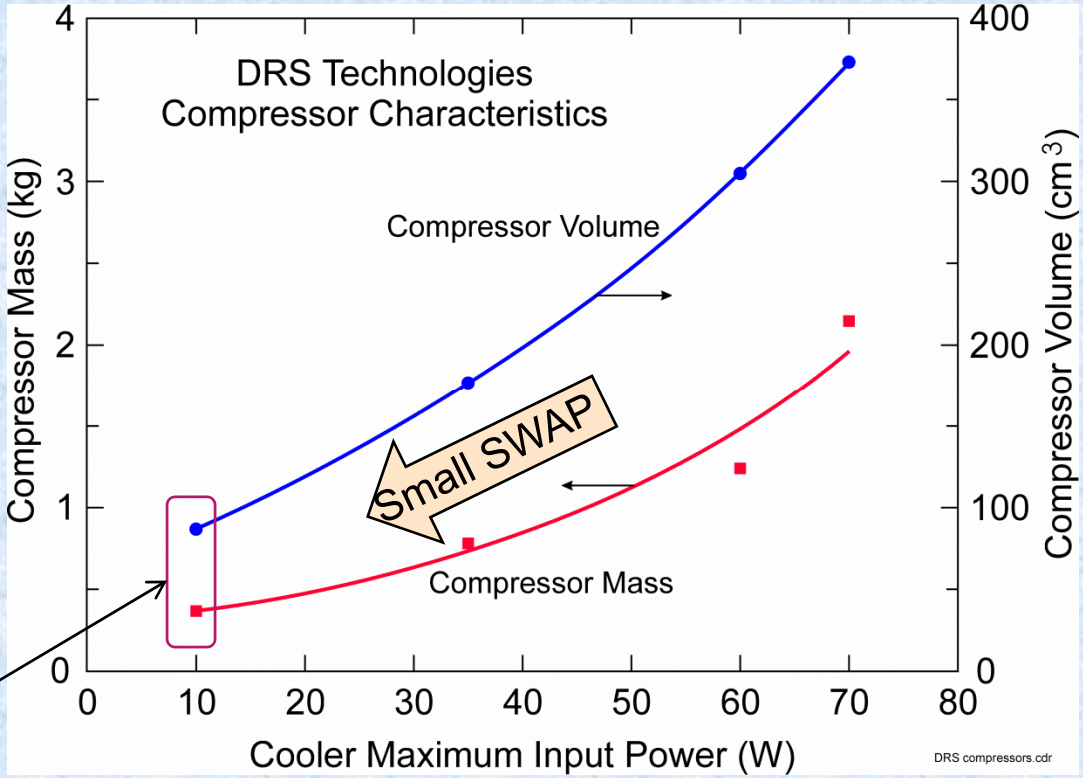
DRS micro-IDCA

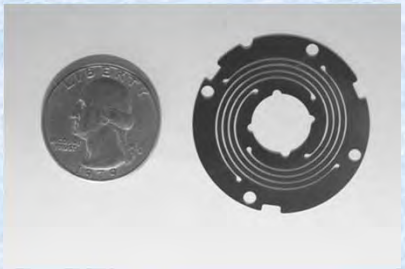


MWIR camera (2 watts)



Stirling cooler





Flexure bearings

Photos and data courtesy:
DRS Technologies

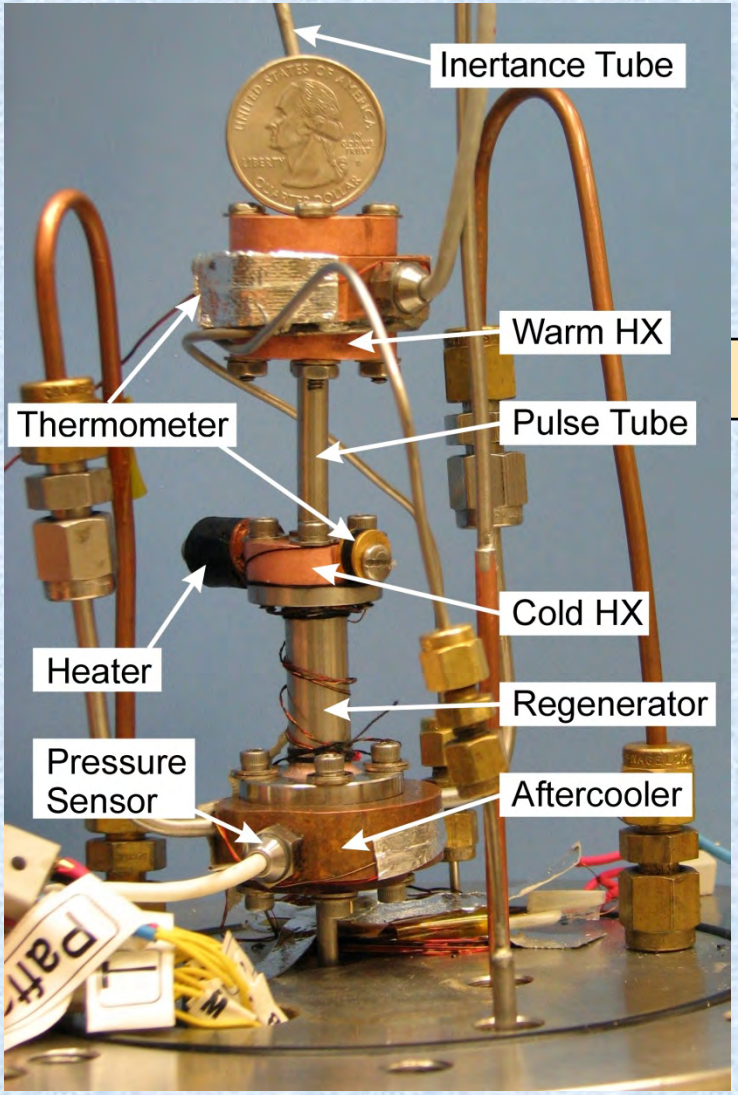
Effect of Increased Frequency

T_c = 80 K, T_h = 300 K

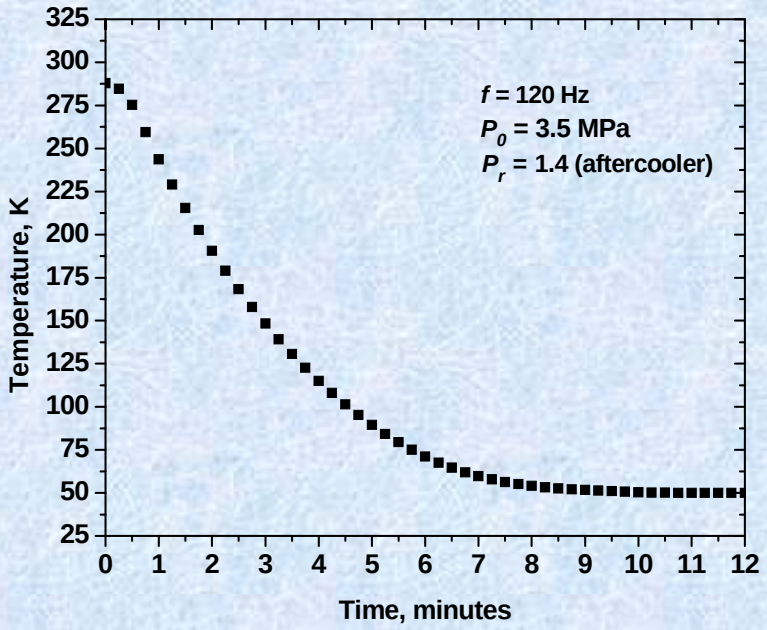
- For f : 60 Hz ⇒ 1000 Hz
  P_0 : 2.5 MPa ⇒ 7.0 MPa
  D_h : 55 μm ⇒ 16 μm

  - Swept volume decreases by 47x
  - Regenerator volume decreases by 39x
  - *COP* 68% of *COP* for 60 Hz
  - Reduced δ_t allows 6.7x smaller pulse tubes
  - Allows micro cryocoolers (MEMS)

120 Hz Pulse Tube Cryocooler

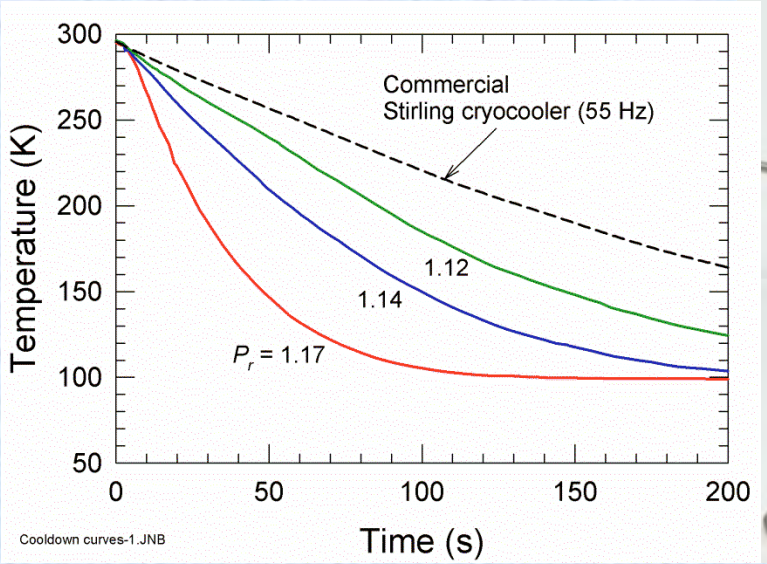


- 49.9 K (no load)
- 3.35 W @ 80 K
- 5.5 minutes to 80 K
- 13.4% Carnot @ 80 K (PV input)
- P_0 = 3.5 MPa; #635 mesh regenerator

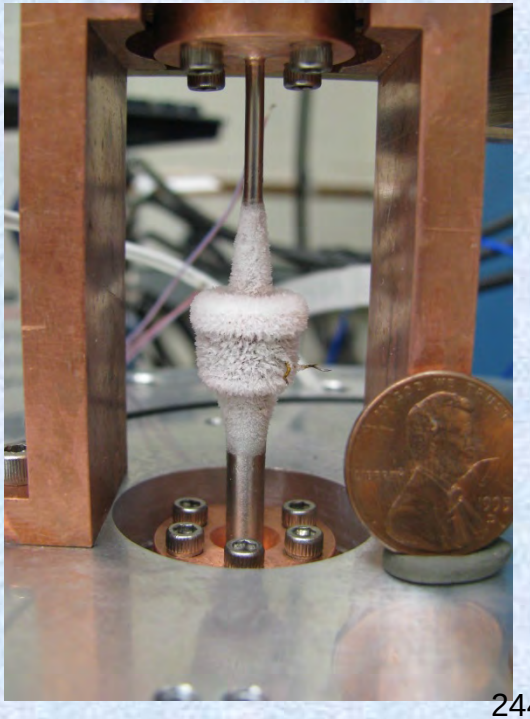


150 Hz Miniature Pulse Tube Cryocooler

- NIST experimental work
  - Garaway, Gan, Bradley, Veprik, Radebaugh, "Development of a Miniature 150 Hz Pulse Tube Cryocooler," *Cryocoolers 15* (2009) pp 105-113
  - Used Ricor K527 compressor, 5 MPa
  - 97 K achieved
  - Poor impedance match to K527

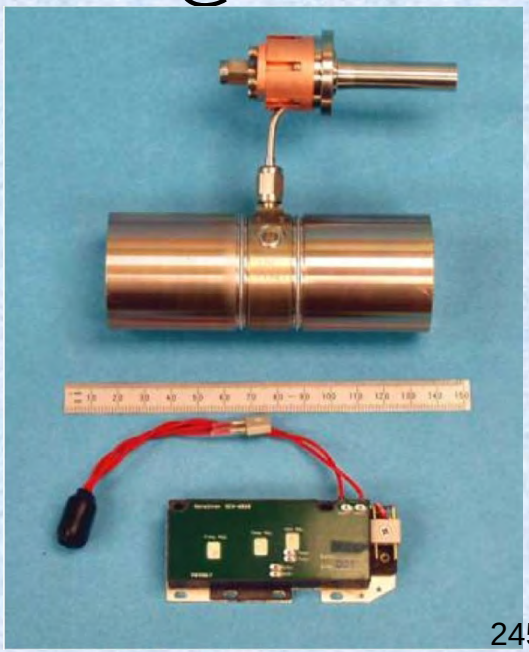




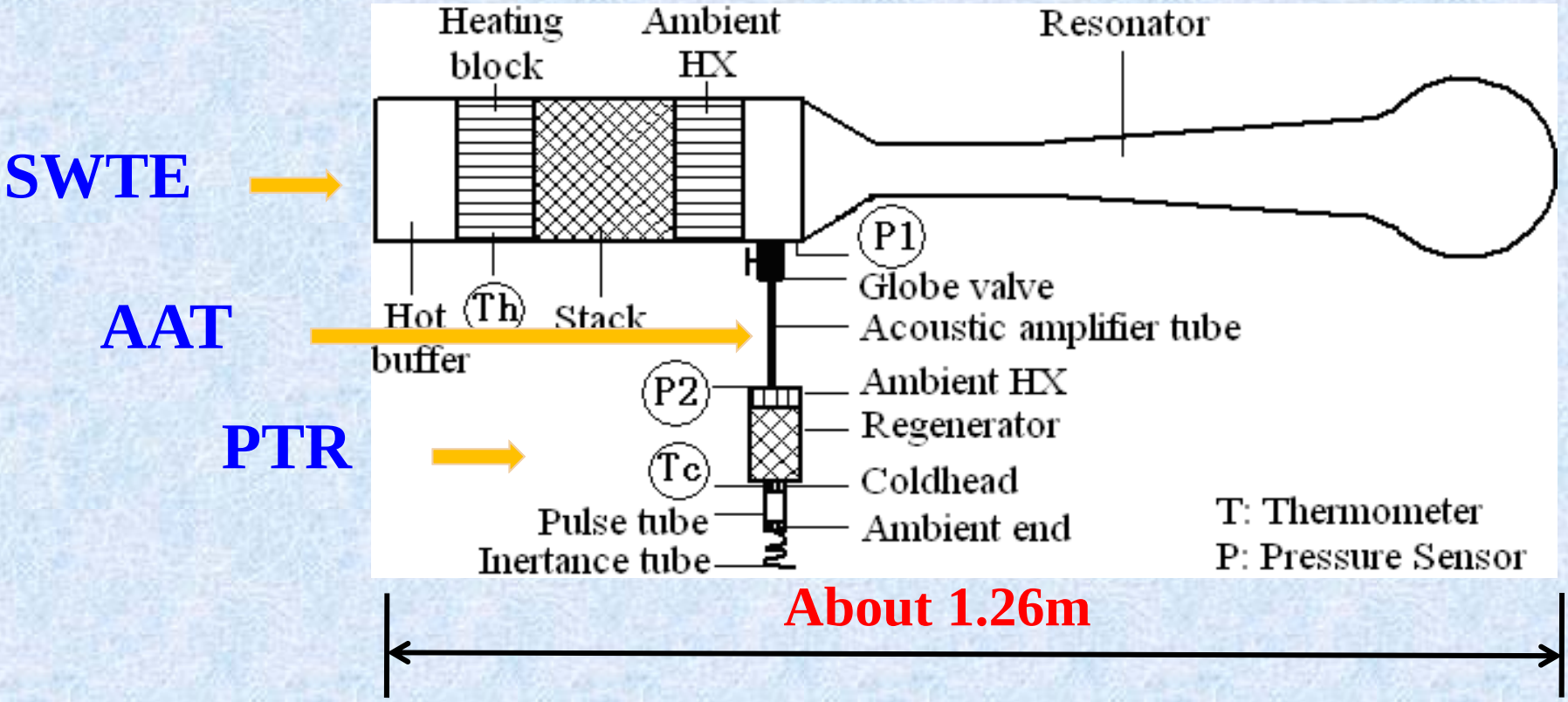


Other High Frequency PT Work

- Northrop Grumman experimental work
  - Petach, Waterman, Pruit, Tward, “High Frequency Coaxial Pulse Tube Microcooler,” *Cryocoolers 15*, 2009, pp. 97-103.
  - 100 Hz: 1.5 W @ 80 K, 144 Hz: 1.35 W @ 80 K
  - 10 W/W at 150 K; 4.0 W max cooling
  - 0.65 cc compressor swept volume

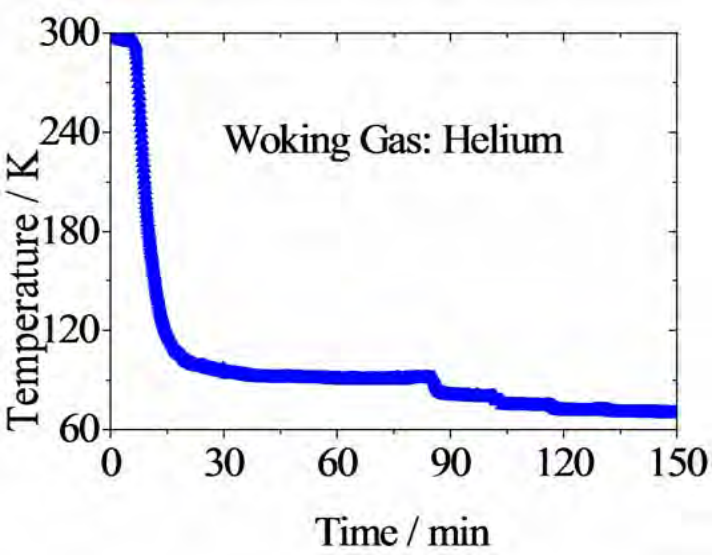


300 Hz Pulse Tube
(Thermoacoustically Driven)



Courtesy: Ercang Luo
Technical Institute of Physics and Chemistry
Chinese Academy of Science, Beijing

300 Hz Pulse Tube: Performance
(Thermoacoustically Driven)



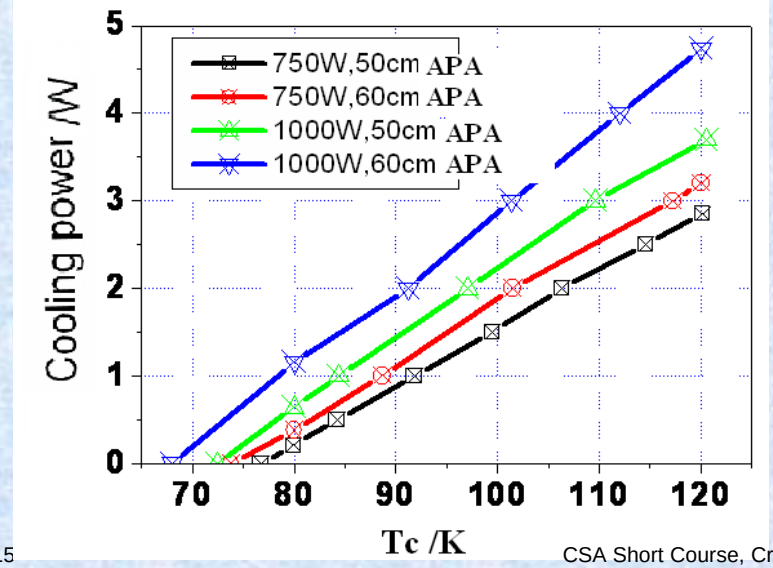
Typical operating conditions

P0=4.0MPa

f=295-300Hz

Pressure ratio=1.20-1.25

#635 mesh S.S. screen regen.

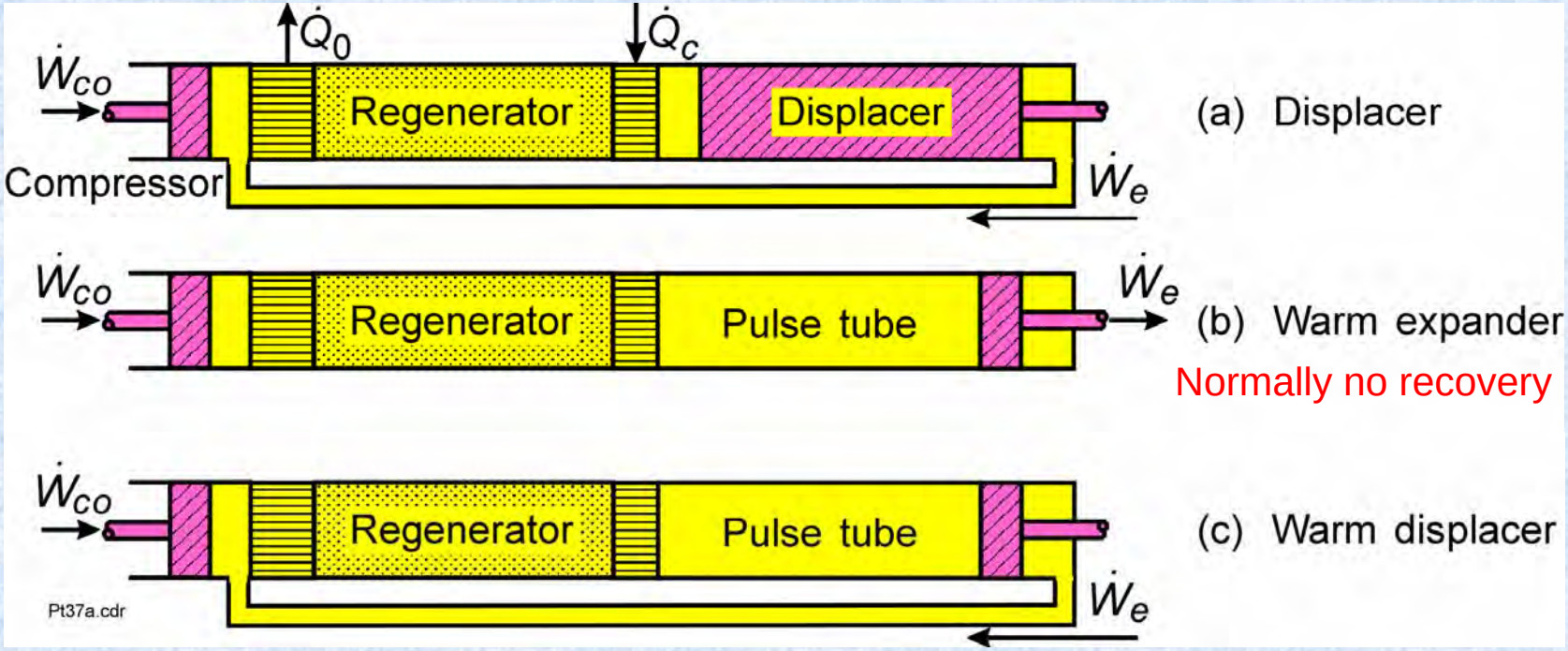


Cooling Performance:

1. Lowest no-load Tc: 68K;
2. 1.2W@80K, 4.8W@120K (Qh=1000W,Th=630°C)

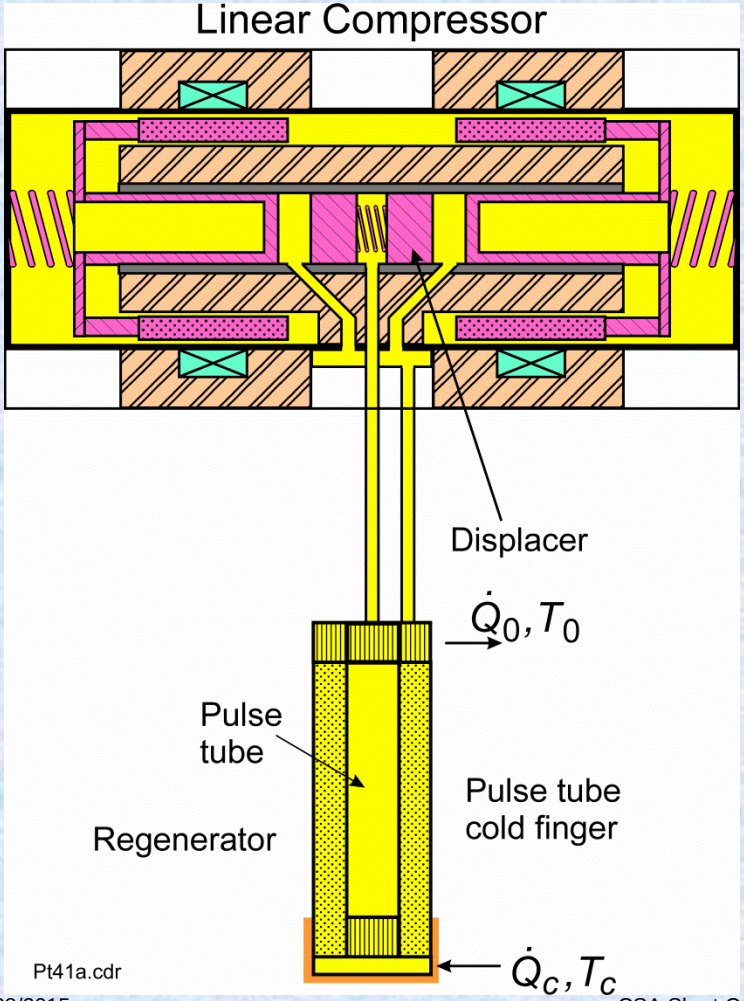
Courtesy: Ercang Luo
Technical Institute of Physics and Chemistry
Chinese Academy of Science, Beijing

Work Recovery Mechanisms

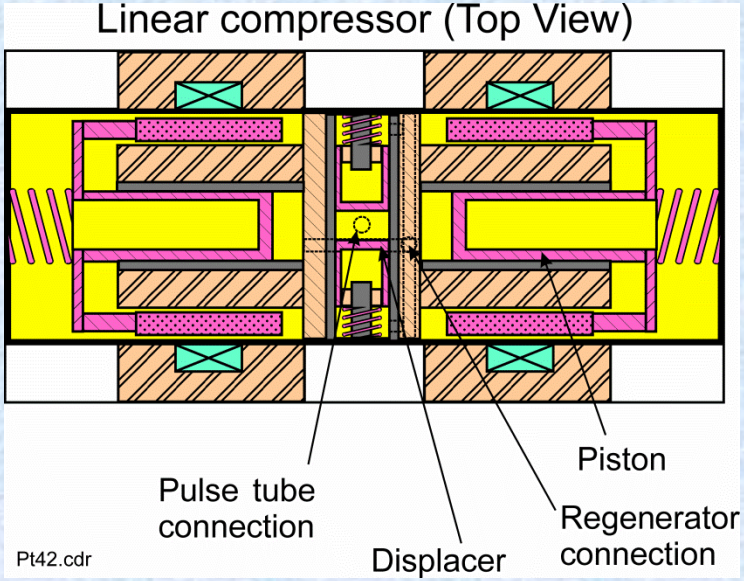


Integral Compressor and Displacer

Parallel arrangement



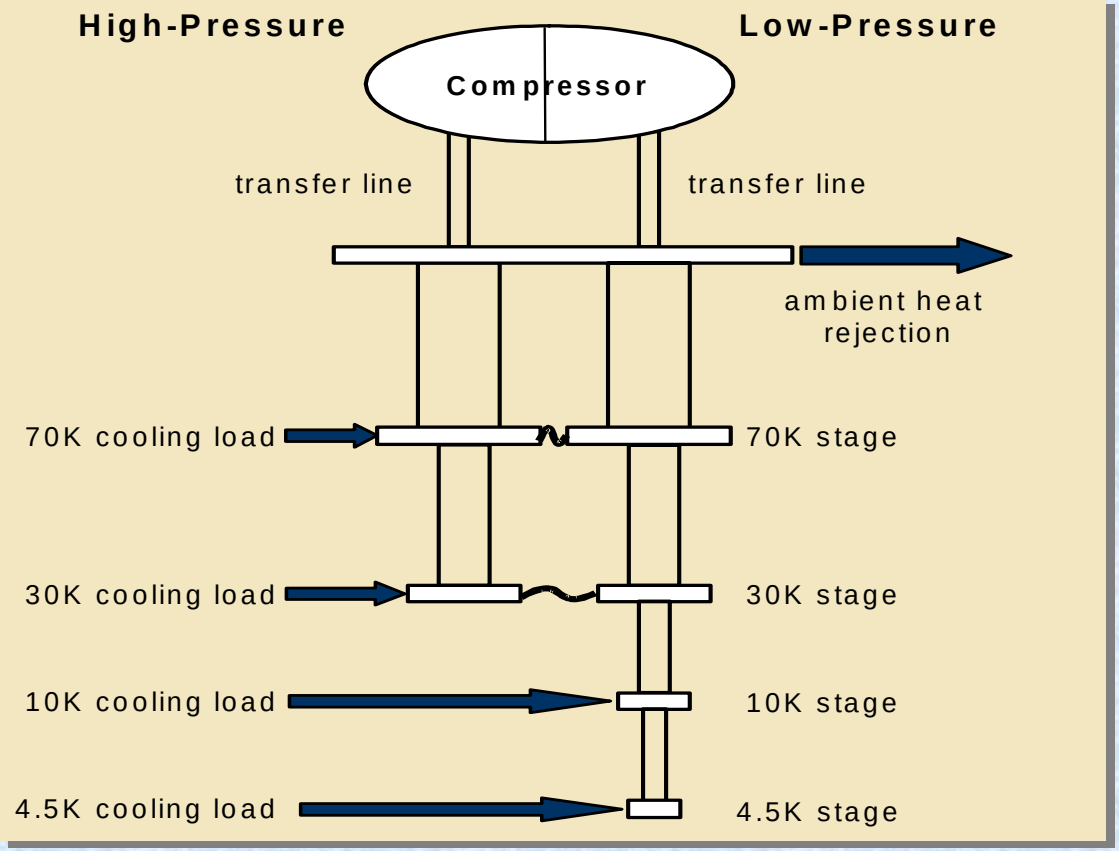
Perpendicular arrangement



With backside area reduction

4-Stage Pulse Tube Cryocooler for LTS Electronics

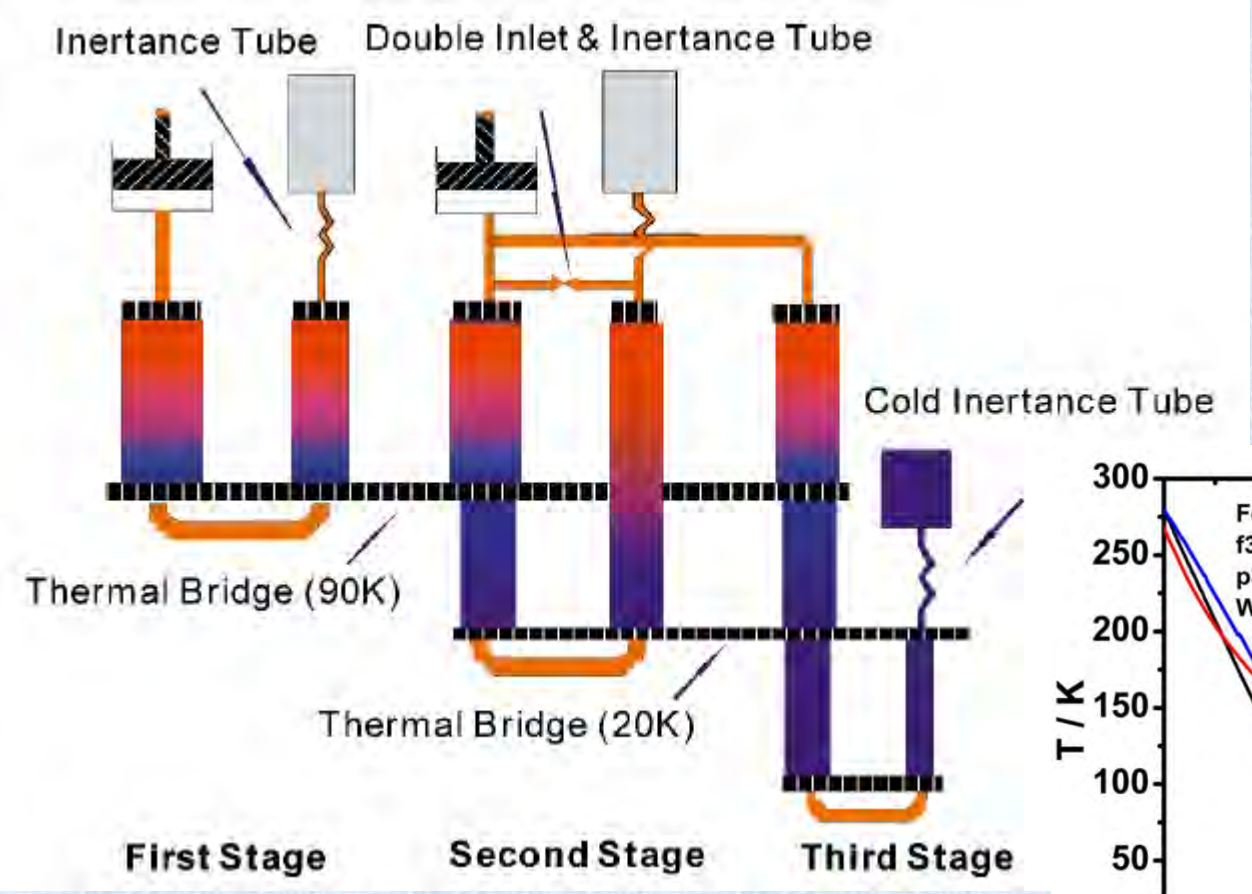
Developed by Lockheed Martin for Hypres LTS communication system



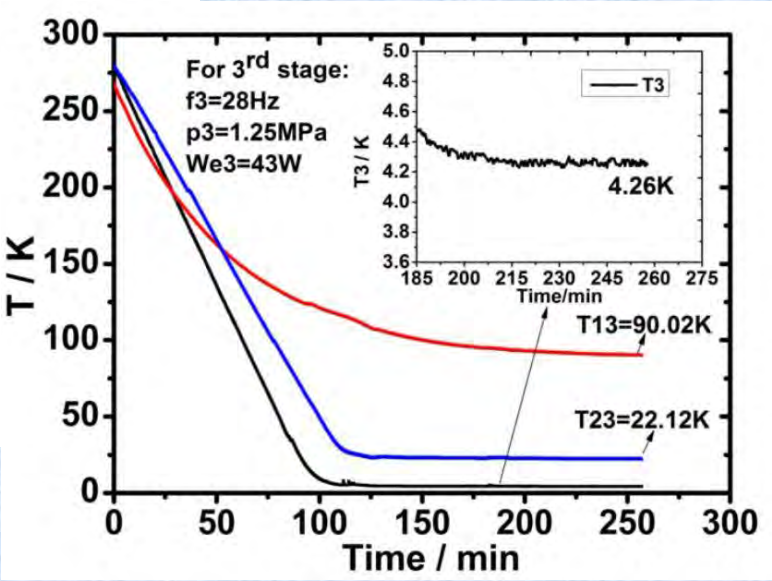
From: CEC2007, Lockheed Martin

- Lockheed Martin
- From CEC 2007
- 40 mW at 4.5 K
- 5.5 W at 68 K
- 685 W input power
- About 10 yr life (space-like)
- 30 Hz frequency
- He-3 (low P); He-4 (hi P)

Zhejiang Univ. 4 K Stirling-Type PT



- Three stages
- Helium-4
- 28 Hz
- 4.26 K



Cryocooler Fundamentals Summary

- Regenerative cycles developed mostly in last 50 years
- Pulse tubes most studied of all cryocoolers since 1990
- Fluid dynamics inside pulse tubes needs further study
- Space applications led to many advances in cryocoolers
- Reduced cost important for application development
- Active research areas
  - Joule-Thomson
  - Miniature Brayton for space
  - Stirling
  - Pulse tube